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United States Patent	12398384
Kind Code	B2
Date of Patent	August 26, 2025
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DNA modifying enzymes and active fragments and variants thereof and methods of use

Abstract

Compositions and methods comprising novel deaminase polypeptides for targeted editing of nucleic acids are provided. Compositions comprise deaminase polypeptides. Also provided are fusion proteins comprising a DNA-binding polypeptide and a deaminase of the invention. The fusion proteins include RNA-guided nucleases fused to deaminases, optionally in complex with guide RNAs. Compositions also include nucleic acid molecules encoding the deaminases or the fusion proteins. Vectors and host cells comprising the nucleic acid molecules encoding the deaminases or the fusion proteins are also provided.

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Appl. No.: 17/851880

Filed: June 28, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220348894 A1	Nov. 03, 2022

Related U.S. Application Data

continuation parent-doc WO PCT/US2021/049853 20210910 PENDING child-doc US 17851880
us-provisional-application US 63146840 20210208
us-provisional-application US 63077089 20200911

Publication Classification

Int. Cl.: C12N9/22 (20060101); C12N15/63 (20060101)

U.S. Cl.:

CPC C12N9/22 (20130101); C12N15/63 (20130101); C12N2310/20 (20170501); C12N2800/80 (20130101)

Field of Classification Search

CPC: C12N (9/22); C12N (15/63); C12N (2310/20); C12N (2800/80); C12N (9/78); C12N (15/52); C12N (15/62); C12N (9/80); C12N (15/102); C12N (15/113); C12Y (305/04002); C12Y (305/01004); C07K (2319/81); C07K (2319/85); A61K (48/00)

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FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3 125 175	12/2019	CA	N/A

OTHER PUBLICATIONS

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/US2021/049853, filed Sep. 10, 2021, which claims priority to U.S. Provisional Application Nos. 63/077,089, filed Sep. 11, 2020, and 63/146,840, filed Feb. 8, 2021, each of which is incorporated by reference herein in its entirety.

STATEMENT REGARDING THE SEQUENCE LISTING

(1) The Sequence Listing associated with this application is provided in ASCII format in lieu of a paper copy, and is hereby incorporated by reference into the specification. The ASCII copy named L103438_1230WO_0108_1_SL.txt is 1,071,246 bytes in size, was created on Sep. 9, 2021, and is being submitted electronically via EFS-Web.

FIELD OF THE INVENTION

(2) The present invention relates to the field of molecular biology and gene editing.

BACKGROUND OF THE INVENTION

(3) Targeted genome editing or modification is rapidly becoming an important tool for basic and applied research. Initial methods involved engineering nucleases such as meganucleases, zinc

finger fusion proteins or TALENs, requiring the generation of chimeric nucleases with engineered, programmable, sequence-specific DNA-binding domains specific for each particular target sequence. RNA-guided nucleases (RGNs), such as the Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR)-associated (Cas) proteins of the CRISPR-Cas bacterial system, allow for the targeting of specific sequences by complexing the nucleases with guide RNA that specifically hybridizes with a particular target sequence. Producing target-specific guide RNAs is less costly and more efficient than generating chimeric nucleases for each target sequence. Such RNA-guided nucleases can be used to edit genomes through the introduction of a sequence-specific, double-stranded break that is repaired via error-prone non-homologous end-joining (NHEJ) to introduce a mutation at a specific genomic location.

(4) Additionally, RGNs are useful for targeted DNA editing approaches. Targeted editing of nucleic acid sequences, for example targeted cleavage, to allow for introduction of a specific modification into genomic DNA, enables a highly nuanced approach to studying gene function and gene expression. RGNs may also be used to generate chimeric proteins which use the RNA-guided activity of the RGN in combination with a DNA modifying enzyme, such as a deaminase, for targeted base editing. Targeted editing may be deployed for targeting genetic diseases in humans or for introducing agronomically beneficial mutations in the genomes of crop plants. The development of genome editing tools provides new approaches to gene editing-based mammalian therapeutics and agrobiotechnology.

BRIEF SUMMARY OF THE INVENTION

(5) Compositions and methods for modifying a target DNA molecule are provided. The compositions find use in modifying a target DNA molecule of interest. Compositions provided comprise deaminase polypeptides. Also provided are fusion proteins comprising a nucleic acid molecule-binding polypeptide (e.g., DNA-binding polypeptide) and a deaminase polypeptide, and ribonucleoprotein complexes comprising a fusion protein comprising an RNA-guided nuclease and a deaminase polypeptide and ribonucleic acids. Compositions provided also include nucleic acid molecules encoding the deaminase polypeptides or the fusion proteins, and vectors and host cells comprising the nucleic acid molecules. The methods disclosed herein are drawn to binding a target sequence of interest within a target DNA molecule of interest and modifying the target DNA molecule of interest.

Description

DETAILED DESCRIPTION

(1) Many modifications and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims.

(2) Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

I. Overview

(3) This disclosure provides novel adenine deaminases and fusion proteins that comprise a nucleic acid molecule-binding polypeptide, such as a DNA-binding polypeptide, and a novel deaminase polypeptide. In certain embodiments, the DNA-binding polypeptide is a sequence-specific DNA-binding polypeptide, in that the DNA-binding polypeptide binds to a target sequence at a greater frequency than binding to a randomized background sequence. In some embodiments, the DNA-binding polypeptide is or is derived from a meganuclease, zinc finger fusion protein, or TALEN. In some embodiments, the fusion protein comprises an RNA-guided DNA-binding polypeptide and a

deaminase polypeptide. In some embodiments, the RNA-guided DNA-binding polypeptide is an RNA-guided nuclease, such as a Cas9 polypeptide domain that binds to a guide RNA (also referred to as gRNA), which, in turn, binds a target nucleic acid sequence via strand hybridization.

(4) The deaminase polypeptides disclosed herein can deaminate a nucleobase, such as, for example, adenine. The deamination of a nucleobase by a deaminase can lead to a point mutation at the respective residue, which is referred to herein as “nucleic acid editing”, or “base editing”. Fusion proteins comprising an RNA-guided nuclease (RGN) polypeptide and a deaminase can thus be used for the targeted editing of nucleic acid sequences.

(5) Such fusion proteins are useful for targeted editing of DNA in vitro, e.g., for the generation of genetically modified cells. These genetically modified cells may be plant cells or animal cells. Such fusion proteins may also be useful for the introduction of targeted mutations, e.g., for the correction of genetic defects in mammalian cells ex vivo, e.g., in cells obtained from a subject that are subsequently re-introduced into the same or another subject; and for the introduction of targeted mutations, e.g., the correction of genetic defects or the introduction of deactivating mutations in disease-associated genes in a mammalian subject. Such fusion proteins may also be useful for the introduction of targeted mutations in plant cells, e.g., for the introduction of beneficial or agronomically valuable traits or alleles.

(6) The terms “protein,” “peptide,” and “polypeptide” are used interchangeably herein, and refer to a polymer of amino acid residues linked together by peptide (amide) bonds. The terms refer to a protein, peptide, or polypeptide of any size, structure, or function. Typically, a protein, peptide, or polypeptide will be at least three amino acids long. A protein, peptide, or polypeptide may refer to an individual protein or a collection of proteins. One or more of the amino acids in a protein, peptide, or polypeptide may be modified, for example, by the addition of a chemical entity such as a carbohydrate group, a hydroxyl group, a phosphate group, a farnesyl group, an isofarnesyl group, a fatty acid group, a linker for conjugation, functionalization, or other modification, etc. A protein, peptide, or polypeptide may also be a single molecule or may be a multi-molecular complex. A protein, peptide, or polypeptide may be just a fragment of a naturally occurring protein or peptide. A protein, peptide, or polypeptide may be naturally occurring, recombinant, or synthetic, or any combination thereof.

(7) Any of the proteins provided herein may be produced by any method known in the art. For example, the proteins provided herein may be produced via recombinant protein expression and purification, which is especially suited for fusion proteins comprising a peptide linker. Methods for recombinant protein expression and purification are well known, and include those described by Green and Sambrook, *Molecular Cloning: A Laboratory Manual* (4th ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (2012)), the entire contents of which are incorporated herein by reference.

II. Deaminases

(8) The term “deaminase” refers to an enzyme that catalyzes a deamination reaction. The deaminases of the invention are nucleobase deaminases and the terms “deaminase” and “nucleobase deaminase” are used interchangeably herein. The deaminase may be a naturally-occurring deaminase enzyme or an active fragment or variant thereof. A deaminase may be active on single-stranded nucleic acids, such as ssDNA or ssRNA, or on double-stranded nucleic acids, such as dsDNA or dsRNA. In some embodiments, the deaminase is only capable of deaminating ssDNA and does not act on dsDNA.

(9) The presently disclosed methods and compositions comprise an adenine deaminase. In some embodiments, the deaminase is an ADAT family deaminase or a variant thereof. Deamination of adenine, adenosine, or deoxyadenosine yields inosine, which is treated as guanine by polymerases. To date there are no known naturally occurring adenine deaminases that deaminate adenine in DNA. Several methods have been employed to evolve and optimize adenine deaminase acting on tRNA (ADAT) proteins to be active on DNA molecules in mammalian cells (Gaudelli et al, 2017;

Koblan, L. W. et al, 2018, *Nat Biotechnol* 36, 843-846; Richter, M. F. et al, 2020, *Nat Biotechnol*, doi:10.1038/s41587-020-0562-8, each of which are incorporated by reference in their entirety herein). One such method uses a bacterial selection assay where only cells with the ability to activate antibiotic resistance through A:T>G:C conversions are able to survive.

(10) The present invention relates to novel adenine deaminase polypeptides which were produced through evolution and optimization of bacterial deaminases. Novel adenine deaminases are presently disclosed and set forth as SEQ ID NOs: 1-10 and 399-441. The deaminases of the invention may be used for editing of DNA or RNA molecules. In some embodiments, the deaminases of the invention may be used for editing of ssDNA or ssRNA molecules. The adenine deaminases described herein are useful as deaminases alone or as components in fusion proteins. A fusion protein comprising a DNA-targeting polypeptide and an adenine deaminase polypeptide is referred to herein as an “A-based editor”, “adenine base editor”, or an “ABE” and can be used for the targeted editing of nucleic acid sequences.

(11) “Base editors” are fusion proteins comprising a DNA-targeting polypeptide, such as an RGN, and a deaminase. Adenine base editors (ABEs) comprise a DNA-targeting protein, such as an RGN, and an adenine deaminase. ABEs function through the deamination of adenine into inosine on a DNA target molecule (Gaudelli, N. M. et al. 2017). Inosine is recognized as a guanine by polymerases and allows for the incorporation of a cytosine on the complementary DNA strand across from the inosine. After a round of replication post-deamination, there is a resulting A:T to G:C base pair change in the genome. In some embodiments, the presently disclosed adenine deaminases or active variants or fragments thereof introduce A>N mutations in a DNA molecule, wherein N is C, G, or T. In further embodiments, they introduce A>G mutations in a DNA molecule.

(12) In those embodiments wherein the deaminase has been targeted to a specific region of a nucleic acid molecule via fusion with a DNA-binding polypeptide, the mutation rate of adenines within or adjacent to the target sequence to which the DNA-binding polypeptide binds can be measured using any method known in the art, including polymerase chain reaction (PCR), restriction fragment length polymorphism (RFLP), or DNA sequencing.

(13) The presently disclosed novel deaminases or active variants or fragments thereof that retain deaminase activity may be introduced into the cell as part of a deaminase-DNA-binding polypeptide fusion, and/or may be co-expressed with a DNA-binding polypeptide-deaminase fusion, to increase the efficiency of introducing the desired A>G mutation in a target DNA molecule. The presently disclosed deaminases have the amino acid sequence of any of SEQ ID NOs: 1-10 and 399-441 or a variant or fragment thereof retaining deaminase activity. In some embodiments, the deaminase has an amino acid sequence having at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identity to the amino acid sequence of any of SEQ ID NOs: 1-10 and 399-441. In particular embodiments, the deaminase comprises an amino acid sequence having at least 80% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. In some embodiments, the deaminase comprises an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 407. For example, the deaminase comprises an amino acid sequence having at least about 80% identity, at least about 90% identity, at least about 95% identity, at least about 96% identity, at least about 97% identity, at least about 98% identity, at least about 99% identity, at least about 99.5% identity, or at least about 99.9% identity to SEQ ID NO: 407. In some embodiments, the deaminase comprises an amino acid sequence having at least 80% identity, at least 90% identity, at least 95% identity, at least 96% identity, at least 97% identity, at least 98% identity, at least 99% identity, at least 99.5% identity, or at least 99.9% identity to SEQ ID NO: 407. In some embodiments, the deaminase comprises the amino acid sequence of SEQ ID NO: 407.

In some embodiments, the deaminase comprises an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 399. For example, the deaminase comprises an amino acid sequence having at least about 80% identity, at least about 90% identity, at least about 95% identity, at least about 96% identity, at least about 97% identity, at least about 98% identity, at least about 99% identity, at least about 99.5% identity, or at least about 99.9% identity to SEQ ID NO: 399. In some embodiments, the deaminase comprises an amino acid sequence having at least 80% identity, at least 90% identity, at least 95% identity, at least 96% identity, at least 97% identity, at least 98% identity, at least 99% identity, at least 99.5% identity, or at least 99.9% identity to SEQ ID NO: 399. In some embodiments, the deaminase comprises the amino acid sequence of SEQ ID NO: 399. In some embodiments, the deaminase comprises an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 405. For example, the deaminase comprises an amino acid sequence having at least about 80% identity, at least about 90% identity, at least about 95% identity, at least about 96% identity, at least about 97% identity, at least about 98% identity, at least about 99% identity, at least about 99.5% identity, or at least about 99.9% identity to SEQ ID NO: 405. In some embodiments, the deaminase comprises an amino acid sequence having at least 80% identity, at least 90% identity, at least 95% identity, at least 96% identity, at least 97% identity, at least 98% identity, at least 99% identity, at least 99.5% identity, or at least 99.9% identity to SEQ ID NO: 405. In some embodiments, the deaminase comprises the amino acid sequence of SEQ ID NO: 405.

III. Nucleic Acid Molecule-Binding Polypeptides

(14) Some aspects of this disclosure provide fusion proteins that comprise a nucleic acid molecule-binding polypeptide and a deaminase polypeptide. While binding to and targeted editing of RNA molecules is contemplated by the present invention, in some embodiments, the nucleic acid molecule-binding polypeptide of the fusion protein is a DNA-binding polypeptide. Such fusion proteins are useful for targeted editing of DNA in vitro, ex vivo, or in vivo. These novel fusion proteins are active in mammalian cells and are useful for targeted editing of DNA molecules.

(15) The term “fusion protein” as used herein refers to a hybrid polypeptide which comprises protein domains from at least two different proteins. A fusion protein may comprise more than one different domain, for example, a DNA-binding domain and a deaminase. In some embodiments, a fusion protein is in a complex with, or is in association with, a nucleic acid, e.g., RNA.

(16) In some embodiments, the presently disclosed fusion proteins comprise a DNA-binding polypeptide. As used herein, the term “DNA-binding polypeptide” refers to any polypeptide which is capable of binding to DNA. In certain embodiments, the DNA-binding polypeptide portion of the presently disclosed fusion proteins binds to double-stranded DNA. In particular embodiments, the DNA-binding polypeptide binds to DNA in a sequence-specific manner. As used herein, the terms “sequence-specific” or “sequence-specific manner” refer to the selective interaction with a specific nucleotide sequence.

(17) Two polynucleotide sequences can be considered to be substantially complementary when the two sequences hybridize to each other under stringent conditions. Likewise, a DNA-binding polypeptide is considered to bind to a particular target sequence in a sequence-specific manner if the DNA-binding polypeptide binds to its sequence under stringent conditions. By “stringent conditions” or “stringent hybridization conditions” is intended conditions under which the two polynucleotide sequences (or the polypeptide binds to its specific target sequence) will bind to each other to a detectably greater degree than to other sequences (e.g., at least 2-fold over background). Stringent conditions are sequence-dependent and will be different in different circumstances. Typically, stringent conditions will be those in which the salt concentration is less than 1.5 M Na ion, typically about 0.01 to 1.0 M Na ion concentration (or other salts) at pH 7.0 to 8.3, and the temperature is at least 30° C. for short sequences (e.g., 10 to 50 nucleotides) and at least 60° C. for long sequences (e.g., greater than 50 nucleotides). Stringent conditions may also be achieved with the addition of destabilizing agents such as formamide. Exemplary low stringency conditions include hybridization with a buffer solution of 30 to 35% formamide, 1 M NaCl, 1% SDS (sodium

dodecyl sulfate) at 37° C., and a wash in 1× to 2×SSC (20×SSC=3.0 M NaCl/0.3 M trisodium citrate) at 50 to 55° C. Exemplary moderate stringency conditions include hybridization in 40 to 45% formamide, 1.0 M NaCl, 1% SDS at 37° C., and a wash in 0.5× to 1×SSC at 55 to 60° C. Exemplary high stringency conditions include hybridization in 50% formamide, 1 M NaCl, 1% SDS at 37° C., and a wash in 0.1×SSC at 60 to 65° C. Optionally, wash buffers may comprise about 0.1% to about 1% SDS. Duration of hybridization is generally less than about 24 hours, usually about 4 to about 12 hours. The duration of the wash time will be at least a length of time sufficient to reach equilibrium.

(18) The T_m is the temperature (under defined ionic strength and pH) at which 50% of a complementary target sequence hybridizes to a perfectly matched sequence. For DNA-DNA hybrids, the T_m can be approximated from the equation of Meinkoth and Wahl (1984) *Anal. Biochem.* 138:267-284: $T_m = 81.5^\circ \text{C.} + 16.6 (\log M) + 0.41 (\% \text{ GC}) - 0.61 (\% \text{ form}) - 500/L$; where M is the molarity of monovalent cations, % GC is the percentage of guanosine and cytosine nucleotides in the DNA, % form is the percentage of formamide in the hybridization solution, and L is the length of the hybrid in base pairs. Generally, stringent conditions are selected to be about 5° C. lower than the thermal melting point (T_m) for the specific sequence and its complement at a defined ionic strength and pH. However, severely stringent conditions can utilize a hybridization and/or wash at 1, 2, 3, or 4° C. lower than the thermal melting point (T_m); moderately stringent conditions can utilize a hybridization and/or wash at 6, 7, 8, 9, or 10° C. lower than the thermal melting point (T_m); low stringency conditions can utilize a hybridization and/or wash at 11, 12, 13, 14, 15, or 20° C. lower than the thermal melting point (T_m). Using the equation, hybridization and wash compositions, and desired T_m , those of ordinary skill will understand that variations in the stringency of hybridization and/or wash solutions are inherently described. An extensive guide to the hybridization of nucleic acids is found in Tijssen (1993) *Laboratory Techniques in Biochemistry and Molecular Biology—Hybridization with Nucleic Acid Probes*, Part I, Chapter 2 (Elsevier, New York); and Ausubel et al., eds. (1995) *Current Protocols in Molecular Biology*, Chapter 2 (Greene Publishing and Wiley-Interscience, New York). See Sambrook et al. (1989) *Molecular Cloning: A Laboratory Manual* (2d ed., Cold Spring Harbor Laboratory Press, Plainview, New York).

(19) In certain embodiments, the sequence-specific DNA-binding polypeptide is an RNA-guided, DNA-binding polypeptide (RGDBP). As used herein, the terms “RNA-guided, DNA-binding polypeptide” and “RGDBP” refer to polypeptides capable of binding to DNA through the hybridization of an associated RNA molecule with the target DNA sequence.

(20) In some embodiments, the DNA-binding polypeptide of the fusion protein is a nuclease, such as a sequence-specific nuclease. As used herein, the term “nuclease” refers to an enzyme that catalyzes the cleavage of phosphodiester bonds between nucleotides in a nucleic acid molecule. In some embodiments, the DNA-binding polypeptide is an endonuclease, which is capable of cleaving phosphodiester bonds between nucleotides within a nucleic acid molecule, whereas in certain embodiments, the DNA-binding polypeptide is an exonuclease that is capable of cleaving the nucleotides at either end (5' or 3') of a nucleic acid molecule. In some embodiments, the sequence-specific nuclease is selected from the group consisting of a meganuclease, a zinc finger nuclease, a TAL-effector DNA binding domain-nuclease fusion protein (TALEN), and an RNA-guided nuclease (RGN) or variants thereof wherein the nuclease activity has been reduced or inhibited.

(21) As used herein, the term “meganuclease” or “homing endonuclease” refers to endonucleases that bind a recognition site within double-stranded DNA that is 12 to 40 bp in length. Non-limiting examples of meganucleases are those that belong to the LAGLIDADG family that comprise the conserved amino acid motif LAGLIDADG (SEQ ID NO: 49). The term “meganuclease” can refer to a dimeric or single-chain meganuclease.

(22) As used herein, the term “zinc finger nuclease” or “ZFN” refers to a chimeric protein comprising a zinc finger DNA-binding domain and a nuclease domain.

(23) As used herein, the term “TAL-effector DNA binding domain-nuclease fusion protein” or “TALEN” refers to a chimeric protein comprising a TAL effector DNA-binding domain and a nuclease domain.

(24) As used herein, the term “RNA-guided nuclease” or “RGN” refers to an RNA-guided, DNA-binding polypeptide that has nuclease activity. RGNs are considered “RNA-guided” because guide RNAs form a complex with the RNA-guided nucleases to direct the RNA-guided nuclease to bind to a target sequence and in some embodiments, introduce a single-stranded or double-stranded break at the target sequence. The RGN may be a CasX, a CasY, a C2c1, a C2c2, a C2c3, a GeoCas9, aSpCas9, a SaCas9, a Nme2Cas9, a CjCas9, a Cas12a (formerly known as Cpf1), a Cas12b, a Cas12g, a Cas12h, a Cas12i, aLbCas12a, a AsCas12a, a CasMINI, a Cas13b, a Cas13c, a Cas13d, a Cas14, a Csn2, an xCas9, an SpCas9-NG, an LbCas12a, an AsCas12a, a Cas9-KKH, a circularly permuted Cas9, an Argonaute (Ago), a SmacCas9, or a Spy-macCas9, a Spy-macCas9 domain, or a RGN with an amino acid sequence set forth in any one of SEQ ID NOs: 41, 60, 366, or 368. In some embodiments, as described below, the RGNs provided herein are RGN nickases.

(25) According to the present invention, an RGN protein that has been mutated to become nuclease-inactive or “dead”, such as for example dCas9, can be referred to as an RNA-guided, DNA-binding polypeptide or a nuclease-inactive RGN or nuclease-dead RGN. Additionally, suitable nuclease-inactive Cas9 domains of other known RNA guided nucleases (RGNs) can be determined (for example, a nuclease-inactive variant of the RGN APG08290.1 disclosed in U.S. Patent Publication No. 2019/0367949, the entire contents of which are incorporated herein by reference herein).

(26) In some embodiments, the fusion protein comprises an RGN fused to a deaminase described herein. In those embodiments of fusion proteins described above, the deaminase is selected from deaminases comprising an amino acid sequence having at least 80% sequence identity to any one of SEQ ID NOs: 1-10 and 399-441. In some embodiments, the deaminase comprises an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 407. In some embodiments, the deaminase comprises an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 399. In some embodiments, the deaminase comprises an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 405. In those embodiments of fusion proteins described above, the RGN is selected from a CasX, a CasY, a C2c1, a C2c2, a C2c3, a GeoCas9, aSpCas9, a SaCas9, a Nme2Cas9, a CjCas9, a Cas12a (formerly known as Cpf1), a Cas12b, a Cas12g, a Cas12h, a Cas12i, aLbCas12a, a AsCas12a, a CasMINI, a Cas13b, a Cas13c, a Cas13d, a Cas14, a Csn2, an xCas9, an SpCas9-NG, an LbCas12a, an AsCas12a, a Cas9-KKH, a circularly permuted Cas9, an Argonaute (Ago), a SmacCas9, a Spy-macCas9 domain, or an RGN with an amino acid sequence set forth in any one of SEQ ID NOs: 41, 60, 366, or 368. In particular embodiments, the fusion protein comprises a Cas9 nickase fused to a deaminase comprising an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 407. In some embodiments, the fusion protein comprises a Cas9 nickase fused to a deaminase comprising an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 399. In particular embodiments, the fusion protein comprises a Cas9 nickase fused to a deaminase comprising an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 405. The Cas9 nickase, can be any Cas9 nickase disclosed in PCT Patent Publication No. WO2020181195, the entire contents of which is incorporated herein by reference herein.

(27) The term “RGN polypeptide” encompasses RGN polypeptides that only cleave a single strand of a target nucleotide sequence, which is referred to herein as a nickase. Such RGNs have a single functioning nuclease domain. RGN nickases can be naturally-occurring nickases or can be RGN proteins that naturally cleave both strands of a double-stranded nucleic acid molecule that have been mutated within one or more nuclease domains such that the nuclease activity of these mutated domains is reduced or eliminated, to become a nickase. In some embodiments, the nickase RGN of the fusion protein comprises a mutation (e.g., a D10A mutation) which renders the RGN capable of

cleaving only the non-base edited, target strand (the strand which comprises the PAM and is base paired to a gRNA) of a nucleic acid duplex. This D10A mutation mutates the first aspartic acid residue in the split RuvC nuclease domain of the RGN. The present application discloses several D10A nickase variants or homologous nickase variants of described RGNs (see Example 4). nAPG07433.1 and nAPG08290.1 (set forth as SEQ ID NOs: 42 and 61, respectively) are nickase variants of APG07433.1 and APG08290.1, which are set forth as SEQ ID NO: 41 and 60, respectively, and are described in WO 2019/236566 (incorporated by reference in its entirety herein). nAPG00969 (set forth as SEQ ID NO: 52) and nAPG09748 (set forth as SEQ ID NO: 54) are nickase variants of APG00969 and APG09748, respectively, which are described in WO 2020/139783 (incorporated by reference in its entirety herein). nAPG06646 (set forth as SEQ ID NO: 53) and nAPG09882 (set forth as SEQ ID NO: 55) are nickase variants of APG06646 and APG09882, respectively, which are described in PCT publication WO 2021/030344 (incorporated by reference in its entirety herein). nAPG03850, nAPG07553, nAPG055886, and nAPG01604 are set forth as SEQ ID NOs: 56-59, respectively, and are nickase variants of APG03850, APG07553, APG055886, and APG01604 which are described in the pending PCT Application No. PCT/US2021/028843 (incorporated by reference in its entirety herein). Various RGN nickases, their variants and their sequences are disclosed in PCT Patent Publication No. WO2020181195, the entire contents of which are incorporated herein by reference herein. One exemplary suitable nuclease-inactive Cas9 is the D10A/H840A Cas9 mutant (see, e.g., Qi et al., *Cell*. 2013; 152(5): 1173-83, the entire contents of which are incorporated herein by reference).

(28) In some embodiments, the nickase RGN of the fusion protein comprises a mutation (e.g., a H840A mutation), which renders the RGN capable of cleaving only the base-edited, non-targeted strand (the strand which does not comprise the PAM and is not base paired to a gRNA) of a nucleic acid duplex. The H840A mutation mutates the first histidine of the HNH nuclease domain. A nickase RGN comprising an H840A mutation, or an equivalent mutation, has an inactivated HNH domain. A nickase RGN with an H840A mutation cleaves the non-targeted strand. A nickase comprising a D10A mutation, or an equivalent mutation, has an inactivated RuvC nuclease domain and cleaves the targeted strand. D10A nickases are not able to cleave the non-targeted strand of the DNA, i.e., the strand where base editing is desired.

(29) Other additional exemplary suitable nuclease inactive Cas9 domains include, but are not limited to, D10A/D839A/H840A, and D10A/D839A/H840A/N863A mutant domains (See, e.g., Mali et al., *Nature Biotechnology*. 2013; 31(9): 833-838, the entire contents of which are incorporated herein by reference). Additional suitable RGN proteins mutated to be nickases will be apparent to those of skill in the art based on this disclosure and knowledge in the field (such as for example the RGNs disclosed in PCT Publication Nos. WO 2019/236566, WO2020181195, which are herein incorporated by reference in their entirety) and are within the scope of this disclosure. In preferred embodiments, an RGN which has nickase activity on the target strand nicks the target strand, while the complementary, non-target strand is modified by the deaminase. Cellular DNA-repair machinery may repair the nicked, target strand using the modified non-target strand as a template, thereby introducing a mutation in the DNA.

(30) In some embodiments the RGN nickase retaining nickase activity comprises an amino acid sequence that has at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5% identity to SEQ ID NO: 42 or any one of SEQ ID NOs: 52-59, 61, 397, and 398.

(31) Any method known in the art for introducing mutations into an amino acid sequence, such as PCR-mediated mutagenesis and site-directed mutagenesis, can be used for generating nickases or nuclease-dead RGNs. See, e.g., U.S. Publ. No. 2014/0068797 and U.S. Pat. No. 9,790,490; each of which is incorporated herein by reference in its entirety. RNA-guided nucleases (RGNs) allow for the targeted manipulation of a single site within a genome and are useful in the context of gene targeting for therapeutic and research applications. In a variety of organisms, including mammals,

RNA-guided nucleases have been used for genome engineering by stimulating either non-homologous end joining or homologous recombination. RGNs include CRISPR-Cas proteins, which are RNA-guided nucleases directed to the target sequence by a guide RNA (gRNA) as part of a Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) RNA-guided nuclease system, or active variants or fragments thereof.

(32) Further provided herein are RGN polypeptides (and nucleic acid molecules encoding RGN polypeptides) that comprise the amino acid sequence set forth as SEQ ID NO: 41 or 60, but lacking amino acid residues 590 to 597 of SEQ ID NO: 41 or 60, or an active variant or fragment thereof. In certain embodiments, the RGN polypeptide comprises the amino acid sequence set forth as SEQ ID NO: 366, 368, 397, or 398 or an active variant or fragment thereof.

(33) Some aspects of this disclosure provide fusion proteins that comprise an RNA-guided DNA-binding polypeptide and a deaminase polypeptide, specifically an adenine deaminase polypeptide. In some embodiments, the RNA-guided DNA-binding polypeptide is an RNA-guided nuclease. In further embodiments, the RNA-guided nuclease is a naturally-occurring CRISPR-Cas protein or an active variant or fragment thereof. CRISPR-Cas systems are classified into Class 1 or Class 2 systems. Class 2 systems comprise a single effector nuclease and include Types II, V, and VI. The Class 1 and 2 systems are subdivided into types (Types I, II, III, IV, V, VI), with some types further divided into subtypes (e.g., Type II-A, Type II-B, Type II-C, Type V-A, Type V-B).

(34) In certain embodiments, the CRISPR-Cas protein is a naturally-occurring Type II CRISPR-Cas protein or an active variant or fragment thereof. As used herein, the term “Type II CRISPR-Cas protein,” “Type II CRISPR-Cas effector protein,” or “Cas9” refers to a CRISPR-Cas effector protein that requires a trans-activating RNA (tracrRNA) and comprises two nuclease domains (i.e., RuvC and HNH), each of which is responsible for cleaving a single strand of a double-stranded DNA molecule. In some embodiments, the present invention provides a fusion protein comprising a presently disclosed deaminase fused to *Streptococcus pyogenes* Cas9 (SpCas9) or a SpCas9 nickase, the sequences of which are set forth as SEQ ID NOs: 555 and 556, respectively, and are described in U.S. Pat. Nos. 10,000,772 and 8,697,359, each of which is herein incorporated by reference in its entirety. In some embodiments, the present invention provides a fusion protein comprising a presently disclosed deaminase fused to *Streptococcus thermophilus* Cas9 (StCas9) or a StCas9 nickase, the sequences of which are set forth as SEQ ID NOs: 557 and 558, respectively, and are disclosed in U.S. Pat. No. 10,113,167, which is herein incorporated by reference in its entirety. In some embodiments, the present invention provides a fusion protein comprising a presently disclosed deaminase fused to *Streptococcus aureus* Cas9 (SaCas9) or a SaCas9 nickase, the sequences of which are set forth as SEQ ID NOs: 559 and 560, respectively, and are disclosed in U.S. Pat. No. 9,752,132, which is herein incorporated by reference in its entirety.

(35) In some embodiments, the CRISPR-Cas protein is a naturally-occurring Type V CRISPR-Cas protein or an active variant or fragment thereof. As used herein, the term “Type V CRISPR-Cas protein,” “Type V CRISPR-Cas effector protein,” or “Cas12” refers to a CRISPR-Cas effector protein that cleaves dsDNA and comprises a single RuvC nuclease domain or a split-RuvC nuclease domain and lacks an HNH domain (Zetsche et al 2015, *Cell*

doi:10.1016/j.cell.2015.09.038; Shmakov et al 2017, *Nat Rev Microbiol*

doi:10.1038/nrmicro.2016.184; Yan et al 2018, *Science* doi:10.1126/science.aav7271; Harrington et al 2018, *Science* doi:10.1126/science.aav4294). It is to be noted that Cas12a is also referred to as Cpf1, and does not require a tracrRNA, although other Type V CRISPR-Cas proteins, such as Cas12b, do require a tracrRNA. Most Type V effectors can also target ssDNA (single-stranded DNA), often without a PAM requirement (Zetsche et al 2015; Yan et al 2018; Harrington et al 2018). The term “Type V CRISPR-Cas protein” encompasses the unique RGNs comprising split RuvC nuclease domains, such as those disclosed in U.S. Provisional Appl. Nos. 62/955,014 filed Dec. 30, 2019 and 63/058,169 filed Jul. 29, 2020, and PCT International Appl. No.

PCT/US2020/067138 filed Dec. 28, 2020, the contents of each of which are incorporated herein by

reference in its entirety. In some embodiments, the present invention provides a fusion protein comprising a presently disclosed deaminase fused to *Francisella novicida* Cas12a (FnCas12a), the sequence of which is set forth as SEQ ID NOs: 561 and is disclosed in U.S. Pat. No. 9,790,490, which is herein incorporated by reference in its entirety, or any of the nuclease-inactivating mutants of FnCas12a disclosed within U.S. Pat. No. 9,790,490.

(36) In some embodiments, the CRISPR-Cas protein is a naturally-occurring Type VI CRISPR-Cas protein or an active variant or fragment thereof. As used herein, the term “Type VI CRISPR-Cas protein,” “Type VI CRISPR-Cas effector protein,” or “Cas13” refers to a CRISPR-Cas effector protein that does not require a tracrRNA and comprises two HEPN domains that cleave RNA.

(37) The term “guide RNA” refers to a nucleotide sequence having sufficient complementarity with a target nucleotide sequence to hybridize with the target sequence and direct sequence-specific binding of an associated RGN to the target nucleotide sequence. For CRISPR-Cas RGNs, the respective guide RNA is one or more RNA molecules (generally, one or two), that can bind to the RGN and guide the RGN to bind to a particular target nucleotide sequence, and in those instances wherein the RGN has nickase or nuclease activity, also cleave the target nucleotide sequence. A guide RNA comprises a CRISPR RNA (crRNA) and in some embodiments, a trans-activating CRISPR RNA (tracrRNA).

(38) A CRISPR RNA comprises a spacer sequence and a CRISPR repeat sequence. The “spacer sequence” is the nucleotide sequence that directly hybridizes with the target nucleotide sequence of interest. The spacer sequence is engineered to be fully or partially complementary with the target sequence of interest. In various embodiments, the spacer sequence comprises from about 8 nucleotides to about 30 nucleotides, or more. For example, the spacer sequence can be about 8, about 9, about 10, about 11, about 12, about 13, about 14, about 15, about 16, about 17, about 18, about 19, about 20, about 21, about 22, about 23, about 24, about 25, about 26, about 27, about 28, about 29, about 30, or more nucleotides in length. In some embodiments, the spacer sequence is 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, or more nucleotides in length. In some embodiments, the spacer sequence is about 10 to about 26 nucleotides in length, or about 12 to about 30 nucleotides in length. In some embodiments, the spacer sequence is 10 to 26 nucleotides in length, or 12 to 30 nucleotides in length. In particular embodiments, the spacer sequence is about 30 nucleotides in length. In particular embodiments, the spacer sequence is 30 nucleotides in length. In some embodiments, the degree of complementarity between a spacer sequence and its corresponding target sequence, when optimally aligned using a suitable alignment algorithm, is between 50% and 99% or more, including but not limited to about or more than about 50%, about 60%, about 70%, about 75%, about 80%, about 81%, about 82%, about 83%, about 84%, about 85%, about 86%, about 87%, about 88%, about 89%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or more. In particular embodiments, the degree of complementarity between a spacer sequence and its corresponding target sequence, when optimally aligned using a suitable alignment algorithm, is 50%, 60%, 70%, 75%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more. In particular embodiments, the spacer sequence is free of secondary structure, which can be predicted using any suitable polynucleotide folding algorithm known in the art, including but not limited to mFold (see, e.g., Zuker and Stiegler (1981) *Nucleic Acids Res.* 9:133-148) and RNAfold (see, e.g., Gruber et al. (2008) *Cell* 106(1):23-24).

(39) The CRISPR RNA repeat sequence comprises a nucleotide sequence that forms a structure, either on its own or in concert with a hybridized tracrRNA, that is recognized by the RGN molecule. In various embodiments, the CRISPR RNA repeat sequence comprises from about 8 nucleotides to about 30 nucleotides, or more. In particular embodiments, the CRISPR RNA repeat sequence comprises from 8 nucleotides to 30 nucleotides, or more. For example, the CRISPR repeat sequence can be about 8, about 9, about 10, about 11, about 12, about 13, about 14, about 15,

about 16, about 17, about 18, about 19, about 20, about 21, about 22, about 23, about 24, about 25, about 26, about 27, about 28, about 29, about 30, or more nucleotides in length. In particular embodiments, the CRISPR repeat sequence is 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, or more nucleotides in length. In some embodiments, the degree of complementarity between a CRISPR repeat sequence and its corresponding tracrRNA sequence, when optimally aligned using a suitable alignment algorithm, is between 50% and 99%, or more, including but not limited to about or more than about 50%, about 60%, about 70%, about 75%, about 80%, about 81%, about 82%, about 83%, about 84%, about 85%, about 86%, about 87%, about 88%, about 89%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or more. In particular embodiments, the degree of complementarity between a CRISPR repeat sequence and its corresponding tracrRNA sequence, when optimally aligned using a suitable alignment algorithm, is 50%, 60%, 70%, 75%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more.

(40) In some embodiments, the guide RNA further comprises a tracrRNA molecule. A trans-activating CRISPR RNA or tracrRNA molecule comprises a nucleotide sequence comprising a region that has sufficient complementarity to hybridize to a CRISPR repeat sequence of a crRNA, which is referred to herein as the anti-repeat region. In some embodiments, the tracrRNA molecule further comprises a region with secondary structure (e.g., stem-loop) or forms secondary structure upon hybridizing with its corresponding crRNA. In particular embodiments, the region of the tracrRNA that is fully or partially complementary to a CRISPR repeat sequence is at the 5' end of the molecule and the 3' end of the tracrRNA comprises secondary structure. This region of secondary structure generally comprises several hairpin structures, including the nexus hairpin, which is found adjacent to the anti-repeat sequence. There are often terminal hairpins at the 3' end of the tracrRNA that can vary in structure and number, but often comprise a GC-rich Rho-independent transcriptional terminator hairpin followed by a string of Us at the 3' end. See, for example, Briner et al. (2014) *Molecular Cell* 56:333-339, Briner and Barrangou (2016) *Cold Spring Harb Protoc*; doi: 10.1101/pdb.top090902, and U.S. Publication No. 2017/0275648, each of which is herein incorporated by reference in its entirety.

(41) In various embodiments, the anti-repeat region of the tracrRNA that is fully or partially complementary to the CRISPR repeat sequence comprises from about 6 nucleotides to about 30 nucleotides, or more. For example, the region of base pairing between the tracrRNA anti-repeat sequence and the CRISPR repeat sequence can be about 6, about 7, about 8, about 9, about 10, about 11, about 12, about 13, about 14, about 15, about 16, about 17, about 18, about 19, about 20, about 21, about 22, about 23, about 24, about 25, about 26, about 27, about 28, about 29, about 30, or more nucleotides in length. In particular embodiments, the region of base pairing between the tracrRNA anti-repeat sequence and the CRISPR repeat sequence is 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, or more nucleotides in length. In particular embodiments, the anti-repeat region of the tracrRNA that is fully or partially complementary to a CRISPR repeat sequence is about 10 nucleotides in length. In particular embodiments, the anti-repeat region of the tracrRNA that is fully or partially complementary to a CRISPR repeat sequence is 10 nucleotides in length. In some embodiments, the degree of complementarity between a CRISPR repeat sequence and its corresponding tracrRNA anti-repeat sequence, when optimally aligned using a suitable alignment algorithm, is between 50% and 99% or more, including but not limited to about or more than about 50%, about 60%, about 70%, about 75%, about 80%, about 81%, about 82%, about 83%, about 84%, about 85%, about 86%, about 87%, about 88%, about 89%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or more. In particular embodiments, the degree of complementarity between a CRISPR repeat sequence and its corresponding tracrRNA anti-repeat sequence, when optimally aligned using a suitable alignment algorithm, is 50%, 60%, 70%, 75%, 80%, 81%, 82%,

83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more.

(42) In various embodiments, the entire tracrRNA comprises from about 60 nucleotides to more than about 210 nucleotides. In particular embodiments, the entire tracrRNA comprises from 60 nucleotides to more than 210 nucleotides. For example, the tracrRNA can be about 60, about 65, about 70, about 75, about 80, about 85, about 90, about 95, about 100, about 105, about 110, about 115, about 120, about 125, about 130, about 135, about 140, about 150, about 160, about 170, about 180, about 190, about 200, about 210 or more nucleotides in length. In particular embodiments, the tracrRNA is 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 150, 160, 170, 180, 190, 200, 210 or more nucleotides in length. In particular embodiments, the tracrRNA is about 100 to about 210 nucleotides in length, including about 95, about 96, about 97, about 98, about 99, about 100, about 105, about 106, about 107, about 108, about 109, and about 110 nucleotides in length. In particular embodiments, the tracrRNA is 100 to 110 nucleotides in length, including 95, 96, 97, 98, 99, 100, 105, 106, 107, 108, 109, and 110 nucleotides in length.

(43) Guide RNAs form a complex with an RNA-guided, DNA-binding polypeptide or an RNA-guided nuclease to direct the RNA-guided nuclease to bind to a target sequence. If the guide RNA complexes with an RGN, the bound RGN introduces a single-stranded or double-stranded break at the target sequence. After the target sequence has been cleaved, the break can be repaired such that the DNA sequence of the target sequence is modified during the repair process. Provided herein are methods for using mutant variants of RNA-guided nucleases, which are either nuclease inactive or nickases, which are linked to deaminases to modify a target sequence in the DNA of host cells. The mutant variants of RNA-guided nucleases in which the nuclease activity is inactivated or significantly reduced may be referred to as RNA-guided, DNA-binding polypeptides, as the polypeptides are capable of binding to, but not necessarily cleaving, a target sequence. RNA-guided nucleases only capable of cleaving a single strand of a double-stranded nucleic acid molecule are referred to herein as nickases.

(44) A target nucleotide sequence is bound by an RNA-guided, DNA-binding polypeptide and hybridizes with the guide RNA associated with the RGDBP. The target sequence can then be subsequently cleaved if the RGDBP possesses nuclease activity (i.e., is an RGN), which encompasses activity as a nickase.

(45) The guide RNA can be a single guide RNA or a dual-guide RNA system. A single guide RNA comprises the crRNA and optionally tracrRNA on a single molecule of RNA, whereas a dual-guide RNA system comprises a crRNA and a tracrRNA present on two distinct RNA molecules, hybridized to one another through at least a portion of the CRISPR repeat sequence of the crRNA and at least a portion of the tracrRNA, which may be fully or partially complementary to the CRISPR repeat sequence of the crRNA. In some of those embodiments wherein the guide RNA is a single guide RNA, the crRNA and optionally tracrRNA are separated by a linker nucleotide sequence.

(46) In general, the linker nucleotide sequence is one that does not include complementary bases in order to avoid the formation of secondary structure within or comprising nucleotides of the linker nucleotide sequence. In some embodiments, the linker nucleotide sequence between the crRNA and tracrRNA is at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, or more nucleotides in length. In particular embodiments, the linker nucleotide sequence between the crRNA and tracrRNA is 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or more nucleotides in length. In particular embodiments, the linker nucleotide sequence of a single guide RNA is at least 4 nucleotides in length. In particular embodiments, the linker nucleotide sequence of a single guide RNA is 4 nucleotides in length.

(47) In certain embodiments, the guide RNA can be introduced into a target cell, organelle, or embryo as an RNA molecule. The guide RNA can be transcribed in vitro or chemically

synthesized. In some embodiments, a nucleotide sequence encoding the guide RNA is introduced into the cell, organelle, or embryo. In some embodiments, the nucleotide sequence encoding the guide RNA is operably linked to a promoter (e.g., an RNA polymerase III promoter). The promoter can be a native promoter or heterologous to the guide RNA-encoding nucleotide sequence.

(48) In various embodiments, the guide RNA can be introduced into a target cell, organelle, or embryo as a ribonucleoprotein complex, as described herein, wherein the guide RNA is bound to an RNA-guided nuclease polypeptide.

(49) The guide RNA directs an associated RNA-guided nuclease to a particular target nucleotide sequence of interest through hybridization of the guide RNA to the target nucleotide sequence. A target nucleotide sequence can comprise DNA, RNA, or a combination of both and can be single-stranded or double-stranded. A target nucleotide sequence can be genomic DNA (i.e., chromosomal DNA), plasmid DNA, or an RNA molecule (e.g., messenger RNA, ribosomal RNA, transfer RNA, micro RNA, small interfering RNA). The target nucleotide sequence can be bound (and in some embodiments, cleaved) by an RNA-guided, DNA-binding polypeptide in vitro or in a cell. The chromosomal sequence targeted by the RGDBP can be a nuclear, plastid or mitochondrial chromosomal sequence. In some embodiments, the target nucleotide sequence is unique in the target genome.

(50) In some embodiments, the target nucleotide sequence is adjacent to a protospacer adjacent motif (PAM). A PAM is generally within about 1 to about 10 nucleotides from the target nucleotide sequence, including about 1, about 2, about 3, about 4, about 5, about 6, about 7, about 8, about 9, or about 10 nucleotides from the target nucleotide sequence. In particular embodiments, a PAM is within 1 to 10 nucleotides from the target nucleotide sequence, including 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 nucleotides from the target nucleotide sequence. The PAM can be 5' or 3' of the target sequence. In some embodiments, the PAM is 3' of the target sequence. Generally, the PAM is a consensus sequence of about 2-6 nucleotides, but in particular embodiments, is 1, 2, 3, 4, 5, 6, 7, 8, 9, or more nucleotides in length.

(51) The PAM restricts which sequences a given RGDBP or RGN can target, as its PAM needs to be proximal to the target nucleotide sequence. Upon recognizing its corresponding PAM sequence, the RGN can cleave the target nucleotide sequence at a specific cleavage site. As used herein, a cleavage site is made up of the two particular nucleotides within a target nucleotide sequence between which the nucleotide sequence is cleaved by an RGN. The cleavage site can comprise the 1.sup.st and 2.sup.nd, 2.sup.nd and 3.sup.rd, 3.sup.rd and 4.sup.th, 4.sup.th and 5.sup.th, 5.sup.th and 6.sup.th, 7.sup.th and 8.sup.th, or 8.sup.th and 9.sup.th nucleotides from the PAM in either the 5' or 3' direction. As RGNs can cleave a target nucleotide sequence resulting in staggered ends, in some embodiments, the cleavage site is defined based on the distance of the two nucleotides from the PAM on the positive (+) strand of the polynucleotide and the distance of the two nucleotides from the PAM on the negative (-) strand of the polynucleotide.

(52) RGDBPs and RGNs can be used to deliver a fused polypeptide, polynucleotide, or small molecule payload to a particular genomic location.

(53) In those embodiments wherein the DNA-binding polypeptide comprises a meganuclease, a target sequence can comprise a pair of inverted, 9 basepair "half sites" which are separated by four basepairs. In the case of a single-chain meganuclease, the N-terminal domain of the protein contacts a first half-site and the C-terminal domain of the protein contacts a second half-site. Cleavage by a meganuclease produces four basepair 3' overhangs. In those embodiments wherein the DNA-binding polypeptide comprises a compact TALEN, the recognition sequence comprises a first CNNNGN sequence that is recognized by the I-TevI domain, followed by a non-specific spacer 4-16 basepairs in length, followed by a second sequence 16-22 bp in length that is recognized by the TAL-effector domain (this sequence typically has a 5' T base). In those embodiments wherein the DNA-binding polypeptide comprises a zinc finger, the DNA binding domains typically recognize an 18-bp recognition sequence comprising a pair of nine basepair

“half-sites” separated by 2-10 basepairs and cleavage by the nuclease creates a blunt end or a 5’ overhang of variable length (frequently four basepairs).

IV. Fusion Proteins

(54) In some embodiments, a DNA-binding polypeptide (e.g., nuclease-inactive or a nickase RGN) is operably linked to a deaminase of the invention. In some embodiments, a DNA-binding polypeptide (e.g., nuclease inactive RGN or nickase RGN) fused to a deaminase of the invention can be targeted to a particular location of a nucleic acid molecule (i.e., target nucleic acid molecule), which in some embodiments is a particular genomic locus, to alter the expression of a desired sequence. In some embodiments, the binding of a fusion protein to a target sequence results in deamination of a nucleobase, resulting in conversion from one nucleobase to another. In some embodiments, the binding of this fusion protein to a target sequence results in deamination of a nucleobase adjacent to the target sequence. The nucleobase adjacent to the target sequence that is deaminated and mutated using the presently disclosed compositions and methods may be 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, or 100 base pairs from the 5’ or 3’ end of the target sequence (bound by the gRNA) within the target nucleic acid molecule. Some aspects of this disclosure provide fusion proteins comprising (i) a DNA-binding polypeptide (e.g., a nuclease-inactive or nickase RGN polypeptide); (ii) a deaminase polypeptide; and optionally (iii) a second deaminase. The second deaminase may be the same deaminase as the first or may be a different deaminase. In some embodiments, both the first and the second deaminase are adenine deaminases of the invention.

(55) The instant disclosure provides fusion proteins of various configurations. In some embodiments, the deaminase polypeptide is fused to the N-terminus of the DNA-binding polypeptide (e.g., RGN polypeptide). In some embodiments, the deaminase polypeptide is fused to the C-terminus of the DNA-binding polypeptide (e.g., RGN polypeptide).

(56) In some embodiments, the deaminase and DNA-binding polypeptide (e.g., RNA-guided, DNA-binding polypeptide) are fused to each other via a peptide linker. The linker between the deaminase and DNA-binding polypeptide (e.g., RNA-guided, DNA-binding polypeptide) can determine the editing window of the fusion protein, thereby increasing deaminase specificity and reducing off-target mutations. Various linker lengths and flexibilities can be employed, ranging from very flexible linkers of the form (GGGGS).sub.n and (G).sub.n to more rigid linkers of the form (EAAAK).sub.n and (XP).sub.n, to achieve the optimal length and rigidity for deaminase activity for the specific applications. The term “linker,” as used herein, refers to a chemical group or a molecule linking two molecules or moieties, e.g., a binding domain and a cleavage domain of a nuclease. In some embodiments, a linker joins an RNA guided nuclease and a deaminase. In some embodiments, a linker joins a dead or inactive RGN and a deaminase. In further embodiments, a linker joins two deaminases. Typically, the linker is positioned between, or flanked by, two groups, molecules, or other moieties and connected to each one via a covalent bond, thus connecting the two. In some embodiments, the linker is an amino acid or a plurality of amino acids (e.g., a peptide or protein). In some embodiments, the linker is an organic molecule, group, polymer, or chemical moiety. In some embodiments, the linker is 3-100 amino acids in length, for example, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 30-35, 35-40, 40-45, 45-50, 50-60, 60-70, 70-80, 80-90, 90-100, 100-150, or 150-200 amino acids in length. Longer or shorter linkers are also contemplated. In some embodiments, a shorter linker is preferred to decrease the overall size or length of the fusion protein or its coding sequence.

(57) In some embodiments, the linker comprises a (GGGGS).sub.n, a (G).sub.n an (EAAAK).sub.n, or an (XP).sub.n motif, or a combination of any of these, wherein n is independently an integer between 1 and 30. In some embodiments, n is independently 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30, or, if more than one linker or more than one linker motif is present, any combination thereof. Additional suitable linker motifs and linker configurations will be apparent to those of skill in the art. In some

embodiments, suitable linker motifs and configurations include those described in Chen et al., 2013 (*Adv Drug Deliv Rev.* 65(10):1357-69, the entire contents of which are incorporated herein by reference). Additional suitable linker sequences will be apparent to those of skill in the art. In some embodiments, the linker sequence comprises the amino acid sequence set forth as SEQ ID NO: 45 or 442.

(58) In some embodiments, the general architecture of exemplary fusion proteins provided herein comprises the structure: [NH.sub.2]-[deaminase]-[DBP]-[COOH]; [NH.sub.2]-[DBP]-[deaminase]-[COOH]; [NH.sub.2]-[DBP]-[deaminase]-[deaminase]-[COOH]; [NH.sub.2]-[deaminase]-[DBP]-[deaminase]-[COOH]; or [NH.sub.2]-[deaminase]-[deaminase]-[DBP]-[COOH], wherein DBP is a DNA-binding polypeptide, NH.sub.2 is the N-terminus of the fusion protein and COOH is the C-terminus of the fusion protein. In some embodiments, the fusion protein comprises more than two deaminase polypeptides.

(59) In certain embodiments, the general architecture of exemplary fusion proteins provided herein comprises the structure: [NH.sub.2]-[deaminase]-[RGN]-[COOH]; [NH.sub.2]-[RGN]-[deaminase]-[COOH]; [NH.sub.2]-[RGN]-[deaminase]-[deaminase]-[COOH]; [NH.sub.2]-[deaminase]-[RGN]-[deaminase]-[COOH]; or [NH.sub.2]-[deaminase]-[deaminase]-[RGN]-[COOH], wherein NH.sub.2 is the N-terminus of the fusion protein and COOH is the C-terminus of the fusion protein. In some embodiments, the fusion protein comprises more than two deaminase polypeptides.

(60) In some embodiments, the fusion protein comprises the structure: [NH.sub.2]-[deaminase]-[nuclease-inactive RGN]-[COOH]; [NH.sub.2]-[deaminase]-[deaminase]-[nuclease-inactive RGN]-[COOH]; [NH.sub.2]-[nuclease-inactive RGN]-[deaminase]-[COOH]; [NH.sub.2]-[deaminase]-[nuclease-inactive RGN]-[deaminase]-[COOH]; or [NH.sub.2]-[nuclease-inactive RGN]-[deaminase]-[deaminase]-[COOH]. It should be understood that “nuclease-inactive RGN” represents any RGN, including any CRISPR-Cas protein, which has been mutated to be nuclease-inactive. In some embodiments, the fusion protein comprises more than two deaminase polypeptides.

(61) In some embodiments, the fusion protein comprises the structure: [NH.sub.2]-[deaminase]-[RGN nickase]-[COOH]; [NH.sub.2]-[deaminase]-[deaminase]-[RGN nickase]-[COOH]; [NH.sub.2]-[RGN nickase]-[deaminase]-[COOH]; [NH.sub.2]-[deaminase]-[RGN nickase]-[deaminase]-[COOH]; or [NH.sub.2]-[RGN nickase]-[deaminase]-[deaminase]-[COOH]. It should be understood that “RGN nickase” represents any RGN, including any CRISPR-Cas protein, which has been mutated to be active as a nickase.

(62) In some embodiments, the “-” used in the general architecture above indicates the presence of an optional linker sequence. In some embodiments, the fusion proteins provided herein do not comprise a linker sequence. In some embodiments, at least one of the optional linker sequences are present.

(63) Other exemplary features that may be present are localization sequences, such as nuclear localization sequences, cytoplasmic localization sequences, export sequences, such as nuclear export sequences, or other localization sequences, as well as sequence tags that are useful for solubilization, purification or detection of the fusion proteins. Suitable localization signal sequences and sequences of protein tags that are provided herein, and include, but are not limited to, biotin carboxylase carrier protein (BCCP) tags, myc-tags, calmodulin-tags, FLAG-tags, hemagglutinin (HA)-tags, polyhistidine tags, also referred to as histidine tags or His-tags, maltose binding protein (MBP)-tags, nus-tags, glutathione-S-transferase (GST)-tags, green fluorescent protein (GFP)-tags, thioredoxin-tags, S-tags, Softags (e.g., Softag 1, Softag 3), streptags, biotin ligase tags, FlAsH tags, V5 tags, and SBP-tags. Additional suitable sequences will be apparent to those of skill in the art.

(64) In certain embodiments, the presently disclosed fusion proteins comprise at least one cell-penetrating domain that facilitates cellular uptake of the fusion protein. Cell-penetrating domains

are known in the art and generally comprise stretches of positively charged amino acid residues (i.e., polycationic cell-penetrating domains), alternating polar amino acid residues and non-polar amino acid residues (i.e., amphipathic cell-penetrating domains), or hydrophobic amino acid residues (i.e., hydrophobic cell-penetrating domains) (see, e.g., Milletti F. (2012) *Drug Discov Today* 17:850-860). A non-limiting example of a cell-penetrating domain is the trans-activating transcriptional activator (TAT) from the human immunodeficiency virus 1.

(65) In some embodiments, deaminases or fusion proteins provided herein further comprise a nuclear localization sequence (NLS). The nuclear localization signal, plastid localization signal, mitochondrial localization signal, dual-targeting localization signal, and/or cell-penetrating domain can be located at the amino-terminus (N-terminus), the carboxyl-terminus (C-terminus), or in an internal location of the fusion protein.

(66) In some embodiments, the NLS is fused to the N-terminus of the fusion protein or deaminase. In some embodiments, the NLS is fused to the C-terminus of the fusion protein or deaminase. In some embodiments, the NLS is fused to the N-terminus of the deaminase of the fusion protein. In some embodiments, the NLS is fused to the C-terminus of the deaminase of the fusion protein. In some embodiments, the NLS is fused to the N-terminus of the DNA-binding polypeptide (e.g., RGN polypeptide) of the fusion protein. In some embodiments, the NLS is fused to the C-terminus of the DNA-binding polypeptide (e.g., RGN polypeptide) of the fusion protein. In some embodiments, the NLS is fused to the N-terminus of the deaminase polypeptide of the fusion protein. In some embodiments, the NLS is fused to the C-terminus of the deaminase polypeptide of the fusion protein. In some embodiments, the NLS is fused to the fusion protein via one or more linkers. In some embodiments, the NLS is fused to the fusion protein without a linker. In some embodiments, the NLS comprises an amino acid sequence of any one of the NLS sequences provided or referenced herein. In some embodiments, the NLS comprises an amino acid sequence as set forth in SEQ ID NO: 43 or SEQ ID NO: 46. In some embodiments, the fusion protein or deaminase comprises SEQ ID NO: 43 on its N-terminus and SEQ ID NO: 46 on its C-terminus.

(67) In some embodiments, fusion proteins as provided herein comprise the full-length sequence of a deaminase, e.g., any one of SEQ ID NO: 1-10 and 399-441. In some embodiments, however, fusion proteins as provided herein do not comprise a full-length sequence of a deaminase, but only a fragment thereof. For example, in some embodiments, a fusion protein provided herein further comprises a DNA-binding polypeptide (e.g., an RNA-guided, DNA-binding) domain and a deaminase domain.

(68) In some embodiments, a fusion protein of the invention comprises a DNA-binding polypeptide (e.g., an RGN) and a deaminase, wherein the deaminase has an amino acid sequence having at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to any of SEQ ID NOs: 1-10 and 399-441. Examples of such fusion proteins are described in the Examples section herein.

(69) In some embodiments, the fusion protein comprises one deaminase polypeptide. In some embodiments, the fusion protein comprises at least two deaminase polypeptides, operably linked either directly or via a peptide linker. In some embodiments, the fusion protein comprises one deaminase polypeptide, and a second deaminase polypeptide is co-expressed with the fusion protein.

(70) Also provided herein is a ribonucleoprotein complex comprising a fusion protein comprising a deaminase and an RGDBP and the guide RNA, either as a single guide or as a dual guide RNA (also collectively referred to as gRNA).

V. Nucleotides Encoding Deaminases, Fusion Proteins, and/or gRNA

(71) The present disclosure provides polynucleotides (SEQ ID NOs: 11-20 and 443-485) encoding the presently disclosed deaminase polypeptides. The present disclosure further provides polynucleotides encoding for fusion proteins which comprise a deaminase and DNA-binding

polypeptide, for example a meganuclease, a zinc finger fusion protein, or a TALEN. The present disclosure further provides polynucleotides encoding for fusion proteins which comprise a deaminase domain and an RNA-guided, DNA-binding polypeptide. Such RNA-guided, DNA-binding polypeptides may be an RGN or RGN variant. The protein variant may be nuclease-inactive or a nickase. The RGN may be a CRISPR-Cas protein or active variant or fragment thereof. SEQ ID NOs: 41 and 42 are non-limiting examples of an RGN and a nickase RGN variant, respectively. Examples of CRISPR-Cas nucleases are well-known in the art, and similar corresponding mutations can create mutant variants which are also nickases or are nuclease inactive.

(72) An embodiment of the invention provides a polynucleotide encoding a fusion protein which comprises an RGDBP and a deaminase described herein (SEQ ID NO: 1-10 and 399-441, or a variant thereof). In some embodiments, a second polynucleotide encodes the guide RNA required by the RGDBP for targeting to the nucleotide sequence of interest. In some embodiments, the guide RNA and the fusion protein are encoded by the same polynucleotide.

(73) The use of the term “polynucleotide” is not intended to limit the present disclosure to polynucleotides comprising DNA, though such DNA polynucleotides are contemplated. Those of ordinary skill in the art will recognize that polynucleotides can comprise ribonucleotides (RNA) and combinations of ribonucleotides and deoxyribonucleotides. Such deoxyribonucleotides and ribonucleotides include both naturally occurring molecules and synthetic analogues. The polynucleotides disclosed herein also encompass all forms of sequences including, but not limited to, single-stranded forms, double-stranded forms, stem-and-loop structures, circular forms (e.g., including circular RNA), and the like.

(74) An embodiment of the invention is a nucleic acid molecule comprising a sequence having at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% or 100% identity to any of SEQ ID NOs: 11-20 and 443-485, wherein the nucleic acid molecule encodes a deaminase having adenine deaminase activity. The nucleic acid molecule may further comprise a heterologous promoter or terminator. The nucleic acid molecule may encode a fusion protein, where the encoded deaminase is operably linked to a DNA-binding polypeptide, and optionally a second deaminase. In some embodiments, the nucleic acid molecule encodes a fusion protein, where the encoded deaminase is operably linked to an RGN and optionally a second deaminase.

(75) In some embodiments, nucleic acid molecules comprising a polynucleotide which encodes a deaminase of the invention are codon optimized for expression in an organism of interest. A “codon-optimized” coding sequence is a polynucleotide coding sequence having its frequency of codon usage designed to mimic the frequency of preferred codon usage or transcription conditions of a particular host cell. Expression in the particular host cell or organism is enhanced as a result of the alteration of one or more codons at the nucleic acid level such that the translated amino acid sequence is not changed. Nucleic acid molecules can be codon optimized, either wholly or in part. Codon tables and other references providing preference information for a wide range of organisms are available in the art (see, e.g., Campbell and Gown (1990) *Plant Physiol.* 92:1-11 for a discussion of plant-preferred codon usage). Methods are available in the art for synthesizing plant-preferred genes. See, for example, U.S. Pat. Nos. 5,380,831, and 5,436,391, and Murray et al. (1989) *Nucleic Acids Res.* 17:477-498, herein incorporated by reference.

(76) In some embodiments, polynucleotides encoding the deaminases, fusion proteins, and/or gRNAs described herein are provided in expression cassettes for in vitro expression or expression in a cell, organelle, embryo, or organism of interest. The cassette may include 5' and 3' regulatory sequences operably linked to a polynucleotide encoding a deaminase and/or a fusion protein comprising a deaminase, an RNA-guided DNA-binding polypeptide and optionally a second deaminase, and/or gRNA provided herein that allows for expression of the polynucleotide. The cassette may additionally contain at least one additional gene or genetic element to be

cotransformed into the organism. Where additional genes or elements are included, the components are operably linked. The term “operably linked” is intended to mean a functional linkage between two or more elements. For example, an operable linkage between a promoter and a coding region of interest (e.g., a region coding for a deaminase, RNA-guided DNA-binding polypeptide, and/or gRNA) is a functional link that allows for expression of the coding region of interest. Operably linked elements may be contiguous or non-contiguous. When used to refer to the joining of two protein coding regions, by operably linked is intended that the coding regions are in the same reading frame. In some embodiments, the additional gene(s) or element(s) are provided on multiple expression cassettes. For example, the nucleotide sequence encoding a presently disclosed deaminase, either alone or as a component of a fusion protein, can be present on one expression cassette, whereas the nucleotide sequence encoding a gRNA can be on a separate expression cassette. Another example may have the nucleotide sequence encoding a presently disclosed deaminase alone on a first expression cassette, a second expression cassette encoding a fusion protein comprising a deaminase, and a nucleotide sequence encoding a gRNA on third expression cassette. Such an expression cassette is provided with a plurality of restriction sites and/or recombination sites for insertion of the polynucleotides to be under the transcriptional regulation of the regulatory regions. Expression cassettes which comprise a selectable marker gene may also be present.

(77) The expression cassette may include in the 5'-3' direction of transcription, a transcriptional (and, in some embodiments, translational) initiation region (i.e., a promoter), a deaminase-encoding polynucleotide of the invention, and a transcriptional (and in some embodiments, translational) termination region (i.e., termination region) functional in the organism of interest. The promoters of the invention are capable of directing or driving expression of a coding sequence in a host cell. The regulatory regions (e.g., promoters, transcriptional regulatory regions, and translational termination regions) may be endogenous or heterologous to the host cell or to each other. As used herein, “heterologous” in reference to a sequence is a sequence that originates from a foreign species, or, if from the same species, is substantially modified from its native form in composition and/or genomic locus by deliberate human intervention. As used herein, a chimeric gene comprises a coding sequence operably linked to a transcription initiation region that is heterologous to the coding sequence.

(78) Convenient termination regions are available from the Ti-plasmid of *A. tumefaciens*, such as the octopine synthase and nopaline synthase termination regions. See also Guerineau et al. (1991) *Mol. Gen. Genet.* 262:141-144; Proudfoot (1991) *Cell* 64:671-674; Sanfacon et al. (1991) *Genes Dev.* 5:141-149; Mogen et al. (1990) *Plant Cell* 2:1261-1272; Munroe et al. (1990) *Gene* 91:151-158; Ballas et al. (1989) *Nucleic Acids Res.* 17:7891-7903; and Joshi et al. (1987) *Nucleic Acids Res.* 15:9627-9639.

(79) Additional regulatory signals include, but are not limited to, transcriptional initiation start sites, operators, activators, enhancers, other regulatory elements, ribosomal binding sites, an initiation codon, termination signals, and the like. See, for example, U.S. Pat. Nos. 5,039,523 and 4,853,331; EPO 0480762A2; Sambrook et al. (1992) *Molecular Cloning: A Laboratory Manual*, ed. Maniatis et al. (Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y.), hereinafter “Sambrook 11”; Davis et al., eds. (1980) *Advanced Bacterial Genetics* (Cold Spring Harbor Laboratory Press), Cold Spring Harbor, N.Y., and the references cited therein.

(80) In preparing the expression cassette, the various DNA fragments may be manipulated, so as to provide for the DNA sequences in the proper orientation and, as appropriate, in the proper reading frame. Toward this end, adapters or linkers may be employed to join the DNA fragments or other manipulations may be involved to provide for convenient restriction sites, removal of superfluous DNA, removal of restriction sites, or the like. For this purpose, in vitro mutagenesis, primer repair, restriction, annealing, resubstitutions, e.g., transitions and transversions, may be involved.

(81) A number of promoters can be used in the practice of the invention. The promoters can be

selected based on the desired outcome. The nucleic acids can be combined with constitutive, inducible, growth stage-specific, cell type-specific, tissue-preferred, tissue-specific, or other promoters for expression in the organism of interest. See, for example, promoters set forth in WO 99/43838 and in U.S. Pat. Nos. 8,575,425; 7,790,846; 8,147,856; 8,586,832; 7,772,369; 7,534,939; 6,072,050; 5,659,026; 5,608,149; 5,608,144; 5,604,121; 5,569,597; 5,466,785; 5,399,680; 5,268,463; 5,608,142; and 6,177,611; herein incorporated by reference.

(82) For expression in plants, constitutive promoters also include CaMV 35S promoter (Odell et al. (1985) *Nature* 313:810-812); rice actin (McElroy et al. (1990) *Plant Cell* 2:163-171); ubiquitin (Christensen et al. (1989) *Plant Mol. Biol.* 12:619-632 and Christensen et al. (1992) *Plant Mol. Biol.* 18:675-689); pEMU (Last et al. (1991) *Theor. Appl. Genet.* 81:581-588); and MAS (Velten et al. (1984) *EMBO J.* 3:2723-2730).

(83) Examples of inducible promoters are the Adh1 promoter which is inducible by hypoxia or cold stress, the Hsp70 promoter which is inducible by heat stress, the PPKK promoter and the pepcarboxylase promoter which are both inducible by light. Also useful are promoters which are chemically inducible, such as the In2-2 promoter which is safener induced (U.S. Pat. No. 5,364,780), the Axig1 promoter which is auxin induced and tapetum specific but also active in callus (PCT US01/22169), the steroid-responsive promoters (see, for example, the ERE promoter which is estrogen induced, and the glucocorticoid-inducible promoter in Schena et al. (1991) *Proc. Natl. Acad. Sci. USA* 88:10421-10425 and McNellis et al. (1998) *Plant J.* 14(2):247-257) and tetracycline-inducible and tetracycline-repressible promoters (see, for example, Gatz et al. (1991) *Mol. Gen. Genet.* 227:229-237, and U.S. Pat. Nos. 5,814,618 and 5,789,156), herein incorporated by reference.

(84) In some embodiments, tissue-specific or tissue-preferred promoters are utilized to target expression of an expression construct within a particular tissue. In certain embodiments, the tissue-specific or tissue-preferred promoters are active in plant tissue. Examples of promoters under developmental control in plants include promoters that initiate transcription preferentially in certain tissues, such as leaves, roots, fruit, seeds, or flowers. A “tissue specific” promoter is a promoter that initiates transcription only in certain tissues. Unlike constitutive expression of genes, tissue-specific expression is the result of several interacting levels of gene regulation. As such, promoters from homologous or closely related plant species can be preferable to use to achieve efficient and reliable expression of transgenes in particular tissues. In some embodiments, the expression comprises a tissue-preferred promoter. A “tissue preferred” promoter is a promoter that initiates transcription preferentially, but not necessarily entirely or solely in certain tissues.

(85) In some embodiments, the nucleic acid molecules encoding a deaminase described herein comprise a cell type-specific promoter. A “cell type specific” promoter is a promoter that primarily drives expression in certain cell types in one or more organs. Some examples of plant cells in which cell type specific promoters functional in plants may be primarily active include, for example, BETL cells, vascular cells in roots, leaves, stalk cells, and stem cells. The nucleic acid molecules can also include cell type preferred promoters. A “cell type preferred” promoter is a promoter that primarily drives expression mostly, but not necessarily entirely or solely in certain cell types in one or more organs. Some examples of plant cells in which cell type preferred promoters functional in plants may be preferentially active include, for example, BETL cells, vascular cells in roots, leaves, stalk cells, and stem cells.

(86) In some embodiments, the nucleic acid sequences encoding the deaminases, fusion proteins, and/or gRNAs are operably linked to a promoter sequence that is recognized by a phage RNA polymerase for example, for in vitro mRNA synthesis. In such embodiments, the in vitro-transcribed RNA can be purified for use in the methods described herein. For example, the promoter sequence can be a T7, T3, or SP6 promoter sequence or a variation of a T7, T3, or SP6 promoter sequence. In such embodiments, the expressed protein and/or RNAs can be purified for use in the methods of genome modification described herein.

(87) In certain embodiments, the polynucleotide encoding the deaminase, fusion protein, and/or gRNA is linked to a polyadenylation signal (e.g., SV40 polyA signal and other signals functional in plants) and/or at least one transcriptional termination sequence. In some embodiments, the sequence encoding the deaminase or fusion protein is linked to sequence(s) encoding at least one nuclear localization signal, at least one cell-penetrating domain, and/or at least one signal peptide capable of trafficking proteins to particular subcellular locations, as described elsewhere herein.

(88) In some embodiments, the polynucleotide encoding the deaminase, fusion protein, and/or gRNA is present in a vector or multiple vectors. A “vector” refers to a polynucleotide composition for transferring, delivering, or introducing a nucleic acid into a host cell. Suitable vectors include plasmid vectors, phagemids, cosmids, artificial/mini-chromosomes, transposons, and viral vectors (e.g., lentiviral vectors, adeno-associated viral vectors, baculoviral vector). In some embodiments, the vector comprises additional expression control sequences (e.g., enhancer sequences, Kozak sequences, polyadenylation sequences, transcriptional termination sequences), selectable marker sequences (e.g., antibiotic resistance genes), origins of replication, and the like. Additional information can be found in “Current Protocols in Molecular Biology” Ausubel et al., John Wiley & Sons, New York, 2003 or “Molecular Cloning: A Laboratory Manual” Sambrook & Russell, Cold Spring Harbor Press, Cold Spring Harbor, N.Y., 3rd edition, 2001.

(89) In some embodiments, the vector comprises a selectable marker gene for the selection of transformed cells. Selectable marker genes are utilized for the selection of transformed cells or tissues. Marker genes include genes encoding antibiotic resistance, such as those encoding neomycin phosphotransferase II (NEO) and hygromycin phosphotransferase (HPT), as well as genes conferring resistance to herbicidal compounds, such as glufosinate ammonium, bromoxynil, imidazolinones, and 2,4-dichlorophenoxyacetate (2,4-D).

(90) In some embodiments, the expression cassette or vector comprising the sequence encoding a fusion protein comprising an RNA-guided DNA-binding polypeptide, such as an RGN, further comprises a sequence encoding a gRNA. In some embodiments, the sequence(s) encoding the gRNA are operably linked to at least one transcriptional control sequence for expression of the gRNA in the organism or host cell of interest. For example, the polynucleotide encoding the gRNA can be operably linked to a promoter sequence that is recognized by RNA polymerase III (Pol III). Examples of suitable Pol III promoters include, but are not limited to, mammalian U6, U3, H1, and 7SL RNA promoters and rice U6 and U3 promoters.

(91) As indicated, expression constructs comprising nucleotide sequences encoding the deaminases, fusion proteins, and/or gRNAs can be used to transform organisms of interest. Methods for transformation involve introducing a nucleotide construct into an organism of interest. By “introducing” is intended to introduce the nucleotide construct to the host cell in such a manner that the construct gains access to the interior of the host cell. The methods of the invention do not require a particular method for introducing a nucleotide construct to a host organism, only that the nucleotide construct gains access to the interior of at least one cell of the host organism. The host cell can be a eukaryotic or prokaryotic cell. In particular embodiments, the eukaryotic host cell is a plant cell, a mammalian cell, or an insect cell. Methods for introducing nucleotide constructs into plants and other host cells are known in the art including, but not limited to, stable transformation methods, transient transformation methods, and virus-mediated methods.

(92) The methods result in a transformed organism, such as a plant, including whole plants, as well as plant organs (e.g., leaves, stems, roots, etc.), seeds, plant cells, propagules, embryos and progeny of the same. Plant cells can be differentiated or undifferentiated (e.g. callus, suspension culture cells, protoplasts, leaf cells, root cells, phloem cells, pollen).

(93) “Transgenic organisms” or “transformed organisms” or “stably transformed” organisms or cells or tissues refers to organisms that have incorporated or integrated a polynucleotide encoding a deaminase of the invention. It is recognized that other exogenous or endogenous nucleic acid sequences or DNA fragments may also be incorporated into the host cell. *Agrobacterium*—and

biolistic-mediated transformation remain the two predominantly employed approaches for transformation of plant cells. However, transformation of a host cell may be performed by infection, transfection, microinjection, electroporation, microprojection, biolistics or particle bombardment, electroporation, silica/carbon fibers, ultrasound mediated, PEG mediated, calcium phosphate co-precipitation, polycation DMSO technique, DEAE dextran procedure, and viral mediated, liposome mediated and the like. Viral-mediated introduction of a polynucleotide encoding a deaminase, fusion protein, and/or gRNA includes retroviral, lentiviral, adenoviral, and adeno-associated viral mediated introduction and expression, as well as the use of Caulimoviruses (e.g., cauliflower mosaic virus), Geminiviruses (e.g., bean golden yellow mosaic virus or maize streak virus), and RNA plant viruses (e.g., tobacco mosaic virus).

(94) Transformation protocols as well as protocols for introducing polypeptides or polynucleotide sequences into plants may vary depending on the type of host cell (e.g., monocot or dicot plant cell) targeted for transformation. Methods for transformation are known in the art and include those set forth in U.S. Pat. Nos. 8,575,425; 7,692,068; 8,802,934; 7,541,517; each of which is herein incorporated by reference. See, also, Rakoczy-Trojanowska, M. (2002) *Cell Mol Biol Lett.* 7:849-858; Jones et al. (2005) *Plant Methods* 1:5; Rivera et al. (2012) *Physics of Life Reviews* 9:308-345; Bartlett et al. (2008) *Plant Methods* 4:1-12; Bates, G. W. (1999) *Methods in Molecular Biology* 111:359-366; Binns and Thomashow (1988) *Annual Reviews in Microbiology* 42:575-606; Christou, P. (1992) *The Plant Journal* 2:275-281; Christou, P. (1995) *Euphytica* 85:13-27; Tzfira et al. (2004) *TRENDS in Genetics* 20:375-383; Yao et al. (2006) *Journal of Experimental Botany* 57:3737-3746; Zupan and Zambryski (1995) *Plant Physiology* 107:1041-1047; Jones et al. (2005) *Plant Methods* 1:5;

(95) Transformation may result in stable or transient incorporation of the nucleic acid into the cell. “Stable transformation” is intended to mean that the nucleotide construct introduced into a host cell integrates into the genome of the host cell and is capable of being inherited by the progeny thereof “Transient transformation” is intended to mean that a polynucleotide is introduced into the host cell and does not integrate into the genome of the host cell.

(96) Methods for transformation of chloroplasts are known in the art. See, for example, Svab et al. (1990) *Proc. Natl. Acad. Sci. USA* 87:8526-8530; Svab and Maliga (1993) *Proc. Natl. Acad. Sci. USA* 90:913-917; Svab and Maliga (1993) *EMBO J.* 12:601-606. The method relies on particle gun delivery of DNA containing a selectable marker and targeting of the DNA to the plastid genome through homologous recombination. Additionally, plastid transformation can be accomplished by transactivation of a silent plastid-borne transgene by tissue-preferred expression of a nuclear-encoded and plastid-directed RNA polymerase. Such a system has been reported in McBride et al. (1994) *Proc. Natl. Acad. Sci. USA* 91:7301-7305.

(97) The cells that have been transformed may be grown into a transgenic organism, such as a plant, in accordance with conventional ways. See, for example, McCormick et al. (1986) *Plant Cell Reports* 5:81-84. These plants may then be grown, and either pollinated with the same transformed strain or different strains, and the resulting hybrid having the deaminase or fusion protein polynucleotide identified. Two or more generations may be grown to ensure that the deaminase or fusion protein polynucleotide is stably maintained and inherited and the seeds harvested to ensure the presence of the deaminase or fusion protein polynucleotide. In this manner, the present invention provides transformed seed (also referred to as “transgenic seed”) having a nucleotide construct of the invention, for example, an expression cassette of the invention, stably incorporated into their genome.

(98) In some embodiments, cells that have been transformed are introduced into an organism. These cells could have originated from the organism, wherein the cells are transformed in an ex vivo approach.

(99) The sequences provided herein may be used for transformation of any plant species, including, but not limited to, monocots and dicots. Examples of plants of interest include, but are not limited

to, corn (maize), sorghum, wheat, sunflower, tomato, crucifers, peppers, potato, cotton, rice, soybean, sugarbeet, sugarcane, tobacco, barley, and oilseed rape, *Brassica* sp., alfalfa, rye, millet, safflower, peanuts, sweet potato, cassava, coffee, coconut, pineapple, citrus trees, cocoa, tea, banana, avocado, fig, guava, mango, olive, papaya, cashew, macadamia, almond, oats, vegetables, ornamentals, and conifers.

(100) Vegetables include, but are not limited to, tomatoes, lettuce, green beans, lima beans, peas, and members of the genus *Curcumis* such as cucumber, cantaloupe, and musk melon. Ornamentals include, but are not limited to, azalea, hydrangea, hibiscus, roses, tulips, daffodils, petunias, carnation, poinsettia, and chrysanthemum. Preferably, plants of the present invention are crop plants (for example, maize, sorghum, wheat, sunflower, tomato, crucifers, peppers, potato, cotton, rice, soybean, sugarbeet, sugarcane, tobacco, barley, oilseed rape, etc.).

(101) As used herein, the term plant includes plant cells, plant protoplasts, plant cell tissue cultures from which plants can be regenerated, plant calli, plant clumps, and plant cells that are intact in plants or parts of plants such as embryos, pollen, ovules, seeds, leaves, flowers, branches, fruit, kernels, ears, cobs, husks, stalks, roots, root tips, anthers, and the like. Grain is intended to mean the mature seed produced by commercial growers for purposes other than growing or reproducing the species. Progeny, variants, and mutants of the regenerated plants are also included within the scope of the invention, provided that these parts comprise the introduced polynucleotides. Further provided is a processed plant product or byproduct that retains the sequences disclosed herein, including for example, soymeal.

(102) In some embodiments, the polynucleotides encoding the deaminases, fusion proteins, and/or gRNAs are used to transform any eukaryotic species, including but not limited to animals (e.g., mammals, insects, fish, birds, and reptiles), fungi, amoeba, algae, and yeast. In some embodiments, the polynucleotides encoding the deaminases, fusion proteins, and/or gRNAs are used to transform any prokaryotic species, including but not limited to, archaea and bacteria (e.g., *Bacillus* spp., *Klebsiella* spp., *Streptomyces* spp., *Rhizobium* spp., *Escherichia* spp., *Pseudomonas* spp., *Salmonella* spp., *Shigella* spp., *Vibrio* spp., *Yersinia* spp., *Mycoplasma* spp., *Agrobacterium* spp., and *Lactobacillus* spp.).

(103) In some embodiments, conventional viral and non-viral based gene transfer methods are used to introduce nucleic acids in mammalian cells or target tissues. Such methods can be used to administer nucleic acids encoding a deaminase or fusion protein of the invention and optionally a gRNA to cells in culture, or in a host organism. Non-viral vector delivery systems include DNA plasmids, RNA (e.g., a transcript of a vector described herein), naked nucleic acid, and nucleic acid complexed with a delivery vehicle, such as a liposome. Viral vector delivery systems include DNA and RNA viruses, which have either episomal or integrated genomes after delivery to the cell. Non-limiting examples include vectors utilizing Caulimoviruses (e.g., cauliflower mosaic virus), Geminiviruses (e.g., bean golden yellow mosaic virus or maize streak virus), and RNA plant viruses (e.g., tobacco mosaic virus). For a review of gene therapy procedures, see Anderson, *Science* 256: 808-813 (1992); Nabel & Feigner, *TIBTECH* 11:211-217 (1993); Mitani & Caskey, *TIBTECH* 11:162-166 (1993); Dillon, *TIBTECH* 11:167-175 (1993); Miller, *Nature* 357:455-460 (1992); Van Brunt, *Biotechnology* 6(10): 1149-1154 (1988); Vigne, *Restorative Neurology and Neuroscience* 8:35-36 (1995); Kremer & Perricaudet, *British Medical Bulletin* 51(1):31-44 (1995); Haddada et al., in *Current Topics in Microbiology and Immunology*, Doerfler and Bohm (eds) (1995); and Yu et al., *Gene Therapy* 1:13-26 (1994).

(104) Methods of non-viral delivery of nucleic acids include lipofection, *Agrobacterium*-mediated transformation, nucleofection, microinjection, biolistics, virosomes, liposomes, immunoliposomes, polycation or lipid: nucleic acid conjugates, naked DNA, artificial virions, and agent-enhanced uptake of DNA. Lipofection is described in e.g., U.S. Pat. Nos. 5,049,386, 4,946,787; and 4,897,355) and lipofection reagents are sold commercially (e.g., Transfectam™ and Lipofectin™). Cationic and neutral lipids that are suitable for efficient receptor-recognition lipofection of

polynucleotides include those of Feigner, WO 91/17424; WO 91/16024. Delivery can be to cells (e.g. in vitro or ex vivo administration) or target tissues (e.g. in vivo administration). The preparation of lipid:nucleic acid complexes, including targeted liposomes such as immunolipid complexes, is well known to one of skill in the art (see, e.g., Crystal, *Science* 270:404-410 (1995); Blaese et al., *Cancer Gene Ther.* 2:291-297 (1995); Behr et al., *Bioconjugate Chem.* 5:382-389 (1994); Remy et al., *Bioconjugate Chem.* 5:647-654 (1994); Gao et al., *Gene Therapy* 2:710-722 (1995); Ahmad et al., *Cancer Res.* 52:4817-4820 (1992); U.S. Pat. Nos. 4,186,183, 4,217,344, 4,235,871, 4,261,975, 4,485,054, 4,501,728, 4,774,085, 4,837,028, and 4,946,787).

(105) The use of RNA or DNA viral based systems for the delivery of nucleic acids takes advantage of highly evolved processes for targeting a virus to specific cells in the body and trafficking the viral payload to the nucleus. Viral vectors can be administered directly to patients (in vivo) or they can be used to treat cells in vitro, and the modified cells may optionally be administered to patients (ex vivo). Conventional viral based systems could include retroviral, lentivirus, adenoviral, adeno-associated and herpes simplex virus vectors for gene transfer. Integration in the host genome is possible with the retrovirus, lentivirus, and adeno-associated virus gene transfer methods, often resulting in long term expression of the inserted transgene. Additionally, high transduction efficiencies have been observed in many different cell types and target tissues.

(106) The tropism of a retrovirus can be altered by incorporating foreign envelope proteins, expanding the potential target population of target cells. Lentiviral vectors are retroviral vectors that are able to transduce or infect non-dividing cells and typically produce high viral titers. Selection of a retroviral gene transfer system would therefore depend on the target tissue. Retroviral vectors are comprised of cis-acting long terminal repeats with packaging capacity for up to 6-10 kb of foreign sequence. The minimum cis-acting LTRs are sufficient for replication and packaging of the vectors, which are then used to integrate the therapeutic gene into the target cell to provide permanent transgene expression. Widely used retroviral vectors include those based upon murine leukemia virus (MuLV), gibbon ape leukemia virus (GaLV), Simian Immuno deficiency virus (SIV), human immuno deficiency virus (HIV), and combinations thereof (see, e.g., Buchscher et al., *J. Virol.* 66:2731-2739 (1992); Johann et al., *J. Virol.* 66:1635-1640 (1992); Sommerfelt et al., *Viol.* 176:58-59 (1990); Wilson et al., *J. Virol.* 63:2374-2378 (1989); Miller et al., *J. Virol.* 65:2220-2224 (1991); PCT/US94/05700).

(107) In applications where transient expression is preferred, adenoviral based systems may be used. Adenoviral based vectors are capable of very high transduction efficiency in many cell types and do not require cell division. With such vectors, high titer and levels of expression have been obtained. This vector can be produced in large quantities in a relatively simple system. Adeno-associated virus ("AAV") vectors may also be used to transduce cells with target nucleic acids, e.g., in the in vitro production of nucleic acids and peptides, and for in vivo and ex vivo gene therapy procedures (see, e.g., West et al., *Virology* 160:38-47 (1987); U.S. Pat. No. 4,797,368; WO 93/24641; Katin, *Human Gene Therapy* 5:793-801 (1994); Muzyczka, *J. Clin. Invest.* 94:1351 (1994). Construction of recombinant AAV vectors are described in a number of publications, including U.S. Pat. No. 5,173,414; Tratschin et al., *Mol. Cell. Biol.* 5:3251-3260 (1985); Tratschin, et al., *Mol. Cell. Biol.* 4:2072-2081 (1984); Hermonat & Muzyczka, *PNAS* 81:6466-6470 (1984); and Samulski et al., *J. Virol.* 63:03822-3828 (1989). Packaging cells are typically used to form virus particles that are capable of infecting a host cell. Such cells include 293 cells, which package adenovirus, and ψ J2 cells or PA317 cells, which package retrovirus.

(108) Viral vectors used in gene therapy are usually generated by producing a cell line that packages a nucleic acid vector into a viral particle. The vectors typically contain the minimal viral sequences required for packaging and subsequent integration into a host, other viral sequences being replaced by an expression cassette for the polynucleotide(s) to be expressed. The missing viral functions are typically supplied in trans by the packaging cell line. For example, AAV vectors

used in gene therapy typically only possess ITR sequences from the AAV genome which are required for packaging and integration into the host genome. Viral DNA is packaged in a cell line, which contains a helper plasmid encoding the other AAV genes, namely rep and cap, but lacking ITR sequences.

(109) The cell line may also be infected with adenovirus as a helper. The helper virus promotes replication of the AAV vector and expression of AAV genes from the helper plasmid. The helper plasmid is not packaged in significant amounts due to a lack of ITR sequences. Contamination with adenovirus can be reduced by, e.g., heat treatment to which adenovirus is more sensitive than AAV. Additional methods for the delivery of nucleic acids to cells are known to those skilled in the art. See, for example, US20030087817, incorporated herein by reference.

(110) In some embodiments, a host cell is transiently or non-transiently transfected with one or more vectors described herein. In some embodiments, a cell is transfected as it naturally occurs in a subject. In some embodiments, a cell that is transfected is taken from a subject.

(111) In some embodiments, a cell that is transfected is a eukaryotic cell. In some embodiments, the eukaryotic cell is an animal cell (e.g., mammals, insects, fish, birds, and reptiles). In some embodiments, a cell that is transfected is a human cell. In some embodiments, a cell that is transfected is a cell of hematopoietic origin, such as an immune cell (i.e., a cell of the innate or adaptive immune system) including but not limited to a B cell, a T cell, a natural killer (NK) cell, a pluripotent stem cell, an induced pluripotent stem cell, a chimeric antigen receptor T (CAR-T) cell, a monocyte, a macrophage, and a dendritic cell.

(112) In some embodiments, the cell is derived from cells taken from a subject, such as a cell line. In some embodiments, the cell or cell line is prokaryotic. In some embodiments, the cell or cell line is eukaryotic. In further embodiments, the cell or cell line is derived from insect, avian, plant, or fungal species. In some embodiments, the cell or cell line may be mammalian, such as for example human, monkey, mouse, cow, swine, goat, hamster, rat, cat, or dog. A wide variety of cell lines for tissue culture are known in the art. Examples of cell lines include, but are not limited to, C8161, CCRF-CEM, MOLT, mIMCD-3, NHDF, HeLaS3, Huhl, Huh4, Huh7, HUVEC, HASMC, HEK_n, HEK_a, MiaPaCell, Panel, PC-3, TF1, CTLL-2, CIR, Rath, CVI, RPTE, A10, T24, 182, A375, ARH-77, Calul, SW480, SW620, SKOV3, SK-UT, CaCo2, P388D1, SEM-K2, WEHI-231, HB56, TIB55, lurkat, 145.01, LRMB, Bcl-1, BC-3, IC21, DLD2, Raw264.7, NRK, NRK-52E, MRCS, MEF, Hep G2, HeLa B, HeLa T4. COS, COS-1, COS-6, COS-M6A, BS-C-1 monkey kidney epithelial, BALB/3T3 mouse embryo fibroblast, 3T3 Swiss, 3T3-L1, 132-d5 human fetal fibroblasts; 10.1 mouse fibroblasts, 293-T, 3T3, 721, 9L, A2780, A2780ADR, A2780cis, A172, A20, A253, A431, A-549, ALC, B16, B35, BCP-I cells, BEAS-2B, bEnd.3, BHK-21, BR 293, BxPC3, C3H-10T1/2, C6/36, Cal-27, CHO, CHO-7, CHO—IR, CHO-K1, CHO-K2, CHO-T, CHO Dhfr^{-/-}, COR-L23, COR-L23/CPR, COR-L235010, CORL23/R23, COS-7, COV-434, CML T1, CMT, CT26, D17, DH82, DU145, DuCaP, EL4, EM2, EM3, EMT6/AR1, EMT6/AR10.0, FM3, H1299, H69, HB54, HB55, HCA2, HEK-293, HeLa, Hepalclc7, HL-60, HMEC, HT-29, lurkat, IY cells, K562 cells, Ku812, KCL22, KG1, KYO1, LNCap, Ma-Mel 1-48, MC-38, MCF-7, MCF-10A, MDA-MB-231, MDA-MB-468, MDA-MB-435, MDCKII, MDCKII, MOR/0.2R, MONO-MAC 6, MTD-1A, MyEnd, NCI-H69/CPR, NCI-H69/LX10, NCI-H69/LX20, NCI-H69/LX4, NIH-3T3, NALM-1, NW-145, OPCN/OPCT cell lines, Peer, PNT-1A/PNT 2, RenCa, RIN-5F, RMA/RMAS, Saos-2 cells, Sf-9, SkBr3, T2, T-47D, T84, THP1 cell line, U373, U87, U937, VCaP, Vero cells, WM39, WT-49, X63, YAC-1, YAR, and transgenic varieties thereof. Cell lines are available from a variety of sources known to those with skill in the art (see, e.g., the American Type Culture Collection (ATCC) (Manassas, Va.)).

(113) In some embodiments, a cell transfected with one or more vectors described herein is used to establish a new cell line comprising one or more vector-derived sequences. In some embodiments, a cell transiently transfected with a fusion protein of the invention and optionally a gRNA, or with a ribonucleoprotein complex of the invention, and modified through the activity of a fusion protein

or ribonucleoprotein complex, is used to establish a new cell line comprising cells containing the modification but lacking any other exogenous sequence. In some embodiments, cells transiently or non-transiently transfected with one or more vectors described herein, or cell lines derived from such cells are used in assessing one or more test compounds.

(114) In some embodiments, one or more vectors described herein are used to produce a non-human transgenic animal or transgenic plant. In some embodiments, the transgenic animal is an insect. In further embodiments, the insect is an insect pest, such as a mosquito or tick. In some embodiments, the insect is a plant pest, such as a corn rootworm or a fall armyworm. In some embodiments, the transgenic animal is a bird, such as a chicken, turkey, goose, or duck. In some embodiments, the transgenic animal is a mammal, such as a human, mouse, rat, hamster, monkey, ape, rabbit, swine, cow, horse, goat, sheep, cat, or dog.

VI. Variants and Fragments of Polypeptides and Polynucleotides

(115) The present disclosure provides novel adenine deaminases which are active on DNA molecules, the amino acid sequence of which are set forth as SEQ ID NO: 1-10 and 399-441, active variants or fragments thereof, and polynucleotides encoding the same.

(116) While the activity of a variant or fragment may be altered compared to the polynucleotide or polypeptide of interest, the variant and fragment should retain the functionality of the polynucleotide or polypeptide of interest. For example, a variant or fragment may have increased activity, decreased activity, different spectrum of activity or any other alteration in activity when compared to the polynucleotide or polypeptide of interest.

(117) Fragments and variants of deaminases of the invention which have adenine deaminase activity will retain said activity if they are part of a fusion protein further comprising a DNA-binding polypeptide or a fragment thereof.

(118) The term “fragment” refers to a portion of a polynucleotide or polypeptide sequence of the invention. “Fragments” or “biologically active portions” include polynucleotides comprising a sufficient number of contiguous nucleotides to retain the biological activity (i.e., deaminase activity on nucleic acids). “Fragments” or “biologically active portions” include polypeptides comprising a sufficient number of contiguous amino acid residues to retain the biological activity. Fragments of the deaminases disclosed herein include those that are shorter than the full-length sequences due to the use of an alternate downstream start site. In some embodiments, a biologically active portion of a deaminase is a polypeptide that comprises, for example, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, or more contiguous amino acid residues of any of SEQ ID NOs: 1-10 and 399-441, or a variant thereof. Such biologically active portions can be prepared by recombinant techniques and evaluated for activity.

(119) In general, “variants” is intended to mean substantially similar sequences. For polynucleotides, a variant comprises a deletion and/or addition of one or more nucleotides at one or more internal sites within the native polynucleotide and/or a substitution of one or more nucleotides at one or more sites in the native polynucleotide. As used herein, a “native” or “wild type” polynucleotide or polypeptide comprises a naturally occurring nucleotide sequence or amino acid sequence, respectively. For polynucleotides, conservative variants include those sequences that, because of the degeneracy of the genetic code, encode the native amino acid sequence of the gene of interest. Naturally occurring allelic variants such as these can be identified with the use of well-known molecular biology techniques, as, for example, with polymerase chain reaction (PCR) and hybridization techniques as outlined below. Variant polynucleotides also include synthetically derived polynucleotides, such as those generated, for example, by using site-directed mutagenesis but which still encode the polypeptide or the polynucleotide of interest. Generally, variants of a particular polynucleotide disclosed herein will have at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% or more sequence identity to that particular polynucleotide as determined

by sequence alignment programs and parameters described elsewhere herein.

(120) Variants of a particular polynucleotide disclosed herein (i.e., the reference polynucleotide) can also be evaluated by comparison of the percent sequence identity between the polypeptide encoded by a variant polynucleotide and the polypeptide encoded by the reference polynucleotide. Percent sequence identity between any two polypeptides can be calculated using sequence alignment programs and parameters described elsewhere herein. Where any given pair of polynucleotides disclosed herein is evaluated by comparison of the percent sequence identity shared by the two polypeptides they encode, the percent sequence identity between the two encoded polypeptides is at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% or more sequence identity.

(121) In particular embodiments, the presently disclosed polynucleotides encode an adenine deaminase comprising an amino acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or greater identity to an amino acid sequence of any of SEQ ID NOs: 1-10 and 399-441.

(122) A biologically active variant of an adenine deaminase of the invention may differ by as few as 1-15 amino acid residues, as few as 1-10, such as 6-10, as few as 5, as few as 4, as few as 3, as few as 2, or as few as 1 amino acid residue. In specific embodiments, the polypeptides comprise an N-terminal or a C-terminal truncation, which can comprise at least a deletion of 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60 amino acids or more from either the N or C terminus of the polypeptide. In some embodiments, the polypeptides comprise an internal deletion which can comprise at least a deletion of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 25, 30, 35, 40, 45, 50, 55, 60 amino acids or more.

(123) It is recognized that modifications may be made to the deaminases provided herein creating variant proteins and polynucleotides. Changes designed by man may be introduced through the application of site-directed mutagenesis techniques. In some embodiments, native, as yet-unknown or as yet unidentified polynucleotides and/or polypeptides structurally and/or functionally-related to the sequences disclosed herein may also be identified that fall within the scope of the present invention. Conservative amino acid substitutions may be made in nonconserved regions that do not alter the function of the polypeptide as an adenine deaminase. In some embodiments, modifications are made that improve the adenine deaminase activity of the deaminase.

(124) Variant polynucleotides and proteins also encompass sequences and proteins derived from a mutagenic and recombinogenic procedure such as DNA shuffling. With such a procedure, one or more different deaminases disclosed herein (e.g., SEQ ID NO: 1-10 and 399-441) is manipulated to create a new adenine deaminase possessing the desired properties. In this manner, libraries of recombinant polynucleotides are generated from a population of related sequence polynucleotides comprising sequence regions that have substantial sequence identity and can be homologously recombined in vitro or in vivo. For example, using this approach, sequence motifs encoding a domain of interest may be shuffled between the deaminase sequences provided herein and other subsequently identified deaminase genes to obtain a new gene coding for a protein with an improved property of interest, such as an increased K_{sub}.m in the case of an enzyme. Strategies for such DNA shuffling are known in the art. See, for example, Stemmer (1994) *Proc. Natl. Acad. Sci. USA* 91:10747-10751; Stemmer (1994) *Nature* 370:389-391; Cramer et al. (1997) *Nature Biotech.* 15:436-438; Moore et al. (1997) *J Mol. Biol.* 272:336-347; Zhang et al. (1997) *Proc. Natl. Acad. Sci. USA* 94:4504-4509; Cramer et al. (1998) *Nature* 391:288-291; and U.S. Pat. Nos. 5,605,793 and 5,837,458. A “shuffled” nucleic acid is a nucleic acid produced by a shuffling

procedure such as any shuffling procedure set forth herein. Shuffled nucleic acids are produced by recombining (physically or virtually) two or more nucleic acids (or character strings), for example in an artificial, and optionally recursive, fashion. Generally, one or more screening steps are used in shuffling processes to identify nucleic acids of interest; this screening step can be performed before or after any recombination step. In some (but not all) shuffling embodiments, it is desirable to perform multiple rounds of recombination prior to selection to increase the diversity of the pool to be screened. The overall process of recombination and selection are optionally repeated recursively. Depending on context, shuffling can refer to an overall process of recombination and selection, or, alternately, can simply refer to the recombinational portions of the overall process.

(125) As used herein, “sequence identity” or “identity” in the context of two polynucleotides or polypeptide sequences makes reference to the residues in the two sequences that are the same when aligned for maximum correspondence over a specified comparison window. When percentage of sequence identity is used in reference to proteins it is recognized that residue positions which are not identical often differ by conservative amino acid substitutions, where amino acid residues are substituted for other amino acid residues with similar chemical properties (e.g., charge or hydrophobicity) and therefore do not change the functional properties of the molecule. When sequences differ in conservative substitutions, the percent sequence identity may be adjusted upwards to correct for the conservative nature of the substitution. Sequences that differ by such conservative substitutions are said to have “sequence similarity” or “similarity”. Means for making this adjustment are well known to those of skill in the art. Typically, this involves scoring a conservative substitution as a partial rather than a full mismatch, thereby increasing the percentage sequence identity. Thus, for example, where an identical amino acid is given a score of 1 and a non-conservative substitution is given a score of zero, a conservative substitution is given a score between zero and 1. The scoring of conservative substitutions is calculated, e.g., as implemented in the program PC/GENE (Intelligenetics, Mountain View, California).

(126) As used herein, “percentage of sequence identity” means the value determined by comparing two optimally aligned sequences over a comparison window, wherein the portion of the polynucleotide sequence in the comparison window may comprise additions or deletions (i.e., gaps) as compared to the reference sequence (which does not comprise additions or deletions) for optimal alignment of the two sequences. The percentage is calculated by determining the number of positions at which the identical nucleic acid base or amino acid residue occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison, and multiplying the result by 100 to yield the percentage of sequence identity.

(127) Unless otherwise stated, sequence identity/similarity values provided herein refer to the value obtained using GAP Version 10 using the following parameters: % identity and % similarity for a nucleotide sequence using GAP Weight of 50 and Length Weight of 3, and the nwsgapdna.cmp scoring matrix; % identity and % similarity for an amino acid sequence using GAP Weight of 8 and Length Weight of 2, and the BLOSUM62 scoring matrix; or any equivalent program thereof. By “equivalent program” is intended any sequence comparison program that, for any two sequences in question, generates an alignment having identical nucleotide or amino acid residue matches and an identical percent sequence identity when compared to the corresponding alignment generated by GAP Version 10.

(128) Two sequences are “optimally aligned” when they are aligned for similarity scoring using a defined amino acid substitution matrix (e.g., BLOSUM62), gap existence penalty and gap extension penalty so as to arrive at the highest score possible for that pair of sequences. Amino acid substitution matrices and their use in quantifying the similarity between two sequences are well-known in the art and described, e.g., in Dayhoff et al. (1978) “A model of evolutionary change in proteins.” In “Atlas of Protein Sequence and Structure,” Vol. 5, Suppl. 3 (ed. M. O. Dayhoff), pp. 345-352. Natl. Biomed. Res. Found., Washington, D.C. and Henikoff et al. (1992) *Proc. Natl.*

Acad. Sci. USA 89:10915-10919. The BLOSUM62 matrix is often used as a default scoring substitution matrix in sequence alignment protocols. The gap existence penalty is imposed for the introduction of a single amino acid gap in one of the aligned sequences, and the gap extension penalty is imposed for each additional empty amino acid position inserted into an already opened gap. The alignment is defined by the amino acids positions of each sequence at which the alignment begins and ends, and optionally by the insertion of a gap or multiple gaps in one or both sequences, so as to arrive at the highest possible score. While optimal alignment and scoring can be accomplished manually, the process is facilitated by the use of a computer-implemented alignment algorithm, e.g., gapped BLAST 2.0, described in Altschul et al. (1997) *Nucleic Acids Res.* 25:3389-3402, and made available to the public at the National Center for Biotechnology Information Website (World Wide Web at ncbi.nlm.nih.gov). Optimal alignments, including multiple alignments, can be prepared using, e.g., PSI-BLAST, available through World Wide Web at ncbi.nlm.nih.gov and described by Altschul et al. (1997) *Nucleic Acids Res.* 25:3389-3402. (129) With respect to an amino acid sequence that is optimally aligned with a reference sequence, an amino acid residue "corresponds to" the position in the reference sequence with which the residue is paired in the alignment. The "position" is denoted by a number that sequentially identifies each amino acid in the reference sequence based on its position relative to the N-terminus. Owing to deletions, insertion, truncations, fusions, etc., that must be taken into account when determining an optimal alignment, in general the amino acid residue number in a test sequence as determined by simply counting from the N-terminal will not necessarily be the same as the number of its corresponding position in the reference sequence. For example, in a case where there is a deletion in an aligned test sequence, there will be no amino acid that corresponds to a position in the reference sequence at the site of deletion. Where there is an insertion in an aligned reference sequence, that insertion will not correspond to any amino acid position in the reference sequence. In the case of truncations or fusions there can be stretches of amino acids in either the reference or aligned sequence that do not correspond to any amino acid in the corresponding sequence.

VII. Antibodies

(130) Antibodies to the deaminases, fusion proteins, or ribonucleoproteins comprising the deaminases of the present invention, including those having the amino acid sequence set forth as any one of SEQ ID NOs: 1-10 and 399-441 or active variants or fragments thereof, are also encompassed. Methods for producing antibodies are well known in the art (see, for example, Harlow and Lane (1988) *Antibodies: A Laboratory Manual*, Cold Spring Harbor Laboratory, Cold Spring Harbor, N.Y.; and U.S. Pat. No. 4,196,265). These antibodies can be used in kits for the detection and isolation of deaminases or fusion proteins or ribonucleoproteins comprising deaminases described herein. Thus, this disclosure provides kits comprising antibodies that specifically bind to the polypeptides or ribonucleoproteins described herein, including, for example, polypeptides comprising a sequence of at least 85% identity to any of SEQ ID NOs: 1-10 and 399-441.

VIII. Systems and Ribonucleoprotein Complexes for Binding and/or Modifying a Target Sequence of Interest and Methods of Making the Same

(131) The present disclosure provides a system which targets to a nucleic acid sequence and modifies a target nucleic acid sequence. In some embodiments, an RNA-guided, DNA-binding polypeptide, such as an RGN, and the gRNA are responsible for targeting the ribonucleoprotein complex to a nucleic acid sequence of interest; the deaminase polypeptide fused to the RGDBP is responsible for modifying the targeted nucleic acid sequence from A>N. In some embodiments, the deaminase converts A>G. The guide RNA hybridizes to the target sequence of interest and also forms a complex with the RNA-guided, DNA-binding polypeptide, thereby directing the RNA-guided, DNA-binding polypeptide to bind to the target sequence. The RNA-guided, DNA-binding polypeptide is one domain of a fusion protein; the second domain is a deaminase described herein.

In some embodiments, the RNA-guided, DNA-binding polypeptide is an RGN, such as a Cas9. Other examples of RNA-guided, DNA-binding polypeptides include RGNs such as those described in International Patent Application Publication Nos. WO 2019/236566 and WO 2020/139783. In some embodiments, the RNA-guided, DNA-binding polypeptide is a Type II CRISPR-Cas polypeptide, or an active variant or fragment thereof. In some embodiments, the RNA-guided, DNA-binding polypeptide is a Type V CRISPR-Cas polypeptide, or an active variant or fragment thereof. In some embodiments, the RNA-guided, DNA-binding polypeptide is a Type VI CRISPR-Cas polypeptide. In some embodiments, the DNA-binding domain of the fusion protein does not require an RNA guide, such as a zinc finger nuclease, TALEN, or meganuclease polypeptide. In some embodiments, the nuclease activity of a DNA-binding domain has been partially or completely inactivated. In further embodiments, the RNA-guided, DNA-binding polypeptide comprises an amino acid sequence of an RGN, such as for example APG07433.1 (SEQ ID NO: 41), or an active variant or fragment thereof such as nickase nAPG07433.1 (SEQ ID NO: 42) or other nickase RGN variants described in the Examples (SEQ ID NOs: 52-59, 61, 397, and 398). (132) In some embodiments, the system for binding and modifying a target sequence of interest provided herein is a ribonucleoprotein complex, which is at least one molecule of an RNA bound to at least one protein. The ribonucleoprotein complexes provided herein comprise at least one guide RNA as the RNA component and a fusion protein comprising a deaminase of the invention and an RNA-guided, DNA-binding polypeptide as the protein component. In some embodiments, the ribonucleoprotein complex is purified from a cell or organism that has been transformed with polynucleotides that encode the fusion protein and a guide RNA and cultured under conditions to allow for the expression of the fusion protein and guide RNA.

(133) In various embodiments, ribonucleoprotein complexes comprising any of the fusion proteins described herein and a guide RNA bound to the DNA-binding polypeptide of the fusion protein, are provided. For example, provided herein is a ribonucleoprotein complex comprising a fusion protein with a deaminase comprising an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 407. In another instance, a ribonucleoprotein complex comprising a fusion protein with a deaminase comprising an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 399, is provided. In yet another example, a ribonucleoprotein complex comprising a fusion protein with a deaminase comprising an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 405, is provided. In some of those embodiments of the ribonucleoprotein complexes described above, the fusion protein comprises an RGN selected from a CasX, a CasY, a C2c1, a C2c2, a C2c3, a GeoCas9, a SpCas9, a SaCas9, a Nme2Cas9, a CjCas9, a Cas12a (formerly known as Cpf1), a Cas12b, a Cas12g, a Cas12h, a Cas12i, a LbCas12a, a AsCas12a, a CasMINI, a Cas13b, a Cas13c, a Cas13d, a Cas14, a Csn2, an xCas9, an SpCas9-NG, an LbCas12a, an AsCas12a, a Cas9-KKH, a circularly permuted Cas9, an Argonaute (Ago), a SmacCas9, a Spy-macCas9 domain, or an RGN with an amino acid sequence set forth in any one of SEQ ID NOs: 41, 60, 366, or 368. In some embodiments, the ribonucleoprotein complex comprises a nickase having an amino acid sequence with at least 95% sequence identity to any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398, fused to a deaminase comprising an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 407. In some embodiments, the ribonucleoprotein complex comprises a nickase having an amino acid sequence with at least 95% sequence identity to any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398, fused to a deaminase comprising an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 399. In some embodiments, the ribonucleoprotein complex comprises a nickase having an amino acid sequence with at least 95% sequence identity to any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398, fused to a deaminase comprising an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 405. In some embodiments, the ribonucleoprotein complex comprises a Cas9 nickase fused to a deaminase comprising an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 407. In some embodiments, the ribonucleoprotein complex comprises a Cas9 nickase fused to a deaminase

comprising an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 399. In some embodiments, the ribonucleoprotein complex comprises a Cas9 nickase fused to a deaminase comprising an amino acid sequence having at least 80% sequence identity to SEQ ID NO: 405. The Cas9 nickase, can be any Cas9 nickase disclosed in PCT Patent Publication No. WO2020181195, the entire contents of which is incorporated herein by reference herein. In various embodiments described herein, the ribonucleoprotein complex may also contain the gRNAs described herein.

(134) Methods are provided for making a deaminase, a fusion protein, or a fusion protein ribonucleoprotein complex. Such methods comprise culturing a cell comprising a nucleotide sequence encoding a deaminase, a fusion protein, and in some embodiments a nucleotide sequence encoding a guide RNA, under conditions in which the deaminase or fusion protein (and in some embodiments, the guide RNA) is expressed. The deaminase, fusion protein, or fusion ribonucleoprotein can then be purified from a lysate of the cultured cells.

(135) Methods for purifying a deaminase, fusion protein, or fusion ribonucleoprotein complex from a lysate of a biological sample are known in the art (e.g., size exclusion and/or affinity chromatography, 2D-PAGE, HPLC, reversed-phase chromatography, immunoprecipitation). In particular methods, the deaminase or fusion protein is recombinantly produced and comprises a purification tag to aid in its purification, including but not limited to, glutathione-S-transferase (GST), chitin binding protein (CBP), maltose binding protein, thioredoxin (TRX), poly(NANP), tandem affinity purification (TAP) tag, myc, AcV5, AU1, AU5, E, ECS, E2, FLAG, HA, nus, Softag 1, Softag 3, Strep, SBP, Glu-Glu, HSV, KT3, S, 51, T7, V5, VSV-G, 6×His, biotin carboxyl carrier protein (BCCP), and calmodulin. Generally, the tagged deaminase, fusion protein, or fusion ribonucleoprotein complex is purified using immunoprecipitation or other similar methods known in the art.

(136) An “isolated” or “purified” polypeptide, or biologically active portion thereof, is substantially or essentially free from components that normally accompany or interact with the polypeptide as found in its naturally occurring environment. Thus, an isolated or purified polypeptide is substantially free of other cellular material, or culture medium when produced by recombinant techniques, or substantially free of chemical precursors or other chemicals when chemically synthesized. A protein that is substantially free of cellular material includes preparations of protein having less than 30%, less than 20%, less than 10%, less than 5%, or less than 1% (by dry weight) of contaminating protein. When the protein of the invention or biologically active portion thereof is recombinantly produced, optimally culture medium represents less than 30%, less than 20%, less than 10%, less than 5%, or less than 1% (by dry weight) of chemical precursors or non-protein-of-interest chemicals.

(137) Particular methods provided herein for binding and/or cleaving a target sequence of interest involve the use of a ribonucleoprotein complex. In some embodiments, the ribonucleoprotein complex is assembled in vitro. In vitro assembly of a ribonucleoprotein complex can be performed using any method known in the art in which an RGDBP polypeptide or a fusion protein comprising the same is contacted with a guide RNA under conditions to allow for binding of the RGDBP polypeptide or fusion protein comprising the same to the guide RNA. As used herein, “contact”, “contacting”, “contacted,” refer to placing the components of a desired reaction together under conditions suitable for carrying out the desired reaction. In some embodiments of the described methods for modifying a target DNA molecule, the step of contacting is performed in vitro. In some embodiments, the step of contacting is performed in vivo. In some embodiments, the step of contacting is performed in a subject (e.g., a human subject or a non-human animal subject). In some embodiments, the step of contacting is performed in a cell, such as a human or non-human animal cell. The RGDBP polypeptide or fusion protein comprising the same can be purified from a biological sample, cell lysate, or culture medium, produced via in vitro translation, or chemically synthesized. The guide RNA can be purified from a biological sample, cell lysate, or culture medium, transcribed in vitro, or chemically synthesized. The RGDBP polypeptide or fusion protein

comprising the same and guide RNA can be brought into contact in solution (e.g., buffered saline solution) to allow for in vitro assembly of the ribonucleoprotein complex.

IX. Methods of Modifying a Target Sequence

(138) The present disclosure provides methods for modifying a target nucleic acid molecule (e.g., target DNA molecule) of interest. The methods include delivering a fusion protein comprising a DNA-binding polypeptide and at least one deaminase of the invention or a polynucleotide encoding the same to a target sequence or a cell, organelle, or embryo comprising a target sequence. In certain embodiments, the methods include delivering a system comprising at least one guide RNA or a polynucleotide encoding the same, and at least one fusion protein comprising at least one deaminase of the invention and an RNA-guided, DNA-binding polypeptide or a polynucleotide encoding the same to the target sequence or a cell, organelle, or embryo comprising the target sequence. In some embodiments, the fusion protein comprises any one of the amino acid sequences of SEQ ID NOs: 1-10 and 399-441, or an active variant or fragment thereof.

(139) In some embodiments, the methods comprise contacting a DNA molecule with (a) a fusion protein comprising a deaminase and an RNA-guided, DNA-binding polypeptide, such as for example a nuclease-inactive or a nickase Cas9 domain; and (b) a gRNA targeting the fusion protein of (a) to a target nucleotide sequence of the DNA molecule; wherein the DNA molecule is contacted with the fusion protein and the gRNA in an amount effective and under conditions suitable for the deamination of a nucleobase. In some embodiments, the target DNA molecule comprises a sequence associated with a disease or disorder, and wherein the deamination of the nucleobase results in a sequence that is not associated with a disease or disorder. In some embodiments, the disease or disorder affects animals. In further embodiments, the disease or disorder affects mammals, such as humans, cows, horses, dogs, cats, goats, sheep, swine, monkeys, rats, mice, or hamsters. In some embodiments, the target DNA sequence resides in an allele of a crop plant, wherein the particular allele of the trait of interest results in a plant of lesser agronomic value. The deamination of the nucleobase results in an allele that improves the trait and increases the agronomic value of the plant.

(140) In those embodiments wherein the method comprises delivering a polynucleotide encoding a guide RNA and/or a fusion protein, the cell or embryo can then be cultured under conditions in which the guide RNA and/or fusion protein are expressed. In various embodiments, the method comprises contacting a target sequence with a ribonucleoprotein complex comprising a gRNA and a fusion protein (which comprises a deaminase of the invention and an RNA-guided DNA-binding polypeptide). In certain embodiments, the method comprises introducing into a cell, organelle, or embryo comprising a target sequence a ribonucleoprotein complex of the invention. The ribonucleoprotein complex of the invention can be one that has been purified from a biological sample, recombinantly produced and subsequently purified, or in vitro-assembled as described herein. In those embodiments wherein the ribonucleoprotein complex that is contacted with the target sequence or a cell organelle, or embryo has been assembled in vitro, the method can further comprise the in vitro assembly of the complex prior to contact with the target sequence, cell, organelle, or embryo.

(141) A purified or in vitro assembled ribonucleoprotein complex of the invention can be introduced into a cell, organelle, or embryo using any method known in the art, including, but not limited to electroporation. In some embodiments, a fusion protein comprising a deaminase of the invention and an RNA-guided, DNA-binding polypeptide, and a polynucleotide encoding or comprising the guide RNA is introduced into a cell, organelle, or embryo using any method known in the art (e.g., electroporation).

(142) Upon delivery to or contact with the target sequence or cell, organelle, or embryo comprising the target sequence, the guide RNA directs the fusion protein to bind to the target sequence in a sequence-specific manner. The target sequence can subsequently be modified via the deaminase domain of the fusion protein. In some embodiments, the binding of this fusion protein to a target

sequence results in modification of a nucleotide adjacent to the target sequence. The nucleobase adjacent to the target sequence that is modified by the deaminase may be 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, or 100 base pairs from the 5' or 3' end of the target sequence. A fusion protein comprising a deaminase of the invention and an RNA-guided, DNA-binding polypeptide can introduce targeted A>N, and preferably targeted A>G, mutations in the targeted DNA molecule.

(143) In some embodiments of the described methods for modifying a target DNA molecule, the step of contacting is performed in vitro. In particular embodiments, the step of contacting is performed in vivo. In some embodiments, the step of contacting is performed in a subject (e.g., a human subject or a non-human animal subject). In some embodiments, the step of contacting is performed in a cell, such as a human or non-human animal cell.

(144) Methods to measure binding of the fusion protein to a target sequence are known in the art and include chromatin immunoprecipitation assays, gel mobility shift assays, DNA pull-down assays, reporter assays, microplate capture and detection assays. Likewise, methods to measure cleavage or modification of a target sequence are known in the art and include in vitro or in vivo cleavage assays wherein cleavage is confirmed using PCR, sequencing, or gel electrophoresis, with or without the attachment of an appropriate label (e.g., radioisotope, fluorescent substance) to the target sequence to facilitate detection of degradation products. In some embodiments, the nicking triggered exponential amplification reaction (NTEXPAR) assay is used (see, e.g., Zhang et al. (2016) *Chem. Sci.* 7:4951-4957). In vivo cleavage can be evaluated using the Surveyor assay (Guschin et al. (2010) *Methods Mol Biol* 649:247-256).

(145) In some embodiments, the methods involve the use of an RNA-binding, DNA-guided domain, as part of the fusion protein, complexed with more than one guide RNA. The more than one guide RNA can target different regions of a single gene or can target multiple genes. This multiple targeting enables the deaminase domain of the fusion protein to modify nucleic acids, thereby introducing multiple mutations in the target nucleic acid molecule (e.g., genome) of interest.

(146) In those embodiments wherein the method involves the use of an RNA-guided nuclease (RGN), such as a nickase RGN (i.e., is only able to cleave a single strand of a double-stranded polynucleotide, for example nAPG07433.1 (SEQ ID NO: 42 or SEQ ID NOs: 50-57), the method can comprise introducing two different RGNs or RGN variants that target identical or overlapping target sequences and cleave different strands of the polynucleotide. For example, an RGN nickase that only cleaves the positive (+) strand of a double-stranded polynucleotide can be introduced along with a second RGN nickase that only cleaves the negative (−) strand of a double-stranded polynucleotide. In some embodiments, two different fusion proteins are provided, where each fusion protein comprises a different RGN with a different PAM recognition sequence, so that a greater diversity of nucleotide sequences may be targeted for mutation.

(147) One of ordinary skill in the art will appreciate that any of the presently disclosed methods can be used to target a single target sequence or multiple target sequences. Thus, methods comprise the use of a fusion protein comprising a single RNA-guided, DNA-binding polypeptide in combination with multiple, distinct guide RNAs, which can target multiple, distinct sequences within a single gene and/or multiple genes. The deaminase domain of the fusion protein would then introduce mutations at each of the targeted sequences. Also encompassed herein are methods wherein multiple, distinct guide RNAs are introduced in combination with multiple, distinct RNA-guided, DNA binding polypeptides. Such RNA-guided, DNA-binding polypeptides may be multiple RGN or RGN variants. These guide RNAs and guide RNA/fusion protein systems can target multiple, distinct sequences within a single gene and/or multiple genes.

(148) In some embodiments, a fusion protein comprising an RNA-guided, DNA-binding polypeptide and a deaminase polypeptide of the invention may be used for generating mutations in a targeted gene or targeted region of a gene of interest. In some embodiments, a fusion protein of

the invention may be used for saturation mutagenesis of a targeted gene or region of a targeted gene of interest followed by high-throughput forward genetic screening to identify novel mutations and/or phenotypes. In some embodiments, a fusion protein described herein may be used for generating mutations in a targeted genomic location, which may or may not comprise coding DNA sequence. Libraries of cell lines generated by the targeted mutagenesis described above may also be useful for study of gene function or gene expression.

X. Target Polynucleotides

(149) In one aspect, the invention provides for methods of modifying a target polynucleotide in a eukaryotic cell, which may be in vivo, ex vivo or in vitro. In some embodiments, the method comprises sampling a cell or population of cells from a human or non-human animal or plant (including microalgae) and modifying the cell or cells. Culturing may occur at any stage ex vivo. The cell or cells may even be re-introduced into the human, non-human animal or plant (including micro-algae).

(150) Using natural variability, plant breeders combine most useful genes for desirable qualities, such as yield, quality, uniformity, hardiness, and resistance against pests. These desirable qualities also include growth, day length preferences, temperature requirements, initiation date of floral or reproductive development, fatty acid content, insect resistance, disease resistance, nematode resistance, fungal resistance, herbicide resistance, tolerance to various environmental factors including drought, heat, wet, cold, wind, and adverse soil conditions including high salinity. The sources of these useful genes include native or foreign varieties, heirloom varieties, wild plant relatives, and induced mutations, e.g., treating plant material with mutagenic agents. Using the present invention, plant breeders are provided with a new tool to induce mutations. Accordingly, one skilled in the art can employ the present invention to induce the rise of useful genes, with more precision than previous mutagenic agents and hence accelerate and improve plant breeding programs.

(151) The target polynucleotide of a deaminase or a fusion protein of the invention can be any polynucleotide endogenous or exogenous to the eukaryotic cell. For example, the target polynucleotide can be a polynucleotide residing in the nucleus of the eukaryotic cell. In some embodiments, the target polynucleotide is a sequence coding a gene product (e.g., a protein) or a non-coding sequence (e.g., a regulatory polynucleotide or a junk DNA). In some embodiments, the target sequence for a fusion protein of the invention is associated with a PAM (protospacer adjacent motif); that is, a short sequence recognized by the RNA-guided DNA-binding polypeptide. The precise sequence and length requirements for the PAM differ depending on the RNA-guided DNA-binding polypeptide used, but PAMs are typically 2-5 base pair sequences adjacent the protospacer (that is, the target sequence).

(152) The target polynucleotide of a fusion protein of the invention may include a number of disease-associated genes and polynucleotides as well as signaling biochemical pathway-associated genes and polynucleotides. Examples of target polynucleotides include a sequence associated with a signaling biochemical pathway, e.g., a signaling biochemical pathway-associated gene or polynucleotide. Examples of target polynucleotides include a disease associated gene or polynucleotide. A “disease-associated” gene or polynucleotide refers to any gene or polynucleotide which is yielding transcription or translation products at an abnormal level or in an abnormal form in cells derived from a disease-affected tissues compared with tissues or cells of a non-disease control. It may be a gene that becomes expressed at an abnormally high level; it may be a gene that becomes expressed at an abnormally low level, where the altered expression correlates with the occurrence and/or progression of the disease. A disease-associated gene also refers to a gene possessing mutation(s) or genetic variation that is directly responsible or is in linkage disequilibrium with a gene(s) that is responsible for the etiology of a disease (e.g., a causal mutation). The transcribed or translated products may be known or unknown, and further may be at a normal or abnormal level.

(153) Non-limiting examples of disease-associated genes that can be targeted using the presently disclosed methods and compositions are provided in Table 34. In some embodiments, the disease-associated gene that is targeted are those disclosed in Table 34 having a G>A mutation. Additional examples of disease-associated genes and polynucleotides are available from McKusick-Nathans Institute of Genetic Medicine, Johns Hopkins University (Baltimore, Md.) and National Center for Biotechnology Information, National Library of Medicine (Bethesda, Md.), available on the World Wide Web.

(154) In some embodiments, the target polynucleotide comprises a cystic fibrosis transmembrane conductance regulator (5) gene.

(155) As used herein, the term “cystic fibrosis transmembrane conductance regulator” or “CFTR” refers to a cAMP regulated chloride channel located in the apical membrane of epithelial cells that catalyze the passage of small ions through the membrane. A non-limiting example of a CFTR gene is set forth as SEQ ID NO: 51.

(156) As used herein, the term “target” or “targets,” in relation to a spacer sequence and a target sequence, refers to the localization of an RNA-guided nuclease to a target sequence based on the ability of a spacer sequence within an associated guide RNA to hybridize sufficiently with a target sequence.

(157) CRISPR RNAs (crRNAs) or nucleic acid molecules encoding the same, wherein the crRNA comprises a spacer sequence that targets a CFTR target sequence are provided. Guide RNAs comprising such crRNAs, one or more nucleic acid molecules encoding a guide RNA comprising such crRNAs, vectors comprising one or more nucleic acid molecules encoding a guide RNA comprising such crRNAs, and systems comprising such crRNAs are also provided. Methods of using such crRNAs or nucleic molecules encoding the same, guide RNAs comprising such crRNAs, one or more nucleic acid molecules encoding a guide RNA comprising such crRNAs, vectors comprising one or more nucleic acid molecules encoding a guide RNA comprising such crRNAs, and systems comprising such crRNAs to bind to, cleave, and/or modulate the target sequence are also provided.

(158) In some embodiments, the CFTR target sequence of a crRNA or a guide RNA has the sequence set forth in any one of SEQ ID NOs: 98-115, 140-151, 186-202, 235-250, 287-304, 345-364, 562, and 563, or the complement thereof. In some embodiments, a single guide RNA (sgRNA) comprising a crRNA having a spacer sequence that targets a CFTR target sequence has at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to any one of SEQ ID NOs: 98-115, 140-151, 186-202, 235-250, 287-304, 345-364, and 564.

(159) In some embodiments, the CFTR target sequence of a crRNA or a guide RNA has the sequence set forth in any one of SEQ ID NOs: 62-68, 80-85, 116-119, 128-131, 163, 164, 180, 181, 203-209, 219-225, 256-258, 274-276, 310-313, and 330-333, or the complement thereof, and the associated RGN polypeptide has an amino acid sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to SEQ ID NO: 53. In some embodiments, a sgRNA comprising a crRNA having a spacer sequence that targets a CFTR target sequence has at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to any one of SEQ ID NOs: 98-104, 140-143, 197, 198, 235-241, 292-294, and 350-353, and the associated RGN polypeptide has an amino acid sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to SEQ ID NO: 53.

(160) In some embodiments, the CFTR target sequence of a crRNA or a guide RNA has the sequence set forth in any one of SEQ ID NOs: 68-71, 86-89, 120-122, 132-134, 152-156, 169-173, 213-215, 229-231, 251-255, 269-273, 305-309, and 325-329, or the complement thereof and the associated RGN polypeptide has an amino acid sequence having at least 40%, 45%, 50%, 55%,

60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to SEQ ID NO: 55. In some embodiments, a sgRNA comprising a crRNA having a spacer sequence that targets a CFTR target sequence has at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to any one of SEQ ID NOs: 104-107, 144-146, 186-190, 245-247, 287-291, and 345-349, and the associated RGN polypeptide has an amino acid sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to SEQ ID NO: 55.

(161) In some embodiments, the CFTR target sequence of a crRNA or a guide RNA has the sequence set forth in any one of SEQ ID NOs: 72, 73, 90, 91, 161, 162, 178, 179, 265, 266, 283, and 284, or the complement thereof and the associated RGN polypeptide has an amino acid sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to SEQ ID NO: 52. In some embodiments, a sgRNA comprising a crRNA having a spacer sequence that targets a CFTR target sequence has at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to any one of SEQ ID NOs: 108, 109, 195, 196, 301, and 302, and the associated RGN polypeptide has an amino acid sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to SEQ ID NO: 52.

(162) In some embodiments, the CFTR target sequence of a crRNA or a guide RNA has the sequence set forth in any one of SEQ ID NOs: 74, 75, 92, 93, 123, 124, 135, 136, 167, 184, 216-218, 232-234, 259-261, 277-279, 314-317, and 334-337, or the complement thereof and the associated RGN polypeptide has an amino acid sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to SEQ ID NO: 56. In some embodiments, a sgRNA comprising a crRNA having a spacer sequence that targets a CFTR target sequence has at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to any one of SEQ ID NOs: 110, 111, 147, 148, 201, 248-250, 295-297, and 354-357, and the associated RGN polypeptide has an amino acid sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to SEQ ID NO: 56.

(163) In some embodiments, the CFTR target sequence of a crRNA or a guide RNA has the sequence set forth in any one of SEQ ID NOs: 76, 94, 210-212, 226-228, 322, 342, 562, and 563, or the complement thereof and the associated RGN polypeptide has an amino acid sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to SEQ ID NO: 42. In some embodiments, a sgRNA comprising a crRNA having a spacer sequence that targets a CFTR target sequence has at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to any one of SEQ ID NOs: 112, 242-244, 362, and 564, and the associated RGN polypeptide has an amino acid sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to SEQ ID NO: 42.

(164) In some embodiments, the CFTR target sequence of a crRNA or a guide RNA has the sequence set forth in any one of SEQ ID NOs: 77, 95, 125, 137, 157-160, 174-177, 323, and 343, or the complement thereof and the associated RGN polypeptide has an amino acid sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to SEQ ID NO: 54. In some embodiments, a sgRNA comprising a crRNA having a spacer sequence that targets a CFTR target sequence has at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to any one of SEQ ID NOs: 113, 149, 191-194,

and 363, and the associated RGN polypeptide has an amino acid sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to SEQ ID NO: 54.

(165) In some embodiments, the CFTR target sequence of a crRNA or a guide RNA has the sequence set forth in any one of SEQ ID NOs: 78, 96, 126, 138, 168, 185, 267, 285, 318, 319, 338, and 339, or the complement thereof and the associated RGN polypeptide has an amino acid sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to SEQ ID NO: 57. In some embodiments, a sgRNA comprising a crRNA having a spacer sequence that targets a CFTR target sequence has at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to any one of SEQ ID NOs: 114, 150, 202, 303, 358, and 359, and the associated RGN polypeptide has an amino acid sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to SEQ ID NO: 57.

(166) In some embodiments, the CFTR target sequence of a crRNA or a guide RNA has the sequence set forth in any one of SEQ ID NOs: 79, 97, 127, 139, 262-264, 280-282, 324, and 344, or the complement thereof and the associated RGN polypeptide has an amino acid sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to SEQ ID NO: 58. In some embodiments, a sgRNA comprising a crRNA having a spacer sequence that targets a CFTR target sequence has at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to any one of SEQ ID NOs: 115, 151, 298-300, and 364, and the associated RGN polypeptide has an amino acid sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to SEQ ID NO: 58.

(167) In some embodiments, the CFTR target sequence of a crRNA or a guide RNA has the sequence set forth in any one of SEQ ID NOs: 165, 166, 182, 183, 268, 286, 320, 321, 340, and 341, or the complement thereof and the associated RGN polypeptide has an amino acid sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to SEQ ID NO: 59. In some embodiments, a sgRNA comprising a crRNA having a spacer sequence that targets a CFTR target sequence has at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to any one of SEQ ID NOs: 199, 200, 304, 360, and 361, and the associated RGN polypeptide has an amino acid sequence having at least 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to SEQ ID NO: 59.

(168) In some embodiments, the methods comprise contacting a DNA molecule comprising a target DNA sequence with a DNA-binding polypeptide-deaminase fusion protein of the invention, wherein the DNA molecule is contacted with the fusion protein in an amount effective and under conditions suitable for the deamination of a nucleobase. In certain embodiments, the methods comprise contacting a DNA molecule comprising a target DNA sequence with (a) an RGN-deaminase fusion protein of the invention; and (b) a gRNA targeting the fusion protein of (a) to a target nucleotide sequence of the DNA strand; wherein the DNA molecule is contacted with the fusion protein and the gRNA in an amount effective and under conditions suitable for the deamination of a nucleobase. In some embodiments, the target DNA sequence comprises a sequence associated with a disease or disorder, and wherein the deamination of the nucleobase results in a sequence that is not associated with a disease or disorder. In some embodiments, the target DNA sequence resides in an allele of a crop plant, wherein the particular allele of the trait of interest results in a plant of lesser agronomic value. The deamination of the nucleobase results in an allele that improves the trait and increases the agronomic value of the plant.

(169) In some embodiments, the target DNA sequence comprises a G>A point mutation associated with a disease or disorder, and wherein the deamination of the mutant A base results in a sequence that is not associated with a disease or disorder. In some embodiments, the deamination corrects a point mutation in the sequence associated with the disease or disorder. In some embodiments, the sequence associated with the disease or disorder encodes a protein, and the deamination introduces a stop codon into the sequence associated with the disease or disorder, resulting in a truncation of the encoded protein. In some embodiments, the contacting is performed in vivo in a subject susceptible to having, having, or diagnosed with the disease or disorder. In some embodiments, the disease or disorder is a disease associated with a point mutation, or a single-base mutation, in the genome. In some embodiments, the disease is a genetic disease, a cancer, a metabolic disease, or a lysosomal storage disease.

XI. Pharmaceutical Compositions and Methods of Treatment

(170) Methods of treating a disease in a subject in need thereof are provided herein. The methods comprise administering to a subject in need thereof an effective amount of a presently disclosed fusion protein or a polynucleotide encoding the same, a presently disclosed gRNA or a polynucleotide encoding the same, a presently disclosed fusion protein system, a presently disclosed ribonucleoprotein complex, or a cell modified by or comprising any one of these compositions.

(171) In some embodiments, the treatment comprises in vivo gene editing by administering to a subject in need thereof a presently disclosed fusion protein, gRNA, or a presently disclosed fusion protein system or polynucleotide(s) encoding the same. In some embodiments, the treatment comprises ex vivo gene editing wherein cells are genetically modified ex vivo with a presently disclosed fusion protein, gRNA, or a presently disclosed fusion protein system or polynucleotide(s) encoding the same and then the modified cells are administered to a subject. In some embodiments, the genetically modified cells originate from the subject that is then administered the modified cells, and the transplanted cells are referred to herein as autologous. In some embodiments, the genetically modified cells originate from a different subject (i.e., donor) within the same species as the subject that is administered the modified cells (i.e., recipient), and the transplanted cells are referred to herein as allogeneic. In some examples described herein, the cells can be expanded in culture prior to administration to a subject in need thereof.

(172) For example, in some embodiments, a method is provided that comprises administering to a subject having such a disease, e.g., a genetic defect associated with the CFTR gene, an effective amount of ribonucleoprotein complex comprising a fusion protein with a deaminase having an amino acid sequence that is at least 80% identical to sequence set forth in any one of the SEQ ID NOs: 399, and 405-407. In the embodiments described herein, the administration of the ribonucleoprotein complex corrects the point mutation or introduces a deactivating mutation into a disease-associated CFTR gene. Other diseases that can be treated by correcting a point mutation or introducing a deactivating mutation into a disease-associated gene will be known to those of skill in the art, and the disclosure is not limited in this respect.

(173) In some embodiments, the disease to be treated with the presently disclosed compositions is one that can be treated with immunotherapy, such as with a chimeric antigen receptor (CAR) T cell. Such diseases include but are not limited to cancer.

(174) In some embodiments, the deamination of the target nucleobase results in the correction of a genetic defect, e.g., to correct the CFTR gene, or in the correction of a point mutation that leads to a loss of function in a gene product. In some embodiments, the genetic defect is associated with a disease or disorder, e.g., a lysosomal storage disorder or a metabolic disease, such as, for example, type I diabetes. Thus, in some embodiments, the disease to be treated with the presently disclosed compositions is associated with a sequence (i.e., the sequence is causal for the disease or disorder or causal for symptoms associated with the disease or disorder) that is mutated in order to treat the disease or disorder or the reduction of symptoms associated with the disease or disorder.

(175) In some embodiments, the disease to be treated with the presently disclosed compositions is associated with a causal mutation. As used herein, a “causal mutation” refers to a particular nucleotide, nucleotides, or nucleotide sequence in the genome that contributes to the severity or presence of a disease or disorder in a subject. The correction of the causal mutation leads to the improvement of at least one symptom resulting from a disease or disorder. In some embodiments, the correction of the causal mutation leads to the improvement of at least one symptom resulting from a disease or disorder. In some embodiments, the causal mutation is adjacent to a PAM site recognized by the RGDBP (e.g., RGN) fused to a deaminase disclosed herein. The causal mutation can be corrected with a fusion polypeptide comprising a RGDBP (e.g., RGN) and a presently disclosed deaminase. Non-limiting examples of diseases associated with a causal mutation include cystic fibrosis, Hurler syndrome, Friedreich's Ataxia, Huntington's Disease, and sickle cell disease. Additional non-limiting examples of disease-associated genes and mutations are available from McKusick-Nathans Institute of Genetic Medicine, Johns Hopkins University (Baltimore, Md.) and National Center for Biotechnology Information, National Library of Medicine (Bethesda, Md.), available on the World Wide Web.

(176) In some embodiments, the methods provided herein are used to introduce a deactivating point mutation into a gene or allele that encodes a gene product that is associated with a disease or disorder. For example, in some embodiments, methods are provided herein that employ a fusion protein to introduce a deactivating point mutation into an oncogene (e.g., in the treatment of a proliferative disease). A deactivating mutation may, in some embodiments, generate a premature stop codon in a coding sequence, which results in the expression of a truncated gene product, e.g., a truncated protein lacking the function of the full-length protein. In some embodiments, the purpose of the methods provided herein is to restore the function of a dysfunctional gene via genome editing. The fusion proteins provided herein can be validated for gene editing-based human therapeutics in vitro, e.g., by correcting a disease associated mutation in human cell culture. It will be understood by the skilled artisan that the fusion proteins provided herein, e.g., the fusion proteins comprising an RNA-guided, DNA-binding polypeptide and deaminase polypeptide can be used to correct any single point G>A mutation. Deamination of the mutant A to G leads to a correction of the mutation.

(177) As used herein, “treatment” or “treating,” or “palliating” or “ameliorating” are used interchangeably. These terms refer to an approach for obtaining beneficial or desired results including but not limited to a therapeutic benefit and/or a prophylactic benefit. By therapeutic benefit is meant any therapeutically relevant improvement in or effect on one or more diseases, conditions, or symptoms under treatment. For prophylactic benefit, the compositions may be administered to a subject at risk of developing a particular disease, condition, or symptom, or to a subject reporting one or more of the physiological symptoms of a disease, even though the disease, condition, or symptom may not have yet been manifested. In some embodiments, treatment may be administered after one or more symptoms have developed and/or after a disease has been diagnosed. In particular embodiments, treatment may be administered in the absence of symptoms, e.g., to prevent or delay onset of a symptom or inhibit onset or progression of a disease. For example, treatment may be administered to a susceptible individual prior to the onset of symptoms (e.g., in light of a history of symptoms and/or in light of genetic or other susceptibility factors). Treatment may also be continued after symptoms have resolved, for example, to prevent or delay their prevention or recurrence.

(178) The term “effective amount” or “therapeutically effective amount” refers to the amount of an agent that is sufficient to effect beneficial or desired results. The therapeutically effective amount may vary depending upon one or more of: the subject and disease condition being treated, the weight and age of the subject, the severity of the disease condition, the manner of administration and the like, which can readily be determined by one of ordinary skill in the art. The specific dose may vary depending on one or more of: the particular agent chosen, the dosing regimen to be

followed, whether it is administered in combination with other compounds, timing of administration, and the delivery system in which it is carried.

(179) The term “administering” refers to the placement of an active ingredient into a subject, by a method or route that results in at least partial localization of the introduced active ingredient at a desired site, such as a site of injury or repair, such that a desired effect(s) is produced. In some embodiments, the disclosure provides methods comprising delivering any of the isolated polypeptides, nucleic acid molecules fusion proteins, ribonucleoprotein complexes, vectors, pharmaceutical compositions and/or gRNAs described herein. In some embodiments, the disclosure further provides cells produced by such methods, and organisms (such as animals or plants) comprising or produced from such cells. In some embodiments, a deaminase, fusion protein and/or nucleic acid molecules as described herein in combination with (and optionally complexed with) a guide sequence is delivered to a cell.

(180) In some embodiments, the administering comprises administering by viral delivery. Viral vectors comprising a nucleic acid encoding the fusion proteins, ribonucleoprotein complexes, or vectors disclosed herein may be administered directly to patients (i.e., in vivo) or they may be used to treat cells in vitro, and the modified cells may optionally be administered to patients (i.e., ex vivo). Conventional viral based systems may include, without limitation, retroviral, lentivirus, adenoviral, adeno-associated and herpes simplex virus vectors for gene transfer. Integration in the host genome is possible with the retrovirus, lentivirus, and adeno-associated virus gene transfer methods, often resulting in long term expression of the inserted transgene. Lentiviral vectors are retroviral vectors that are able to transduce or infect non-dividing cells and typically produce high viral titers. In applications where transient expression is preferred, adenoviral based systems may be used. Adenoviral based vectors are capable of very high transduction efficiency in many cell types and do not require cell division.

(181) In some embodiments, the administering comprises administering by electroporation. In some embodiments, the administering comprises administering by nanoparticle delivery. In some embodiments, the administering comprises administering by liposome delivery. Any effective route of administration can be used to administer an effective amount of a pharmaceutical composition described herein.

(182) In some embodiments, the administering comprises administering by other non-viral delivery of nucleic acids. Exemplary non-viral delivery methods, without limitation, include RNP complexes, lipofection, nucleofection, microinjection, biolistics, virosomes, liposomes, immunoliposomes, polycation or lipid-nucleic acid conjugates, naked DNA, artificial virions, and agent-enhanced uptake of DNA. Lipofection is described in e.g., U.S. Pat. Nos. 5,049,386, 4,946,787; and 4,897,355) and lipofection reagents are sold commercially (e.g., Transfectam™ and Lipofectin™). Cationic and neutral lipids that are suitable for efficient receptor-recognition lipofection of polynucleotides include those of Feigner, WO1991/17424; WO 1991/16024. Delivery can be to cells (e.g. in vitro or ex vivo administration) or target tissues (e.g. in vivo administration).

(183) As used herein, the term “subject” refers to any individual for whom diagnosis, treatment or therapy is desired. In some embodiments, the subject is an animal. In some embodiments, the subject is a mammal. In some embodiments, the subject is a human being.

(184) The efficacy of a treatment can be determined by the skilled clinician. However, a treatment is considered an “effective treatment,” if any one or all of the signs or symptoms of a disease or disorder are altered in a beneficial manner (e.g., decreased by at least 10%), or other clinically accepted symptoms or markers of disease are improved or ameliorated. Efficacy can also be measured by failure of an individual to worsen as assessed by hospitalization or need for medical interventions (e.g., progression of the disease is halted or at least slowed). Methods of measuring these indicators are known to those of skill in the art. Treatment includes: (1) inhibiting the disease, e.g., arresting, or slowing the progression of symptoms; or (2) relieving the disease, e.g., causing

regression of symptoms; and (3) preventing or reducing the likelihood of the development of symptoms.

(185) Pharmaceutical compositions comprising the presently disclosed RGN polypeptides or polynucleotides encoding the same, the presently disclosed gRNAs or polynucleotides encoding the same, the presently disclosed deaminases or polynucleotides encoding the same, the presently disclosed fusion proteins, the presently disclosed systems (such as those comprising a fusion protein), the presently disclosed ribonucleoprotein complex or cells comprising any of the RGN polypeptides or RGN-encoding polynucleotides, gRNA or gRNA-encoding polynucleotides, fusion protein-encoding polynucleotides, or the systems, and a pharmaceutically acceptable carrier are provided.

(186) As used herein, a “pharmaceutically acceptable carrier” refers to a material that does not cause significant irritation to an organism and does not abrogate the activity and properties of the active ingredient (e.g., a deaminase or fusion protein or nucleic acid molecule encoding the same). Carriers must be of sufficiently high purity and of sufficiently low toxicity to render them suitable for administration to a subject being treated. The carrier can be inert, or it can possess pharmaceutical benefits. In some embodiments, a pharmaceutically acceptable carrier comprises one or more compatible solid or liquid filler, diluents or encapsulating substances which are suitable for administration to a human or other vertebrate animal. In some embodiments, the pharmaceutical composition comprises a pharmaceutically acceptable carrier that is non-naturally occurring. In some embodiments, the pharmaceutically acceptable carrier and the active ingredient are not found together in nature and are thus, heterologous.

(187) Pharmaceutical compositions used in the presently disclosed methods can be formulated with suitable carriers, excipients, and other agents that provide suitable transfer, delivery, tolerance, and the like. A multitude of appropriate formulations are known to those skilled in the art. See, e.g., Remington, *The Science and Practice of Pharmacy* (21st ed. 2005). Non-limiting examples include a sterile diluent such as water for injection, saline solution, fixed oils, polyethylene glycols, glycerine, propylene glycol or other synthetic solvents; antibacterial agents such as benzyl alcohol or methyl parabens; antioxidants such as ascorbic acid or sodium bisulfate; chelating agents such as ethylenediaminetetraacetic acid; buffers such as acetates, citrates or phosphates and agents for the adjustment of tonicity such as sodium chloride or dextrose. Administered intravenously, particular carriers are physiological saline or phosphate buffered saline (PBS). Pharmaceutical compositions for oral or parenteral use may be prepared into dosage forms in a unit dose suited to fit a dose of the active ingredients. Such dosage forms in a unit dose include, for example, tablets, pills, capsules, injections (ampoules), suppositories, etc. These compositions also may contain adjuvants including preservative agents, wetting agents, emulsifying agents, and dispersing agents. Prevention of the action of microorganisms may be ensured by various antibacterial and antifungal agents, for example, parabens, chlorobutanol, phenol, sorbic acid, and the like. It also may be desirable to include isotonic agents, for example, sugars, sodium chloride and the like. Prolonged absorption of the injectable pharmaceutical form may be brought about by the use of agents delaying absorption, for example, aluminum monostearate and gelatin.

(188) In some embodiments wherein cells comprising or modified with the presently disclosed RGNs, gRNAs, deaminases, fusion proteins, systems (including those comprising fusion proteins) or polynucleotides encoding the same are administered to a subject, the cells are administered as a suspension with a pharmaceutically acceptable carrier. One of skill in the art will recognize that a pharmaceutically acceptable carrier to be used in a cell composition will not include buffers, compounds, cryopreservation agents, preservatives, or other agents in amounts that substantially interfere with the viability of the cells to be delivered to the subject. A formulation comprising cells can include e.g., osmotic buffers that permit cell membrane integrity to be maintained, and optionally, nutrients to maintain cell viability or enhance engraftment upon administration. Such formulations and suspensions are known to those of skill in the art and/or can be adapted for use

with the cells described herein using routine experimentation.

(189) A cell composition can also be emulsified or presented as a liposome composition, provided that the emulsification procedure does not adversely affect cell viability. The cells and any other active ingredient can be mixed with excipients that are pharmaceutically acceptable and compatible with the active ingredient, and in amounts suitable for use in the therapeutic methods described herein.

(190) Additional agents included in a cell composition can include pharmaceutically acceptable salts of the components therein. Pharmaceutically acceptable salts include the acid addition salts (formed with the free amino groups of the polypeptide) that are formed with inorganic acids, such as, for example, hydrochloric or phosphoric acids, or such organic acids as acetic, tartaric, mandelic and the like. Salts formed with the free carboxyl groups can also be derived from inorganic bases, such as, for example, sodium, potassium, ammonium, calcium or ferric hydroxides, and such organic bases as isopropylamine, trimethylamine, 2-ethylamino ethanol, histidine, procaine and the like.

(191) Suitable routes of administering the pharmaceutical composition described herein include, without limitation: topical, subcutaneous, transdermal, intradermal, intralesional, intraarticular, intraperitoneal, intravesical, transmucosal, gingival, intradental, intracochlear, transtympanic, intraorgan, epidural, intrathecal, intramuscular, intravenous, intravascular, intraosseous, periocular, intratumoral, intracerebral, and intracerebroventricular administration.

(192) In some embodiments, the pharmaceutical composition described herein is administered locally to a diseased site (e.g., the lung). In some embodiments, the pharmaceutical composition described herein is administered to a subject by injection, inhalation (e.g., of an aerosol), by means of a catheter, by means of a suppository, or by means of an implant, the implant being of a porous, non-porous, or gelatinous material, including a membrane, such as a sialastic membrane, or a fiber. In some embodiments, the pharmaceutical composition is formulated for delivery to a subject, e.g., for gene editing.

(193) In some embodiments, the pharmaceutical composition is formulated in accordance with routine procedures as a composition adapted for intravenous or subcutaneous administration to a subject, e.g., a human. In some embodiments, pharmaceutical composition for administration by injection are solutions in sterile isotonic aqueous buffer. Where necessary, the pharmaceutical can also include a solubilizing agent and a local anesthetic such as lignocaine to ease pain at the site of the injection. Generally, the ingredients are supplied either separately or mixed together, for example, as a dry lyophilized powder or water free concentrate in a hermetically sealed container such as an ampoule or sachette indicating the quantity of active agent. Where the pharmaceutical is to be administered by infusion, it can be dispensed with an infusion bottle containing sterile pharmaceutical grade water or saline. Where the pharmaceutical composition is administered by injection, an ampoule of sterile water for injection or saline can be provided so that the ingredients can be mixed prior to administration.

(194) In some embodiments, the pharmaceutical composition can be contained within a lipid particle or vesicle, such as a liposome or microcrystal, which is also suitable for parenteral administration.

(195) Although the descriptions of pharmaceutical compositions provided herein are principally directed to pharmaceutical compositions which are suitable for administration to humans, it will be understood by the skilled artisan that such compositions are generally suitable for administration to animals or organisms of all sorts.

(196) Modifying Causal Mutations Using Base-Editing

(197) An example of a genetically inherited disease which could be corrected using an approach that relies on an RGN-deaminase fusion protein of the invention is Cystic Fibrosis. Cystic fibrosis (CF) is an autosomal recessive disease caused by mutations in the cystic fibrosis transmembrane regulator (CFTR) gene (set forth as SEQ ID NO: 51). CFTR encodes for a cAMP regulated

chloride channel located in the apical membrane of epithelial cells that catalyze the passage of small ions through the membrane. Dysregulation of this mechanism causes an impairment of salt and fluid homeostasis that results in multiorgan dysfunctions and ultimately mortality from respiratory failure.

(198) Almost 2,000 mutations in the CFTR gene have been found to cause CF. CFTR mutations are divided into six classes based on the functional defect in either CFTR protein synthesis, trafficking, function, or stability, although it is acknowledged that many CFTR mutants present multiple defects. Class I mutations lead to severely defective protein production. They are primarily nonsense or frameshift mutations which introduce a premature termination codon (PTC), leading to unstable messenger RNA (mRNA) degraded by the mRNA decay pathway (NMD). Nonsense mutations due to single nucleotide changes comprise a major subset of Class I mutations (Marangi, M. and Pistrutto, G, 2018, *Front Pharmacol* 9, 396, doi:10.3389/fphar.2018.00396; Pranke, I., et al., 2019, *Front Pharmacol* 10, 121, doi:10.3389/fphar.2019.00121, both of which are incorporated by reference herein). Treatment for patients with Class I cystic fibrosis can be difficult, as no functional CFTR protein is produced. Notably, a significant fraction of these nonsense mutations are potentially addressable with A to G base editors (Geurts, M. H. et al, 2020, *Cell Stem Cell* 26, 503-510 e507, doi:10.1016/j.stem.2020.01.019 incorporated by reference herein).

(199) Geurts et al. were the first group to perform precise base editing in cultured lung epithelial cells with Class I mutations from cystic fibrosis patients, using a fusion protein comprising an adenine deaminase operably linked an RGN, namely either SpyCas9 or the xSpyCas9 variant. SpyCas9 recognizes a 5'-nGG-3'PAM, while the xSpyCas9 variant recognized the reduced 5'-nG-3'. The authors state that a major limitation of the base editing technology is the PAM requirement of the Cas protein being used. They find that the majority of nonsense mutations identified in the CFTR gene are not in the required targeting window for a fusion protein comprising the RGN SpyCas9. The PAM is a short motif, generally one to four nucleotides, on the target DNA sequence that is recognized by the RGN. The PAM sequence is intrinsic to each RGN protein, such that an RGN can only access the genomic space around a suitable PAM. Additionally, the base editing window for base editors is limited, frequently to just a portion of the nucleotides in the target sequence. If the nucleotide of interest is too close to the PAM, the RGN blocks access to the nucleotide. If the nucleotide is too far away from the PAM, the deaminase tethered to the RGN is unable to reach the nucleotide. Also, the amount of ssDNA exposed by the R-loop limits the accessibility of the deaminases. The present invention includes RGN-deaminase fusion proteins where the RGN recognizes a PAM which is proximal to a Class I mutation of the CFTR gene and the deaminase is able to successfully modify the targeted causal mutation.

(200) Another limitation to RGN-deaminase fusion proteins known in the art is that the vector construct encoding for the fusion protein is too large for methods of in vivo delivery. AAV delivery of these fusion proteins is not an option for SpyCas9-based fusion proteins because their size exceeds the limit for efficient AAV packaging. The RGN component of the fusion proteins described herein are smaller in size and are therefore viable candidates for AAV vector delivery strategies. The present invention also discloses guide RNAs which are specific for the RGNs described herein and which guide the fusion proteins of the invention to target sites of nonsense mutations in the CFTR gene which were previously unreachable. The present invention also teaches methods of using said fusion proteins for targeted base editing through in vivo AAV vector delivery.

(201) Ideally, the coding sequence of an RGN-deaminase fusion protein of the invention and a corresponding guide RNA for targeting the fusion protein to the CFTR gene may all be packaged into a single AAV vector. The generally accepted size limit for AAV vectors is 4.7 kb, although larger sizes may be contemplated at the expense of reduced packing efficiency. The RGN nickases in Table 28 have a coding sequence length of about 3.15-3.45 kB. To ensure that the expression cassettes for both the fusion protein and its corresponding guide RNA could fit into an AAV vector,

novel, active deletion variants of RGNs are described herein. In addition to shortening the amino acid sequence and therefore the coding sequence of the RGN of the fusion protein, the peptide linker which links the RGN and the deaminase may also be shortened. Finally, the genetic elements, such as the promoters, enhancers, and/or terminators, may also be engineered via deletion analysis to determine the minimal size required for each to be functional.

(202) Some embodiments of the disclosure provide methods for editing a nucleic acid using the deaminases or the RGN complexes described herein to achieve the nucleobase change, e.g., an A:T base pair to G:C base pair. In some embodiments, the method is a method for editing a nucleobase of a nucleic acid (e.g., a base pair of a double-stranded DNA sequence). In some embodiments, the deaminases or the RGN complexes described herein are used to introduce a point mutation into a nucleic acid by deaminating and excising a target “A” nucleobase. In some embodiments, the deamination-and-excision of the target nucleobase results in the correction of a genetic defect, e.g., in the correction of a point mutation in a CFTR gene. In some embodiments, the genetic defect is associated with a disease, disorder, or condition, e.g., Cystic Fibrosis. For example, in some embodiments, methods are provided herein employ a base editing RGN complexes comprising a fusion protein with a deaminase having an amino acid sequence that is at least 80% identical to sequence set forth in any one of the SEQ ID NOs: 399, and 405-407, to correct a gene associated with a genetic defect, e.g., to correct a point mutation in a CFTR gene (e.g., in the treatment of a proliferative disease). In specific embodiments, the target sequence in the CFTR gene is 62-97, 116-139, 152-185, 203-234, 251-286, 305-344, 562, or 563.

(203) In some embodiments, the purpose of the methods provided herein is to restore the function of a dysfunctional gene via genome editing. The base editor proteins provided herein may be validated for gene editing-based human therapeutics in vitro, e.g., by correcting a disease-associated mutation in human cell culture. It will be understood by the skilled artisan that the fusion proteins and/or the RGN complexes provided herein comprising a nucleic acid binding protein (e.g., nCas9) and a nucleobase modification domain (e.g., deaminase with an amino acid sequence set forth in SEQ ID NO.: 407, 399, or 405 may be used to correct any single point of T to G or change a pairing of T:A to G:C.

(204) In some embodiments, provided herein are the methods for the treatment of a subject diagnosed with a disease associated with or caused by a point mutation (e.g., mutation in CFTR gene) that can be corrected by a fusion protein or the RGN complexes described herein. For example, in some embodiments, a method is provided that comprises administering to a subject having such a disease, e.g., cystic fibrosis, an effective amount of a fusion protein or RGN complex disclosed herein that corrects the point mutation or introduces a deactivating mutation into a disease-associated gene. In some embodiments, a method is provided that comprises administering to a subject having such a disease, e.g., a cancer associated with a point mutation as described above, an effective amount of a fusion protein, RGN complex, or pharmaceutical composition disclosed herein that corrects the point mutation or introduces a deactivating mutation into a disease-associated gene. In specific embodiments, methods of treating cystic fibrosis are provided along with methods of reducing at least one symptom of cystic fibrosis by administering an effective amount of a pharmaceutical composition disclosed herein. An effective amount of a pharmaceutical composition for treating or reducing a symptom of cystic fibrosis can reduce a symptom (i.e., treat) of cystic fibrosis by about 5%, 10%, 15%, 20%, 25%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, or more; or about 10-20%, 15-25%, 20-40%, 30-50%, 40-60%, 50-70%, 60-80%, 70-90%, 80-95%, or 90-95% when compared to a control patient. In specific embodiments, the control patient can be the same patient before administration of the effective amount of the pharmaceutical composition disclosed herein. Symptoms of cystic fibrosis can include, but are not limited to: sneezing, a persistent cough that produces mucus or phlegm, shortness of breath, especially when exercising, recurrent lung infections, a stuffy nose, stuffy sinuses, greasy foul-smelling stools, constipation, nausea, swollen abdomen, loss of appetite, among others. Methods of

identifying and measuring symptoms of cystic fibrosis are known in the art.

(205) In some embodiments of the described methods for modifying a target DNA molecule, the step of contacting is performed in vitro. In particular embodiments, the step of contacting is performed in vivo. In some embodiments, the step of contacting is performed in a subject (e.g., a human subject or a non-human animal subject). In some embodiments, the step of contacting is performed in a cell, such as a human or non-human animal cell.

XII. Cells Comprising a Polynucleotide Genetic Modification

(206) Provided herein are cells and organisms comprising a target nucleic acid molecule of interest that has been modified using a process mediated by a fusion protein, optionally with a gRNA, as described herein. In some embodiments, the fusion protein comprises a deaminase polypeptide comprising an amino acid sequence of any of SEQ ID NOs: 1-10 and 399-441, or an active variant or fragment thereof. In some embodiments, the fusion protein comprises an adenine deaminase comprising an amino acid sequence having at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identity to any of SEQ ID NOs: 1-10 and 399-441. In some embodiments, the fusion protein comprises a deaminase and a DNA-binding polypeptide (e.g., an RNA-guided, DNA-binding polypeptide). In further embodiments, the fusion protein comprises a deaminase and an RGN or a variant thereof, such as for example APG07433.1 (SEQ ID NO: 41) or its nickase variant nAPG07433.1 (SEQ ID NO: 42). In some embodiments, the fusion protein comprises a deaminase and a Cas9 or a variant thereof, such as for example dCas9 or nickase Cas9. In some embodiments, the fusion protein comprises a nuclease-inactive or nickase variant of a Type II CRISPR-Cas polypeptide. In some embodiments, the fusion protein comprises a nuclease-inactive or nickase variant of a Type V CRISPR-Cas polypeptide. In some embodiments, the fusion protein comprises a nuclease-inactive or nickase variant of a Type VI CRISPR-Cas polypeptide.

(207) The modified cells can be eukaryotic (e.g., mammalian, plant, insect, avian cell) or prokaryotic. Also provided are organelles and embryos comprising at least one nucleotide sequence that has been modified by a process utilizing a fusion protein as described herein. The genetically modified cells, organisms, organelles, and embryos can be heterozygous or homozygous for the modified nucleotide sequence. The mutation(s) introduced by the deaminase domain of the fusion protein can result in altered expression (up-regulation or down-regulation), inactivation, or the expression of an altered protein product or an integrated sequence. In those instances wherein the mutation(s) results in either the inactivation of a gene or the expression of a non-functional protein product, the genetically modified cell, organism, organelle, or embryo is referred to as a “knock out”. The knock out phenotype can be the result of a deletion mutation (i.e., deletion of at least one nucleotide), an insertion mutation (i.e., insertion of at least one nucleotide), or a nonsense mutation (i.e., substitution of at least one nucleotide such that a stop codon is introduced).

(208) In some embodiments, the mutation(s) introduced by the deaminase domain of the fusion protein results in the production of a variant protein product. The expressed variant protein product can have at least one amino acid substitution and/or the addition or deletion of at least one amino acid. The variant protein product can exhibit modified characteristics or activities when compared to the wild-type protein, including but not limited to altered enzymatic activity or substrate specificity.

(209) In some embodiments, the mutation(s) introduced by the deaminase domain of the fusion protein result in an altered expression pattern of a protein. As a non-limiting example, mutation(s) in the regulatory regions controlling the expression of a protein product can result in the overexpression or downregulation of the protein product or an altered tissue or temporal expression pattern.

(210) The cells that have been modified can be grown into an organism, such as a plant, in accordance with conventional ways. See, for example, McCormick et al. (1986) *Plant Cell Reports* 5:81-84. These plants may then be grown, and either pollinated with the same modified strain or

different strains, and the resulting hybrid having the genetic modification. The present invention provides genetically modified seed. Progeny, variants, and mutants of the regenerated plants are also included within the scope of the invention, provided that these parts comprise the genetic modification. Further provided is a processed plant product or byproduct that retains the genetic modification, including for example, soymeal.

(211) The methods provided herein may be used for modification of any plant species, including, but not limited to, monocots and dicots. Examples of plants of interest include, but are not limited to, corn (maize), sorghum, wheat, sunflower, tomato, crucifers, peppers, potato, cotton, rice, soybean, sugarbeet, sugarcane, tobacco, barley, and oilseed rape, *Brassica* sp., alfalfa, rye, millet, safflower, peanuts, sweet potato, cassava, coffee, coconut, pineapple, citrus trees, cocoa, tea, banana, avocado, fig, guava, mango, olive, papaya, cashew, macadamia, almond, oats, vegetables, ornamentals, and conifers.

(212) Vegetables include, but are not limited to, tomatoes, lettuce, green beans, lima beans, peas, and members of the genus *Curcumis* such as cucumber, cantaloupe, and musk melon. Ornamentals include, but are not limited to, azalea, hydrangea, hibiscus, roses, tulips, daffodils, petunias, carnation, poinsettia, and chrysanthemum. Preferably, plants of the present invention are crop plants (for example, maize, sorghum, wheat, sunflower, tomato, crucifers, peppers, potato, cotton, rice, soybean, sugarbeet, sugarcane, tobacco, barley, oilseed rape, etc.).

(213) The methods provided herein can also be used to genetically modify any prokaryotic species, including but not limited to, archaea and bacteria (e.g., *Bacillus* sp., *Klebsiella* sp., *Streptomyces* sp., *Rhizobium* sp., *Escherichia* sp., *Pseudomonas* sp., *Salmonella* sp., *Shigella* sp., *Vibrio* sp., *Yersinia* sp., *Mycoplasma* sp., *Agrobacterium*, *Lactobacillus* sp.).

(214) The methods provided herein can be used to genetically modify any eukaryotic species or cells therefrom, including but not limited to animals (e.g., mammals, insects, fish, birds, and reptiles), fungi, amoeba, algae, and yeast. In some embodiments, the cell that is modified by the presently disclosed methods include cells of hematopoietic origin, such as immune cells (i.e., a cell of the innate or adaptive immune system) including but not limited to B cells, T cells, natural killer (NK) cells, pluripotent stem cells, induced pluripotent stem cells, chimeric antigen receptor T (CAR-T) cells, monocytes, macrophages, and dendritic cells.

(215) Cells that have been modified may be introduced into an organism. These cells could have originated from the same organism (e.g., person) in the case of autologous cellular transplants, wherein the cells are modified in an ex vivo approach. In some embodiments, the cells originated from another organism within the same species (e.g., another person) in the case of allogeneic cellular transplants.

XIII. Kits

(216) Some aspects of this disclosure provide kits comprising a deaminase of the invention. In certain embodiments, the disclosure provides kits comprising a fusion protein comprising a deaminase of the invention and a DNA-binding polypeptide (e.g., an RNA-guided, DNA-binding polypeptide, such as an RGN polypeptide, for example a nuclease-inactive Cas9 domain), and, optionally, a linker positioned between the DNA-binding polypeptide domain and the deaminase. In addition, in some embodiments, the kit comprises suitable reagents, buffers, and/or instructions for using the fusion protein, e.g., for in vitro or in vivo DNA or RNA editing. In some embodiments, the kit comprises instructions regarding the design and use of suitable gRNAs for targeted editing of a nucleic acid sequence.

(217) In some embodiments, the pharmaceutical composition may be provided as a pharmaceutical kit comprising (a) a container containing a composition of the disclosure in lyophilized form and (b) a second container containing a pharmaceutically acceptable diluent (e.g., sterile water) for injection. The pharmaceutically acceptable diluent can be used for reconstitution or dilution of the lyophilized compound of the disclosure. Optionally associated with such container(s) can be a notice in the form prescribed by a governmental agency regulating the manufacture, use or sale of

pharmaceuticals or biological products, which notice reflects approval by the agency of manufacture, use or sale for human administration.

(218) The article “a” and “an” are used herein to refer to one or more than one (i.e., to at least one) of the grammatical object of the article. By way of example, “a polypeptide” means one or more polypeptides.

(219) All publications and patent applications mentioned in the specification are indicative of the level of those skilled in the art to which this disclosure pertains. All publications and patent applications are herein incorporated by reference to the same extent as if each individual publication or patent application was specifically and individually indicated to be incorporated herein by reference.

(220) Although the foregoing invention has been described in some detail by way of illustration and example for purposes of clarity of understanding, it will be obvious that certain changes and modifications may be practiced within the scope of the appended claims.

(221) Non-limiting embodiments include: 1. An isolated polypeptide comprising an amino acid sequence having at least 90% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441, wherein said polypeptide has deaminase activity. 2. The isolated polypeptide of embodiment 1, comprising an amino acid sequence having at least 95% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 3. The isolated polypeptide of embodiment 1, comprising an amino acid sequence having 100% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 4. A nucleic acid molecule comprising a polynucleotide encoding a deaminase polypeptide, wherein the deaminase is encoded by a nucleotide sequence that: a) has at least 80% sequence identity to any one of SEQ ID NOs: 451, 449, 443, 11-20, 444-448, 450, and 452-485, or b) encodes an amino acid sequence having at least 90% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 5. The nucleic acid molecule of embodiment 4, wherein the deaminase is encoded by a nucleotide sequence that has at least 90% sequence identity to any one of SEQ ID NOs: 451, 449, 443, 11-20, 444-448, 450, and 452-485. 6. The nucleic acid molecule of embodiment 4, wherein the deaminase is encoded by a nucleotide sequence that has at least 95% sequence identity to any one of SEQ ID NOs: 451, 449, 443, 11-20, 444-448, 450, and 452-485. 7. The nucleic acid molecule of embodiment 4, wherein the deaminase is encoded by a nucleotide sequence that has 100% sequence identity to any one of SEQ ID NOs: 451, 449, 443, 11-20, 444-448, 450, and 452-485. 8. The nucleic acid molecule of embodiment 4, wherein the deaminase polypeptide has an amino acid sequence having at least 95% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 9. The nucleic acid molecule of embodiment 4, wherein the deaminase polypeptide has an amino acid sequence having 100% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 10. The nucleic acid molecule of any one of embodiments 4-9, wherein said nucleic acid molecule further comprises a heterologous promoter operably linked to said polynucleotide. 11. A pharmaceutical composition comprising a pharmaceutically acceptable carrier and the polypeptide of any one of embodiments of 1-3 or the nucleic acid molecule of any one of embodiments 4-10. 12. The pharmaceutical composition of embodiment 11, wherein the pharmaceutically acceptable carrier is heterologous to said polypeptide or said nucleic acid molecule. 13. The pharmaceutical composition of embodiment 11 or 12, wherein the pharmaceutically acceptable carrier is not naturally-occurring. 14. A fusion protein comprising a DNA-binding polypeptide and a deaminase having at least 90% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 15. The fusion protein of embodiment 14, wherein said deaminase has at least 95% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 16. The fusion protein of embodiment 14, wherein said deaminase has 100% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 17. The fusion protein of any one of embodiments 14-16, wherein the deaminase is an adenine deaminase. 18. The fusion

protein of any one of embodiments 14-17, wherein the DNA-binding polypeptide is a meganuclease, zinc finger fusion protein, or a TALEN. 19. The fusion protein of any one of embodiments 14-17, wherein the DNA-binding polypeptide is an RNA-guided, DNA-binding polypeptide. 20. The fusion protein of embodiment 19, wherein the RNA-guided, DNA-binding polypeptide is an RNA-guided nuclease (RGN) polypeptide. 21. The fusion protein of embodiment 20, wherein the RGN is a Type II CRISPR-Cas polypeptide. 22. The fusion protein of embodiment 20, wherein the RGN is a Type V CRISPR-Cas polypeptide. 23. The fusion protein of any one of embodiments 20-22, wherein the RGN is an RGN nickase. 24. The fusion protein of embodiment 20, wherein the RGN has an amino acid sequence having at least 95% sequence identity to any one of SEQ ID NOs: 41, 60, 366, and 368. 25. The fusion protein of embodiment 20, wherein the RGN has an amino acid sequence of any one of SEQ ID NOs: 41, 60, 366, and 368. 26. The fusion protein of embodiment 23, wherein the RGN nickase is any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 27. The fusion protein of any of embodiments 14-26, wherein the fusion protein further comprises at least one nuclear localization signal (NLS). 28. A nucleic acid molecule comprising a polynucleotide encoding a fusion protein comprising a DNA-binding polypeptide and a deaminase, wherein the deaminase is encoded by a nucleotide sequence that: a) has at least 80% sequence identity to any one of SEQ ID NOs: 451, 449, 443, 11-20, 444-448, 450, and 452-485, or b) encodes an amino acid sequence having at least 90% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 29. The nucleic acid molecule of embodiment 28, wherein said nucleotide sequence has at least 90% sequence identity to any one of SEQ ID NOs: 451, 449, 443, 11-20, 444-448, 450, and 452-485. 30. The nucleic acid molecule of embodiment 28, wherein said nucleotide sequence has at least 95% sequence identity to any one of SEQ ID NOs: 451, 449, 443, 11-20, 444-448, 450, and 452-485. 31. The nucleic acid molecule of embodiment 28, wherein said nucleotide sequence has 100% sequence identity to any one of SEQ ID NOs: 451, 449, 443, 11-20, 444-448, 450, and 452-485. 32. The nucleic acid molecule of embodiment 28, wherein said nucleotide sequence encodes an amino acid sequence having at least 95% sequence identity to any one of SEQ ID NOs 407, 405, 399, 1-10, 400-404, 406, and 408-441. 33. The nucleic acid molecule of embodiment 28, wherein said nucleotide sequence encodes an amino acid sequence having 100% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 34. The nucleic acid molecule of any one of embodiments 28-33, wherein the deaminase is an adenine deaminase. 35. The nucleic acid molecule of any one of embodiments 28-34, wherein the DNA-binding polypeptide is a meganuclease, zinc finger fusion protein, or a TALEN. 36. The nucleic acid molecule of any one of embodiments 28-34, wherein the DNA-binding polypeptide is an RNA-guided, DNA-binding polypeptide. 37. The nucleic acid molecule of embodiment 36, wherein the RNA-guided, DNA-binding polypeptide is an RNA-guided nuclease (RGN) polypeptide. 38. The nucleic acid molecule of embodiment 37, wherein the RGN is a Type II CRISPR-Cas polypeptide. 39. The nucleic acid molecule of embodiment 37, wherein the RGN is a Type V CRISPR-Cas polypeptide. 40. The nucleic acid molecule of any one of embodiments 37-39, wherein the RGN is an RGN nickase. 41. The nucleic acid molecule of embodiment 37, wherein the RGN has an amino acid sequence having at least 95% sequence identity to any one of SEQ ID NOs: 41, 60, 366, and 368. 42. The nucleic acid molecule of embodiment 37, wherein the RGN is SEQ ID NO: 41, 60, 366, or 368. 43. The nucleic acid molecule of embodiment 40, wherein the RGN nickase is any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 44. The nucleic acid molecule of any of embodiments 28-43, wherein the polynucleotide encoding the fusion protein is operably linked at its 5' end to a heterologous promoter. 45. The nucleic acid molecule of any of embodiments 28-44, wherein the polynucleotide encoding the fusion protein is operably linked at its 3' end to a heterologous terminator. 46. The nucleic acid molecule of any of embodiments 28-45, wherein the fusion protein comprises one or more nuclear localization signals. 47. The nucleic acid molecule of any of embodiments 28-46, wherein the fusion protein is codon optimized for expression in a eukaryotic cell. 48. The nucleic

acid molecule of any of embodiments 28-46, wherein the fusion protein is codon optimized for expression in a prokaryotic cell. 49. A vector comprising the nucleic acid molecule of any one of embodiments 28-48. 50. A vector comprising the nucleic acid molecule of any one of embodiments 28-48, further comprising at least one nucleotide sequence encoding a guide RNA (gRNA) capable of hybridizing to a target sequence. 51. The vector of embodiment 50, wherein the gRNA is a single guide RNA. 52. The vector of embodiment 50, wherein the gRNA is a dual guide RNA. 53. A cell comprising the fusion protein of any of embodiments 14-27. 54. A cell comprising the fusion protein of any one of embodiments 14-27, wherein the cell further comprises a guide RNA. 55. A cell comprising the nucleic acid molecule of any one of embodiments 28-48. 56. A cell comprising the vector of embodiments of any one of embodiments 49-52. 57. The cell of any one of embodiments 53-56, wherein the cell is a prokaryotic cell. 58. The cell of any one of embodiments 53-56, wherein the cell is a eukaryotic cell. 59. The cell of embodiment 58, wherein the eukaryotic cell is a mammalian cell. 60. The cell of embodiment 59, wherein the mammalian cell is a human cell. 61. The cell of embodiment 60, wherein the human cell is an immune cell. 62. The cell of embodiment 61, wherein the immune cell is a stem cell. 63. The cell of embodiment 62, wherein the stem cell is an induced pluripotent stem cell. 64. The cell of embodiment 58, wherein the eukaryotic cell is an insect or avian cell. 65. The cell of embodiment 58, wherein the eukaryotic cell is a fungal cell. 66. The cell of embodiment 58, wherein the eukaryotic cell is a plant cell. 67. A plant comprising the cell of embodiment 66. 68. A seed comprising the cell of embodiment 66. 69. A pharmaceutical composition comprising a pharmaceutically acceptable carrier and the fusion protein of any one of embodiments 14-27, the nucleic acid molecule of any one of embodiments 28-48, the vector of any one of embodiments 49-52, or the cell of any one of embodiments 59-63. 70. A method for making a fusion protein comprising culturing the cell of any one of embodiments 53-66 under conditions in which the fusion protein is expressed. 71. A method for making a fusion protein comprising introducing into a cell the nucleic acid molecule of any of embodiments 28-48 or a vector of any one of embodiments 49-52 and culturing the cell under conditions in which the fusion protein is expressed. 72. The method of embodiment 70 or 71, further comprising purifying said fusion protein. 73. A method for making an RGN fusion ribonucleoprotein complex, comprising introducing into a cell the nucleic acid molecule of any one of embodiments 37-43 and a nucleic acid molecule comprising an expression cassette encoding for a guide RNA, or the vector of any of embodiments 50-52, and culturing the cell under conditions in which the fusion protein and the gRNA are expressed and form an RGN fusion ribonucleoprotein complex. 74. The method of embodiment 73, further comprising purifying said RGN fusion ribonucleoprotein complex. 75. A system for modifying a target DNA molecule comprising a target DNA sequence, said system comprising: a) a fusion protein comprising an RNA-guided nuclease polypeptide (RGN) and a deaminase, wherein the deaminase has an amino acid sequence having at least 90% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441, or a nucleotide sequence encoding said fusion protein; and b) one or more guide RNAs capable of hybridizing to said target DNA sequence or one or more nucleotide sequences encoding the one or more guide RNAs (gRNAs); and wherein the one or more guide RNAs are capable of forming a complex with the fusion protein in order to direct said fusion protein to bind to said target DNA sequence and modify the target DNA molecule. 76. The system of embodiment 75, wherein said deaminase has an amino acid sequence having at least 95% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 77. The system of embodiment 75, wherein said deaminase has an amino acid sequence having 100% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 78. The system of any one of embodiments 75-77, wherein at least one of said nucleotide sequence encoding the one or more guide RNAs and said nucleotide sequence encoding the fusion protein is operably linked to a promoter heterologous to said nucleotide sequence. 79. The system of any one of embodiments 75-78, wherein the target DNA sequence is a eukaryotic target DNA sequence. 80. The system of any one of embodiments

75-79, wherein the target DNA sequence is located adjacent to a protospacer adjacent motif (PAM) that is recognized by the RGN. 81. The system of any one of embodiments 75-80, wherein the target DNA molecule is within a cell. 82. The system of embodiment 81, wherein the cell is a eukaryotic cell. 83. The system of embodiment 82, wherein the eukaryotic cell is a plant cell. 84. The system of embodiment 82, wherein the eukaryotic cell is a mammalian cell. 85. The system of embodiment 84, wherein the mammalian cell is a human cell. 86. The system of embodiment 85, wherein the human cell is an immune cell. 87. The system of embodiment 86, wherein the immune cell is a stem cell. 88. The system of embodiment 87, wherein the stem cell is an induced pluripotent stem cell. 89. The system of embodiment 82, wherein the eukaryotic cell is an insect cell. 90. The system of embodiment 81, wherein the cell is a prokaryotic cell. 91. The system of any one of embodiments 75-90, wherein the RGN of the fusion protein is a Type II CRISPR-Cas polypeptide. 92. The system of any one of embodiments 75-90, wherein the RGN of the fusion protein is a Type V CRISPR-Cas polypeptide. 93. The system of any one of embodiments 75-90, wherein the RGN of the fusion protein has an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 41, 60, 366, or 368. 94. The system of any one of embodiments 75-90, wherein the RGN of the fusion protein has an amino acid sequence of any one of SEQ ID NOs: 41, 60, 366, and 368. 95. The system of any one of embodiments 75-90, wherein the RGN of the fusion protein is an RGN nickase. 96. The system of embodiment 95, wherein the RGN nickase is any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 97. The system of any of embodiments 75-96, wherein the fusion protein comprises one or more nuclear localization signals. 98. The system of any of embodiments 75-97, wherein the fusion protein is codon optimized for expression in a eukaryotic cell. 99. The system of any of embodiments 75-98, wherein nucleotide sequences encoding the one or more guide RNAs and the nucleotide sequence encoding a fusion protein are located on one vector. 100. A pharmaceutical composition comprising a pharmaceutically acceptable carrier and the system of any one of embodiments 75-99. 101. A method for modifying a target DNA molecule comprising a target DNA sequence, said method comprising delivering a system according to any one of embodiments 75-99 to said target DNA molecule or a cell comprising the target DNA molecule. 102. The method of embodiment 101, wherein said modified target DNA molecule comprises an A>N mutation of at least one nucleotide within the target DNA molecule, wherein N is C, G, or T. 103. The method of embodiment 102, wherein said modified target DNA molecule comprises an A>G mutation of at least one nucleotide within the target DNA molecule. 104. A method for modifying a target DNA molecule comprising a target sequence comprising: a) assembling an RGN-deaminase ribonucleotide complex in vitro by combining: i) one or more guide RNAs capable of hybridizing to the target DNA sequence; and ii) a fusion protein comprising an RNA-guided nuclease polypeptide (RGN), and at least one deaminase, wherein the deaminase has an amino acid sequence having at least 90% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441; under conditions suitable for formation of the RGN-deaminase ribonucleotide complex; and b) contacting said target DNA molecule or a cell comprising said target DNA molecule with the in vitro-assembled RGN-deaminase ribonucleotide complex; wherein the one or more guide RNAs hybridize to the target DNA sequence, thereby directing said fusion protein to bind to said target DNA sequence and modification of the target DNA molecule occurs. 105. The method of embodiment 104, wherein said deaminase has an amino acid sequence having at least 95% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 106. The method of embodiment 104, wherein said deaminase has an amino acid sequence having 100% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 107. The method of any one of embodiments 104-106, wherein said modified target DNA molecule comprises an A>N mutation of at least one nucleotide within the target DNA molecule, wherein N is C, G, or T. 108. The method of embodiment 107, wherein said modified target DNA molecule comprises an A>G mutation of at least one nucleotide within the target DNA molecule. 109. The method of any one of

embodiments 104-108, wherein the RGN of the fusion protein is a Type II CRISPR-Cas polypeptide. 110. The method of any of embodiments 104-108, wherein the RGN of the fusion protein is a Type V CRISPR-Cas polypeptide. 111. The method of any of embodiments 104-108, wherein the RGN of the fusion protein has an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 41, 60, 366, or 368. 112. The method of any one of embodiments 104-108, wherein the RGN of the fusion protein has an amino acid sequence of any one of SEQ ID NOs: 41, 60, 366, and 368. 113. The method of any of embodiments 104-108, wherein the RGN of the fusion protein is an RGN nickase. 114. The method of embodiment 113, wherein the RGN nickase is any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 115. The method of any of embodiments 104-114, wherein the fusion protein comprises one or more nuclear localization signals. 116. The method of any of embodiments 104-115, wherein the fusion protein is codon optimized for expression in a eukaryotic cell. 117. The method of any one of embodiments 104-116, wherein said target DNA sequence is a eukaryotic target DNA sequence. 118. The method of any of embodiments 104-117, wherein said target DNA sequence is located adjacent to a protospacer adjacent motif (PAM). 119. The method of any of embodiments 104-118, wherein the target DNA molecule is within a cell. 120. The method of embodiment 119, wherein the cell is a eukaryotic cell. 121. The method of embodiment 120, wherein the eukaryotic cell is a plant cell. 122. The method of embodiment 120, wherein the eukaryotic cell is a mammalian cell. 123. The method of embodiment 122, wherein the mammalian cell is a human cell. 124. The method of embodiment 123, wherein the human cell is an immune cell. 125. The method of embodiment 124, wherein the immune cell is a stem cell. 126. The method of embodiment 125, wherein the stem cell is an induced pluripotent stem cell. 127. The method of embodiment 120, wherein the eukaryotic cell is an insect cell. 128. The method of embodiment 119, wherein the cell is a prokaryotic cell. 129. The method of any one of embodiments 119-128, further comprising selecting a cell comprising said modified DNA molecule. 130. A cell comprising a modified target DNA sequence according to the method of embodiment 129. 131. The cell of embodiment 130, wherein the cell is a eukaryotic cell. 132. The cell of embodiment 131, wherein the eukaryotic cell is a plant cell. 133. A plant comprising the cell of embodiment 132. 134. A seed comprising the cell of embodiment 132. 135. The cell of embodiment 131, wherein the eukaryotic cell is a mammalian cell. 136. The cell of embodiment 135, wherein the mammalian cell is a human cell. 137. The cell of embodiment 136, wherein the human cell is an immune cell. 138. The cell of embodiment 137, wherein the immune cell is a stem cell. 139. The cell of embodiment 138, wherein the stem cell is an induced pluripotent stem cell. 140. The cell of embodiment 131, wherein the eukaryotic cell is an insect cell. 141. The cell of embodiment 130, wherein the cell is a prokaryotic cell. 142. A pharmaceutical composition comprising the cell of any one of embodiments 135-139, and a pharmaceutically acceptable carrier. 143. A method for producing a genetically modified cell with a correction in a causal mutation for a genetically inherited disease, the method comprising introducing into the cell: a) a fusion protein comprising an RNA-guided nuclease polypeptide (RGN) and a deaminase, wherein the deaminase has an amino acid sequence having at least 90% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441, or a polynucleotide encoding said fusion protein, wherein said polynucleotide encoding the fusion protein is operably linked to a promoter to enable expression of the fusion protein in the cell; and b) one or more guide RNAs (gRNA) capable of hybridizing to a target DNA sequence, or a polynucleotide encoding said gRNA, wherein said polynucleotide encoding the gRNA is operably linked to a promoter to enable expression of the gRNA in the cell; whereby the fusion protein and gRNA target to the genomic location of the causal mutation and modify the genomic sequence to remove the causal mutation. 144. The method of embodiment 143, wherein said deaminase has an amino acid sequence having at least 95% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 145. The method of embodiment 143, wherein said deaminase has an amino acid sequence having 100% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-

441. 146. The method of any one of embodiments 143-145, wherein said RGN of the fusion protein is an RGN nickase. 147. The method of embodiment 146, wherein the RGN nickase is any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 148. The method of any one of embodiments 143-147, wherein the genome modification comprises introducing an A>G mutation of at least one nucleotide within the target DNA sequence. 149. The method of any of embodiments 143-148, wherein the cell is an animal cell. 150. The method of embodiment 149, wherein the animal cell is a mammalian cell. 151. The method of embodiment 150, wherein the cell is derived from a dog, cat, mouse, rat, rabbit, horse, sheep, goat, cow, pig, or human. 152. The method of any one of embodiments 143-151, wherein the correction of the causal mutation comprises correcting a nonsense mutation. 153. The method of embodiment 149, wherein the genetically inherited disease is a disease listed in Table 34. 154. The method of embodiment 149, wherein the genetically inherited disease is cystic fibrosis. 155. The method of embodiment 154, wherein the gRNA further comprises a spacer sequence that targets any one of SEQ ID NOs: 62-97, 116-139, 152-185, 203-234, 251-286, 305-344, 562, and 563, or the complement thereof. 156. The method of embodiment 155, wherein the gRNA comprises any one of SEQ ID NOs: 98-115, 140-151, 186-202, 235-250, 287-304, 345-364, and 564. 157. A CRISPR RNA (crRNA) or a nucleic acid molecule encoding the same, wherein said CRISPR RNA comprises a spacer sequence that targets a target DNA sequence within a cystic fibrosis transmembrane conductance regulator (CFTR) gene, wherein said target sequence has the sequence set forth as any one of SEQ ID NOs: 98-115, 140-151, 186-202, 235-250, 287-304, 345-364, 562, and 563, or the complement thereof. 158. A guide RNA comprising the crRNA of embodiment 157. 159. The guide RNA of embodiment 158, wherein said guide RNA is a dual-guide RNA. 160. The guide RNA of embodiment 158, wherein said guide RNA is a single guide RNA (sgRNA). 161. The guide RNA of embodiment 160, wherein said sgRNA has at least 90% sequence identity to any one of SEQ ID NOs: 98-115, 140-151, 186-202, 235-250, 287-304, 345-364, and 564. 162. The guide RNA of embodiment 160, wherein said sgRNA has at least 95% sequence identity to any one of SEQ ID NOs: 98-115, 140-151, 186-202, 235-250, 287-304, 345-364, and 564. 163. The guide RNA of embodiment 160, wherein said sgRNA has the sequence set forth as any one of SEQ ID NOs: 98-115, 140-151, 186-202, 235-250, 287-304, 345-364, and 564. 164. A vector comprising one or more nucleic acid molecules encoding said guide RNA of any one of embodiments 158-163. 165. A system for binding a target DNA sequence of a DNA molecule, said system comprising: a) one or more guide RNAs capable of hybridizing to said target DNA sequence or one or more polynucleotides comprising one or more nucleotide sequences encoding the one or more guide RNAs (gRNAs); and b) a fusion protein comprising an RNA-guided nuclease polypeptide (RGN) and an adenine deaminase, or a polynucleotide comprising a nucleotide sequence encoding the fusion protein; wherein the one or more guide RNAs are capable of hybridizing to the target DNA sequence, wherein the one or more guide RNAs are capable of forming a complex with the RGN polypeptide in order to direct said RGN polypeptide to bind to said target DNA sequence of the DNA molecule, and wherein at least one guide RNA comprises a CRISPR RNA (crRNA) comprising a spacer sequence that targets a target DNA sequence within a cystic fibrosis transmembrane conductance regulator (CFTR) gene, wherein said target sequence has the sequence set forth as any one of SEQ ID NOs: 98-115, 140-151, 186-202, 235-250, 287-304, 345-364, 562, and 563, or the complement thereof. 166. The system of embodiment 165, wherein at least one of said nucleotide sequences encoding the one or more guide RNAs and said nucleotide sequence encoding the fusion protein is operably linked to a promoter heterologous to said nucleotide sequence. 167. A system for binding a target DNA sequence of a DNA molecule, said system comprising: a) one or more guide RNAs capable of hybridizing to said target DNA sequence or one or more polynucleotides comprising one or more nucleotide sequences encoding the one or more guide RNAs (gRNAs); and b) a fusion protein comprising an RNA-guided nuclease polypeptide (RGN) and an adenine deaminase; wherein the one or more guide RNAs are capable of hybridizing to the target DNA sequence, wherein the one or more guide RNAs are

capable of forming a complex with the RGN polypeptide in order to direct said RGN polypeptide to bind to said target DNA sequence of the DNA molecule, and wherein at least one guide RNA comprises a CRISPR RNA (crRNA) comprising a spacer sequence that targets a target DNA sequence within a cystic fibrosis transmembrane conductance regulator (CFTR) gene, wherein said target sequence has the sequence set forth as any one of SEQ ID NOs: 98-115, 140-151, 186-202, 235-250, 287-304, 345-364, 562, and 563, or the complement thereof. 168. The system of embodiment 167, wherein at least one of said nucleotide sequences encoding the one or more guide RNAs is operably linked to a promoter heterologous to said nucleotide sequence. 169. The system of any one of embodiments 165-168, wherein the deaminase has an amino acid sequence having at least 90% sequence identity to any one of SEQ ID NOs: 1-10 and 399-441. 170. The system of any one of embodiments 165-168, wherein the deaminase has an amino acid sequence having at least 95% sequence identity to any one of SEQ ID NOs: 1-10 and 399-441. 171. The system of any one of embodiments 165-168, wherein the deaminase has an amino acid sequence having the sequence set forth in any one of SEQ ID NOs: 1-10 and 399-441. 172. The system of any one of embodiments 165-171, wherein said RGN polypeptide and said one or more guide RNAs are not found complexed to one another in nature. 173. The system of any one of embodiments 165-172, wherein: a) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 62-68, 80-85, 116-119, 128-131, 163, 164, 180, 181, 203-209, 219-225, 256-258, 274-276, 310-313, and 330-333, or the complement thereof, and wherein said RGN polypeptide has a sequence having at least 90% sequence identity to SEQ ID NO: 53; b) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 68-71, 86-89, 120-122, 132-134, 152-156, 169-173, 213-215, 229-231, 251-255, 269-273, 305-309, and 325-329, or the complement thereof, and wherein said RGN polypeptide has a sequence having at least 90% sequence identity to SEQ ID NO: 55; c) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 72, 73, 90, 91, 161, 162, 178, 179, 265, 266, 283, and 284 or the complement thereof, and wherein said RGN polypeptide has a sequence having at least 90% sequence identity to SEQ ID NO: 52; d) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 74, 75, 92, 93, 123, 124, 135, 136, 167, 184, 216-218, 232-234, 259-261, 277-279, 314-317, and 334-337, or the complement thereof, and wherein said RGN polypeptide has a sequence having at least 90% sequence identity to SEQ ID NO: 56; e) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 76, 94, 210-212, 226-228, 322, 342, 562, and 563, or the complement thereof, and wherein said RGN polypeptide has a sequence having at least 90% sequence identity to SEQ ID NO: 42; f) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 77, 95, 125, 137, 157-160, 174-177, 323, and 343, or the complement thereof, and wherein said RGN polypeptide has a sequence having at least 90% sequence identity to SEQ ID NO: 54; g) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 78, 96, 126, 138, 168, 185, 267, 285, 318, 319, 338, and 339, or the complement thereof, and wherein said RGN polypeptide has a sequence having at least 90% sequence identity to SEQ ID NO: 57; h) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 79, 97, 127, 139, 262-264, 280-282, 324, and 344, or the complement thereof, and wherein said RGN polypeptide has a sequence having at least 90% sequence identity to SEQ ID NO: 58; and i) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 165, 166, 182, 183, 268, 286, 320, 321, 340, and 341, or the complement thereof, and wherein said RGN polypeptide has a sequence having at least 90% sequence identity to SEQ ID NO: 59. 174. The system of any one of embodiments 165-172, wherein: a) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 62-68, 80-85, 116-119, 128-131, 163, 164, 180, 181, 203-209, 219-225, 256-258, 274-276, 310-313, and 330-333, or the complement thereof, and wherein said RGN polypeptide has a sequence having at least 95% sequence identity to SEQ ID NO: 53; b) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 68-71, 86-89, 120-122, 132-134, 152-156, 169-173, 213-215, 229-231, 251-255, 269-273, 305-309, and 325-329, or the complement thereof, and wherein said

RGN polypeptide has a sequence having at least 95% sequence identity to SEQ ID NO: 55; c) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 72, 73, 90, 91, 161, 162, 178, 179, 265, 266, 283, and 284 or the complement thereof, and wherein said RGN polypeptide has a sequence having at least 95% sequence identity to SEQ ID NO: 52; d) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 74, 75, 92, 93, 123, 124, 135, 136, 167, 184, 216-218, 232-234, 259-261, 277-279, 314-317, and 334-337, or the complement thereof, and wherein said RGN polypeptide has a sequence having at least 95% sequence identity to SEQ ID NO: 56; e) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 76, 94, 210-212, 226-228, 322, 342, 562, and 563, or the complement thereof, and wherein said RGN polypeptide has a sequence having at least 95% sequence identity to SEQ ID NO: 42; f) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 77, 95, 125, 137, 157-160, 174-177, 323, and 343, or the complement thereof, and wherein said RGN polypeptide has a sequence having at least 95% sequence identity to SEQ ID NO: 54; g) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 78, 96, 126, 138, 168, 185, 267, 285, 318, 319, 338, and 339, or the complement thereof, and wherein said RGN polypeptide has a sequence having at least 95% sequence identity to SEQ ID NO: 57; h) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 79, 97, 127, 139, 262-264, 280-282, 324, and 344, or the complement thereof, and wherein said RGN polypeptide has a sequence having at least 95% sequence identity to SEQ ID NO: 58; and i) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 165, 166, 182, 183, 268, 286, 320, 321, 340, and 341, or the complement thereof, and wherein said RGN polypeptide has a sequence having at least 95% sequence identity to SEQ ID NO: 59.

175. The system of any one of embodiments 165-172, wherein: a) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 62-68, 80-85, 116-119, 128-131, 163, 164, 180, 181, 203-209, 219-225, 256-258, 274-276, 310-313, and 330-333, or the complement thereof, and wherein said RGN polypeptide has a sequence having 100% sequence identity to SEQ ID NO: 53; b) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 68-71, 86-89, 120-122, 132-134, 152-156, 169-173, 213-215, 229-231, 251-255, 269-273, 305-309, and 325-329, or the complement thereof, and wherein said RGN polypeptide has a sequence having 100% sequence identity to SEQ ID NO: 55; c) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 72, 73, 90, 91, 161, 162, 178, 179, 265, 266, 283, and 284 or the complement thereof, and wherein said RGN polypeptide has a sequence having 100% sequence identity to SEQ ID NO: 52; d) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 74, 75, 92, 93, 123, 124, 135, 136, 167, 184, 216-218, 232-234, 259-261, 277-279, 314-317, and 334-337, or the complement thereof, and wherein said RGN polypeptide has a sequence having 100% sequence identity to SEQ ID NO: 56; e) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 76, 94, 210-212, 226-228, 322, 342, 562, and 563, or the complement thereof, and wherein said RGN polypeptide has a sequence having 100% sequence identity to SEQ ID NO: 42; f) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 77, 95, 125, 137, 157-160, 174-177, 323, and 343, or the complement thereof, and wherein said RGN polypeptide has a sequence having 100% sequence identity to SEQ ID NO: 54; g) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 78, 96, 126, 138, 168, 185, 267, 285, 318, 319, 338, and 339, or the complement thereof, and wherein said RGN polypeptide has a sequence having 100% sequence identity to SEQ ID NO: 57; h) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 79, 97, 127, 139, 262-264, 280-282, 324, and 344, or the complement thereof, and wherein said RGN polypeptide has a sequence having 100% sequence identity to SEQ ID NO: 58; and i) said target DNA sequence has the sequence set forth as any one of SEQ ID NOs: 165, 166, 182, 183, 268, 286, 320, 321, 340, and 341, or the complement thereof, and wherein said RGN polypeptide has a sequence having 100% sequence identity to SEQ ID NO: 59.

176. The system of any one of embodiments 165-175, wherein at least one guide RNA is a dual-guide RNA.

177. The

system of any one of embodiments 165-175, wherein at least one guide RNA is a single guide RNA (sgRNA). 178. The system of embodiment 177, wherein: a) said sgRNA has at least 90% sequence identity to any one of SEQ ID NOs: 98-104, 140-143, 197, 198, 235-241, 292-294, and 350-353, and wherein said RGN polypeptide has a sequence having at least 90% sequence identity to SEQ ID NO: 53; b) said sgRNA has at least 90% sequence identity to any one of SEQ ID NOs: 104-107, 144-146, 186-190, 245-247, 287-291, and 345-349, and wherein said RGN polypeptide has a sequence having at least 90% sequence identity to SEQ ID NO: 55; c) said sgRNA has at least 90% sequence identity to any one of SEQ ID NOs: 108, 109, 195, 196, 301, and 302, and wherein said RGN polypeptide has a sequence having at least 90% sequence identity to SEQ ID NO: 52; d) said sgRNA has at least 90% sequence identity to any one of SEQ ID NOs: 110, 111, 147, 148, 201, 248-250, 295-297, and 354-357, and wherein said RGN polypeptide has a sequence having at least 90% sequence identity to SEQ ID NO: 56; e) said sgRNA has at least 90% sequence identity to any one of SEQ ID NOs: 112, 242-244, 362, and 564, and wherein said RGN polypeptide has a sequence having at least 90% sequence identity to SEQ ID NO: 42; f) said sgRNA has at least 90% sequence identity to any one of SEQ ID NOs: 113, 149, 191-194, and 363, and wherein said RGN polypeptide has a sequence having at least 90% sequence identity to SEQ ID NO: 54; g) said sgRNA has at least 90% sequence identity to any one of SEQ ID NOs: 114, 150, 202, 303, 358, and 359, and wherein said RGN polypeptide has a sequence having at least 90% sequence identity to SEQ ID NO: 57; h) said sgRNA has at least 90% sequence identity to any one of SEQ ID NOs: 115, 151, 298-300, and 364, and wherein said RGN polypeptide has a sequence having at least 90% sequence identity to SEQ ID NO: 58; and i) said sgRNA has at least 90% sequence identity to any one of SEQ ID NOs: 199, 200, 304, 360, and 361, and wherein said RGN polypeptide has a sequence having at least 90% sequence identity to SEQ ID NO: 59. 179. The system of embodiment 177, wherein: a) said sgRNA has at least 95% sequence identity to any one of SEQ ID NOs: 98-104, 140-143, 197, 198, 235-241, 292-294, and 350-353, and wherein said RGN polypeptide has a sequence having at least 95% sequence identity to SEQ ID NO: 53; b) said sgRNA has at least 95% sequence identity to any one of SEQ ID NOs: 104-107, 144-146, 186-190, 245-247, 287-291, and 345-349, and wherein said RGN polypeptide has a sequence having at least 95% sequence identity to SEQ ID NO: 55; c) said sgRNA has at least 95% sequence identity to any one of SEQ ID NOs: 108, 109, 195, 196, 301, and 302, and wherein said RGN polypeptide has a sequence having at least 95% sequence identity to SEQ ID NO: 52; d) said sgRNA has at least 95% sequence identity to any one of SEQ ID NOs: 110, 111, 147, 148, 201, 248-250, 295-297, and 354-357, and wherein said RGN polypeptide has a sequence having at least 95% sequence identity to SEQ ID NO: 56; e) said sgRNA has at least 95% sequence identity to any one of SEQ ID NOs: 112, 242-244, 362, and 564, and wherein said RGN polypeptide has a sequence having at least 95% sequence identity to SEQ ID NO: 42; f) said sgRNA has at least 95% sequence identity to any one of SEQ ID NOs: 113, 149, 191-194, and 363, and wherein said RGN polypeptide has a sequence having at least 95% sequence identity to SEQ ID NO: 54; g) said sgRNA has at least 95% sequence identity to any one of SEQ ID NOs: 114, 150, 202, 303, 358, and 359, and wherein said RGN polypeptide has a sequence having at least 95% sequence identity to SEQ ID NO: 57; h) said sgRNA has at least 95% sequence identity to any one of SEQ ID NOs: 115, 151, 298-300, and 364, and wherein said RGN polypeptide has a sequence having at least 95% sequence identity to SEQ ID NO: 58; and i) said sgRNA has at least 95% sequence identity to any one of SEQ ID NOs: 199, 200, 304, 360, and 361, and wherein said RGN polypeptide has a sequence having at least 95% sequence identity to SEQ ID NO: 59. 180. The system of embodiment 177, wherein: a) said sgRNA has 100% sequence identity to any one of SEQ ID NOs: 98-104, 140-143, 197, 198, 235-241, 292-294, and 350-353, and wherein said RGN polypeptide has a sequence having 100% sequence identity to SEQ ID NO: 53; b) said sgRNA has 100% sequence identity to any one of SEQ ID NOs: 104-107, 144-146, 186-190, 245-247, 287-291, and 345-349, and wherein said RGN polypeptide has a sequence having 100% sequence identity to SEQ ID NO: 55; c) said sgRNA has 100%

sequence identity to any one of SEQ ID NOs: 108, 109, 195, 196, 301, and 302, and wherein said RGN polypeptide has a sequence having 100% sequence identity to SEQ ID NO: 52; d) said sgRNA has 100% sequence identity to any one of SEQ ID NOs: 110, 111, 147, 148, 201, 248-250, 295-297, and 354-357, and wherein said RGN polypeptide has a sequence having 100% sequence identity to SEQ ID NO: 56; e) said sgRNA has 100% sequence identity to any one of SEQ ID NOs: 112, 242-244, 362, and 564, and wherein said RGN polypeptide has a sequence having 100% sequence identity to SEQ ID NO: 42; f) said sgRNA has 100% sequence identity to any one of SEQ ID NOs: 113, 149, 191-194, and 363, and wherein said RGN polypeptide has a sequence having 100% sequence identity to SEQ ID NO: 54; g) said sgRNA has 100% sequence identity to any one of SEQ ID NOs: 114, 150, 202, 303, 358, and 359, and wherein said RGN polypeptide has a sequence having 100% sequence identity to SEQ ID NO: 57; h) said sgRNA has 100% sequence identity to any one of SEQ ID NOs: 115, 151, 298-300, and 364, and wherein said RGN polypeptide has a sequence having 100% sequence identity to SEQ ID NO: 58; and i) said sgRNA has 100% sequence identity to any one of SEQ ID NOs: 199, 200, 304, 360, and 361, and wherein said RGN polypeptide has a sequence having 100% sequence identity to SEQ ID NO: 59. 181. A cell comprising the crRNA or nucleic acid molecule of embodiment 157, the guide RNA of any one of embodiments 158-163, the vector of embodiment 164 or the system of any one of embodiments 165-180. 182. A pharmaceutical composition comprising the crRNA or nucleic acid molecule of embodiment 157, the guide RNA of any one of embodiments 158-163, the vector of embodiment 164, the cell of embodiment 181, or the system of any one of embodiments 165-180, and a pharmaceutically acceptable carrier. 183. A composition comprising: a) a fusion protein comprising a DNA-binding polypeptide and an adenine deaminase, or a nucleic acid molecule encoding the fusion protein; and b) a second adenine deaminase having at least 90% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441; or a nucleic acid molecule encoding the deaminase. 184. The composition of embodiment 183, wherein said second adenine deaminase has at least 90% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 185. The composition of embodiment 183, wherein said second adenine deaminase has 100% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 186. The composition of any one of embodiments 183-185, wherein the first adenine deaminase has at least 90% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 187. The composition of any one of embodiments 183-186, wherein the first adenine deaminase has at least 95% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 188. The composition of any one of embodiments 183-186, wherein the first adenine deaminase has 100% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 189. The composition of any one of embodiments 183-188, wherein the DNA-binding polypeptide is a meganuclease, zinc finger fusion protein, or a TALEN. 190. The composition of any one of embodiments 183-189, wherein the DNA-binding polypeptide is an RNA-guided, DNA-binding polypeptide. 191. The composition of embodiment 190, wherein the RNA-guided, DNA-binding polypeptide is an RNA-guided nuclease (RGN) polypeptide. 192. The composition of embodiment 191, wherein the RGN is an RGN nickase. 193. A vector comprising a nucleic acid molecule encoding a fusion protein and a nucleic acid molecule encoding a second deaminase, wherein said fusion protein comprises a DNA-binding polypeptide and a first adenine deaminase, and wherein said second adenine deaminase has at least 90% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 194. The vector of embodiment 193, wherein said second adenine deaminase has at least 90% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 195. The vector of embodiment 193, wherein said second adenine deaminase has 100% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 196. The vector of any one of embodiments 193-195, wherein the first adenine deaminase has at least 90% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-

404, 406, and 408-441. 197. The vector of any one of embodiments 193-195, wherein the first adenine deaminase has at least 95% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 198. The vector of any one of embodiments 193-195, wherein the first adenine deaminase has 100% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 199. The vector of any one of embodiments 193-198, wherein the DNA-binding polypeptide is a meganuclease, zinc finger fusion protein, or a TALEN. 200. The vector of any one of embodiments 193-198, wherein the DNA-binding polypeptide is an RNA-guided, DNA-binding polypeptide. 201. The vector of embodiment 200, wherein the RNA-guided, DNA-binding polypeptide is an RNA-guided nuclease (RGN) polypeptide. 202. The vector of embodiment 201, wherein the RGN is an RGN nickase. 203. A cell comprising the vector of any one of embodiments 193-202. 204. A cell comprising: a) a fusion protein comprising a DNA-binding polypeptide and a first adenine deaminase; or a nucleic acid molecule encoding the fusion protein; and b) a second adenine deaminase having at least 90% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441; or a nucleic acid molecule encoding the second adenine deaminase. 205. The cell of embodiment 204, wherein said second adenine deaminase has at least 90% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 206. The cell of embodiment 204, wherein said second adenine deaminase has 100% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 207. The cell of any one of embodiments 204-206, wherein the first adenine deaminase has at least 90% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 208. The cell of any one of embodiments 204-206, wherein the first adenine deaminase has at least 95% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 209. The cell of any one of embodiments 204-206, wherein the first adenine deaminase has 100% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441. 210. The cell of any one of embodiments 204-209, wherein the DNA-binding polypeptide is a meganuclease, zinc finger fusion protein, or a TALEN. 211. The cell of any one of embodiments 204-209, wherein the DNA-binding polypeptide is an RNA-guided, DNA-binding polypeptide. 212. The cell of embodiment 211, wherein the RNA-guided, DNA-binding polypeptide is an RNA-guided nuclease (RGN) polypeptide. 213. The cell of embodiment 212, wherein the RGN is an RGN nickase. 214. A pharmaceutical composition comprising a pharmaceutically acceptable carrier and the composition of any one of embodiments 183-192, the vector of any one of embodiments 193-202, or the cell of any one of embodiments 203-213. 215. A method for treating a disease, said method comprising administering to a subject in need thereof an effective amount of a pharmaceutical composition of any one of embodiments 69, 100, 142, and 214. 216. The method of embodiment 215, wherein said disease is associated with a causal mutation and said effective amount of said pharmaceutical composition corrects said causal mutation. 217. Use of the fusion protein of any one of embodiments 14-27, the nucleic acid molecule of any one of embodiments 28-48, the vector of any one of embodiments 49-52 and 193-202, the cell of any one of embodiments 59-63, 135-139, and 203-213, the system of any one of embodiments 75-99, or the composition of any one of embodiments 183-192 for the treatment of a disease in a subject. 218. The use of embodiment 217, wherein said disease is associated with a causal mutation and said treating comprises correcting said causal mutation. 219. Use of the fusion protein of any one of embodiments 14-27, the nucleic acid molecule of any one of embodiments 28-48, the vector of any one of embodiments 49-52 and 193-202, the cell of any one of embodiments 59-63, 135-139, and 203-213, the system of any one of embodiments 75-99, or the composition of any one of embodiments 183-192 for the manufacture of a medicament useful for treating a disease. 220. The use of embodiment 219, wherein said disease is associated with a causal mutation and an effective amount of said medicament corrects said causal mutation. 221. A nucleic acid molecule comprising a polynucleotide encoding an RNA-guided nuclease (RGN) polypeptide, wherein said polynucleotide comprises a nucleotide sequence encoding an RGN

polypeptide comprising an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 41 or 60, but lacking amino acid residues 590 to 597 of SEQ ID NO: 41 or 60; wherein said RGN polypeptide is capable of binding a target DNA sequence in an RNA-guided sequence specific manner when bound to a guide RNA (gRNA) capable of hybridizing to said target DNA sequence. 222. The nucleic acid molecule of embodiment 221, wherein said polynucleotide encoding an RGN polypeptide is operably linked to a promoter heterologous to said polynucleotide. 223. The nucleic acid molecule of embodiment 221 or 222, wherein said RGN polypeptide comprises an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 366 or 368. 224. The nucleic acid molecule of embodiment 221 or 222, wherein said RGN polypeptide comprises an amino acid sequence of SEQ ID NO: 366 or 368. 225. The nucleic acid molecule of any one of embodiments 221-223, wherein said RGN polypeptide is nuclease dead or functions as a nickase. 226. The nucleic acid molecule of embodiment 225, wherein said nickase has the amino acid sequence set forth in SEQ ID NO: 397 or 398. 227. The nucleic acid molecule of any one of embodiments 221-226, wherein the RGN polypeptide is operably fused to a base-editing polypeptide. 228. A vector comprising the nucleic acid molecule of any one of claims 221-227. 229. An isolated polypeptide comprising an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 41 or 60, but lacking amino acid residues 590 to 597 of SEQ ID NO: 41 or 60, wherein said polypeptide is an RNA-guided nuclease. 230. The isolated polypeptide of embodiment 229, wherein said RGN polypeptide comprises an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 366 or 368. 231. The isolated polypeptide of embodiment 230, wherein said RGN polypeptide comprises an amino acid sequence of SEQ ID NO: 366 or 368. 232. The isolated polypeptide of embodiment 229 or 230, wherein said RGN polypeptide is nuclease dead or functions as a nickase. 233. The isolated polypeptide of embodiment 232, wherein said nickase has the amino acid sequence set forth in SEQ ID NO: 397 or 398. 234. The isolated polypeptide of any one of embodiments 229-233, wherein the RGN polypeptide is operably fused to a base-editing polypeptide. 235. A cell comprising the nucleic acid molecule of any one of embodiments 221-227, the vector of claim 228, or the polypeptide of any one of claims 229-234. 236. An isolated polypeptide comprising an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 407, wherein said polypeptide has deaminase activity. 237. The isolated polypeptide of embodiment 236 comprising an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 407, wherein said polypeptide has deaminase activity. 238. The isolated polypeptide of embodiment 236, wherein the polypeptide comprises an amino acid sequence set forth in SEQ ID NO: 407. 239. A nucleic acid molecule comprising a polynucleotide encoding a deaminase polypeptide, wherein the deaminase is encoded by a nucleotide sequence that: a) has at least 80% sequence identity to SEQ ID NO: 451, or b) encodes an amino acid sequence having at least 90% sequence identity to any one of SEQ ID NO: 407. 240. The nucleic acid molecule of embodiment 239, wherein the deaminase is encoded by a nucleotide sequence that has at least 90% sequence identity to SEQ ID NO: 451. 241. The nucleic acid molecule of embodiment 239, wherein the deaminase is encoded by a nucleotide sequence that has at least 95% sequence identity to SEQ ID NO: 451. 242. The nucleic acid molecule of embodiment 239, wherein the deaminase is encoded by a nucleotide sequence that has at least 100% sequence identity to SEQ ID NO: 451. 243. The nucleic acid molecule of embodiments 239-242, wherein said nucleic acid molecule further comprises a heterologous promoter operably linked to said polynucleotide. 244. A pharmaceutical composition comprising a pharmaceutically acceptable carrier and the polypeptide of any one of embodiments 236-238 or the nucleic acid molecule of any one of embodiments 239-242. 245. A fusion protein comprising a DNA-binding polypeptide and a deaminase having at least 90% sequence identity to SEQ ID NO: 407. 246. A fusion protein of embodiment 245 comprising a DNA-binding polypeptide and a deaminase having at least 95% sequence identity to SEQ ID NO: 407. 247. A fusion protein of embodiment 245 comprising a DNA-binding polypeptide and a deaminase having 100% sequence identity to SEQ ID NO: 407.

248. The fusion protein of any one of embodiments 245-247, wherein the DNA-binding polypeptide is a RNA-guided nuclease (RGN) polypeptide. 249. The fusion protein of embodiment 248, wherein the RGN polypeptide is a Type II CRISPR-Cas polypeptide or a Type V CRISPR-Cas polypeptide. 250. The fusion protein of any one of embodiments 248-249, wherein the RGN polypeptide is a Cas9, a CasX, a CasY, a Cpf1, a C2c1, a C2c2, a C2c3, a GeoCas9, a CjCas9, a Cas12a, a Cas12b, a Cas12g, a Cas12h, a Cas12i, a Cas13b, a Cas13c, a Cas13d, a Cas14, a Csn2, an xCas9, an SpCas9-NG, an LbCas12a, an AsCas12a, a Cas9-KKH, a circularly permuted Cas9, an Argonaute (Ago), a SmacCas9, a Spy-macCas9 domain, or a RGN polypeptide with an amino acid sequence set forth in any one of SEQ ID NOs: 41, 60, 366, or 368. 251. The fusion protein of any one of embodiments 248-250, wherein the RGN polypeptide is a nickase. 252. The fusion protein of embodiment 251, wherein the nickase has an amino acid sequence having at least 95% sequence identity to any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 253. The fusion protein of embodiment 251, wherein the nickase has an amino acid sequence having 100% sequence identity to any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 254. A nucleic acid molecule comprising a polynucleotide encoding a fusion protein comprising a DNA-binding polypeptide and a deaminase, wherein the deaminase is encoded by a nucleotide sequence that: a) has at least 80% sequence identity to SEQ ID NO: 451, or b) encodes an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 407. 255. The nucleic acid molecule of embodiment 254, wherein the deaminase is encoded by a nucleotide sequence that has at least 90% sequence identity to SEQ ID NO: 451. 256. The nucleic acid molecule of embodiment 254, wherein the deaminase is encoded by a nucleotide sequence that has at least 95% sequence identity to SEQ ID NO: 451. 257. The nucleic acid molecule of embodiment 254, wherein the deaminase is encoded by a nucleotide sequence that has at least 100% sequence identity to SEQ ID NO: 451. 258. The nucleic acid molecule of any one of embodiments 254-257, wherein the DNA-binding polypeptide is a RGN polypeptide. 259. The nucleic acid molecule of embodiment 258, wherein the RGN is a Type II CRISPR-Cas polypeptide or a Type V CRISPR-Cas polypeptide. 260. The nucleic acid molecule of any one of embodiments 258-259, wherein the RGN polypeptide is a Cas9, a CasX, a CasY, a Cpf1, a C2c1, a C2c2, a C2c3, a GeoCas9, a CjCas9, a Cas12a, a Cas12b, a Cas12g, a Cas12h, a Cas12i, a Cas13b, a Cas13c, a Cas13d, a Cas14, a Csn2, an xCas9, an SpCas9-NG, an LbCas12a, an AsCas12a, a Cas9-KKH, a circularly permuted Cas9, an Argonaute (Ago), a SmacCas9, a Spy-macCas9 domain, or a RGN polypeptide with an amino acid sequence set forth in any one of SEQ ID NOs: 41, 60, 366, or 368. 261. The nucleic acid molecule of any one of embodiments 258-260, wherein the RGN polypeptide is a nickase. 262. The nucleic acid molecule of embodiment 261, wherein the nickase has an amino acid sequence having at least 95% sequence identity to any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 263. The nucleic acid molecule of embodiment 262, wherein the nickase has an amino acid sequence having 100% sequence identity to any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 264. A vector comprising the nucleic acid molecule of any one of embodiments 254-263. 265. The vector of embodiment 264, further comprising at least one nucleotide sequence encoding a guide RNA (gRNA) capable of hybridizing to a target sequence. 266. A ribonucleoprotein (RNP) complex comprising the fusion protein of any one of embodiments 245-253 and a guide RNA bound to the DNA-binding polypeptide of the fusion protein. 267. A cell comprising the fusion protein of any of embodiments 245-253, the nucleic acid molecule of any one of embodiments 254-263, the vector of any one of embodiments 264-265, or the RNP complex of embodiment 266. 268. A system for modifying a target DNA molecule comprising a target DNA sequence, said system comprising: a) a fusion protein comprising an RNA-guided nuclease (RGN) polypeptide and a deaminase, wherein the deaminase has an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 407, or a nucleotide sequence encoding said fusion protein; and b) one or more guide RNAs capable of hybridizing to said target DNA sequence or one or more nucleotide sequences encoding the one or more guide RNAs (gRNAs); and wherein the one or more guide RNAs are capable of forming a complex with the fusion protein in order to direct said

fusion protein to bind to said target DNA sequence and modify the target DNA molecule. 269. The system of embodiment 268, wherein said deaminase has an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 407. 270. The system of embodiment 268, wherein said deaminase has an amino acid sequence having 100% sequence identity to SEQ ID NO: 407. 271. The system of any one of embodiments 268-270, wherein at least one of said nucleotide sequence encoding the one or more guide RNAs and said nucleotide sequence encoding the fusion protein is operably linked to a promoter heterologous to said nucleotide sequence. 272. The system of any one of embodiments 268-271, wherein the target DNA sequence is located adjacent to a protospacer adjacent motif (PAM) that is recognized by the RGN polypeptide. 273. The system of any one of embodiments 268-272, wherein the target DNA sequence comprises a nucleic acid sequence selected from the group consisting of SEQ ID NOs: 62-97, 116-139, 152-185, 203-234, 251-286, 305-344, 562, and 563, or the complement thereof. 274. The system of any one of embodiments 268-273, wherein the gRNA sequence comprises a nucleic acid sequence selected from the group consisting of SEQ ID NOs: 98-115, 140-151, 186-202, 235-250, 287-304, 345-364, and 564. 275. The system of any one of embodiments 268-274, wherein the RGN polypeptide of the fusion protein is a Type II CRISPR-Cas polypeptide or a Type V CRISPR-Cas polypeptide. 276. The system of any one of embodiments 272-275, wherein the RGN polypeptide is a Cas9, a CasX, a CasY, a Cpf1, a C2c1, a C2c2, a C2c3, a GeoCas9, a CjCas9, a Cas12a, a Cas12b, a Cas12g, a Cas12h, a Cas12i, a Cas13b, a Cas13c, a Cas13d, a Cas14, a Csn2, an xCas9, an SpCas9-NG, an LbCas12a, an AsCas12a, a Cas9-KKH, a circularly permuted Cas9, an Argonaute (Ago), a SmacCas9, a Spy-macCas9 domain, or a RGN with an amino acid sequence set forth in any one of SEQ ID NOs: 41, 60, 366, or 368. 277. The system of embodiment 276, wherein the RGN polypeptide is a nickase. 278. The system of embodiment 277, wherein the nickase has an amino acid sequence having at least 95% sequence identity to any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 279. A pharmaceutical composition comprising a pharmaceutically acceptable carrier and the fusion protein of any of embodiments 245-253, the nucleic acid molecule of any one of embodiments 254-263, the vector of any one of embodiments 264-265, the RNP complex of embodiment 266, the cell of embodiment 267, or the system of any one of embodiments 268-28. 280. A method for modifying a target DNA molecule comprising a target sequence comprising: a) assembling an RGN-deaminase ribonucleotide complex by combining: i) one or more guide RNAs capable of hybridizing to the target DNA sequence; and ii) a fusion protein comprising an RNA-guided nuclease polypeptide (RGN), and at least one deaminase, wherein the deaminase has an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 407; under conditions suitable for formation of the RGN-deaminase ribonucleotide complex; and b) contacting said target DNA molecule or a cell comprising said target DNA molecule with the assembled RGN-deaminase ribonucleotide complex; wherein the one or more guide RNAs hybridize to the target DNA sequence, thereby directing said fusion protein to bind to said target DNA sequence and modification of the target DNA molecule occurs. 281. The method of embodiment 280, wherein the target DNA sequence comprises a nucleic acid sequence selected from the group consisting of SEQ ID NOs: 62-97, 116-139, 152-185, 203-234, 251-286, 305-344, 562, and 563, or the complement thereof. 282. The method of any one of embodiments 280-281, wherein the gRNA sequence comprises a nucleic acid sequence selected from the group consisting of SEQ ID NOs: 98-115, 140-151, 186-202, 235-250, 287-304, 345-364, and 564. 283. The method of any one of embodiments 280-283, wherein the method is performed in vitro, in vivo, or ex vivo. 284. A method of treating a subject having or at risk of developing a disease, disorder, or condition, the method comprising: administering to the subject the fusion protein of any of embodiments 245-253, the nucleic acid molecule of any one of embodiments 254-263, the vector of any one of embodiments 264-265, the RNP complex of embodiment 266, the cell of embodiment 267, the system of any one of embodiments 268-28, or the pharmaceutical composition of embodiment 279. 285. The method of embodiment 284, further comprising administering any one of a gRNA

comprising a nucleic acid sequence selected from the group consisting of SEQ ID NOs: 98-115, 140-151, 186-202, 235-250, 287-304, 345-364, and 564. 286. An isolated polypeptide comprising an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 405, wherein said polypeptide has deaminase activity. 287. The isolated polypeptide of embodiment 286 comprising an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 405, wherein said polypeptide has deaminase activity. 288. The isolated polypeptide of embodiment 286, wherein the polypeptide comprises an amino acid sequence set forth in SEQ ID NO: 407. 289. A nucleic acid molecule comprising a polynucleotide encoding a deaminase polypeptide, wherein the deaminase is encoded by a nucleotide sequence that: a) has at least 80% sequence identity to SEQ ID NO: 449, or b) encodes an amino acid sequence having at least 90% sequence identity to any one of SEQ ID NO: 405. 290. The nucleic acid molecule of embodiment 289, wherein the deaminase is encoded by a nucleotide sequence that has at least 90% sequence identity to SEQ ID NO: 449. 291. The nucleic acid molecule of embodiment 289, wherein the deaminase is encoded by a nucleotide sequence that has at least 95% sequence identity to SEQ ID NO: 449. 292. The nucleic acid molecule of embodiment 289, wherein the deaminase is encoded by a nucleotide sequence that has at least 100% sequence identity to SEQ ID NO: 449. 293. The nucleic acid molecule of embodiments 289-292, wherein said nucleic acid molecule further comprises a heterologous promoter operably linked to said polynucleotide. 294. A pharmaceutical composition comprising a pharmaceutically acceptable carrier and the polypeptide of any one of embodiments 286-288 or the nucleic acid molecule of any one of embodiments 289-293. 295. A fusion protein comprising a DNA-binding polypeptide and a deaminase having at least 90% sequence identity to SEQ ID NO: 405. 296. A fusion protein of embodiment 295 comprising a DNA-binding polypeptide and a deaminase having at least 95% sequence identity to SEQ ID NO: 405. 297. A fusion protein of embodiment 295 comprising a DNA-binding polypeptide and a deaminase having 100% sequence identity to SEQ ID NO: 405. 298. The fusion protein of any one of embodiments 295-297, wherein the DNA-binding polypeptide is a RNA-guided nuclease (RGN) polypeptide. 299. The fusion protein of embodiment 298, wherein the RGN polypeptide is a Type II CRISPR-Cas polypeptide or a Type V CRISPR-Cas polypeptide. 300. The fusion protein of any one of embodiments 298-299, wherein the RGN polypeptide is a Cas9, a CasX, a CasY, a Cpf1, a C2c1, a C2c2, a C2c3, a GeoCas9, a CjCas9, a Cas12a, a Cas12b, a Cas12g, a Cas12h, a Cas12i, a Cas13b, a Cas13c, a Cas13d, a Cas14, a Csn2, an xCas9, an SpCas9-NG, an LbCas12a, an AsCas12a, a Cas9-KKH, a circularly permuted Cas9, an Argonaute (Ago), a SmacCas9, a Spy-macCas9 domain, or a RGN polypeptide with an amino acid sequence set forth in any one of SEQ ID NOs: 41, 60, 366, or 368. 301. The fusion protein of any one of embodiments 298-300, wherein the RGN polypeptide is a nickase. 302. The fusion protein of embodiment 301, wherein the nickase has an amino acid sequence having at least 95% sequence identity to any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 303. The fusion protein of embodiment 301, wherein the nickase has an amino acid sequence having 100% sequence identity to any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 304. A nucleic acid molecule comprising a polynucleotide encoding a fusion protein comprising a DNA-binding polypeptide and a deaminase, wherein the deaminase is encoded by a nucleotide sequence that: a) has at least 80% sequence identity to SEQ ID NO: 449, or b) encodes an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 405. 305. The nucleic acid molecule of embodiment 304, wherein the deaminase is encoded by a nucleotide sequence that has at least 90% sequence identity to SEQ ID NO: 449. 306. The nucleic acid molecule of embodiment 304, wherein the deaminase is encoded by a nucleotide sequence that has at least 95% sequence identity to SEQ ID NO: 449. 307. The nucleic acid molecule of embodiment 304, wherein the deaminase is encoded by a nucleotide sequence that has at least 100% sequence identity to SEQ ID NO: 449. 308. The nucleic acid molecule of any one of embodiments 304-307, wherein the DNA-binding polypeptide is a RGN polypeptide. 309. The nucleic acid molecule of embodiment 308, wherein the RGN is a Type II CRISPR-Cas polypeptide or a Type V CRISPR-Cas polypeptide.

310. The nucleic acid molecule of any one of embodiments 308-309, wherein the RGN polypeptide is a Cas9, a CasX, a CasY, a Cpf1, a C2c1, a C2c2, a C2c3, a GeoCas9, a CjCas9, a Cas12a, a Cas12b, a Cas12g, a Cas12h, a Cas12i, a Cas13b, a Cas13c, a Cas13d, a Cas14, a Csn2, an xCas9, an SpCas9-NG, an LbCas12a, an AsCas12a, a Cas9-KKH, a circularly permuted Cas9, an Argonaute (Ago), a SmacCas9, a Spy-macCas9 domain, or a RGN polypeptide with an amino acid sequence set forth in any one of SEQ ID NOs: 41, 60, 366, or 368. 311. The nucleic acid molecule of any one of embodiments 308-310, wherein the RGN polypeptide is a nickase. 312. The nucleic acid molecule of embodiment 311, wherein the nickase has an amino acid sequence having at least 95% sequence identity to any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 313. The nucleic acid molecule of embodiment 312, wherein the nickase has an amino acid sequence having 100% sequence identity to any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 314. A vector comprising the nucleic acid molecule of any one of embodiments 304-313. 315. The vector of embodiment 314, further comprising at least one nucleotide sequence encoding a guide RNA (gRNA) capable of hybridizing to a target sequence. 316. A ribonucleoprotein (RNP) complex comprising the fusion protein of any one of embodiments 295-303 and a guide RNA bound to the DNA-binding polypeptide of the fusion protein. 317. A cell comprising the fusion protein of any of embodiments 295-303, the nucleic acid molecule of any one of embodiments 304-313, the vector of any one of embodiments 314-315, or the RNP complex of embodiment 316. 318. A system for modifying a target DNA molecule comprising a target DNA sequence, said system comprising: a) a fusion protein comprising an RNA-guided nuclease (RGN) polypeptide and a deaminase, wherein the deaminase has an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 405, or a nucleotide sequence encoding said fusion protein; and b) one or more guide RNAs capable of hybridizing to said target DNA sequence or one or more nucleotide sequences encoding the one or more guide RNAs (gRNAs); and wherein the one or more guide RNAs are capable of forming a complex with the fusion protein in order to direct said fusion protein to bind to said target DNA sequence and modify the target DNA molecule. 319. The system of embodiment 318, wherein said deaminase has an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 405. 320. The system of embodiment 318, wherein said deaminase has an amino acid sequence having 100% sequence identity to SEQ ID NO: 405. 321. The system of any one of embodiments 318-320, wherein at least one of said nucleotide sequence encoding the one or more guide RNAs and said nucleotide sequence encoding the fusion protein is operably linked to a promoter heterologous to said nucleotide sequence. 322. The system of any one of embodiments 318-321, wherein the target DNA sequence is located adjacent to a protospacer adjacent motif (PAM) that is recognized by the RGN polypeptide. 323. The system of any one of embodiments 318-322, wherein the target DNA sequence comprises a nucleic acid sequence selected from the group consisting of SEQ ID NOs: 62-97, 116-139, 152-185, 203-234, 251-286, 305-344, 562, and 563, or the complement thereof. 324. The system of any one of embodiments 318-323, wherein the gRNA sequence comprises a nucleic acid sequence selected from the group consisting of SEQ ID NOs: 98-115, 140-151, 186-202, 235-250, 287-304, 345-364, and 564. 325. The system of any one of embodiments 318-324, wherein the RGN polypeptide of the fusion protein is a Type II CRISPR-Cas polypeptide or a Type V CRISPR-Cas polypeptide. 326. The system of any one of embodiments 322-325, wherein the RGN polypeptide is a Cas9, a CasX, a CasY, a Cpf1, a C2c1, a C2c2, a C2c3, a GeoCas9, a CjCas9, a Cas12a, a Cas12b, a Cas12g, a Cas12h, a Cas12i, a Cas13b, a Cas13c, a Cas13d, a Cas14, a Csn2, an xCas9, an SpCas9-NG, an LbCas12a, an AsCas12a, a Cas9-KKH, a circularly permuted Cas9, an Argonaute (Ago), a SmacCas9, a Spy-macCas9 domain, or a RGN with an amino acid sequence set forth in any one of SEQ ID NOs: 41, 60, 366, or 368. 327. The system of embodiment 326, wherein the RGN polypeptide is a nickase. 328. The system of embodiment 327, wherein the nickase has an amino acid sequence having at least 95% sequence identity to any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 329. A pharmaceutical composition comprising a pharmaceutically acceptable carrier and the fusion protein of any of

embodiments 295-303, the nucleic acid molecule of any one of embodiments 304-313, the vector of any one of embodiments 314-315, the RNP complex of embodiment 316, the cell of embodiment 317, or the system of any one of embodiments 318-328. 330. A method for modifying a target DNA molecule comprising a target sequence comprising: a) assembling an RGN-deaminase ribonucleotide complex by combining: i) one or more guide RNAs capable of hybridizing to the target DNA sequence; and ii) a fusion protein comprising an RNA-guided nuclease polypeptide (RGN), and at least one deaminase, wherein the deaminase has an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 405; under conditions suitable for formation of the RGN-deaminase ribonucleotide complex; and b) contacting said target DNA molecule or a cell comprising said target DNA molecule with the assembled RGN-deaminase ribonucleotide complex; wherein the one or more guide RNAs hybridize to the target DNA sequence, thereby directing said fusion protein to bind to said target DNA sequence and modification of the target DNA molecule occurs. 331. The method of embodiment 330, wherein the target DNA sequence comprises a nucleic acid sequence selected from the group consisting of SEQ ID NOs: 62-97, 116-139, 152-185, 203-234, 251-286, 305-344, 562, and 563, or the complement thereof. 332. The method of any one of embodiments 330-331, wherein the gRNA sequence comprises a nucleic acid sequence selected from the group consisting of SEQ ID NOs: 98-115, 140-151, 186-202, 235-250, 287-304, 345-364, and 564. 333. The method of any one of embodiments 330-332, wherein the method is performed in vitro, in vivo, or ex vivo. 334. A method of treating a subject having or at risk of developing a disease, disorder, or condition, the method comprising: administering to the subject the fusion protein of any of embodiments 295-303, the nucleic acid molecule of any one of embodiments 304-313, the vector of any one of embodiments 314-315, the RNP complex of embodiment 316, the cell of embodiment 317, the system of any one of embodiments 318-328, or the pharmaceutical composition of embodiment 329. 335. The method of embodiment 334, further comprising administering any one of a gRNA comprising a nucleic acid sequence selected from the group consisting of SEQ ID NOs: 98-115, 140-151, 186-202, 235-250, 287-304, 345-364, and 564. 336. An isolated polypeptide comprising an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 399, wherein said polypeptide has deaminase activity. 337. The isolated polypeptide of embodiment 336 comprising an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 399, wherein said polypeptide has deaminase activity. 338. The isolated polypeptide of embodiment 336, wherein the polypeptide comprises an amino acid sequence set forth in SEQ ID NO: 399. 339. A nucleic acid molecule comprising a polynucleotide encoding a deaminase polypeptide, wherein the deaminase is encoded by a nucleotide sequence that: a) has at least 80% sequence identity to SEQ ID NO: 443, or b) encodes an amino acid sequence having at least 90% sequence identity to any one of SEQ ID NO: 399. 340. The nucleic acid molecule of embodiment 339, wherein the deaminase is encoded by a nucleotide sequence that has at least 90% sequence identity to SEQ ID NO: 443. 341. The nucleic acid molecule of embodiment 339, wherein the deaminase is encoded by a nucleotide sequence that has at least 95% sequence identity to SEQ ID NO: 443. 342. The nucleic acid molecule of embodiment 339, wherein the deaminase is encoded by a nucleotide sequence that has at least 100% sequence identity to SEQ ID NO: 443. 343. The nucleic acid molecule of embodiments 339-342, wherein said nucleic acid molecule further comprises a heterologous promoter operably linked to said polynucleotide. 344. A pharmaceutical composition comprising a pharmaceutically acceptable carrier and the polypeptide of any one of embodiments 336-338 or the nucleic acid molecule of any one of embodiments 339-342. 345. A fusion protein comprising a DNA-binding polypeptide and a deaminase having at least 90% sequence identity to SEQ ID NO: 399. 346. A fusion protein of embodiment 345 comprising a DNA-binding polypeptide and a deaminase having at least 95% sequence identity to SEQ ID NO: 399. 347. A fusion protein of embodiment 345 comprising a DNA-binding polypeptide and a deaminase having 100% sequence identity to SEQ ID NO: 399. 348. The fusion protein of any one of embodiments 345-347, wherein the DNA-binding polypeptide is a RNA-guided nuclease (RGN)

polypeptide. 349. The fusion protein of embodiment 348, wherein the RGN polypeptide is a Type II CRISPR-Cas polypeptide or a Type V CRISPR-Cas polypeptide. 350. The fusion protein of any one of embodiments 348-349, wherein the RGN polypeptide is a Cas9, a CasX, a CasY, a Cpf1, a C2c1, a C2c2, a C2c3, a GeoCas9, a CjCas9, a Cas12a, a Cas12b, a Cas12g, a Cas12h, a Cas12i, a Cas13b, a Cas13c, a Cas13d, a Cas14, a Csn2, an xCas9, an SpCas9-NG, an LbCas12a, an AsCas12a, a Cas9-KKH, a circularly permuted Cas9, an Argonaute (Ago), a SmacCas9, a Spy-macCas9 domain, or a RGN polypeptide with an amino acid sequence set forth in any one of SEQ ID NOs: 41, 60, 366, or 368. 351. The fusion protein of any one of embodiments 348-350, wherein the RGN polypeptide is a nickase. 352. The fusion protein of embodiment 351, wherein the nickase has an amino acid sequence having at least 95% sequence identity to any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 353. The fusion protein of embodiment 351, wherein the nickase has an amino acid sequence having 100% sequence identity to any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 354. A nucleic acid molecule comprising a polynucleotide encoding a fusion protein comprising a DNA-binding polypeptide and a deaminase, wherein the deaminase is encoded by a nucleotide sequence that: a) has at least 80% sequence identity to SEQ ID NO: 443, or b) encodes an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 399. 355. The nucleic acid molecule of embodiment 354, wherein the deaminase is encoded by a nucleotide sequence that has at least 90% sequence identity to SEQ ID NO: 443. 356. The nucleic acid molecule of embodiment 354, wherein the deaminase is encoded by a nucleotide sequence that has at least 95% sequence identity to SEQ ID NO: 443. 357. The nucleic acid molecule of embodiment 354, wherein the deaminase is encoded by a nucleotide sequence that has at least 100% sequence identity to SEQ ID NO: 443. 358. The nucleic acid molecule of any one of embodiments 354-357, wherein the DNA-binding polypeptide is a RGN polypeptide. 359. The nucleic acid molecule of embodiment 358, wherein the RGN is a Type II CRISPR-Cas polypeptide or a Type V CRISPR-Cas polypeptide. 360. The nucleic acid molecule of any one of embodiments 358-359, wherein the RGN polypeptide is a Cas9, a CasX, a CasY, a Cpf1, a C2c1, a C2c2, a C2c3, a GeoCas9, a CjCas9, a Cas12a, a Cas12b, a Cas12g, a Cas12h, a Cas12i, a Cas13b, a Cas13c, a Cas13d, a Cas14, a Csn2, an xCas9, an SpCas9-NG, an LbCas12a, an AsCas12a, a Cas9-KKH, a circularly permuted Cas9, an Argonaute (Ago), a SmacCas9, a Spy-macCas9 domain, or a RGN polypeptide with an amino acid sequence set forth in any one of SEQ ID NOs: 41, 60, 366, or 368. 361. The nucleic acid molecule of any one of embodiments 358-360, wherein the RGN polypeptide is a nickase. 362. The nucleic acid molecule of embodiment 361, wherein the nickase has an amino acid sequence having at least 95% sequence identity to any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 363. The nucleic acid molecule of embodiment 362, wherein the nickase has an amino acid sequence having 100% sequence identity to any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 364. A vector comprising the nucleic acid molecule of any one of embodiments 354-363. 365. The vector of embodiment 364, further comprising at least one nucleotide sequence encoding a guide RNA (gRNA) capable of hybridizing to a target sequence. 366. A ribonucleoprotein (RNP) complex comprising the fusion protein of any one of embodiments 345-353 and a guide RNA bound to the DNA-binding polypeptide of the fusion protein. 367. A cell comprising the fusion protein of any of embodiments 345-353, the nucleic acid molecule of any one of embodiments 354-363, the vector of any one of embodiments 364-365, or the RNP complex of embodiment 366. 368. A system for modifying a target DNA molecule comprising a target DNA sequence, said system comprising: a) a fusion protein comprising an RNA-guided nuclease (RGN) polypeptide and a deaminase, wherein the deaminase has an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 399, or a nucleotide sequence encoding said fusion protein; and b) one or more guide RNAs capable of hybridizing to said target DNA sequence or one or more nucleotide sequences encoding the one or more guide RNAs (gRNAs); and wherein the one or more guide RNAs are capable of forming a complex with the fusion protein in order to direct said fusion protein to bind to said target DNA sequence and modify the target DNA molecule. 369. The system of embodiment 368,

wherein said deaminase has an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 399. 370. The system of embodiment 368, wherein said deaminase has an amino acid sequence having 100% sequence identity to SEQ ID NO: 399. 371. The system of any one of embodiments 368-370, wherein at least one of said nucleotide sequence encoding the one or more guide RNAs and said nucleotide sequence encoding the fusion protein is operably linked to a promoter heterologous to said nucleotide sequence. 372. The system of any one of embodiments 368-371, wherein the target DNA sequence is located adjacent to a protospacer adjacent motif (PAM) that is recognized by the RGN polypeptide. 373. The system of any one of embodiments 368-372, wherein the target DNA sequence comprises a nucleic acid sequence selected from the group consisting of SEQ ID NOs: 62-97, 116-139, 152-185, 203-234, 251-286, 305-344, 562, and 563, or the complement thereof. 374. The system of any one of embodiments 368-373, wherein the gRNA sequence comprises a nucleic acid sequence selected from the group consisting of SEQ ID NOs: 98-115, 140-151, 186-202, 235-250, 287-304, 345-364, and 564. 375. The system of any one of embodiments 368-374, wherein the RGN polypeptide of the fusion protein is a Type II CRISPR-Cas polypeptide or a Type V CRISPR-Cas polypeptide. 376. The system of any one of embodiments 372-375, wherein the RGN polypeptide is a Cas9, a CasX, a CasY, a Cpf1, a C2c1, a C2c2, a C2c3, a GeoCas9, a CjCas9, a Cas12a, a Cas12b, a Cas12g, a Cas12h, a Cas12i, a Cas13b, a Cas13c, a Cas13d, a Cas14, a Csn2, an xCas9, an SpCas9-NG, an LbCas12a, an AsCas12a, a Cas9-KKH, a circularly permuted Cas9, an Argonaute (Ago), a SmacCas9, a Spy-macCas9 domain, or a RGN with an amino acid sequence set forth in any one of SEQ ID NOs: 41, 60, 366, or 368. 377. The system of embodiment 376, wherein the RGN polypeptide is a nickase. 378. The system of embodiment 377, wherein the nickase has an amino acid sequence having at least 95% sequence identity to any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398. 379. A pharmaceutical composition comprising a pharmaceutically acceptable carrier and the fusion protein of any of embodiments 345-353, the nucleic acid molecule of any one of embodiments 354-363, the vector of any one of embodiments 364-365, the RNP complex of embodiment 366, the cell of embodiment 367, or the system of any one of embodiments 368-378. 380. A method for modifying a target DNA molecule comprising a target sequence comprising: a) assembling an RGN-deaminase ribonucleotide complex by combining: i) one or more guide RNAs capable of hybridizing to the target DNA sequence; and ii) a fusion protein comprising an RNA-guided nuclease polypeptide (RGN), and at least one deaminase, wherein the deaminase has an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 399; under conditions suitable for formation of the RGN-deaminase ribonucleotide complex; and b) contacting said target DNA molecule or a cell comprising said target DNA molecule with the assembled RGN-deaminase ribonucleotide complex; wherein the one or more guide RNAs hybridize to the target DNA sequence, thereby directing said fusion protein to bind to said target DNA sequence and modification of the target DNA molecule occurs. 381. The method of embodiment 380, wherein the target DNA sequence comprises a nucleic acid sequence selected from the group consisting of SEQ ID NOs: 62-97, 116-139, 152-185, 203-234, 251-286, 305-344, 562, and 563, or the complement thereof. 382. The method of any one of embodiments 380-381, wherein the gRNA sequence comprises a nucleic acid sequence selected from the group consisting of SEQ ID NOs: 98-115, 140-151, 186-202, 235-250, 287-304, 345-364, and 564. 383. The method of any one of embodiments 380-382, wherein the method is performed in vitro, in vivo, or ex vivo. 384. A method of treating a subject having or at risk of developing a disease, disorder, or condition, the method comprising: administering to the subject the fusion protein of any of embodiments 345-353, the nucleic acid molecule of any one of embodiments 354-363, the vector of any one of embodiments 364-365, the RNP complex of embodiment 366, the cell of embodiment 367, the system of any one of embodiments 368-378, or the pharmaceutical composition of embodiment 379. 385. The method of embodiment 384, further comprising administering any one of a gRNA comprising a nucleic acid sequence selected from the group consisting of SEQ ID NOs: 98-115, 140-151, 186-202, 235-250, 287-304, 345-364, and 564.

386. A method for producing or reducing at least one symptom of cystic fibrosis, the method comprising administering to a subject in need thereof an effective amount of: a) a fusion protein comprising an RNA-guided nuclease polypeptide (RGN) and a deaminase, wherein the deaminase has an amino acid sequence having at least 90% sequence identity to any one of SEQ ID NOs: 407, 405, 399, 1-10, 400-404, 406, and 408-441, or a polynucleotide encoding said fusion protein, wherein said polynucleotide encoding the fusion protein is operably linked to a promoter to enable expression of the fusion protein in the cell; and b) one or more guide RNAs (gRNA) capable of hybridizing to a target DNA sequence, or a polynucleotide encoding said gRNA, wherein said polynucleotide encoding the gRNA is operably linked to a promoter to enable expression of the gRNA in the cell; whereby the fusion protein and gRNA target to the genomic location of the causal mutation and modify the genomic sequence to remove the causal mutation. 387. The method of embodiment 386, wherein the gRNA comprises a spacer sequence that targets any one of SEQ ID NOs: 62-97, 116-139, 152-185, 203-234, 251-286, 305-344, 562, and 563, or the complement thereof. 388. The method of embodiments 386 or 387, wherein the gRNA comprises any one of SEQ ID NOs: 98-115, 140-151, 186-202, 235-250, 287-304, 345-364, and 564. 389. The method of any one of claims 386-388, wherein said the RGN has an amino acid sequence having at least 90% sequence identity to any one of SEQ ID NOs: 41, 60, 366, and 368. 390. The method of any one of claims 386-389, wherein said the RGN has an amino acid sequence having at least 90% sequence identity to any one of SEQ ID NOs: 42, 52-59, 61, 397, and 398.

(222) The following examples are offered by way of illustration and not by way of limitation.

EXPERIMENTAL

Example 1: Demonstration of Base Editing in Mammalian Cells

(223) The deaminases shown in Table 1 below were produced based on naturally occurring deaminases which were then mutated and selected for adenine deaminase activity in prokaryotic cells.

(224) TABLE-US-00001 TABLE 1 Deaminase sequences Deaminase SEQ ID NO. APG09982 1 APG03724 2 APG09949 3 APG08196 4 APG06333 5 APG06489 6 APG08449 7 APG05174 8 APG09102 9 APG05723 10

(225) To determine if the deaminases of Table 1 are able to perform adenine base editing in mammalian cells, each deaminase was operably fused to an RGN nickase to produce a fusion protein. Residues predicted to deactivate the RuvC domain of the RGN APG07433.1 (set forth as SEQ ID NO: 41; described in PCT publication WO 2019/236566, incorporated by reference herein) were identified and the RGN was modified to a nickase variant (nAPG07433.1; SEQ ID NO: 42). A nickase variant of an RGN is referred to herein as “nRGN”. It should be understood that any nickase variant of an RGN may be used to produce a fusion protein of the invention.

(226) Deaminase and nRGN nucleotide sequences codon optimized for mammalian expression were synthesized as fusion proteins with an N-terminal nuclear localization tag and cloned into the pTwist CMV (Twist Biosciences) expression plasmid. Each fusion protein comprises, starting at the amino terminus, the SV40 NLS (SEQ ID NO: 43) operably linked at the C-terminal end to 3×FLAG Tag (SEQ ID NO: 44), operably linked at the C-terminal end to a deaminase, operably linked at the C-terminal end to a peptide linker (SEQ ID NO: 45), operably linked at the C-terminal end to an nRGN (for example, nAPG07433.1, which is SEQ ID NO: 42), finally operably linked at the C-terminal end to the nucleoplasmin NLS (SEQ ID NO: 45). All fusion proteins comprise at least one NLS and a 3×FLAG Tag, as described above.

(227) Expression plasmids comprising an expression cassette encoding a sgRNA expressed by a human U6 promoter (SEQ ID NO: 50) were also produced. Human genomic target sequences and the sgRNA sequences for guiding the fusion proteins to the genomic targets are indicated in Table 2.

(228) TABLE-US-00002 TABLE 2 Guide RNA sequences Forward Reverse Target sgRNA Primer for Primer for sgRNA ID sequence sequence amplification amplification SGN000930 21 26 31 32

0.1 0.0 APG05174 C 0 0 0 0.1 0 0 0 0 G 0.9 3.2 1 0 0 0.1 0 T 0 0 0 0 0.1 0 0 APG09102 C 0 0 0 0 0 0
0 G 2.3 6.2 2.1 0 0 0 0 T 0 0 0 0 0 0 0 APG05723 C 0 0 0 0 0 0 0 G 1.1 1.9 1.2 0 0 0 0 T 0 0 0 0 0 0
0

(235) All fusion proteins showed base editing of over 1% at multiple locations in target SGN000186. APG09102 showed 6.2% A>G conversion at position A16; it also showed over 2% base editing at positions A9 and A18. For all fusion proteins tested, position A16 was the most highly edited.

(236) TABLE-US-00006 TABLE 6 A > N Editing Rate using guide SGN000194 Deaminase A6
A10 A13 A15 A21 A23 A26 A27 APG09982 C 0 0 0 0 0 0 0 0 G 0 0.3 0.6 1.5 0 0 0 0 T 0 0 0 0 0 0
0 0 APG03724 C 0 0 0 0 0 0 0 0 G 0 0.1 0.3 1 0 0 0 0 T 0 0 0 0 0 0 0 0 APG09949 C 0 0 0 0 0 0 0 0
G 0 0.2 0.3 1.6 0 0 0 0 T 0 0 0 0 0 0 0 0 APG08196 C 0 0 0 0 0 0 0 0 G 0.1 0.4 0.1 0.9 0 0 0 0 T 0 0
0 0 0 0 0 0 APG06333 C 0 0 0 0 0 0 0 0 G 0 0.2 0.3 1 0 0 0 0 T 0 0 0 0 0 0 0 0 APG06489 C 0 0 0 0
0 0 0 0 G 0 0.4 0.2 1.1 0 0 0 0 T 0 0 0 0 0 0 0 0 APG08449 C 0 0 0 0 0 0 0 0 G 0.1 0.3 0.4 1.8 0 0 0
0 T 0 0 0 0 0 0 0 0 APG05174 C 0 0 0 0 0 0 0 0 G 0.1 0.1 0.3 0.9 0 0 0 0 T 0 0 0 0 0 0 0 0
APG09102 C 0 0 0 0 0 0 0 0 G 0 0.2 0.7 1.6 0 0 0 0 T 0 0 0 0 0 0 0 0 APG05723 C 0 0 0 0 0 0 0 0
G 0 0 0.1 0.9 0 0 0 0 T 0 0 0 0 0 0 0 0

(237) With SGN00194, all fusion proteins showed 0.9%-1.8% A>G editing at position A15. No detectable editing was seen in positions A21, A23, A26 and A27.

(238) TABLE-US-00007 TABLE 7 A > N Editing Rate using guide SGN000930 Deaminase A2 A4
A5 A8 A9 A10 A14 A15 A16 A20 A21 A23 A24 A26 A27 A29 A30 APG09982 C 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 G 0 0 0 0 0 0.3 0.7 0.1 0.2 0.5 0.2 0 0 0 0 0 0 T 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
APG03724 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 G 0 0 0 0 0.1 0.1 0.4 0.5 0.2 0.2 0.3 0.1 0 0 0 0 0 0 T 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 APG09949 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 G 0 0 0 0 0 0.1 0.5 0.3
0.3 0.4 0 0 0 0 0 0 0 0 T 0 APG08196 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 G 0 0 0 0 0.1 0 0.2 0.7 0.3 0.2 0.4 0.1 0 0 0 0 0 0 T 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 APG06333 C 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 G 0 0 0 0 0.1 0.1 0 0.3 0.4 0.3 0.9 0.2 0.1 0.1 0 0 0 0 T 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 APG06489 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 G 0 0 0 0 0.3 0.1 0.2 0.8 0.3 0.4 0.6 0
0.1 0 0 0 0 0 0 T 0 APG08449 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 G 0 0
0 0.1 0.1 0.3 0.6 0.4 0.2 0.4 0.1 0 0 0 0 0 0 T 0 APG05174 C 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 G 0 0 0 0 0.1 0.2 0.8 0.3 0.4 0.2 0.2 0 0 0 0 0 0 T 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 APG09102 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 G 0 0 0 0 0 0 0.9 0.1 0.1 0.6 0.3 0 0 0 0 0 0 T 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 APG05723 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 G 0 0 0 0 0 0.1 0.1 1.2
0.6 0.2 0.5 0 0 0 0 0 0 0 0 T 0

(239) A14 was the most highly edited position in SGN000930 with all fusion proteins tested. The editing rate ranged from 0.3%-1.2% for A>G conversions.

Example 2: Fluorescence Assay for Targeted Adenine Base Editing

(240) A vector harboring Enhanced Green Fluorescent Protein (EGFP) containing a W58X mutation which causes a premature stop codon (GFP-STOP, SEQ ID NO: 47) was constructed such that the W58 codon can be reverted from a stop codon (TGA) to the wild-type tryptophan (TGG) residue using an adenine deaminase to alter the third position A to G. Successful A to G conversion results in the expression of EGFP which can be quantified. A second vector capable of expressing a guide RNA which targets the deaminase-RGN fusion protein to the region around the W58X mutation (SEQ ID NO: 48) was also produced.

(241) This GFP-STOP reporter vector, along with the vectors capable of expressing a deaminase-nRGN fusion protein and the corresponding guide RNA, were transfected into HEK293T cells, using either lipofection or electroporation. For lipofection, cells were seeded at 1×10⁵ cells/well in 24-well plates the day prior to transfection in growth medium (DMEM+10% Fetal Bovine Serum+1% Penicillin/streptomycin), 500 ng each of the GFP-STOP reporter vector, deaminase-RGN expression vector, and guide RNA expression vector were transfected using Lipofectamine® 3000 reagent (Thermo Fisher Scientific) following manufacturer's instructions.

For electroporation, cells were electroporated using the Neon® Transfection System (Thermo Fisher Scientific) following manufacturer's instructions.

(242) In addition to transient transfection of the fluorescent GFP-STOP reporter, a stable cell line harboring a chromosomally integrated GFP-STOP cassette was generated. Once the stable line was established, for transfection, cells were seeded at 1×10^5 cells/well in 24-well plates the day prior to transfection in growth medium (DMEM+10% Fetal Bovine Serum+1%

Penicillin/streptomycin). 500 mg each of the deaminase-nRGN expression vector and guide RNA expression vector were transfected using Lipofectamine® 3000 reagent (Thermo Fisher Scientific) following manufacturer's instructions. For electroporation, cells were electroporated using the Neon® Transfection System (Thermo Fisher Scientific) following manufacturer's instructions.

(243) 24-48 hours after lipofection or electroporation, the expression of GFP was determined by microscopically surveying the cells for the presence of GFP+ cells. Following visual inspection, the proportion of GFP+ cells versus GFP- cells may be determined. Fluorescence was observed in mammalian cells expressing each of the deaminase-nRGN fusion proteins, indicating the fusion protein successfully targeted to the GFP-STOP mutation and edited the mutation to restore fluorescence of the GFP protein.

(244) Following microscopic analysis, the cells were lysed in RIPA buffer and the resulting lysate was analyzed on a fluorescence plate reader to determine the fluorescence intensity of GFP (Table 8), A person of skill in the art will appreciate that the cells may be analyzed by flow cytometry or fluorescence activated cell sorting to determine the exact proportions of GFP+ and GFP- cells.

(245) TABLE-US-00008 TABLE 8 GFP-STOP assay results Deaminase of fusion GFP+ cells protein detected APG09982 ++ APG03724 ++ APG09949 ++ APG08196 ++ APG06333 +++ APG06489 ++ APG08449 ++ APG05174 +++ APG09102 ++ APG05723 ++ N.D = None Detected; + = few GFP+ cells detected; ++ = several GFP+ cells detected; +++ = many GFP+ cells detected

Example 3: Demonstration of a Base Editing in Mammalian Cells

(246) The deaminases shown in Table 9 below were produced based on naturally occurring deaminases which were then mutated and selected for adenine deaminase activity in prokaryotic cells.

(247) TABLE-US-00009 TABLE 9 Deaminase sequences Deaminase SEQ ID NO. LPG50140 399 LPG50141 400 LPG50142 401 LPG50143 402 LPG50144 403 LPG50145 404 LPG50146 405 LPG50147 406 LPG50148 407 LPG50149 408 LPG50150 409 LPG50151 410 LPG50152 411 LPG50153 412 LPG50154 413 LPG50155 414 LPG50156 415 LPG50157 416 LPG50158 417 LPG50159 418 LPG50160 419 LPG50161 420 LPG50162 421 LPG50163 422 LPG50164 423 LPG50165 424 LPG50166 425 LPG50167 426 LPG50168 427 LPG50169 428 LPG50170 429 LPG50171 430 LPG50172 431 LPG50173 432 LPG50174 433 LPG50175 434 LPG50176 435 LPG50177 436 LPG50178 437 LPG50179 438 LPG50180 439 LPG50181 440 LPG50182 441

(248) To determine if the deaminases of Table 9 are able to perform adenine base editing in mammalian cells, each deaminase was operably fused to an RGN nickase to produce a fusion protein. Residues predicted to deactivate the RuvC domain of the RGN APG07433.1 (set forth as SEQ ID NO: 41; described in PCT publication WO 2019/236566, incorporated by reference herein) were identified and the RGN was modified to a nickase variant (nAPG07433.1; SEQ ID NO: 42). A nickase variant of an RGN is referred to herein as “nRGN”. It should be understood that any nickase variant of an RGN may be used to produce a fusion protein of the invention.

(249) Deaminase and nRGN nucleotide sequences codon optimized for mammalian expression were synthesized as fusion proteins with an N-terminal nuclear localization tag and cloned into the pTwist CMV (Twist Biosciences) expression plasmid. Each fusion protein comprises, starting at the amino terminus, the SV40 NLS (SEQ ID NO: 43) operably linked at the C-terminal end to 3×FLAG Tag (SEQ ID NO: 44), operably linked at the C-terminal end to a deaminase, operably linked at the C-terminal end to a peptide linker (SEQ ID NO: 442), operably linked at the C-

terminal end to an nRGN (for example, nAPG07433.1, which is SEQ ID NO: 42), finally operably linked at the C-terminal end to the nucleoplasmin NLS (SEQ ID NO: 46). The nAPG07433.1 and peptide linker nucleotide sequences codon optimized for mammalian expression are set forth as SEQ ID NOs: 486 and 487, respectively. Table 10 shows the fusion proteins produced and tested for activity. All fusion proteins comprise at least one NLS and a 3× FLAG Tag, as described above.

(250) TABLE-US-00010 TABLE 10 Fusion protein sequences with N-terminus SV40 NLS, 3X FLAG Tag and C-terminus Nucleoplasmin NLS Fusion Protein SEQ ID LPG50140-nAPG07433.1 488 LPG50141-nAPG07433.1 489 LPG50142-nAPG07433.1 490 LPG50143-nAPG07433.1 491 LPG50144-nAPG07433.1 492 LPG50145-nAPG07433.1 493 LPG50146-nAPG07433.1 494 LPG50147-nAPG07433.1 495 LPG50148-nAPG07433.1 496 LPG50149-nAPG07433.1 497 LPG50150-nAPG07433.1 498 LPG50151-nAPG07433.1 499 LPG50152-nAPG07433.1 500 LPG50153-nAPG07433.1 501 LPG50154-nAPG07433.1 502 LPG50155-nAPG07433.1 503 LPG50156-nAPG07433.1 504 LPG50157-nAPG07433.1 505 LPG50158-nAPG07433.1 506 LPG50159-nAPG07433.1 507 LPG50160-nAPG07433.1 508 LPG50161-nAPG07433.1 509 LPG50162-nAPG07433.1 510 LPG50163-nAPG07433.1 511 LPG50164-nAPG07433.1 512 LPG50165-nAPG07433.1 513 LPG50166-nAPG07433.1 514 LPG50167-nAPG07433.1 515 LPG50168-nAPG07433.1 516 LPG50169-nAPG07433.1 517 LPG50170-nAPG07433.1 518 LPG50171-nAPG07433.1 519 LPG50172-nAPG07433.1 520 LPG50173-nAPG07433.1 521 LPG50174-nAPG07433.1 522 LPG50175-nAPG07433.1 523 LPG50176-nAPG07433.1 524 LPG50177-nAPG07433.1 525 LPG50178-nAPG07433.1 526 LPG50179-nAPG07433.1 527 LPG50180-nAPG07433.1 528 LPG50181-nAPG07433.1 529 LPG50182-nAPG07433.1 530

(251) Expression plasmids comprising an expression cassette encoding for a sgRNA were also produced. Human genomic target sequences and the sgRNA sequences for guiding the fusion proteins to the genomic targets are indicated in Table 11.

(252) TABLE-US-00011 TABLE 11 Guide RNA sequences Forward Reverse Target sgRNA Primer for Primer for sgRNA ID sequence sequence amplification amplification SGN000139 537 531 543 549 SGN000143 538 532 544 550 SGN000186 539 533 545 551 SGN000194 540 534 546 552 SGN000930 541 535 547 553 SGN001681 542 536 548 554

(253) 500 ng of plasmid comprising an expression cassette comprising a coding sequence for a fusion protein shown in Table 10 and 500 ng of plasmid comprising an expression cassette encoding for an sgRNA shown in Table 11 were co-transfected into HEK293FT cells at 75-90% confluency in 24-well plates using Lipofectamine 2000 reagent (Life Technologies). Cells were then incubated at 37° C. for 72 h. Following incubation, genomic DNA was then extracted using NucleoSpin 96 Tissue (Macherey-Nagel) following the manufacturer's protocol. The genomic region flanking the targeted genomic site was PCR amplified using the primers in Table 11 and products were purified using ZR-96 DNA Clean and Concentrator (Zymo Research) following the manufacturer's protocol. The purified PCR products underwent Next Generation Sequencing on Illumina MiSeq. Typically, 100,000 of 250 bp paired-end reads (2×100,000 reads) are generated per amplicon. The reads were analyzed using CRISPResso (Pinello, et al. 2016 *Nature Biotech*, 34:695-697) to calculate the rates of editing. Output alignments were analyzed for INDEL formation or introduction of specific adenine mutations.

(254) Table 12 shows all of the adenine base editing for each adenine deaminase fusion in Table 10 and a guide RNA from Table 12. Tables 13-27 show the specific nucleotide mutation profile for select exemplary samples. The editing rate for adenines within or proximal to the target sequence is indicated. “A5” indicates, for example, an adenine at position 5 of the target sequence. The position of each nucleotide in the target sequence was determined by numbering the first nucleotide in the target sequence closest to the PAM (which is 3’ of the target for APG07433.1) as position 1, and the position number increases in the 5’ direction away from the PAM sequence. The tables also show which nucleotide the adenine was changed to, and at what rate. For example, Table 13 shows that for the LPG50148-nAPG07433.1 fusion protein, the adenine at position 13 was mutated to a

guanine at a rate of 9.7%.

(255) TABLE-US-00012 TABLE 12 Estimate of base editing rates for each adenine deaminase
Deaminase SGN % Mutated Reads LPG50140 SGN001681 30.01% LPG50140 SGN000139
6.91% LPG50140 SGN000143 16.09% LPG50140 SGN000186 18.76% LPG50140 SGN000194
9.77% LPG50140 SGN000930 3.51% LPG50141 SGN001681 21.37% LPG50141 SGN000139
2.43% LPG50141 SGN000143 6.93% LPG50141 SGN000186 9.79% LPG50141 SGN000194
4.45% LPG50141 SGN000930 5.29% LPG50142 SGN001681 34.19% LPG50142 SGN000139
3.10% LPG50142 SGN000143 8.67% LPG50142 SGN000186 14.12% LPG50142 SGN000194
10.04% LPG50142 SGN000930 6.78% LPG50143 SGN001681 20.62% LPG50143 SGN000139
1.99% LPG50143 SGN000143 6.09% LPG50143 SGN000186 10.58% LPG50143 SGN000194
5.60% LPG50143 SGN000930 3.98% LPG50144 SGN001681 28.26% LPG50144 SGN000139
3.55% LPG50144 SGN000143 5.77% LPG50144 SGN000186 12.22% LPG50144 SGN000194
6.40% LPG50144 SGN000930 5.81% LPG50145 SGN001681 29.23% LPG50145 SGN000139
2.53% LPG50145 SGN000143 3.75% LPG50145 SGN000186 9.93% LPG50145 SGN000194
3.98% LPG50145 SGN000930 3.84% LPG50146 SGN001681 32.53% LPG50146 SGN000139
5.95% LPG50146 SGN000143 11.30% LPG50146 SGN000186 17.78% LPG50146 SGN000194
7.38% LPG50146 SGN000930 7.13% LPG50147 SGN001681 49.10% LPG50147 SGN000139
3.26% LPG50147 SGN000143 8.59% LPG50147 SGN000186 12.61% LPG50147 SGN000194
8.80% LPG50147 SGN000930 4.96% LPG50148 SGN001681 49.39% LPG50148 SGN000139
10.80% LPG50148 SGN000143 12.49% LPG50148 SGN000186 32.65% LPG50148
SGN000194 16.60% LPG50148 SGN000930 7.61% LPG50149 SGN001681 27.62% LPG50149
SGN000139 2.83% LPG50149 SGN000143 9.33% LPG50149 SGN000186 22.12% LPG50149
SGN000194 7.94% LPG50149 SGN000930 7.06% LPG50150 SGN001681 28.46% LPG50150
SGN000139 3.06% LPG50150 SGN000143 6.00% LPG50150 SGN000186 23.67% LPG50150
SGN000194 9.47% LPG50150 SGN000930 5.41% LPG50151 SGN001681 3.01% LPG50151
SGN000139 0% LPG50151 SGN000143 1.53% LPG50151 SGN000186 7.76% LPG50151
SGN000194 1.43% LPG50151 SGN000930 0% LPG50152 SGN001681 26.06% LPG50152
SGN000139 2% LPG50152 SGN000143 3% LPG50152 SGN000186 .sup. 18%
LPG50152 SGN000194 3% LPG50152 SGN000930 6% LPG50153 SGN001681 1.12%
LPG50153 SGN000139 0% LPG50153 SGN000143 0% LPG50153 SGN000186 0%
LPG50153 SGN000194 1% LPG50153 SGN000930 0% LPG50154 SGN001681 2.26%
LPG50154 SGN000139 0% LPG50154 SGN000143 0% LPG50154 SGN000186 0%
LPG50154 SGN000194 1% LPG50154 SGN000930 0% LPG50155 SGN001681 14.91%
LPG50155 SGN000139 2% LPG50155 SGN000143 4% LPG50155 SGN000186 .sup.
17% LPG50155 SGN000194 7% LPG50155 SGN000930 5% LPG50156 SGN001681
11.19% LPG50156 SGN000139 3.79% LPG50156 SGN000143 6.44% LPG50156 SGN000186
12.69% LPG50156 SGN000194 6.87% LPG50156 SGN000930 4.10% LPG50157 SGN001681
20.66% LPG50157 SGN000139 3.37% LPG50157 SGN000143 6.91% LPG50157 SGN000186
12.15% LPG50157 SGN000194 9.98% LPG50157 SGN000930 5.55% LPG50158 SGN001681
1.56% LPG50158 SGN000139 0% LPG50158 SGN000143 1.15% LPG50158 SGN000186
4.91% LPG50158 SGN000194 1.73% LPG50158 SGN000930 0% LPG50159 SGN001681
5.85% LPG50159 SGN000139 0% LPG50159 SGN000143 2.78% LPG50159 SGN000186
6.99% LPG50159 SGN000194 4.40% LPG50159 SGN000930 2.60% LPG50160 SGN001681
22.20% LPG50160 SGN000139 4% LPG50160 SGN000143 8% LPG50160 SGN000186
.sup. 16% LPG50160 SGN000194 5% LPG50160 SGN000930 6% LPG50161
SGN001681 1.47% LPG50161 SGN000139 0% LPG50161 SGN000143 0% LPG50161
SGN000186 0% LPG50161 SGN000194 0% LPG50161 SGN000930 0% LPG50162
SGN001681 21.73% LPG50162 SGN000139 2% LPG50162 SGN000143 5% LPG50162
SGN000186 .sup. 14% LPG50162 SGN000194 6% LPG50162 SGN000930 5%
LPG50163 SGN001681 12.80% LPG50163 SGN000139 0% LPG50163 SGN000143 2%

LPG50163 SGN000186 .sup. 10% LPG50163 SGN000194 4% LPG50163 SGN000930
 3% LPG50164 SGN001681 4.28% LPG50164 SGN000139 0% LPG50164 SGN000143 3.36%
 LPG50164 SGN000186 7.38% LPG50164 SGN000194 2.73% LPG50164 SGN000930 1.47%
 LPG50165 SGN001681 25.66% LPG50165 SGN000139 2% LPG50165 SGN000143 5.11%
 LPG50165 SGN000186 9.88% LPG50165 SGN000194 3.97% LPG50165 SGN000930 3.18%
 LPG50166 SGN000139 2% LPG50166 SGN000143 4% LPG50166 SGN000186 8%
 LPG50166 SGN000194 2% LPG50166 SGN000930 4% LPG50167 SGN001681 20.56%
 LPG50167 SGN000139 2% LPG50167 SGN000143 4% LPG50167 SGN000186 8%
 LPG50167 SGN000194 5% LPG50167 SGN000930 4% LPG50168 SGN001681 13.81%
 LPG50168 SGN000139 2% LPG50168 SGN000143 3% LPG50168 SGN000186 7%
 LPG50168 SGN000194 2% LPG50168 SGN000930 3% LPG50169 SGN001681 25.73%
 LPG50169 SGN000139 4% LPG50169 SGN000143 8% LPG50169 SGN000186 .sup.
 13% LPG50169 SGN000194 9% LPG50169 SGN000930 8% LPG50170 SGN001681
 12.87% LPG50170 SGN000139 1.50% LPG50170 SGN000143 3.14% LPG50170 SGN000186
 12.16% LPG50170 SGN000194 2.76% LPG50170 SGN000930 4.10% LPG50171 SGN001681
 27.16% LPG50171 SGN000139 1.75% LPG50171 SGN000143 6.14% LPG50171 SGN000186
 12.65% LPG50171 SGN000194 5.60% LPG50171 SGN000930 4.55% LPG50172 SGN001681
 1.78% LPG50172 SGN000139 0% LPG50172 SGN000143 0% LPG50172 SGN000186
 0% LPG50172 SGN000194 0% LPG50172 SGN000930 0% LPG50173 SGN001681
 12.64% LPG50173 SGN000139 1.00% LPG50173 SGN000143 3.23% LPG50173 SGN000186
 7.88% LPG50173 SGN000194 2.66% LPG50173 SGN000930 1.77% LPG50174 SGN001681
 14.11% LPG50174 SGN000139 0% LPG50174 SGN000143 3% LPG50174 SGN000186
 8% LPG50174 SGN000194 2% LPG50174 SGN000930 3% LPG50175 SGN001681
 22.29% LPG50175 SGN000139 4% LPG50175 SGN000143 9% LPG50175 SGN000186
 .sup. 14% LPG50175 SGN000194 .sup. 13% LPG50175 SGN000930 5% LPG50176
 SGN001681 9.52% LPG50176 SGN000139 0% LPG50176 SGN000143 2% LPG50176
 SGN000186 7% LPG50176 SGN000194 2% LPG50176 SGN000930 0% LPG50177
 SGN001681 7.98% LPG50177 SGN000139 2% LPG50177 SGN000143 4% LPG50177
 SGN000186 .sup. 11% LPG50177 SGN000194 3% LPG50177 SGN000930 9% LPG50178
 SGN000139 2.00% LPG50178 SGN000143 6.19% LPG50178 SGN000186 12.94% LPG50178
 SGN000194 5.51% LPG50178 SGN000930 3.95% LPG50179 SGN001681 23.35% LPG50179
 SGN000139 2.00% LPG50179 SGN000143 5.08% LPG50179 SGN000186 12.50% LPG50179
 SGN000194 4.49% LPG50179 SGN000930 4.62% LPG50180 SGN001681 1.80% LPG50180
 SGN000139 0% LPG50180 SGN000143 0% LPG50180 SGN000186 0% LPG50180
 SGN000194 0% LPG50180 SGN000930 0% LPG50181 SGN001681 7.93% LPG50181
 SGN000139 2.88% LPG50181 SGN000143 3.78% LPG50181 SGN000186 12.56% LPG50181
 SGN000194 3.39% LPG50181 SGN000930 1.20% LPG50182 SGN001681 16.49% LPG50182
 SGN000139 1.00% LPG50182 SGN000143 5% LPG50182 SGN000186 9% LPG50182
 SGN000194 6% LPG50182 SGN000930 3%

(256) TABLE-US-00013 TABLE 13 A > N Editing Rate using deaminase LPG50148 and guide
 SGN000139 SGN000139 A5 A12 A13 A20 A22 LPG50148 C 0 0 0 0.1 0 G 0 2.2 9.7 0.2 0 T 0 0 0
 0 0 LPG50140, LPG50146, and LPG50148 showed detectable A > G conversion at positions A12
 and A13. LPG50148 showed over 9% editing at position A13.

(257) TABLE-US-00014 TABLE 14 A > N Editing Rate using deaminase LPG50148 and guide
 SGN000143 SGN000143 A1 A4 A6 A9 A11 A14 A19 A30 LPG50148 C 0 0 0 0 0 0 0 0 G 0 0 0.1
 1.2 11 6.7 0.1 0 T 0 0 0 0 0 0 0 0 LPG50140, LPG50146, and LPG50148 showed detectable A > G
 conversion at positions A9, A11 and A14. LPG50148 showed over 11% editing at position A11.

(258) TABLE-US-00015 TABLE 15 A > N Editing Rate using deaminase LPG50148 and guide
 SGN000186 SGN000186 A9 A16 A18 A22 A25 A28 A30 LPG50148 C 0 0 0 0 0.4 0 0 G 23.7 29.2
 4.1 0.2 0 0 0 T 0 0 0 0 0 0 0 0 LPG50140, LPG50146, and LPG50148 showed detectable A > G

conversion at positions A9, A16 and A18. LPG50148 showed over 23% editing at positions A9 and A16.

(259) TABLE-US-00016 TABLE 16 A > N Editing Rate using deaminase LPG50148 and guide SGN000194 SGN000194 A6 A10 A13 A15 A21 A23 A26 A27 LPG50148 C 0 0 0 0 0 0 0 0 G 0.3 5.3 13 14 0 0 0 0 T 0 0 0 0 0 0 0 0 LPG50140, LPG50146, and LPG50148 showed detectable A > G conversion at positions A13 and A15. LPG50148 showed over 12% editing at positions A13 and A15.

(260) TABLE-US-00017 TABLE 17 A > N Editing Rate using deaminase LPG50148 and guide SGN000930 SGN000930 A2 A4 A5 A8 A9 A10 A14 A15 A16 A20 A21 A23 A24 A26 A27 A29 A30 LPG50148 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 G 0 0 0 0 0.2 2 2.2 1.1 2.2 2.2 2.5 0 0 0 0 0 0 T 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 LPG50140, LPG50146, and LPG50148 showed detectable A > G conversion at positions A10, A14, A15, A16, A20 and A21. LPG50148 showed over 2% editing at positions A10, A14, A16, A20 and A21.

(261) TABLE-US-00018 TABLE 18 A > N Editing Rate using deaminase LPG50146 and guide SGN000139 SGN000139 A5 A12 A13 A20 A22 LPG50146 C 0 0 0 0.4 0.1 G 0 2.1 4.1 0 0 T 0 0 0 0 0 0 LPG50140, LPG50146, and LPG50148 showed detectable A > G conversion at positions A12 and A13. LPG50146 showed over 4% editing at position A13.

(262) TABLE-US-00019 TABLE 19 A > N Editing Rate using deaminase LPG50146 and guide SGN000143 SGN000143 A1 A4 A6 A9 A11 A14 A19 A30 LPG50146 C 0 0 0 0 0 0 0 0 G 0 0 0 0.8 8.4 5 0 0 T 0 0 0 0 0 0 0 0 0 0 LPG50140, LPG50146, and LPG50148 showed detectable A > G conversion at positions A9, A11 and A14. LPG50146 showed over 8% editing at position All.

(263) TABLE-US-00020 TABLE 20 A > N Editing Rate using deaminase LPG50146 and guide SGN000186 SGN000186 A9 A16 A18 A22 A25 A28 A30 LPG50146 C 0 0 0 0 0.2 0 0 G 7.4 13.4 3.1 0.1 0 0 0 T 0 0 0 0 0 0 0 0 0 0 LPG50140, LPG50146, and LPG50148 showed detectable A > G conversion at positions A9, A16 and A18. LPG50146 showed over 13% editing at position A16.

(264) TABLE-US-00021 TABLE 21 A > N Editing Rate using deaminase LPG50146 and guide SGN000194 SGN000194 A6 A10 A13 A15 A21 A23 A26 A27 LPG50146 C 0 0 0 0 0 0 0 0 G 0 1.8 3.2 4.5 0 0 0 0 T 0 0 0 0 0 0 0 0 0 0 LPG50140, LPG50146, and LPG50148 showed detectable A > G conversion at positions A13 and A15. LPG50146 showed over 3% editing at positions A13 and A15.

(265) TABLE-US-00022 TABLE 22 A > N Editing Rate using deaminase LPG50146 and guide SGN000930 SGN000930 A2 A4 A5 A8 A9 A10 A14 A15 A16 A20 A21 A23 A24 A26 A27 A29 A30 LPG50146 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 G 0 0 0 0.1 0.1 0.7 2.9 2.6 2.4 1 0.8 0 0 0 0 0 0 T 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 LPG50140, LPG50146, and LPG50148 showed detectable A > G conversion at positions A10, A14, A15, A16, A20 and A21. LPG50146 showed over 2% editing at positions A14 and A16.

(266) TABLE-US-00023 TABLE 23 A > N Editing Rate using deaminase LPG50140 and guide SGN000139 SGN000139 A5 A12 A13 A20 A22 LPG50140 C 0 0 0 0.4 0 G 0 0.5 5.5 0 0 T 0 0 0 0 0 0 LPG50140, LPG50146, and LPG50148 showed detectable A > G conversion at positions A12 and A13. LPG50140 showed over 5% editing at position A13.

(267) TABLE-US-00024 TABLE 24 A > N Editing Rate using deaminase LPG50140 and guide SGN000143 SGN000143 A1 A4 A6 A9 A11 A14 A19 A30 LPG50140 C 0 0 0 0 0 0 0 0 G 0 0 0 1.2 14 5.6 0 0 T 0 0 0 0 0 0 0 0 0 0 LPG50140, LPG50146, and LPG50148 showed detectable A > G conversion at positions A9, A11 and A14. LPG50140 showed 14% editing at position All.

(268) TABLE-US-00025 TABLE 25 A > N Editing Rate using deaminase LPG50140 and guide SGN000186 SGN000186 A9 A16 A18 A22 A25 A28 A30 LPG50140 C 0 0 0 0 0.2 0 0 G 9.4 15 1.7 0 0 0 0 T 0 0 0 0 0 0 0 0 0 0 LPG50140, LPG50146, and LPG50148 showed detectable A > G conversion at positions A9, A16 and A18. LPG50140 showed over 9% editing at positions A9 and A16.

(269) TABLE-US-00026 TABLE 26 A > N Editing Rate using deaminase LPG50140 and guide

SGN000194 SGN000194 A6 A10 A13 A15 A21 A23 A26 A27 LPG50140 C 0 0 0 0 0 0 0 G 0 0 6.7 7.8 0 0 0 0 T 0 0 0 0 0 0 0 0 LPG50140, LPG50146, and LPG50148 showed detectable A > G conversion at positions A13 and A15. LPG50140 showed over 6% editing at positions A13 and A15.

(270) TABLE-US-00027 TABLE 27 A > N Editing Rate using deaminase LPG50140 and guide SGN000930 SGN000930 A2 A4 A5 A8 A9 A10 A14 A15 A16 A20 A21 A23 A24 A26 A27 A29 A30 LPG50140 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 G 0 0 0 0 0 0.4 1.4 0.6 1.1 0.4 0.5 0 0 0 0 0 0 T 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 LPG50140, LPG50146, and LPG50148 showed detectable A > G conversion at positions A10, A14, A15, A16, A20 and A21. LPG50140 showed over 1% editing at positions A14 and A16.

(271) Table 28 below shows the average editing rates for LPG50148-nAPG07433.1 at several guides tested in HEK293T cells by lipofection of two plasmids. The base editor was encoded on one plasmid and the guide RNA was encoded on a second plasmid. Total substitution rate in the target is used to measure the base editing rate.

(272) TABLE-US-00028 TABLE 28 Average Editing Rate for LPG50148-nAPG07433.1 Average % Substitution Gene SGN rate N Gene A SGN000139 10.8 1 Gene A SGN000143 29.65 2 Gene B SGN000487 34.68 2 Gene B SGN000488 39.94 1 Gene B SGN001061 9.18 2 Gene B SGN001062 32.77 1 Gene B SGN001270 8.34 3 Gene B SGN001946 5.1 1 Gene B SGN001947 16.43 1 Gene B SGN001948 0.46 1 Gene B SGN001949 1.44 1 Gene B SGN001950 10.96 1 Gene B SGN001951 5.38 1 Gene B SGN001952 6.29 1 Gene B SGN001953 5.28 1 Gene B SGN001954 7.95 1 Gene B SGN001955 7.83 1 Gene B SGN001956 4.78 1 Gene B SGN001959 1.43 1 Gene B SGN001960 17.4 1 Gene B SGN001961 1.46 1 Gene B SGN001962 1.62 1 Gene B SGN001963 11.31 1 Gene B SGN001964 2.03 1 Gene B SGN001965 9.3 1 Gene B SGN001966 1.51 1 CFTR SGN001101 17.06 1 Gene D SGN001196 14.58 1 Gene D SGN001199 42.05 1 Gene E SGN001681 48.85 1 Gene F SGN000169 55.13 2 Gene F SGN000173 47.13 1 Gene G SGN000412 16.58 1 Gene G SGN000414 14.5 2 Gene G SGN001259 24.16 1 Gene G SGN001274 10.45 2 Gene G SGN001275 5.25 1 Gene H SGN000186 32.65 1 Gene I SGN000754 30.76 1 Gene I SGN000909 21.57 2 Gene I SGN000927 3.8 1 Gene I SGN000928 28.77 1 Gene I SGN000929 17.58 2 Gene I SGN000949 26.43 1 Gene I SGN001268 16.64 2 Gene I SGN001269 6.42 1 Gene I SGN001967 1.45 1 Gene I SGN001968 5.61 1 Gene I SGN001973 5.14 1 Gene I SGN001975 0.16 1 Gene I SGN001976 0.62 1 Gene I SGN001977 0.65 1 Gene I SGN001978 3.09 1 Gene I SGN001981 2.34 1 LPG50148-nAPG07433.1 shows editing at many different guides across the genome.

(273) Table 29 shows the editing rates of adenine bases in each guide from LPG50148-nAPG07433.1. Only the adenine positions are shown below. The rate of adenine conversion is the average of multiple experiments when appropriate.

(274) TABLE-US-00029 TABLE 29 Editing rate of A nucleotides in mammalian cells for top 10 guides Position SGN A1 A3 A4 A5 A6 A7 A8 A9 A10 A11 A13 A14 SGN001681 13 47 SGN000169 0.2 1.3 17 22 SGN001199 3.5 42 SGN000186 24 SGN000754 0 0 1.3 6.1 SGN000143 0 0 0.4 4.4 27 17 SGN000928 0.3 0.2 0.3 6.1 SGN000487 0.2 0.2 12 25 SGN001259 0 12 SGN001062 0 0.7 0 Position SGN A15 A16 A17 A18 A19 A20 A21 A22 A24 A25 SGN001681 SGN000169 43 11 1.7 SGN001199 SGN000186 29 4.1 0.2 0.4 SGN000754 29 SGN000143 0.3 SGN000928 26 SGN000487 8.7 7.6 14 SGN001259 16 1 SGN001062 10 5.8 13 2.4 0.1 0

(275) LPG50148-nAPG07433.1 shows adenine base editing in positions 6 through 21 in the target region depending on the guide RNA used. Editing rates vary by guide RNAs used.

Example 4: Correction of Class I Cystic Fibrosis Nonsense Mutations

Example 4.1: Identification of RGNs and Guide RNAs

(276) Cystic fibrosis is generally caused by deleterious mutations in the CFTR gene (SEQ ID NO: 51). Six of the most common nonsense mutations are G542X, W1282X, R553X, R1162X, E60X,

R785X, and Q493X. Each of these stop mutations could be edited to restore a coding codon by an RGN-deaminase fusion protein described herein. To target each mutation, the following must be determined: 1) an RGN which has a PAM recognition site proximal to the nonsense mutation; and 2) a guide RNA which optimally targets the RGN-deaminase fusion protein to the target DNA. Table 30 below shows nickase variants of RGNs which possess PAMs that are proximal to each of the six nonsense mutations and the number of guide RNAs which can be used for each RGN. Table 31 describes the genetic loci for each guide RNA. The PAM recognition site for each genetic locus is underlined. The target sequence for the guide RNA and the guide RNA sequence itself are also indicated.

(277)	TABLE-US-00030	TABLE 30 RGN nickases and number of guide RNAs for nonsense mutations in CFTR SEQ ID NO.	for RGN nickase	RGN nickase	E60X	G542X	Q493X	R1162X	R553X	W1282X	nAPG00969	52	2	2	2	nAPG07433.1	42	1	3	1	nAPG06646	53	6	4	2	3	7	4	nAPG09748	54	1	1	4	1	nAPG09882	55	4	3	5	5	3	5	nAPG03850	56	2	2	1	3	3	4	nAPG07553	57	1	1	1	1	2	nAPG05586	58	1	1	3	1	nAPG01604	59	2	1	2
(278)	TABLE-US-00031	TABLE 31 guide RNAs for nonsense mutations in CFTR Genetic Target gRNA locus (SEQ (SEQ (SEQ ID ID Guide ID Genetic locus ID NO.) NO.) NO.)	E60X	nAPG06646	Target	1	AATGAGTTTAGGATTTTTCTTTGAAGCCAGCTATCTATCCCATTCTCTGCAAAAG 62 80 98																																																													
			AATAAAAAGT	E60X	nAPG06646	Target	2	ATTAATGAGTTTAGGATTTTTCTTTGAAGCCAGCTATCTATCCCATTCTCTGCAA 63 81 99																																																												
			AAGAATAAAA	E60X	nAPG06646	Target	3	GCATTAATGAGTTTAGGATTTTTCTTTGAAGCCAGCTATCTATCCCATTCTCTGC 64 82 100																																																												
			AAAAGAATAA	E60X	nAPG06646	Target	4	AAGGGCATTAAATGAGTTTAGGATTTTTCTTTGAAGCCAGCTATCTATCCCATTCT 65 83																																																												
			101 CTGCAAAAGA	E60X	nAPG06646	Target	5	GAAGGGCATTAAATGAGTTTAGGATTTTTCTTTGAAGCCAGCTATCTATCCCATTCT 66 84																																																												
			102 TCTGCAAAAG	E60X	nAPG06646	Target	6	CGAAGGGCATTAAATGAGTTTAGGATTTTTCTTTGAAGCCAGCTATCTATCCCATT 67 85																																																												
			103 CTCTGCAAAA	E60X	nAPG09882	Target	1	GAGTTTAGGATTTTTCTTTGAAGCCAGCTATCTATCCCATTCTCTGCAAAAGAAT 68 86																																																												
			104 AAAAAGTGGG	E60X	nAPG09882	Target	2	TGAGTTTAGGATTTTTCTTTGAAGCCAGCTATCTATCCCATTCTCTGCAAAAGAA 69 87																																																												
			105 TAAAAAGTGG	E60X	nAPG09882	Target	3	ATGAGTTTAGGATTTTTCTTTGAAGCCAGCTATCTATCCCATTCTCTGCAAAAGA 70 88																																																												
			106 ATAAAAAGTG	E60X	nAPG09882	Target	4	AGGGCATTAAATGAGTTTAGGATTTTTCTTTGAAGCCAGCTATCTATCCCATTCTC 71 89																																																												
			107 TGCAAAAGAA	E60X	nAPG00969	Target	1	GTTTAGGATTTTTCTTTGAAGCCAGCTATCTATCCCATTCTCTGCAAAAGAATAA 72 90																																																												
			108 AAAGTGGGAC	E60X	nAPG00969	Target	2	AGTTTAGGATTTTTCTTTGAAGCCAGCTATCTATCCCATTCTCTGCAAAAGAATA 73 91																																																												
			109 AAAAGTGGGA	E60X	nAPG03850	Target	1	GGATTTTTCTTTGAAGCCAGCTATCTATCCCATTCTCTGCAAAAGAATAAAAAGT 74 92																																																												
			110 GGGAC	E60X	nAPG03850	Target	2	AGTTTAGGATTTTTCTTTGAAGCCAGCTATCTATCCCATTCTCTGCAAAAGAATA 75 93																																																												
			111 AAAAG	E60X	nAPG07433.1	Target	1	GAAGGGCATTAAATGAGTTTAGGATTTTTCTTTGAAGCCAGCTATCTATCCCATTCT 76 94																																																												
			112 TCTGCAAAAG	E60X	nAPG09748	Target	1	GTCCCACTTTTTATTCTTTTGCAGAGAATGGGATAGATAGCTGGCTTCAAAGAAA 77 95																																																												
			113 AATCC	E60X	nAPG07553	Target	1	AGTTTAGGATTTTTCTTTGAAGCCAGCTATCTATCCCATTCTCTGCAAAAGAATA 78 96																																																												

114 AAAG E60X nAPG05586 Target 1
TTTAGGATTTTCTTTGAAGCCAGCTATCTATCCCATTCTCTGCAAAAGAATAAA 79 97
115 AAGTG G542X nAPG06646 Target 1
CGTTGACCTCCACTCAGTGTGATTCCACCTTCTCAAAGAACTATATTGTCTTTCT 116 128
140 CTGCAAACTT G542X nAPG06646 Target 2
GACCTCCACTCAGTGTGATTCCACCTTCTCAAAGAACTATATTGTCTTTCTCTG 117 129
141 AAAC TTGGAG G542X nAPG06646 Target 3
CCTCCACTCAGTGTGATTCCACCTTCTCAAAGAACTATATTGTCTTTCTCTGCAA 118 130
142 ACTTGGAGAT G542X nAPG06646 Target 4
CCACTCAGTGTGATTCCACCTTCTCAAAGAACTATATTGTCTTTCTCTGCAAACT 119 131
143 TGGAGATGTC G542X nAPG09882 Target 1
TCTTGCTCGTTGACCTCCACTCAGTGTGATTCCACCTTCTCAAAGAACTATATTG 120 132
144 TCTTTCTCTG G542X nAPG09882 Target 2
TTGCTCGTTGACCTCCACTCAGTGTGATTCCACCTTCTCAAAGAACTATATTGTC 121 133
145 TTTCTCTGCA G542X nAPG09882 Target 3
CACTCAGTGTGATTCCACCTTCTCAAAGAACTATATTGTCTTTCTCTGCAAACTT 122 134
146 GGAGATGTCC G542X nAPG03850 Target 1
TGACCTCCACTCAGTGTGATTCCACCTTCTCAAAGAACTATATTGTCTTTCTCTG 123 135
147 CAAAC G542X nAPG03850 Target 2
TCAGTGTGATTCCACCTTCTCAAAGAACTATATTGTCTTTCTCTGCAAACTTGGA 124 136
148 GATGT G542X nAPG09748 Target 1
AGAGAAAGACAATATAGTTCTTTGAGAAGGTGGAATCACACTGAGTGGAGGTCAA 125
137 149 CGAGC G542X nAPG07553 Target 1
TCAGTGTGATTCCACCTTCTCAAAGAACTATATTGTCTTTCTCTGCAAACTTGGA 126 138
150 GATGT G542X nAPG05586 Target 1
CGTTGACCTCCACTCAGTGTGATTCCACCTTCTCAAAGAACTATATTGTCTTTCT 127 139
151 CTGCA Q493X nAPG09882 Target 1
GATATTTTCTTTAATGGTGCCAGGCATAATCCAGGAAAAC TAAGAACAGAAATGAA 152
169 186 ATTCTTCCAC Q493X nAPG09882 Target 2
ATATTTTCTTTAATGGTGCCAGGCATAATCCAGGAAAAC TAAGAACAGAAATGAAA 153
170 187 TTCTTCCACT Q493X nAPG09882 Target 3
TTTTCTTTAATGGTGCCAGGCATAATCCAGGAAAAC TAAGAACAGAAATGAAATTC 154
171 188 TTCCACTGTG Q493X nAPG09882 Target 4
TTTCTTTAATGGTGCCAGGCATAATCCAGGAAAAC TAAGAACAGAAATGAAATTCT 155
172 189 TCCACTGTGC Q493X nAPG09882 Target 5
TTCTTTAATGGTGCCAGGCATAATCCAGGAAAAC TAAGAACAGAAATGAAATTCTT 156
173 190 CCACTGTGCT Q493X nAPG09748 Target 1
TAAGCACAGTGGAAGAATTTTCATTCTGTTCTTAGTTTTCTGGATTATGCCTGGC 157 174
191 ACCAT Q493X nAPG09748 Target 2
AAGCACAGTGGAAGAATTTTCATTCTGTTCTTAGTTTTCTGGATTATGCCTGGCA 158 175
192 CCATT Q493X nAPG09748 Target 3
ACAGTGGAAGAATTTTCATTCTGTTCTTAGTTTTCTGGATTATGCCTGGCACCAT 159 176
193 TAAAG Q493X nAPG09748 Target 4
GGAAGAATTTTCATTCTGTTCTTAGTTTTCTGGATTATGCCTGGCACCATTAAG 160 177
194 AAAAT Q493X nAPG00969 Target 1
GATATTTTCTTTAATGGTGCCAGGCATAATCCAGGAAAAC TAAGAACAGAAATGAA 161
178 195 ATTCTTCCAC Q493X nAPG00969 Target 2
TTCTTTAATGGTGCCAGGCATAATCCAGGAAAAC TAAGAACAGAAATGAAATTCTT 162
179 196 CCACTGTGCT Q493X nAPG06646 Target 1
TTTAATGGTGCCAGGCATAATCCAGGAAAAC TAAGAACAGAAATGAAATTCTTCCA 163

180 197 CTGTGTTTAA Q493X nAPG06646 Target 2
AATGGTGCCAGGCATAATCCAGGAAAACTAAGAACAGAATGAAATTCTTCCACTG 164
181 198 TGCTTAATTT Q493X nAPG01604 Target 1
TTCTTTAATGGTGCCAGGCATAATCCAGGAAAACTAAGAACAGAATGAAATTCTT 165
182 199 CCACT Q493X nAPG01604 Target 2
TTAATGGTGCCAGGCATAATCCAGGAAAACTAAGAACAGAATGAAATTCTTCCAC 166
183 200 TGTGC Q493X nAPG03850 Target 1
CTTTAATGGTGCCAGGCATAATCCAGGAAAACTAAGAACAGAATGAAATTCTTCC 167
184 201 ACTGT Q493X nAPG07553 Target 1
CTTTAATGGTGCCAGGCATAATCCAGGAAAACTAAGAACAGAATGAAATTCTTCC 168
185 202 ACTGT R553X nAPG06646 Target 1
CCAATAATTAGTTATTCACCTTGCTAAAGAAATTCTTGCTCATTGACCTCCACTC 203 219
235 AGTGTGATTC R553X nAPG06646 Target 2
CAATAATTAGTTATTCACCTTGCTAAAGAAATTCTTGCTCATTGACCTCCACTCA 204 220
236 GTGTGATTCC R553X nAPG06646 Target 3
ATAATTAGTTATTCACCTTGCTAAAGAAATTCTTGCTCATTGACCTCCACTCAGT 205 221
237 GTGATTCCAC R553X nAPG06646 Target 4
AATTAGTTATTCACCTTGCTAAAGAAATTCTTGCTCATTGACCTCCACTCAGTGT 206 222
238 GATTCCACCT R553X nAPG06646 Target 5
TCACCTTGCTAAAGAAATTCTTGCTCATTGACCTCCACTCAGTGTGATTCCACCT 207 223
239 TCTCCAAGAA R553X nAPG06646 Target 6
CACCTTGCTAAAGAAATTCTTGCTCATTGACCTCCACTCAGTGTGATTCCACCTT 208 224
240 CTCCAAGAAC R553X nAPG06646 Target 7
CCTTGCTAAAGAAATTCTTGCTCATTGACCTCCACTCAGTGTGATTCCACCTTCT 209 225
241 CCAAGAACTA R553X nAPG07433.1 Target 1
CCAATAATTAGTTATTCACCTTGCTAAAGAAATTCTTGCTCATTGACCTCCACTC 210 226
242 AGTGTGATTC R553X nAPG07433.1 Target 2
TCACCTTGCTAAAGAAATTCTTGCTCATTGACCTCCACTCAGTGTGATTCCACCT 211 227
243 TCTCCAAGAA R553X nAPG07433.1 Target 3
CCTTGCTAAAGAAATTCTTGCTCATTGACCTCCACTCAGTGTGATTCCACCTTCT 212 228
244 CCAAGAACTA R553X nAPG09882 Target 1
AATAATTAGTTATTCACCTTGCTAAAGAAATTCTTGCTCATTGACCTCCACTCAG 213 229
245 TGTGATTCCA R553X nAPG09882 Target 2
ATTAGTTATTCACCTTGCTAAAGAAATTCTTGCTCATTGACCTCCACTCAGTGTG 214 230
246 ATTCCACCTT R553X nAPG09882 Target 3
TATTCACCTTGCTAAAGAAATTCTTGCTCATTGACCTCCACTCAGTGTGATTCCA 215 231
247 CCTTCTCCAA R553X nAPG03850 Target 1
TATTCACCTTGCTAAAGAAATTCTTGCTCATTGACCTCCACTCAGTGTGATTCCA 216 232
248 CCTTC R553X nAPG03850 Target 2
TTCACCTTGCTAAAGAAATTCTTGCTCATTGACCTCCACTCAGTGTGATTCCACC 217 233
249 TTCTC R553X nAPG03850 Target 3
CACCTTGCTAAAGAAATTCTTGCTCATTGACCTCCACTCAGTGTGATTCCACCTT 218 234
250 CTCCA R1162X nAPG09882 Target 1
GGTTTACCTTCTGTTGGCATGTCAATGAACTTAAAGACTCAGCTCACAGATCGCA 251
269 287 TCTGAAATAA R1162X nAPG09882 Target 2
ACCTTCTGTTGGCATGTCAATGAACTTAAAGACTCAGCTCACAGATCGCATCTGA 252
270 288 AATAAAAATA R1162X nAPG09882 Target 3
CTGTTGGCATGTCAATGAACTTAAAGACTCAGCTCACAGATCGCATCTGAAATAA 253
271 289 AAATAACAAC R1162X nAPG09882 Target 4
TGTTGGCATGTCAATGAACTTAAAGACTCAGCTCACAGATCGCATCTGAAATAA 254

272 290 ATAACAACAR1162X nAPG09882 Target 5
GTTGGCATGTCAATGAACTTAAAGACTCAGCTCACAGATCGCATCTGAAATAAAA 255
273 291 ATAACAACAT R1162X nAPG06646 Target 1
TTTACCTTCTGTTGGCATGTCAATGAACTTAAAGACTCAGCTCACAGATCGCATC 256
274 292 TGAAATAAAA R1162X nAPG06646 Target 2
TACCTTCTGTTGGCATGTCAATGAACTTAAAGACTCAGCTCACAGATCGCATCTG 257
275 293 AAATAAAAAT R1162X nAPG06646 Target 3
TGGCATGTCAATGAACTTAAAGACTCAGCTCACAGATCGCATCTGAAATAAAAAT 258
276 294 AACAA R1162X nAPG03850 Target 1
TACCTTCTGTTGGCATGTCAATGAACTTAAAGACTCAGCTCACAGATCGCATCTG 259
277 295 AAATA R1162X nAPG03850 Target 2
TTCTGTTGGCATGTCAATGAACTTAAAGACTCAGCTCACAGATCGCATCTGAAAT 260
278 296 AAAAA R1162X nAPG03850 Target 3
TGGCATGTCAATGAACTTAAAGACTCAGCTCACAGATCGCATCTGAAATAAAAAT 261
279 297 AACAA R1162X nAPG05586 Target 1
TTACCTTCTGTTGGCATGTCAATGAACTTAAAGACTCAGCTCACAGATCGCATCT 262 280
298 GAAAT R1162X nAPG05586 Target 2
CTGTTGGCATGTCAATGAACTTAAAGACTCAGCTCACAGATCGCATCTGAAATAA 263
281 299 AAATA R1162X nAPG05586 Target 3
TGTCAATGAACTTAAAGACTCAGCTCACAGATCGCATCTGAAATAAAAATAACAA 264
282 300 CATTT R1162X nAPG00969 Target 1
GGTTTACCTTCTGTTGGCATGTCAATGAACTTAAAGACTCAGCTCACAGATCGCA 265
283 301 TCTGAAATAA R1162X nAPG00969 Target 2
GTTGGCATGTCAATGAACTTAAAGACTCAGCTCACAGATCGCATCTGAAATAAAA 266
284 302 ATAACAACAT R1162X nAPG07553 Target 1
TGGCATGTCAATGAACTTAAAGACTCAGCTCACAGATCGCATCTGAAATAAAAAT 267
285 303 AACAA R1162X nAPG01604 Target 1
GCATGTCAATGAACTTAAAGACTCAGCTCACAGATCGCATCTGAAATAAAAATAA 268
286 304 CAACA W1282X nAPG09882 Target 1
GTGTGTCTTGGGATTCAATAACTTTGCAACAGTGAAGGAAAGCCTTTGGAGTGAT 305
325 345 ACCACAGGTG W1282X nAPG09882 Target 2
GTCTTGGGATTCAATAACTTTGCAACAGTGAAGGAAAGCCTTTGGAGTGATACCA 306
326 346 CAGGTGAGCA W1282X nAPG09882 Target 3
CTTGGGATTCAATAACTTTGCAACAGTGAAGGAAAGCCTTTGGAGTGATACCACA 307
327 347 GGTGAGCAAA W1282X nAPG09882 Target 4
GGGATTCAATAACTTTGCAACAGTGAAGGAAAGCCTTTGGAGTGATACCACAGGT 308
328 348 GAGCAAAAGG W1282X nAPG09882 Target 5
GATTCAATAACTTTGCAACAGTGAAGGAAAGCCTTTGGAGTGATACCACAGGTGA 309
329 349 GCAAAAGGAC W1282X nAPG06646 Target 1
TCGATGGTGTGTCTTGGGATTCAATAACTTTGCAACAGTGAAGGAAAGCCTTTGG 310
330 350 AGTGATACCA W1282X nAPG06646 Target 2
TTGGGATTCAATAACTTTGCAACAGTGAAGGAAAGCCTTTGGAGTGATACCACAG 311
331 351 GTGAGCAAAA W1282X nAPG06646 Target 3
TGGGATTCAATAACTTTGCAACAGTGAAGGAAAGCCTTTGGAGTGATACCACAGG 312
332 352 TGAGCAAAAG W1282X nAPG06646 Target 4
GGATTCAATAACTTTGCAACAGTGAAGGAAAGCCTTTGGAGTGATACCACAGGTG 313
333 353 AGCAAAAGGA W1282X nAPG03850 Target 1
TGTCTTGGGATTCAATAACTTTGCAACAGTGAAGGAAAGCCTTTGGAGTGATACC 314
334 354 ACAGG W1282X nAPG03850 Target 2
GTCTTGGGATTCAATAACTTTGCAACAGTGAAGGAAAGCCTTTGGAGTGATACCA 315

335 355 CTTGGGATTCAATAACTTTGCAACAGTGAAGGAAAGCCTTTGGAGTGATACCACA 316
 336 356 GGTGA W1282X nAPG03850 Target 4
 TGGGATTCAATAACTTTGCAACAGTGAAGGAAAGCCTTTGGAGTGATACCACAGG 317
 337 357 TGAGC W1282X nAPG07553 Target 1
 CTTGGGATTCAATAACTTTGCAACAGTGAAGGAAAGCCTTTGGAGTGATACCACA 318
 338 358 GGTGA W1282X nAPG07553 Target 2
 TGGGATTCAATAACTTTGCAACAGTGAAGGAAAGCCTTTGGAGTGATACCACAGG 319
 339 359 TGAGC W1282X nAPG01604 Target 1
 TCTTGGGATTCAATAACTTTGCAACAGTGAAGGAAAGCCTTTGGAGTGATACCAC 320
 340 360 AGGTG W1282X nAPG01604 Target 2
 CTTGGGATTCAATAACTTTGCAACAGTGAAGGAAAGCCTTTGGAGTGATACCACA 321
 341 361 GGTGA W1282X nAPG07433.1
 TTGGGATTCAATAACTTTGCAACAGTGAAGGAAAGCCTTTGGAGTGATACCACAG 322
 342 362 Target 1 GTGAGCAAAA W1282X nAPG09748 Target 1
 GTATCACTCCAAAGGCTTTCCTTCACTGTTGCAAAGTTATTGAATCCCAAGACAC 323
 343 363 ACCAT W1282X nAPG05586 Target 1
 GATTCAATAACTTTGCAACAGTGAAGGAAAGCCTTTGGAGTGATACCACAGGTGA 324
 344 364 GCAAA F508de nAPG07433.1
 ACCAAAGATGATATTTTCTTTAATGGTGCCAGGCATAATCCAGGAAAACCTGAGAA 562
 563 564 SGN001101 Target1 CAGAATGAAA

(279) Table 28 in Example 3 provides editing data for the SGN001101 sgRNA targeting CFTR.

(280) To assay for activity of the other guide RNAs, a guide RNA of Table 31 is provided with the corresponding nickase variant of each RGN described in Table 30, which is operably linked to a deaminase of the invention to produce a fusion protein. It is recognized that nuclease inactive variants of each RGN may be tested similarly as well. Each guide and fusion protein combination is assayed for the ability to edit at the target location in 16HBE14o– immortalized bronchial epithelial cells. Currently, three HBE cell lines containing the CFTR nonsense mutations are available (Cystic Fibrosis Foundation, Lexington, MA). These cell lines are used to assay the G542X, W1282X, and R1162X nonsense mutation targets and compared to the 16HBE14o– line. The fusion protein and guide RNA is delivered to the cells as ribonucleoproteins (RNPs), which are nucleofected into the 16HBE14o– cell line following culturing and transformation methods provided in Valley et al (Valley et al, 2019. *J Cyst Fibros* 18, 476-483, incorporated by reference herein). The guide RNA is provided as a single guide RNA or as a 1:1 or 1:1.2 molar ratio of tracrRNA:crRNA duplex with RGN proteins. Nucleofection of RNPs into cells is performed on a Lonza 4D-Nucleofector. Cells are then incubated at 37° C. for 72 h. In some embodiments, the fusion protein and gRNA are delivered to the cells as RNA molecules, with the fusion protein encoded in an mRNA.

(281) Because there are no cell lines available for the E60X, R553X, and Q493X, these mutations are assayed in HEK293 cells using a modification of the GFP restoration assay described in Example 2, where the mutant locus containing the nonsense mutation is cloned into the GFP reading frame 2.

(282) Following incubation, genomic DNA is then extracted using NucleoSpin 96 Tissue (Macherey-Nagel) following the manufacturer's protocol. The genomic region flanking the targeted genomic site is PCR amplified and products are purified using ZR-96 DNA Clean and Concentrator (Zymo Research) following the manufacturer's protocol. The purified PCR products are then sent for Next Generation Sequencing on Illumina MiSeq. Typically, 100,000 of 250 bp paired-end reads (2×100,000 reads) are generated per amplicon. The reads are analyzed using CRISPResso (Pinello, et al. 2016) to calculate the rates of editing. Output alignments are hand-curated to confirm introduction of the base-edited mutations of interest and also to screen for undesirable INDEL

formation.

(283) In addition to efficiency of base editing, the protein product of the base-edited CFTR gene is evaluated for function. For two of the nonsense mutations, Glu60X and Gly542X, the base edited change of adenine to guanine does not restore the wildtype sequence, as these mutations are caused by guanine to thymine transversions. The targeted activity of the fusion protein changes the Glu60X to Glu60Gln and Gly452X to Gly542Arg. While these mutations do allow for a full-length protein to be made, the stability and functionality of the CFTR protein is also confirmed.

Example 4.2: Engineering RGNs for Decreased Size

(284) Ideally, the coding sequence of an RGN-deaminase fusion protein of the invention and a corresponding guide RNA for targeting the fusion protein to the CFTR gene is all packaged into a single AAV vector. The generally accepted size limit for AAV vectors is 4.7 kb, although larger sizes may be contemplated at the expense of reduced packing efficiency. The RGN nickases in Table 30 have a coding sequence length of about 3.15-3.45 kB. To ensure that the expression cassettes for both the fusion protein and its corresponding guide RNA could fit into an AAV vector, shortening the length of RGN amino acid and its corresponding nucleic acid coding sequence is desirable.

(285) Through alignment with closely related homologs, a unique 8 amino acid region at positions 590-597 was identified in APG07433.1 and its close homolog APG08290.1 (described in WO 2019/236566 and set forth herein as SEQ ID NO: 60). This region, set forth as SEQ ID NO: 365 for APG07433.1 and SEQ ID NO: 367 for APG08290.1, was removed from both proteins, resulting in variant RGNs APG07433.1-del (SEQ ID NO: 366) and APG08290.1-del (SEQ ID NO: 368). These deletion variants and their corresponding wild-type RGNs were assayed for editing activity in HEK293T cells using the guide RNAs indicated in Tables 32 and 33 following methods similar to those described in Example 1. Rates of editing of the target sequences are shown in Tables 32 and 33 below.

(286) TABLE-US-00032 TABLE 32 Editing Rate for APG07433.1 Protein Deletion Variants Target sgRNA (SEQ ID (SEQ ID APG07433.1- guide RNA NO.) NO.) APG07433.1 del SGN000139 369 383 11.09% 1.00% SGN000143 370 384 2.68% 0.71% SGN000169 371 385 13.37% 15.48% SGN000173 372 386 13.65% 15.37% SGN000186 373 387 14.72% 15.16% SGN000194 374 388 11.91% 7.66% SGN000927 376 390 9.53% 11.47% SGN000929 378 392 6.14% 13.10% SGN000930 379 393 7.52% 9.51% SGN000935 381 395 11.08% 15.99% SGN001101 382 396 6.16% 6.75%

(287) For targets SGN000169, SGN000173, SGN000186, SGN000927, SGN000930, and SGN001101, the editing rate of the wild type APG07433.1 protein and the engineered variant was similar. For targets SGN000139, SGN000143, and SGN000194, the editing rate is decreased when the engineered variant was used compared to the wild type protein. With SGN000929 and SGN000935, the editing rate increased with the engineered APG07433.1 variant compared to the wild type sequence.

(288) TABLE-US-00033 TABLE 33 Editing Rate for APG08290.1 Protein Deletion Variants Target sgRNA (SEQ ID (SEQ ID APG08290.1- sgRNA ID NO.) NO.) APG08290.1 del SGN000926 375 389 N.D. 6.47% SGN000929 378 392 1.83% 0.61% SGN000930 379 393 9.93% 6.47% SGN000928 377 391 N.D. 0.13% SGN000931 380 394 0% 0% N.D. = Not determined

(289) The APG08290.1 deletion variant showed editing in all samples where the wild type APG08290.1 protein also showed editing. The lowest editing rate detected was 0.13% with the engineered protein. Target SGN000926 showed the highest editing rate: 9.17%.

(290) Fusion proteins comprising APG07433.1-del or APG08290.1-del and a deaminase of the invention are produced and assayed for base editing activity using methods similar to Example 1.

(291) A fusion protein comprises an RGN and a deaminase linked by a flexible peptide linker, such as that set forth as SEQ ID NO: 45. The linker of SEQ ID NO: 45 is 16 amino acids in length; this size may be reduced to reduce the size of the coding sequence of the fusion protein. Peptide linkers

of less than 16 amino acids can be produced and operably link RGNs APG07433.1-del or APG08290.1-del and a deaminase of the invention and tested for base editing activity using methods similar to Example 1. Because the peptide linker between the RGN and the deaminase can determine the editing window of the fusion protein, testing of alternative linkers with different lengths and rigidity may also lead to improvements in editing efficiency while reducing off-target mutations. Therefore, fusion proteins with the highest editing rate are then assayed following methods similar to Example 4.1 to determine editing efficiency for each of the CFTR target sequences. Fusion protein-gRNA combinations with the highest editing efficiency are selected as the preferred guide for editing at that location and are used for AAV vector design.

Example 4.3: AAV Delivery

(292) The coding sequences for validated fusion protein/gRNA combinations with the highest editing rate are packaged into AAV vectors. AAV delivery has a number of benefits including a lack of pathogenicity, low immunogenicity, high transduction rates, and a defined path to manufacturing. Also, AAV dosing of the lungs has been shown to be safe and at least to some degree, efficacious with both single and repeat dosing (Guggino et al., 2017, *Expert Opin Biol Ther* 17, 1265-1273). After a fusion protein/gRNA combination has been cloned into an AAV vector, it may be packaged into several different serotypes to optimize tissue specific infectivity. For treatment of CF, the target for base editing is progenitor apical epithelium cells of the lungs, which will allow the correction to persist throughout cell turnover. To target respiratory epithelium, the capsid for serotypes AAV1, AAV5 or AAV6 are utilized, as these serotypes have been shown to have high infectivity in respiratory epithelium cells (Zabner et al., 2000, *J Virol* 74, 3852-3858). (293) Once the AAV vectors are produced, they are transduced into human airway epithelial cells in culture. The three HBE cell lines containing the CFTR G542X, R1162X, and W1282X nonsense mutation targets are used to validate the constructs for correction of those mutations. The 16HBE14o- line is used to test the constructs correcting the other nonsense mutations. A range of multiplicities of infection (MOIs) are tested. In either case, reversion of the nonsense mutation to the wild type CFTR sequence is assessed. After 2-3 days in culture, genomic DNA is harvested, amplicons around the targeted sites are generated by PCR, and NGS is performed to determine editing rates at each locus similar to the methods described in Example 1. Because airway epithelial cells are used, AAV introduction and editing rates are as similar to an in vivo treatment as possible while using a cultured cell system. AAVs with different serotypes are compared to determine which serotype is optimal for delivery of the fusion protein/gRNA into airway cells. The editing rates achieved by AAV introduction of these systems are compared with the RNP editing rates observed in Example 4.2.

(294) Because cell lines for the nonsense mutations R553X, E60X, and Q493X are not available, fusion protein/gRNA systems targeting these mutations are evaluated in wild type 16HBE14o- cells to assay for AAV introduction, base editor expression, and off-target editing rates at the location of interest. To determine the rate of stop codon correction, the mutant locus is cloned into GFP for a GFP restoration assay as described in Example 4.1.

(295) In parallel with determining editing rates by NGS, total protein lysates from cells harboring CFTR mutations edited with fusion protein/gRNA systems are collected and the levels of full-length CFTR protein assessed by western blotting. To test whether functional CFTR protein is formed, forskolin activation assays are performed using methods similar to those described by Devor et al (2000, *Am J Physiol Cell Physiol* 279, C461-479, incorporated by reference herein) and/or Dousmais et al (2002, *J Gen Physiol* 119, 545-559, incorporated by reference herein). In these experiments, edited CFTR mutant cells are treated with forskolin, an activator of adenylate cyclase, to increase intracellular levels of cAMP. Elevated cAMP levels then activate CFTR, and the influx of Cl_{sup}- is measured by either a genetically-encoded yellow fluorescent protein based Cl_{sup}- sensor or a small molecule fluorescent indicator of chloride such as MQAE. The G542X, R1162X, and W1282X edited cell lines are tested in this assay.

(296) To determine the rate of off-target mutations, a bioinformatic approach which is customized with information about the seed region and flexible off-target PAM recognition space of each specific nuclease is used. These pieces of information have been determined bioinformatically for each protein and are used to rank the likelihood of off-target activity for each protein.

(297) To complement bioinformatic prediction of off targets, biochemical detection of off-targets via a modified SITE-seq protocol (Cameron et al., 2017, *Nat Methods* 14, 600-606, herein incorporated by reference) is also performed. Briefly, genomic DNA from human airway epithelial cells is obtained. This DNA is then treated with the RGN of interest at several different concentrations. Any DNA double stranded breaks are labelled, selectively isolated, and PCR amplified with adapter sequences that allow for NGS. Sequencing reads are then mapped to the genome and “pileups” of reads are identified at sites of double stranded breaks, marking putative off target locations. In a subsequent set of experiments, cells are edited with the RGN or RGN-deaminase fusion protein of interest and these putative sites are individually sequenced to confirm if they are bona fide off-targets. Since chromatin context, DNA accessibility, and other factors can impact the efficiency of genome editors in living cells, biochemical methods typically overestimate the number of off-targets. Therefore, both bioinformatic and biochemical methods together provide complementary methods to identify putative off-target sites, but these sites must be verified by amplicon sequencing to get an accurate assessment of off-target editing.

(298) Once putative off-target sites are identified, amplicon sequencing on 16HBE airway epithelial cells edited with the same optimized fusion protein and guide(s) ensures that the off-target profile established for these systems matches the expected profile in patient lungs as closely as possible.

(299) To determine if the fusion proteins described herein induce changes in cellular RNA, careful analysis of the cellular transcriptome following editing is necessary. Fortunately, RNA-seq techniques to assess adenine base-editing off-target effects have become routine (Grunewald et al, 2017, *Nature* 569, 433-437; Zhou et al, *Nature* 571, 275-278, both incorporated by reference herein). Briefly, after editing cells with the fusion protein/gRNA systems determined in Example 4.2, total cellular mRNA is collected and subjected to RNA-seq. Transcriptomes from edited cells are compared to cells transfected with the ABE alone, and significant differences in RNA sequence are identified.

Example 5: Targeted Base-Editing for Correction of Causal Disease Mutations

(300) A database of clinical variants was obtained from NCBI ClinVar database, which is available through the world wide web at the NCBI ClinVar website. Pathogenic Single Nucleotide Polymorphisms (SNPs) were identified from this list. Using the genomic locus information, CRISPR targets in the region overlapping and surrounding each SNP were identified. A selection of SNPs that can be corrected using base editing in combination with an RGN, such as for example an RGN listed in Table 30 or a variant thereof, to target the causal mutation (“Cas1 Mut.”) is listed in Table 34. In Table 34 below, only one alias of each disease is listed. The “RS #” corresponds to the RS accession number through the SNP database at the NCBI website. The “AlleleID” corresponds to a causal allele accession number. The “Name” column contains the genetic locus identifier, the gene name, the location of the mutation in the gene, and the change resulting from the mutation.

(301) TABLE-US-00034

TABLE 34	Disease	Targets for Base Editing	RS#	AlleleID	Name
GeneSymbol	36053993	20333	NM_001128425.1(MUTYH): c.1187G > A (p.Gly396Asp)	MUTYH	
41293455	32714	NM_007294.3(BRCA1): c.4327C > T (p.Arg1443Ter)	BRCA1	62625308	32710
NM_007294.3(BRCA1): c.3607C > T (p.Arg1203Ter)	BRCA1	41293465	70268		
NM_007294.3(BRCA1): c.5503C > T (p.Arg1835Ter)	BRCA1	80357123	70147		
NM_007294.3(BRCA1): c.5251C > T (p.Arg1751Ter)	BRCA1	137929307	171217		
NM_000527.4(LDLR): c.1775G > A (p.Gly592Glu)	LDLR	80356898	45982		
NM_007294.3(BRCA1): c.1687C > T (p.Gln563Ter)	BRCA1	28936415	22745		
NM_000303.2(PMM2): c.422G > A (p.Arg141His)	PMM2	11555217	34125		

NM_01360.2(DHCR7): c.457G > A (p.Trp151Ter) DHCR7 55770810 70063
NM_007294.3(BRCA1): c.5095C > T (p.Arg1699Trp) BRCA1 28934906 26850
NM_004992.3(MECP2): c.473C > T (p.Thr158Met) MECP2 28929474 33006
NM_001127701.1(SERPINA1): c.1096G > A (p.Glu366Lys) SERPINA1 371898076 52045
NM_000257.4(MYH7): c.1988G > A (p.Arg663His) MYH7 5030858 15616 NM_000277.3(PAH):
c.1222C > T (p.Arg408Trp) PAH 80356945 69207 NM_007294.3(BRCA1): c.2338C > T
(p.Gln780Ter) BRCA1 1800553 22927 NM_000350.2(ABCA4): c.5882G > A (p.Gly1961Glu)
ABCA4 80356962 70247 NM_007294.3(BRCA1): c.5444G > A (p.Trp1815Ter) BRCA1
104894396 32041 NM_004004.6(GJB2): c.71G > A (p.Trp24Ter) GJB2 113994095 28535
NM_002693.2(POLG): c.1399G > A (p.Ala467Thr) POLG 61749721 26868
NM_004992.3(MECP2): c.763C > T (p.Arg255Ter) MECP2 137852700 23943
NM_000310.3(PPT1): c.451C > T (p.Arg151Ter) PPT1 75527207 22159 NM_000492.3(CFTR):
c.1652G > A (p.Gly551Asp) CFTR 78655421 22148 NM_000492.3(CFTR): c.350G > A
(p.Arg117His) CFTR 80356885 69888 NM_007294.3(BRCA1): c.4524G > A (p.Trp1508Ter)
BRCA1 113994098 28541 NM_002693.2(POLG): c.2542G > A (p.Gly848Ser) POLG 61750240
26854 NM_004992.3(MECP2): c.808C > T (p.Arg270Ter) MECP2 61751362 26858
NM_001110792.1(MECP2): c.916C > T (p.Arg306Ter) MECP2 80357260 69792
NM_007294.3(BRCA1): c.4183C > T (p.Gln1395Ter) BRCA1 80359071 67203
NM_000059.3(BRCA2): c.8243G > A (p.Gly2748Asp) BRCA2 62625307 69596
NM_007294.3(BRCA1): c.3598C > T (p.Gln1200Ter) BRCA1 76992529 28465
NM_000371.3(TTR): c.424G > A (p.Val142Ile) TTR 77010898 22168 NM_000492.3(CFTR):
c.3846G > A (p.Trp1282Ter) CFTR 80359003 67069 NM_000059.3(BRCA2): c.7757G > A
(p.Trp2586Ter) BRCA2 61750420 22555 NM_000466.2(PEX1): c.2528G > A (p.Gly843Asp)
PEX1 80357284 46214 NM_007294.3(BRCA1): c.5346G > A (p.Trp1782Ter) BRCA1 200411226
174776 NM_000256.3(MYBPC3): c.1484G > A (p.Arg495Gln) MYBPC3 5030857 98638
NM_000277.3(PAH): c.1208C > T (p.Ala403Val) PAH 28935468 26863 NM_004992.3(MECP2):
c.916C > T (p.Arg306Cys) MECP2 62642937 15667 NM_000277.3(PAH): c.1139C > T
(p.Thr380Met) PAH 80356989 69812 NM_007294.3(BRCA1): c.4222C > T (p.Gln1408Ter)
BRCA1 28942080 18735 NM_000527.4(LDLR): c.1567G > A (p.Val523Met) LDLR 121908039
18778 NM_000527.4(LDLR): c.551G > A (p.Cys184Tyr) LDLR 267607213 18780
NM_000527.4(LDLR): c.131G > A (p.Trp44Ter) LDLR 3218716 52071 NM_000257.3(MYH7):
c.2389G > A (p.Ala797Thr) MYH7 104895097 17588 NM_000243.2(MEFV): c.2282G > A
(p.Arg761His) MEFV 397516074 51962 NM_000256.3(MYBPC3): c.772G > A (p.Glu258Lys)
MYBPC3 119455955 17682 NM_000391.3(TPP1): c.622C > T (p.Arg208Ter) TPP1 75184679
16301 NM_024570.3(RNASEH2B): c.529G > A (p.Ala177Thr) RNASEH2B 80338901 26909
NM_000137.2(FAH): c.1062+5G > A FAH 119450941 17501 NM_000026.3(ADSL): c.1277G > A
(p.Arg426His) ADSL 121965019 26947 NM_000203.4(IDUA): c.1205G > A (p.Trp402Ter) IDUA
141659620 21858 NM_003119.3(SPG7): c.1045G > A (p.Gly349Ser) SPG7 41276738 15335
NM_000552.4(VWF): c.2561G > A (p.Arg854Gln) VWF 80338940 32068 NM_004004.5(GJB2):
c.-23 + 1G > A GJB2 80357292 46268 NM_007294.3(BRCA1): c.962G > A (p.Trp321Ter)
BRCA1 121913627 29130 NM_000257.3(MYH7): c.1816G > A (p.Val606Met) MYH7 137854601
24416 NM_198056.2(SCN5A): c.5350G > A (p.Glu1784Lys) SCN5A 80338933 17521
NM_024577.3(SH3TC2): c.2860C > T (p.Arg954Ter) SH3TC2 80338948 32048
NM_004004.5(GJB2): c.427C > T (p.Arg143Trp) GJB2 80356903 69645 NM_007294.3(BRCA1):
c.3718C > T (p.Gln1240Ter) BRCA1 80356969 70213 NM_007294.3(BRCA1): c.5353C > T
(p.Gln1785Ter) BRCA1 80357010 45971 NM_007294.3(BRCA1): c.1480C > T (p.Gln494Ter)
BRCA1 116987552 17337 NM_005609.3(PYGM): c.148C > T (p.Arg50Ter) PYGM 121913625
29128 NM_000257.4(MYH7): c.1357C > T (p.Arg453Cys) MYH7 387907267 45725
NM_000256.3(MYBPC3): c.2827C > T (p.Arg943Ter) MYBPC3 28934897 26968
NM_000431.3(MVK): c.1129G > A (p.Val377Ile) MVK 76713772 22151 NM_000492.3(CFTR):

c.1585-1G > A CFTR 137852959 19587 NM_153638.3(PANK2): c.1561G > A (p.Gly521Arg) PANK2 199682486 101428 NM_013339.4(ALG6): c.257 + 5G > A ALG6 397507389 46666 NM_000059.3(BRCA2): c.7618-1G > A BRCA2 769370816 228176 NM_000527.4(LDLR): c.1618G > A (p.Ala540Thr) LDLR 36211715 29159 NM_000257.4(MYH7): c.2609G > A (p.Arg870His) MYH7 76434661 53916 NM_004004.5(GJB2): c.416G > A (p.Ser139Asn) GJB2 104894368 29104 NM_000432.3(MYL2): c.64G > A (p.Glu22Lys) MYL2 104894635 20146 NM_000199.3(SGSH): c.734G > A (p.Arg245His) SGSH 121913628 29131 NM_000257.3(MYH7): c.2770G > A (p.Glu924Lys) MYH7 193922390 45304 NM_000257.4(MYH7): c.5135G > A (p.Arg1712Gln) MYH7 397515757 51454 NM_000138.4(FBN1): c.1468 + 5G > A FBN1 11549407 30441 NM_000518.5(HBB): c.118C > T (p.Gln40Ter) HBB 61751374 22933 NM_000350.2(ABCA4): c.3113C > T (p.Ala1038Val) ABCA4 121434420 21793 NM_004572.3(PKP2): c.235C > T (p.Arg79Ter) PKP2 137853007 20631 NM_007194.4(CHEK2): c.433C > T (p.Arg145Trp) CHEK2 1137887 18083 NM_000051.3(ATM): c.2250G > A (p.Lys750=) ATM 28934872 27436 NM_000548.3(TSC2): c.1832G > A (p.Arg611Gln) TSC2 80224560 47062 NM_000492.3(CFTR): c.2657 + 5G > A CFTR 80359004 46672 NM_000059.3(BRCA2): c.7758G > A (p.Trp2586Ter) BRCA2 121434274 18627 NM_000016.5(ACADM): c.799G > A (p.Gly267Arg) ACADM 121908529 38436 NM_000030.2(AGXT): c.508G > A (p.Gly170Arg) AGXT 121918007 28709 NM_000478.4(ALPL): c.571G > A (p.Glu191Lys) ALPL 121918243 16464 NM_015506.2(MMACHC): c.482G > A (p.Arg161Gln) MMACHC 397518423 94255 NM_005026.4(PIK3CD): c.3061G > A (p.Glu1021Lys) PIK3CD 587781629 150997 NM_000059.3(BRCA2): c.1909 + 1G > A BRCA2 765696008 228162 NM_000527.4(LDLR): c.1187-10G > A LDLR 3218713 29127 NM_000257.3(MYH7): c.746G > A (p.Arg249Gln) MYH7 5030855 15646 NM_000277.3(PAH): c.1066-11G > A PAH 55851803 69067 NM_007294.3(BRCA1): c.191G > A (p.Cys64Tyr) BRCA1 62508698 15619 NM_000277.1(PAH): c.838G > A (p.Glu280Lys) PAH 62516152 108520 NM_000277.3(PAH): c.688G > A (p.Val230Ile) PAH 62644499 15656 NM_000277.3(PAH): c.1243G > A (p.Asp415Asn) PAH 80338815 18090 NM_000487.5(ARSA): c.465 + 1G > A ARSA 121908987 21885 NM_016203.3(PRKAG2): c.905G > A (p.Arg302Gln) PRKAG2 121964962 15156 NM_000071.2(CBS): c.919G > A (p.Gly307Ser) CBS 5030851 15628 NM_000277.3(PAH): c.842C > T (p.Pro281Leu) PAH 63750871 24273 NM_000535.6(PMS2): c.400C > T (p.Arg134Ter) PMS2 80338853 21822 NM_001360.2(DHCR7): c.278C > T (p.Thr93Met) DHCR7 80356893 68976 NM_007294.3(BRCA1): c.1612C > T (p.Gln538Ter) BRCA1 80357131 46031 NM_007294.3(BRCA1): c.2563C > T (p.Gln855Ter) BRCA1 80357223 69350 NM_007294.3(BRCA1): c.2800C > T (p.Gln934Ter) BRCA1 80357318 46112 NM_007294.3(BRCA1): c.3937C > T (p.Gln1313Ter) BRCA1 104886457 27086 NM_000136.2(FANCC): c.1642C > T (p.Arg548Ter) FANCC 137852944 19147 NM_138694.3(PKHD1): c.107C > T (p.Thr36Met) PKHD1 180177083 132139 NM_024675.3(PALB2): c.196C > T (p.Gln66Ter) PALB2 180177110 152117 NM_024675.3(PALB2): c.2257C > T (p.Arg753Ter) PALB2 199475575 108459 NM_000277.3(PAH): c.526C > T (p.Arg176Ter) PAH 387906843 39241 NM_002878.3(RAD51D): c.556C > T (p.Arg186Ter) RAD51D 529008617 152318 NM_001128425.1(MUTYH): c.1214C > T (p.Pro405Leu) MUTYH 587780021 133177 NM_000465.3(BARD1): c.1690OT (p.Gln564Ter) BARD1 34637584 16979 NM_198578.3(LRRK2): c.6055G > A (p.Gly2019Ser) LRRK2 78802634 22233 NM_000492.3(CFTR): c.3266G > A (p.Trp1089Ter) CFTR 80358809 66611 NM_000059.3(BRCA2): c.581G > A (p.Trp194Ter) BRCA2 80359011 46678 NM_000059.3(BRCA2): c.7857G > A (p.Trp2619Ter) BRCA2 104894503 27495 NM_001018005.1(TPM1): c.523G > A (p.Asp175Asn) TPM1 121908641 21368 NM_000050.4(ASS1): c.1168G > A (p.Gly390Arg) ASS1 121918593 28009

NM_000540.2(RYR1): c.7300G > A (p.Gly2434Arg) RYR1 140108514 100191
NM_003494.3(DYSF): c.2643 + 1G > A DYSF 145138923 98271 NM_000048.3(ASL): c.35G > A
(p.Arg12Gln) ASL 150726175 45795 NM_022787.3(NMNAT1): c.769G > A (p.Glu257Lys)
NMNAT1 267607578 45138 NM_170707.3(LMNA): c.1412G > A (p.Arg471His) LMNA
376607329 48992 NM_002834.4(PTPN11): c.794G > A (p.Arg265Gln) PTPN11 587776934 48407
NM_005027.3(PIK3R2): c.1117G > A (p.Gly373Arg) PIK3R2 62508588 15630
NM_000277.1(PAH): c.728G > A (p.Arg243Gln) PAH 62637014 20604 NM_014336.4(AIPL1):
c.834G > A (p.Trp278Ter) AIPL1 80356860 46194 NM_007294.3(BRCA1): c.5117G > A
(p.Gly1706Glu) BRCA1 80357268 70265 NM_007294.3(BRCA1): c.5497G > A (p.Val1833Met)
BRCA1 80357418 70077 NM_007294.3(BRCA1): c.5136G > A (p.Trp1712Ter) BRCA1
80358145 46229 NM_007294.3(BRCA1): c.5467 + 1G > A BRCA1 121918166 15994
NM_000275.2(OCA2): c.1327G > A (p.Val443Ile) OCA2 140342925 150591
NM_001128425.1(MUTYH): c.734G > A (p.Arg245His) MUTYH 148660051 195093
NM_206933.2(USH2A): c.10073G > A (p.Cys3358Tyr) USH2A 193922672 45341
NM_004572.3(PKP2): c.1613G > A (p.Trp538Ter) PKP2 267607144 20039
NM_021625.4(TRPV4): c.806G > A (p.Arg269His) TRPV4 397516083 51977
NM_000256.3(MYBPC3): c.927-9G > A MYBPC3 397516357 52565 NM_000363.4(TNNI3):
c.557G > A (p.Arg186Gln) TNNI3 587782958 165560 NM_000256.3(MYBPC3): c.3190 + 5G > A
MYBPC3 28934907 26853 NM_004992.3(MECP2): c.316C > T (p.Arg106Trp) MECP2 28934908
26862 NM_004992.3(MECP2): c.419C > T (p.Ala140Val) MECP2 28940893 18091
NM_000487.5(ARSA): c.1283C > T (p.Pro428Leu) ARSA 63751422 96795
NM_000535.5(PMS2): c.1927C > T (p.Gln643Ter) PMS2 74315366 27817
NM_003000.2(SDHB): c.268C > T (p.Arg90Ter) SDHB 80338856 34127
NM_001360.2(DHCR7): c.724C > T (p.Arg242Cys) DHCR7 80357038 69707
NM_007294.3(BRCA1): c.3895C > T (p.Gln1299Ter) BRCA1 80357136 69535
NM_007294.3(BRCA1): c.3403C > T (p.Gln1135Ter) BRCA1 80357208 69682
NM_007294.3(BRCA1): c.3817C > T (p.Gln1273Ter) BRCA1 80357234 69166
NM_007294.3(BRCA1): c.220C > T (p.Gln74Ter) BRCA1 80357262 69729
NM_007294.3(BRCA1): c.3967C > T (p.Gln1323Ter) BRCA1 80357305 69822
NM_007294.3(BRCA1): c.4258C > T (p.Gln1420Ter) BRCA1 80357350 69232
NM_007294.3(BRCA1): c.241C > T (p.Gln81Ter) BRCA1 104894636 20147
NM_000199.3(SGSH): c.220C > T (p.Arg74Cys) SGSH 111401431 44742 NM_000138.4(FBN1):
c.4588C > T (p.Arg1530Cys) FBN1 121918624 27928 NM_006920.5(SCN1A): c.664C > T
(p.Arg222Ter) SCN1A 137852981 19794 NM_014795.3(ZEB2): c.2083C > T (p.Arg695Ter)
ZEB2 137854476 31491 NM_000138.4(FBN1): c.1585C > T (p.Arg529Ter) FBN1 137854480
31500 NM_000138.4(FBN1): c.718C > T (p.Arg240Cys) FBN1 180177100 133574
NM_024675.3(PALB2): c.1240C > T (p.Arg414Ter) PALB2 193922109 44392
NM_000053.3(ATP7B): c.3955C > T (p.Arg1319Ter) ATP7B 200640585 96857
NM_000535.6(PMS2): c.943C > T (p.Arg315Ter) PMS2 201431517 48426
NM_139242.3(MTFMT): c.626C > T (p.Ser209Leu) MTFMT 397516037 51905
NM_000256.3(MYBPC3): c.3697C > T (p.Gln1233Ter) MYBPC3 587780104 133350
NM_002878.3(RAD51D): c.694C > T (p.Arg232Ter) RAD51D 765123255 181726
NM_001128425.1(MUTYH): c.325C > T (p.Arg109Trp) MUTYH 63751657 95331
NM_000249.3(MLH1): c.1731G > A (p.Ser577=) MLH1 75549581 22162 NM_000492.3(CFTR):
c.1675G > A (p.Ala559Thr) CFTR 80338851 16303 NM_194318.3(B3GLCT): c.660 + 1G > A
B3GLCT 80358544 46368 NM_000059.3(BRCA2): c.2979G > A (p.Trp993Ter) BRCA2
111033178 52388 NM_000260.3(MYO7A): c.3719G > A (p.Arg1240Gln) MYO7A 121908188
19535 NM_020451.2(SELENON): c.943G > A (p.Gly315Ser) SELENON 139770721 180483
NM_000051.3(ATM): c.6095G > A (p.Arg2032Lys) ATM 199476315 40542
NM_001018005.1(TPM1): c.574G > A (p.Glu192Lys) TPM1 267607004 15310

NM_001134363.2(RBM20): c.1907G > A (p.Arg636His) RBM20 267608122 94980
NM_000179.2(MSH6): c.4001G > A (p.Arg1334Gln) MSH6 377349459 150947
NM_000051.3(ATM): c.7913G > A (p.Trp2638Ter) ATM 387906303 18745
NM_000527.4(LDLR): c.670G > A (p.Asp224Asn) LDLR 587779227 94719
NM_000179.2(MSH6): c.2057G > A (p.Gly686Asp) MSH6 587780290 134019
NM_000070.2(CAPN3): c.2243G > A (p.Arg748Gln) CAPN3 727504317 49251
NM_002755.3(MAP2K1): c.1199G > A (p.Asp67Asn) MAP2K1 5030869 25402
NM_000402.4(G6PD): c.1093G > A (p.Ala365Thr) G6PD 9332964 18390
NM_000348.3(SRD5A2): c.680G > A (p.Arg227Gln) SRD5A2 36211723 45266
NM_000256.3(MYBPC3): c.2308G > A (p.Asp770Asn) MYBPC3 72549410 78547
NM_000335.4(SCN5A): c.1231G > A (p.Val411Met) SCN5A 80357498 45948
NM_007294.3(BRCA1): c.116G > A (p.Cys39Tyr) BRCA1 80358079 70118
NM_007294.3(BRCA1): c.5194-12G > A BRCA1 121434529 33201 NM_000262.2(NAGA):
c.973G > A (p.Glu325Lys) NAGA 121908627 21067 NM_005476.5(GNE): c.2086G > A
(p.Val696Met) GNE 387906592 38552 NM_001613.2(ACTA2): c.536G > A (p.Arg179His)
ACTA2 397515907 51711 NM_000256.3(MYBPC3): c.1505G > A (p.Arg502Gln) MYBPC3
397516089 51992 NM_000257.4(MYH7): c.1106G > A (p.Arg369Gln) MYH7 397516248 52239
NM_000257.4(MYH7): c.5401G > A (p.Glu1801Lys) MYH7 397516349 52554
NM_000363.4(TNNI3): c.434G > A (p.Arg145Gln) TNNI3 5030846 15627 NM_000277.3(PAH):
c.727C > T (p.Arg243Ter) PAH 28941784 18134 NM_052845.3(MMAB): c.556C > T
(p.Arg186Trp) MMAB 34126013 181693 NM_001128425.1(MUTYH): c.721C > T (p.Arg241Trp)
MUTYH 62541771 21074 NM_001128227.2(GNE): c.1985C > T (p.Ala662Val) GNE 62625303
68931 NM_007294.3(BRCA1): c.1471C > T (p.Gln491Ter) BRCA1 74315379 27453
NM_001001430.2(TNNT2): c.421C > T (p.Arg141Trp) TNNT2 76687508 108539
NM_000277.3(PAH): c.721C > T (p.Arg241Cys) PAH 80338794 20654
NM_012434.4(SLC17A5): c.115C > T (p.Arg39Cys) SLC17A5 80356866 69689
NM_007294.3(BRCA1): c.3841C > T (p.Gln1281Ter) BRCA1 80357134 69569
NM_007294.3(BRCA1): c.34C > T (p.Gln12Ter) BRCA1 80357229 69904
NM_007294.3(BRCA1): c.4609C > T (p.Gln1537Ter) BRCA1 111033260 19972
NM_033056.3(PCDH15): c.733C > T (p.Arg245Ter) PCDH15 121909398 17403
NM_201548.4(CERKL): c.769C > T (p.Arg257Ter) CERKL 121913637 29143
NM_000257.4(MYH7): c.2155C > T (p.Arg719Trp) MYH7 200495564 50200
NM_001128425.1(MUTYH): c.733C > T (p.Arg245Cys) MUTYH 267607203 20760
NM_194456.1(KRIT1): c.1363C > T (p.Gln455Ter) KRIT1 587776527 132239
NM_024675.3(PALB2): c.3256C > T (p.Arg1086Ter) PALB2 587777219 125784
NM_172107.3(KCNQ2): c.794C > T (p.Ala265Val) KCNQ2 587778617 96774
NM_000535.5(PMS2): c.1261C > T (p.Arg421Ter) PMS2 587783057 166274
NM_001128425.1(MUTYH): c.1171C > T (p.Gln391Ter) MUTYH 730880099 178699
NM_000138.4(FBN1): c.1633C > T (p.Arg545Cys) FBN1 2309689 33868
NM_000018.3(ACADVL): c.1322G > A (p.Gly441Asp) ACADVL 28933093 29543
NM_170707.3(LMNA): c.481G > A (p.Glu161Lys) LMNA 28937873 20571
NM_014249.3(NR2E3): c.932G > A (p.Arg311Gln) NR2E3 59332535 77828
NM_170707.3(LMNA): c.746G > A (p.Arg249Gln) LMNA 62645748 48213
NM_201253.2(CRB1): c.2843G > A (p.Cys948Tyr) CRB1 63750828 96748
NM_000251.2(MSH2): c.998G > A (p.Cys333Tyr) MSH2 80358456 65843
NM_000059.3(BRCA2): c.1689G > A (p.Trp563Ter) BRCA2 80359101 67273
NM_000059.3(BRCA2): c.8489G > A (p.Trp2830Ter) BRCA2 80359148 131733
NM_000059.3(BRCA2): c.8969G > A (p.Trp2990Ter) BRCA2 80359149 67384
NM_000059.3(BRCA2): c.8970G > A (p.Trp2990Ter) BRCA2 80359211 46791
NM_000059.3(BRCA2): c.9380G > A (p.Trp3127Ter) BRCA2 111033565 26915

NM_002769.4(PRSS1): c.365G > A (p.Arg122His) PRSS1 113994205 19482
NM_004937.2(CTNS): c.414G > A (p.Trp138Ter) CTNS 116840778 23322 NM_033337.2(CAV3):
c.80G > A (p.Arg27Gln) CAV3; SSUH2 118192158 76835 NM_000540.2(RYR1): c.14818G > A
(p.Ala4940Thr) RYR1 121434278 18633 NM_000016.5(ACADM): c.583G > A (p.Gly195Arg)
ACADM 121434346 17058 NM_001003841.2(SLC6A19): c.517G > A (p.Asp173Asn) SLC6A19
121908011 18814 NM_000372.4(TYR): c.1147G > A (p.Asp383Asn) TYR 121908638 21365
NM_000050.4(ASS1): c.539G > A (p.Ser180Asn) ASS1 121912938 32219
NM_001848.2(COL6A1): c.850G > A (p.Gly284Arg) COL6A1 137853096 22694
NM_000414.3(HSD17B4): c.46G > A (p.Gly16Ser) HSD17B4 151344631 45847
NM_000218.2(KCNQ1): c.613G > A (p.Val205Met) KCNQ1 192838388 98283
NM_000050.4(ASS1): c.787G > A (p.Val263Met) ASS1 267607768 95759 NM_000249.3(MLH1):
c.588 + 5G > A MLH1 376107921 213634 NM_000070.2(CAPN3): c.1319G > A (p.Arg440Gln)
CAPN3 397507981 67234 NM_000059.3(BRCA2): c.8364G > A (p.Trp2788Ter) BRCA2
398124321 101692 NM_017780.3(CHD7): c.5405-7G > A CHD7 730882246 181441
NM_194279.3(ISCA2): c.229G > A (p.Gly77Ser) ISCA2 778906552 195186
NM_000016.5(ACADM): c.443G > A (p.Arg148Lys) ACADM 139428292 39421
NM_005105.4(RBM8A): c.-21G > A RBM8A 28934891 15165 NM_000071.2(CBS): c.1330G >
A (p.Asp444Asn) CBS 28937316 24408 NM_198056.2(SCN5A): c.4931G > A (p.Arg1644His)
SCN5A 33930165 30165 NM_000518.4(HBB): c.19G > A (p.Glu7Lys) HBB 35004220 30493
NM_000518.5(HBB): c.93-21G > A HBB 45546039 48043 NM_198056.2(SCN5A): c.665G > A
(p.Arg222Gln) SCN5A 61751402 105177 NM_000350.2(ABCA4): c.4469G > A (p.Cys1490Tyr)
ABCA4 72549387 22776 NM_000104.3(CYP1B1): c.171G > A (p.Trp57Ter) CYP1B1 75822236
19350 NM_000157.3(GBA): c.1604G > A (p.Arg535His) GBA 79389353 20821
NM_014270.4(SLC7A9): c.544G > A (p.Ala182Thr) SLC7A9 80338862 34124
NM_001360.2(DHCR7): c.1228G > A (p.Gly410Ser) DHCR7 80338892 27366
NM_199292.2(TH): c.698G > A (p.Arg233His) TH 80356935 68777 NM_007294.3(BRCA1):
c.1059G > A (p.Trp353Ter) BRCA1 80357468 68802 NM_007294.3(BRCA1): c.1116G > A
(p.Trp372Ter) BRCA1 104894365 27628 NM_004985.4(KRAS): c.40G > A (p.Val14Ile) KRAS
104894639 20153 NM_000199.3(SGSH): c.1339G > A (p.Glu447Lys) SGSH 111033364 17396
NM_206933.2(USH2A): c.11864G > A (p.Trp3955Ter) USH2A 119103251 17338
NM_005609.3(PYGM): c.613G > A (p.Gly205Ser) PYGM 119455954 17681
NM_000391.3(TPP1): c.1094G > A (p.Cys365Tyr) TPP1 121913638 29144
NM_000257.4(MYH7): c.2146G > A (p.Gly716Arg) MYH7 137854478 31496
NM_000138.4(FBN1): c.3217G > A (p.Glu1073Lys) FBN1 143353451 179937
NM_001128425.1(MUTYH): c.545G > A (p.Arg182His) MUTYH 151045328 20182
NM_005709.3(USH1C): c.216G > A (p.Val72=) USH1C 151344623 24127
NM_001287174.1(ABCC8): c.3992-9G > A ABCC8 193922204 44739 NM_000138.4(FBN1):
c.4460-8G > A FBN1 193922219 51564 NM_000138.4(FBN1): c.5788 + 5G > A FBN1
193922680 33370 NM_005159.4(ACTC1): c.301G > A (p.Glu101Lys) ACTC1 267608172 96804
NM_000535.5(PMS2): c.2174 + 1G > A PMS2 397516202 52163 NM_000257.3(MYH7):
c.4135G > A (p.Ala1379Thr) MYH7 397516209 52176 NM_000257.4(MYH7): c.428G > A
(p.Arg143Gln) MYH7 397517159 49176 NM_005633.3(SOS1): c.2536G > A (p.Glu846Lys)
SOS1 587776576 18532 NM_024426.5(WT1): c.1447 + 5G > A WT1 727503246 175600
NM_000257.4(MYH7): c.4066G > A (p.Glu1356Lys) MYH7 730881687 181107
NM_007194.4(CHEK2): c.793-1G > A CHEK2 748170941 181727 NM_001128425.1(MUTYH):
c.309G > A (p.Trp103Ter) MUTYH 140583 260073 NM_000138.4(FBN1): c.2581C > T
(p.Arg861Ter) FBN1 2754158 175617 NM_000257.3(MYH7): c.2572C > T (p.Arg858Cys)
MYH7 28931570 33013 NM_001127701.1(SERPINA1): c.187C > T (p.Arg63Cys) SERPINA1
34424986 22089 NM_004562.2(PRKN): c.823C > T (p.Arg275Trp) PRKN 61750130 22943
NM_000350.2(ABCA4): c.4139C > T (p.Pro1380Leu) ABCA4 61750200 22937

NM_000535.2(ABCA4): c.634C > T (p.Arg212Cys) ABCA4 63750451 24281
NM_000535.5(PMS2): c.1882C > T (p.Arg628Ter) PMS2 72653706 21598
NM_001171.5(ABCC6): c.3421C > T (p.Arg1141Ter) ABCC6 74503222 108557
NM_000277.3(PAH): c.745C > T (p.Leu249Phe) PAH 76296470 15620 NM_000277.3(PAH):
c.331C > T (p.Arg111Ter) PAH 80338860 21826 NM_001360.2(DHCR7): c.1054C > T
(p.Arg352Trp) DHCR7 80356682 29578 NM_000228.2(LAMB3): c.1903C > T (p.Arg635Ter)
LAMB3 80356771 19334 NM_001005741.2(GBA): c.1504C > T (p.Arg502Cys) GBA 80356904
68978 NM_007294.3(BRCA1): c.1621C > T (p.Gln541Ter) BRCA1 80356932 69850
NM_007294.3(BRCA1): c.4372C > T (p.Gln1458Ter) BRCA1 80356947 70087
NM_007294.3(BRCA1): c.514C > T (p.Gln172Ter) BRCA1 80356992 69906
NM_007294.3(BRCA1): c.4612C > T (p.Gln1538Ter) BRCA1 80357133 70034
NM_007294.3(BRCA1): c.505C > T (p.Gln169Ter) BRCA1 80357215 68781
NM_007294.3(BRCA1): c.1066C > T (p.Gln356Ter) BRCA1 104894419 22712
NM_002312.3(LIG4): c.2440C > T (p.Arg814Ter) LIG4 113871094 44746 NM_000138.4(FBN1):
c.4786C > T (p.Arg1596Ter) FBN1 118203682 58105 NM_000368.4(TSC1): c.2356C > T
(p.Arg786Ter) TSC1 121908177 19611 NM_031885.3(BBS2): c.823C > T (p.Arg275Ter) BBS2
121908715 16998 NM_000022.2(ADA): c.986C > T (p.Ala329Val) ADA 121909122 22411
NM_001083962.1(TCF4): c.1153C > T (p.Arg385Ter) TCF4 121917901 16740
NM_000124.3(ERCC6): c.2203C > T (p.Arg735Ter) ERCC6 121964964 15158
NM_000071.2(CBS): c.341C > T (p.Ala114Val) CBS 137852924 18422 NM_147127.4(EVC2):
c.1195C > T (p.Arg399Ter) EVC2 137854466 31478 NM_000138.4(FBN1): c.8326C > T
(p.Arg2776Ter) FBN1 137854467 31479 NM_000138.4(FBN1): c.364C > T (p.Arg122Cys) FBN1
137854604 24422 NM_000335.4(SCN5A): c.5126C > T (p.Ser1709Leu) SCN5A 150518260
51200 NM_000232.4(SGCB): c.341C > T (p.Ser114Phe) SGCB 200432447 133521
NM_007194.4(CHEK2): c.1555C > T (p.Arg519Ter) CHEK2 201587138 176561
NM_144612.6(LOXHD1): c.4480C > T (p.Arg1494Ter) LOXHD1 367543286 70502
NM_002609.3(PDGFRB): c.1681C > T (p.Arg561Cys) PDGFRB 372827156 54183
NM_004572.3(PKP2): c.1237C > T (p.Arg413Ter) PKP2 374950566 181683
NM_001128425.1(MUTYH): c.884C > T (p.Pro295Leu) MUTYH 397514558 48266
NM_000138.4(FBN1): c.2920C > T (p.Arg974Cys) FBN1 397515992 51839
NM_000256.3(MYBPC3): c.2905C > T (p.Gln969Ter) MYBPC3 397516456 52796
NM_000364.3(TNNT2): c.304C > T (p.Arg102Trp) TNNT2 587780082 133292
NM_001128425.1(MUTYH): c.1012C > T (p.Gln338Ter) MUTYH 587782705 152480
NM_000546.5(TP53): c.455C > T (p.Pro152Leu) TP53 727503974 177432
NM_172107.3(KCNQ2): c.821C > T (p.Thr274Met) KCNQ2 730881864 180279
NM_002485.4(NBN): c.2140C > T (p.Arg714Ter) NBN 767215758 188057 NM_002485.4(NBN):
c.1030C > T (p.Gln344Ter) NBN 45517259 27442 NM_000548.3(TSC2): c.2714G > A
(p.Arg905Gln) TSC2 61195471 57234 NM_170707.3(LMNA): c.607G > A (p.Glu203Lys) LMNA
61753185 18815 NM_000372.4(TYR): c.230G > A (p.Arg77Gln) TYR 63749869 28021
NM_000540.2(RYR1): c.14582G > A (p.Arg4861His) RYR1 63749939 32145
NM_000249.3(MLH1): c.200G > A (p.Gly67Glu) MLH1 63750119 150580
NM_000179.2(MSH6): c.3725G > A (p.Arg1242His) MSH6 72554308 26053
NM_000531.5(OTC): c.119G > A (p.Arg40His) OTC 79891110 32671
NM_000719.6(CACNA1C): c.1216G > A (p.Gly406Arg) CACNA1C 80338707 22758
NM_000303.2(PMM2): c.691G > A (p.Val231Met) PMM2 80338802 32652
NM_000070.2(CAPN3): c.2306G > A (p.Arg769Gln) CAPN3 80356700 32571
NM_000083.2(CLCN1): c.689G > A (p.Gly230Glu) CLCN1 80359803 67339
NM_000059.3(BRCA2): c.8754G > A (p.Glu2918=) BRCA2 81002809 67078
NM_000059.3(BRCA2): c.7805 + 1G > A BRCA2 104886142 35796 NM_000495.4(COL4A5):
c.1871G > A (p.Gly624Asp) COL4A5 104894423 17048 NM_000231.2(SGCG): c.787G > A

(p.Glu263Lys) SGCG104894525 22747 NM_000303.2(PMM2): c.385G > A (p.Val129Met)
PMM2 113994049 20984 NM_003907.3(EIF2B5): c.338G > A (p.Arg113His) EIF2B5 121434372
17127 NM_000159.3(GCDH): c.1198G > A (p.Val400Met) GCDH 121908099 19299
NM_000784.3(CYP27A1): c.1214G > A (p.Arg405Gln) CYP27A1 121908192 23730
NM_005262.2(GFER): c.581G > A (p.Arg194His) GFER 121908753 22237
NM_000492.3(CFTR): c.1055G > A (p.Arg352Gln) CFTR 121918013 28716
NM_000478.4(ALPL): c.346G > A (p.Ala116Thr) ALPL 139729994 68418 NM_000492.3(CFTR):
c.3468G > A (p.Leu1156=) CFTR 142637046 98272 NM_000048.3(ASL): c.446 + 1G > A ASL
142761835 177782 NM_002225.3(IVD): c.367G > A (p.Gly123Arg) IVD 146015592 46845
NM_000060.4(BTD): c.470G > A (p.Arg157His) BTD 150877497 226470 NM_003494.3(DYSF):
c.3113G > A (p.Arg1038Gln) DYSF 199472815 67686 NM_000218.2(KCNQ1): c.1781G > A
(p.Arg594Gln) KCNQ1 199474738 79199 NM_001042492.2(NF1): c.1885G > A (p.Gly629Arg)
NF1 199476112 24747 NC_012920.1: m.11778G > A MT-ND4 199476317 40544
NM_001018005.1(TPM1): c.688G > A (p.Asp230Asn) TPM1 201540674 51186 RTEL1: c.2402G
> A (p.Arg801His) RTEL1 267606640 16147 NM_000642.2(AGL): c.3980G > A (p.Trp1327Ter)
AGL 386834233 76679 NM_183050.3(BCKDHB): c.832G > A (p.Gly278Ser) BCKDHB
397515355 19301 NM_000784.3(CYP27A1): c.1263 + 1G > A CYP27A1 397515404 48194
NM_020822.2(KCNT1): c.1421G > A (p.Arg474His) KCNT1 398123787 100221
NM_003494.3(DYSF): c.4253G > A (p.Gly1418Asp) DYSF 398124641 44139
NM_024531.4(SLC52A2): c.916G > A (p.Gly306Arg) SLC52A2 587776783 132342
NM_000321.2(RB1): c.1215 + 1G > A RB1 587776889 39757 NM_015506.2(MMACHC): c.609G
> A (p.Trp203Ter) MMACHC 587777721 165903 NM_014191.3(SCN8A): c.4850G > A
(p.Arg1617Gln) SCN8A 587779818 132798 NM_000051.3(ATM): c.170G > A (p.Trp57Ter) ATM
587780537 136457 NM_004360.4(CDH1): c.715G > A (p.Gly239Arg) CDH1 587783050 166264
NM_004360.5(CDH1): c.1137G > A (p.Thr379=) CDH1 751995154 200340
NM_000018.4(ACADVL): c.1376G > A (p.Arg459Gln) ACADVL 781404312 186796
NM_000051.3(ATM): c.3G > A (p.Met1Ile) ATM 786202112 184694 NM_001042492.2(NF1):
c.5609G > A (p.Arg1870Gln) NF1 794727152 191718 NM_021007.2(SCN2A): c.2558G > A
(p.Arg853Gln) SCN2A 796051858 18086 NM_000051.3(ATM): c.496 + 5G > A ATM 796052505
201880 NM_000816.3(GABRG2): c.316G > A (p.Ala106Thr) GABRG2 863223408 210238
NM_000020.2(ACVRL1): c.1451G > A (p.Arg484Gln) ACVRL1 863225082 188114
NM_006245.3(PPP2R5D): c.592G > A (p.Glu198Lys) PPP2R5D 875989911 228151
NM_000527.4(LDLR): c.938G > A (p.Cys313Tyr) LDLR 5030852 15638 NM_000277.3(PAH):
c.842 + 1G > A PAH 5030859 15651 NM_000277.3(PAH): c.1223G > A (p.Arg408Gln) PAH
28930068 32662 NM_000069.2(CACNA1S): c.3716G > A (p.Arg1239His) CACNA1S 56264519
55267 NM_024022.2(TMPRSS3): c.1276G > A (p.Ala426Thr) TMPRSS3 61750641 105317
NM_000350.2(ABCA4): c.6089G > A (p.Arg2030Gln) ABCA4 61751276 104715
NM_000329.2(RPE65): c.11 + 5G > A RPE65 62507336 108472 NM_000277.3(PAH): c.561G > A
(p.Trp187Ter) PAH 62508613 108291 NM_000277.2(PAH): c.1199 + 17G > A PAH 72645357
32351 NM_000088.3(COL1A1): c.994G > A (p.Gly332Arg) COL1A1 80338777 32664
NM_000069.2(CACNA1S): c.1583G > A (p.Arg528His) CACNA1S 80356908 68776
NM_007294.3(BRCA1): c.1058G > A (p.Trp353Ter) BRCA1 80357093 69031
NM_007294.3(BRCA1): c.182G > A (p.Cys61Tyr) BRCA1 80357219 70211
NM_007294.3(BRCA1): c.5345G > A (p.Trp1782Ter) BRCA1 104886460 99352
NM_001005741.2(GBA): c.115 + 1G > A GBA 104894129 27501 NM_003289.3(TPM2): c.349G
> A (p.Glu117Lys) TPM2 104894401 32056 NM_004004.5(GJB2): c.428G > A (p.Arg143Gln)
GJB2 104895085 17592 NM_000243.2(MEFV): c.1958G > A (p.Arg653His) MEFV 111033299
53902 NM_004004.5(GJB2): c.283G > A (p.Val95Met) GJB2 113994139 33347
NM_139276.2(STAT3): c.1909G > A (p.Val637Met) STAT3 120074135 18010
NM_000271.4(NPC1): c.2848G > A (p.Val950Met) NPC1 121909334 23512 NM_007126.4(VCP):

c.572G > A (p.Arg191Gln) VCP 121918491 28307 NM_000141.4(FGFR2): c.1032G > A (p.Ala344=) FGFR2 137852314 25406 NM_000402.4(G6PD): c.577G > A (p.Gly193Ser) G6PD 137852327 25425 NM_000402.4(G6PD): c.961G > A (p.Val321Met) G6PD 137853285 166061 NM_000053.3(ATP7B): c.2128G > A (p.Gly710Ser) ATP7B 138213197 133488 NM_006361.5(HOXB13): c.251G > A (p.Gly84Glu) HOXB13 148311934 44907 NM_000162.5(GCK): c.676G > A (p.Val226Met) GCK 199473684 25807 NM_000169.2(GLA): c.639 + 919G > A GLA 200482683 131950 NM_014625.3(NPHS2): c.868G > A (p.Val290Met) NPHS2 371418985 232124 NM_007194.4(CHEK2): c.1232G > A (p.Trp411Ter) CHEK2 387907281 45778 NM_152296.4(ATP1A3): c.2443G > A (p.Glu815Lys) ATP1A3 397509284 70248 NM_007294.3(BRCA1): c.5445G > A (p.Trp1815Ter) BRCA1 397514495 152034 NM_000546.5(TP53): c.542G > A (p.Arg181His) TP53 397514581 48359 NM_172107.3(KCNQ2): c.638G > A (p.Arg213Gln) KCNQ2 397516101 52008 NM_000257.4(MYH7): c.1358G > A (p.Arg453His) MYH7 397516264 52270 NM_000257.3(MYH7): c.715G > A (p.Asp239Asn) MYH7 398122822 48057 NM_001111.5(ADAR): c.3019G > A (p.Gly1007Arg) ADAR 587777446 141325 NM_022168.4(IFIH1): c.2336G > A (p.Arg779His) IFIH1 587782962 165566 NM_000257.4(MYH7): c.3158G > A (p.Arg1053Gln) MYH7 606231435 170985 NM_152296.4(ATP1A3): c.2267G > A (p.Arg756His) ATP1A3 727504247 172354 NM_001001430.2(TNNT2): c.860G > A (p.Trp287Ter) TNNT2 730881833 179933 NM_001128425.1(MUTYH): c.857G > A (p.Gly286Glu) MUTYH 762307622 232266 NM_001128425.1(MUTYH): c.467G > A (p.Trp156Ter) MUTYH 777759523 17038 NM_199242.2(UNC13D): c.1389 + 1G > A UNC13D 794728625 197538 NM_130799.2(MEN1): c.784-9G > A MEN1 1060499814 389282 NM_024675.3(PALB2): c.108 + 1G > A PALB2 25403 51465 NM_000138.4(FBN1): c.184C > T (p.Arg62Cys) FBN1 28931591 32539 NM_000744.6(CHRNA4): c.851C > T (p.Ser284Leu) CHRNA4 28942108 18015 NM_000271.4(NPC1): c.2932C > T (p.Arg978Cys) NPC1 61750152 105192 NM_000350.2(ABCA4): c.4577C > T (p.Thr1526Met) ABCA4 61750654 105349 NM_000350.2(ABCA4): c.6445C > T (p.Arg2149Ter) ABCA4 61751404 105219 NM_000350.2(ABCA4): c.4918C > T (p.Arg1640Trp) ABCA4 61751408 22921 NM_000350.2(ABCA4): c.6079C > T (p.Leu2027Phe) ABCA4 63751466 24276 NM_000535.5(PMS2): c.2404C > T (p.Arg802Ter) PMS2 72552255 44374 NM_000053.3(ATP7B): c.2930C > T (p.Thr977Met) ATP7B 74315369 27822 NM_003000.2(SDHB): c.79C > T (p.Arg27Ter) SDHB 80338680 16726 NM_000528.3(MAN2B1): c.2248C > T (p.Arg750Trp) MAN2B1 80356952 68980 NM_007294.3(BRCA1): c.1630C > T (p.Gln544Ter) BRCA1 80357011 69802 NM_007294.3(BRCA1): c.4186C > T (p.Gln1396Ter) BRCA1 80357296 69580 NM_007294.3(BRCA1): c.3544C > T (p.Gln1182Ter) BRCA1 80357367 70140 NM_007294.3(BRCA1): c.5239C > T (p.Gln1747Ter) BRCA1 80357377 69340 NM_007294.3(BRCA1): c.2761C > T (p.Gln921Ter) BRCA1 80357471 69016 NM_007294.3(BRCA1): c.178C > T (p.Gln60Ter) BRCA1 80357497 69389 NM_007294.3(BRCA1): c.2923C > T (p.Gln975Ter) BRCA1 104893950 18137 NM_005670.3(EPM2A): c.721C > T (p.Arg241Ter) EPM2A 104894787 26252 NM_004006.2(DMD): c.10108C > T (p.Arg3370Ter) DMD 111231312 51536 NM_000138.4(FBN1): c.4615C > T (p.Arg1539Ter) FBN1 112645512 178700 NM_000138.4(FBN1): c.1285C > T (p.Arg429Ter) FBN1 113001196 51577 NM_000138.4(FBN1): c.6658C > T (p.Arg2220Ter) FBN1 113249837 51552 NM_000138.4(FBN1): c.5368C > T (p.Arg1790Ter) FBN1 113812345 51455 NM_000138.4(FBN1): c.1546C > T (p.Arg516Ter) FBN1 116100695 16552 NM_000298.5(PKLR): c.1456C > T (p.Arg486Trp) PKLR 118203631 58047 NM_000368.4(TSC1): c.2074C > T (p.Arg692Ter) TSC1 118203963 16148

NM_025137.3(SPG11): c.6100C > T (p.Arg2034Ter) SPG11 118204437 15739
NM_000512.4(GALNS): c.1156C > T (p.Arg386Cys) GALNS 121434526 33315
NM_001613.3(ACTA2): c.445C > T (p.Arg149Cys) ACTA2 121908547 20943
NM_000334.4(SCN4A): c.3938C > T (p.Thr1313Met) SCN4A 121912504 29459
NM_000238.3(KCNH2): c.1682C > T (p.Ala561Val) KCNH2 121913120 31271
NM_000143.3(FH): c.301C > T (p.Arg101Ter) FH 121913122 31274 NM_000143.3(FH): c.1027C
> T (p.Arg343Ter) FH 121917783 27083 NM_000136.2(FANCC): c.553C > T (p.Arg185Ter)
FANCC 121918775 79496 NM_006920.4(SCN1A): c.2803C > T (p.Arg935Cys) SCN1A
121964972 15170 NM_000071.2(CBS): c.1058C > T (p.Thr353Met) CBS 128627256 26327
NM_004006.2(DMD): c.8713C > T (p.Arg2905Ter) DMD 137854613 24413
NM_198056.2(SCN5A): c.4867C > T (p.Arg1623Ter) SCN5A 137886232 39244
NM_002878.3(RAD51D): c.757C > T (p.Arg253Ter) RAD51D 138996609 181608
NM_003000.2(SDHB): c.688C > T (p.Arg230Cys) SDHB 144500145 202960
NM_002693.2(POLG): c.2554C > T (p.Arg852Cys) POLG 180177111 132156
NM_024675.3(PALB2): c.2323C > T (p.Gln775Ter) PALB2 185492864 99918
NM_001918.3(DBT): c.901C > T (p.Arg301Cys) DBT 193922185 44706 NM_000138.4(FBN1):
c.1948C > T (p.Arg650Cys) FBN1 199472944 38732 NM_000238.3(KCNH2): c.1841C > T
(p.Ala614Val) KCNH2 199472990 78275 NM_000238.3(KCNH2): c.2254C > T (p.Arg752Trp)
KCNH2 199473161 78626 NM_198056.2(SCN5A): c.2440C > T (p.Arg814Trp) SCN5A
199473524 78188 NM_000238.3(KCNH2): c.1838C > T (p.Thr613Met) KCNH2 273898674
69115 NM_007294.3(BRCA1): c.2059C > T (p.Gln687Ter) BRCA1 368796923 151096
NM_032043.2(BRIP1): c.1240C > T (p.Gln414Ter) BRIP1 376128990 215031
NM_052845.3(MMAB): c.571C > T (p.Arg191Trp) MMAB 397509283 70244
NM_007294.3(BRCA1): c.5431C > T (p.Gln1811Ter) BRCA1 397515812 51535
NM_000138.4(FBN1): c.4567C > T (p.Arg1523Ter) FBN1 397516005 51860
NM_000256.3(MYBPC3): c.3181C > T (p.Gln1061Ter) MYBPC3 397516042 51914
NM_000256.3(MYBPC3): c.3811C > T (p.Arg1271Ter) MYBPC3 397516127 52044
NM_000257.3(MYH7): c.1987C > T (p.Arg663Cys) MYH7 397516201 52162
NM_000257.4(MYH7): c.4130C > T (p.Thr1377Met) MYH7 397516435 52758
NM_000546.5(TP53): c.586C > T (p.Arg196Ter) TP53 397517689 56466
NM_001267550.2(TTN): c.71602C > T (p.Arg23868Ter) TTN 398123585 99539
NM_001165963.1(SCN1A): c.1837C > T (p.Arg613Ter) SCN1A 549794342 360820
NM_001271208.1(NEB): c.24094C > T (p.Arg8032Ter) NEB 574660186 178478
NM_001267550.2(TTN): c.67495C > T (p.Arg22499Ter) TTN 575822089 227149
NM_001163435.2(TBCK): c.376C > T (p.Arg126Ter) TBCK 587778618 138806
NM_000535.7(PMS2): c.1687C > T (p.Arg563Ter) PMS2 587779343 96837
NM_000535.5(PMS2): c.697C > T (p.Gln233Ter) PMS2 587780088 133302
NM_001128425.1(MUTYH): c.55C > T (p.Arg19Ter) MUTYH 587781269 150486
NM_007194.4(CHEK2): c.283C > T (p.Arg95Ter) CHEK2 587781756 151166
NM_002878.3(RAD51D): c.451C > T (p.Gln151Ter) RAD51D 672601370 171771
NM_001244008.1(KIF1A): c.946C > T (p.Arg316Trp) KIF1A 727505006 176130
NM_000138.4(FBN1): c.3373C > T (p.Arg1125Ter) FBN1 794728165 197808
NM_000138.4(FBN1): c.1090C > T (p.Arg364Ter) FBN1 794728228 197690
NM_000138.4(FBN1): c.4621C > T (p.Arg1541Ter) FBN1 794728283 197585
NM_000138.4(FBN1): c.8038C > T (p.Arg2680Cys) FBN1 879255678 247653
NM_144997.5(FLCN): c.1429C > T (p.Arg477Ter) FLCN 886041116 263863
NM_015339.4(ADNP): c.2188C > T (p.Arg730Ter) ADNP 1553547838 512805
NM_001172509.1(SATB2): c.1375C > T (p.Arg459Ter) SATB2 45507199 59122
NM_000548.3(TSC2): c.5228G > A (p.Arg1743Gln) TSC2 60458016 29564
NM_170707.3(LMNA): c.1072G > A (p.Glu358Lys) LMNA 61672878 29534

NM_1707.3(LMNA): c.1130G > A (p.Arg377His) LMNA 61750173 24396
NM_000180.3(GUCY2D): c.2513G > A (p.Arg838His) GUCY2D 61753180 18833
NM_000372.4(TYR): c.140G > A (p.Gly47Asp) TYR 61754375 18835 NM_000372.4(TYR):
c.896G > A (p.Arg299His) TYR 62636275 20778 NM_201253.2(CRB1): c.3307G > A
(p.Gly1103Arg) CRB1 63750453 95615 NM_000249.3 (MLH1): c.304G > A (p.Glu102Lys)
MLH1 63750604 95363 NM_000249.3(MLH1): c.1790G > A (p.Trp597Ter) MLH1 63751632
95404 NM_000249.3(MLH1): c.1896G > A (p.Glu632=) MLH1 74315205 19565
NM_006005.3(WFS1): c.2590G > A (p.Glu864Lys) WFS1 74503330 22256
NM_000492.3(CFTR): c.3752G > A (p.Ser1251Asn) CFTR 80282562 57854
NM_000492.3(CFTR): c.532G > A (p.Gly178Arg) CFTR 80356702 32581
NM_000083.2(CLCN1): c.950G > A (p.Arg317Gln) CLCN1 80358543 131539
NM_000059.3(BRCA2): c.2978G > A (p.Trp993Ter) BRCA2 80358810 46556
NM_000059.3(BRCA2): c.582G > A (p.Trp194Ter) BRCA2 80358997 67062
NM_000059.3(BRCA2): c.7721G > A (p.Trp2574Ter) BRCA2 80359205 67482
NM_000059.3(BRCA2): c.9317G > A (p.Trp3106Ter) BRCA2 81002873 67120
NM_000059.3(BRCA2): c.7976 + 1G > A BRCA2 104894317 18840 NM_000372.4(TYR):
c.1336G > A (p.Gly446Ser) TYR 104894590 16599 NM_000263.3(NAGLU): c.2021G > A
(p.Arg674His) NAGLU 111033270 19955 NM_022124.5(CDH23): c.5237G > A (p.Arg1746Gln)
CDH23 111436401 226974 NM_000540.2(RYR1): c.10347 + 1G > A RYR1 112406105 200333
NM_000018.4(ACADVL): c.1097G > A (p.Arg366His) ACADVL 113560320 15440
NM_017841.2(SDHAF2): c.232G > A (p.Gly78Arg) SDHAF2 113690956 16661
NM_000018.2(ACADVL): c.1182 + 1G > A ACADVL 113994171 33871
NM_000018.3(ACADVL): c.1679-6G > A ACADVL 113994207 19490 NM_004937.2(CTNS):
c.589G > A (p.Gly197Arg) CTNS 114925667 260377 NM_024818.4(UBA5): c.1111G > A
(p.Ala371Thr) UBA5 118192122 76888 NM_000540.2(RYR1): c.7361G > A (p.Arg2454His)
RYR1 118192176 28015 NM_000540.2(RYR1): c.6502G > A (p.Val2168Met) RYR1 118203982
16396 NM_001080.3(ALDH5A1): c.612G > A (p.Trp204Ter) ALDH5A1 119462987 18289
NM_007171.3(POMT1): c.2005G > A (p.Ala669Thr) POMT1 120074190 18179
NM_000218.2(KCNQ1): c.1766G > A (p.Gly589Asp) KCNQ1 121434544 32653
NM_000070.2(CAPN3): c.1715G > A (p.Arg572Gln) CAPN3 121434548 32661
NM_000070.2(CAPN3): c.1469G > A (p.Arg490Gln) CAPN3; POMT1 121908153 19416
NM_001243133.1(NLRP3): c.907G > A (p.Asp303Asn) NLRP3 121908185 19531
NM_020451.2(SELENON): c.1397G > A (p.Arg466Gln) SELENON 121908419 20395
NM_014384.2(ACAD8): c.1129G > A (p.Gly377Ser) ACAD8 121908759 44497
NM_000492.3(CFTR): c.1865G > A (p.Gly622Asp) CFTR 121908889 21460
NM_003060.3(SLC22A5): c.506G > A (p.Arg169Gln) SLC22A5 121909013 22181
NM_000492.3(CFTR): c.1651G > A (p.Gly551Ser) CFTR 121909019 22197
NM_000492.3(CFTR): c.3197G > A (p.Arg1066His) CFTR 121909092 22321
NM_001005360.2(DNM2): c.1102G > A (p.Glu368Lys) DNM2 121918009 28711
NM_000478.5(ALPL): c.1001G > A (p.Gly334Asp) ALPL 121918592 28008
NM_000540.2(RYR1): c.1021G > A (p.Gly341Arg) RYR1 137852871 17416
NM_000709.3(BCKDHA): c.868G > A (p.Gly290Arg) BCKDHA 141158996 22214
NM_000492.3(CFTR): c.2490 + 1G > A CFTR 141554661 208401 NM_004287.4(GOSR2): c.336
+ 1G > A GOSR2 148032587 194820 NM_000303.2(PMM2): c.442G > A (p.Asp148Asn) PMM2
193922503 44492 NM_000492.3(CFTR): c.1585-8G > A CFTR 199472687 77968
NM_000218.2(KCNQ1): c.421G > A (p.Val141Met) KCNQ1 201016593 245339
NM_000527.4(LDLR): c.11G > A (p.Trp4Ter) LDLR 267606997 21861 NM_058216.2(RAD51C):
c.773G > A (p.Arg258His) RAD51C 267607914 96367 NM_000251.2(MSH2): c.212-1G > A
MSH2 369560930 98197 NM_000018.4(ACADVL): c.520G > A (p.Val174Met) ACADVL
370523609 227889 NM_000016.5(ACADM): c.600-18G > A ACADM 370950728 186993

NM_000152.3(GAA): c.655G > A (p.Gly219Arg) GAA 374143224 187013 NM_000152.3(GAA): c.1979G > A (p.Arg660His) GAA 397508045 67476 NM_000059.3(BRCA2): c.92G > A (p.Trp31Ter) BRCA2 397508200 67910 NM_000492.3(CFTR): c.1393-1G > A CFTR 397509418 75098 NM_021942.5(TRAPPC11): c.1287 + 5G > A TRAPPC11 397515330 76388 NM_001098512.2(PRKG1): c.530G > A (p.Arg177Gln) PRKG1 398122711 97208 NM_000059.3(BRCA2): c.8633-1G > A BRCA2 398123139 98311 NM_000060.4(BTD): c.626G > A (p.Arg209His) BTD 398123763 100162 NM_003494.3(DYSF): c.1053 + 1G > A DYSF 587777057 77012 NM_020988.2(GNAO1): c.607G > A (p.Gly203Arg) GNAO1 587777570 150453 NM_004522.2(KIF5C): c.709G > A (p.Glu237Lys) KIF5C 587778777 76741 NM_000784.3(CYP27A1): c.1184 + 1G > A CYP27A1 587779110 96248 NM_000251.2(MSH2): c.1760-1G > A MSH2 587780639 139490 NM_000051.3(ATM): c.7788G > A (p.Glu2596=) ATM 587781894 151348 NM_000051.3(ATM): c.9023G > A (p.Arg3008His) ATM 587782719 152505 NM_000051.3(ATM): c.8122G > A (p.Asp2708Asn) ATM 727503030 176785 NM_001278939.1(ELN): c.1150 + 1G > A ELN 730881581 180665 NM_000059.3(BRCA2): c.8174G > A (p.Trp2725Ter) BRCA2 730882035 180121 NM_000551.3(VHL): c.482G > A (p.Arg161Gln) VHL 750663117 234071 NM_000051.3(ATM): c.3078-1G > A ATM 756039188 243266 NM_000527.4(LDLR): c.12G > A (p.Trp4Ter) LDLR 796053216 202741 NM_014191.3(SCN8A): c.4423G > A (p.Gly1475Arg) SCN8A 876661242 231905 NM_000059.3(BRCA2): c.9381G > A (p.Trp3127Ter) BRCA2 879254600 245669 NM_000527.4(LDLR): c.626G > A (p.Cys209Tyr) LDLR 1057519632 362622 NM_003718.4(CDK13): c.2149G > A (p.Gly717Arg) CDK13 10250779 15457 NM_000290.3(PGAM2): c.233G > A (p.Trp78Ter) PGAM2 28928905 29469 NM_000238.3(KCNH2): c.1468G > A (p.Ala490Thr) KCNH2 28931593 32066 NM_004004.5(GJB2): c.224G > A (p.Arg75Gln) GJB2 28937318 24429 NM_198056.2(SCN5A): c.1100G > A (p.Arg367His) SCN5A 61749397 15329 NM_000552.4(VWF): c.3946G > A (p.Val1316Met) VWF 61751403 105220 NM_000350.2(ABCA4): c.4919G > A (p.Arg1640Gln) ABCA4 62514907 15633 NM_000277.3(PAH): c.442-1G > A PAH 62514956 98659 NM_000277.3(PAH): c.912 + 1G > A PAH 62516146 108608 NM_000277.1(PAH): c.842 + 5G > A PAH 62642939 98657 NM_000277.2(PAH): c.890G > A (p.Arg297His) PAH 62644503 108560 NM_000277.3(PAH): c.755G > A (p.Arg252Gln) PAH 63749856 21618 NM_001171.5(ABCC6): c.3904G > A (p.Gly1302Arg) ABCC6 63750783 30442 NM_000518.5(HBB): c.47G > A (p.Trp16Ter) HBB 66555264 414003 NM_000088.3(COL1A1): c.1821 + 1G > A COL1A1 72645321 414022 NM_000088.3(COL1A1): c.769G > A (p.Gly257Arg) COL1A1 74315368 27820 NM_003000.2(SDHB): c.725G > A (p.Arg242His) SDHB 74315471 18113 NM_000487.5(ARSA): c.739G > A (p.Gly247Arg) ARSA 78973108 19367 NM_001005741.2(GBA): c.887G > A (p.Arg296Gln) GBA 80338735 33917 NM_000156.5(GAMT): c.327G > A (p.Lys109=) GAMT 80338857 34128 NM_001360.2(DHCR7): c.725G > A (p.Arg242His) DHCR7 80338864 21831 NM_001360.2(DHCR7): c.1342G > A (p.Glu448Lys) DHCR7 80338944 32040 NM_004004.5(GJB2): c.231G > A (p.Trp77Ter) GJB2 80356914 70276 NM_007294.3(BRCA1): c.5511G > A (p.Trp1837Ter) BRCA1 80357212 70255 NM_007294.3(BRCA1): c.5467G > A (p.Ala1823Thr) BRCA1 80357307 70275 NM_007294.3(BRCA1): c.5510G > A (p.Trp1837Ter) BRCA1 80358252 18013 NM_000271.4(NPC1): c.530G > A (p.Cys177Tyr) NPC1 104894103 19470 NM_175073.2(APTX): c.837G > A (p.Trp279Ter) APTX 104894415 20583 NM_006783.4(GJB6): c.31G > A (p.Gly11Arg) GJB6 104894519 21096 NM_004862.3(LITAF): c.334G > A (p.Gly112Ser) LITAF 104894727 27461 NM_000363.4(TNNI3): c.586G > A (p.Aspl96Asn) TNNI3 104894828 25754 NM_000169.2(GLA): c.902G > A (p.Arg301Gln) GLA 111683277 175150 NM_000256.3(MYBPC3): c.3190 + 1G > A MYBPC3 111984349 258823 NM_000138.4(FBN1): c.7828G > A (p.Glu2610Lys) FBN1 113403872 16550 NM_000298.5(PKLR): c.1529G > A (p.Arg510Gln) PKLR 121434249 18383

NM_000348.3(SRD5A2): c.682G > A (p.Ala228Thr) SRD5A2 121908216 23534
NM_001127221.1(CACNA1A): c.4982G > A (p.Arg1661His) CACNA1A 121908551 20948
NM_000334.4(SCN4A): c.3877G > A (p.Val1293Ile) SCN4A 121908552 20949
NM_000334.4(SCN4A): c.1333G > A (p.Val445Met) SCN4A 121908557 20958
NM_000334.4(SCN4A): c.2024G > A (p.Arg675Gln) SCN4A 121908716 16996
NM_000022.2(ADA): c.632G > A (p.Arg211His) ADA 121908723 17007 NM_000022.3(ADA):
c.646G > A (p.Gly216Arg) ADA 121909768 21834 NM_001360.2(DHCR7): c.1055G > A
(p.Arg352Gln) DHCR7 121913039 31702 NM_001953.4(TYMP): c.622G > A (p.Val208Met)
TYMP 137853050 22116 NM_006009.3(TUBA1A): c.1265G > A (p.Arg422His) TUBA1A
137853283 166064 NM_000053.3(ATP7B): c.2336G > A (p.Trp779Ter) ATP7B 137854612 24434
NM_198056.2(SCN5A): c.4222G > A (p.Gly1408Arg) SCN5A 139751448 187031
NM_000271.4(NPC1): c.1211G > A (p.Arg404Gln) NPC1 150038620 187049
NM_004646.3(NPHS1): c.2335-1G > A NPHS1 180177122 132185 NM_024675.3(PALB2):
c.2718G > A (p.Trp906Ter) PALB2 181087667 40103 NM_007055.3(POLR3A): c.2617-1G > A
POLR3A 193922110 44393 NM_000053.3(ATP7B): c.4058G > A (p.Trp1353Ter) ATP7B
199473565 78528 NM_198056.2(SCN5A): c.1066G > A (p.Asp356Asn) SCN5A 199474703
40437 NM_000258.2(MYL3): c.281G > A (p.Arg94His) MYL3 199971687 216058
NM_052845.3(MMAB): c.291-1G > A MMAB 201188361 40345 NM_014714.3(IFT140): c.634G
> A (p.Gly212Arg) IFT140 202160208 75126 NM_013334.3(GMPPB): c.860G > A (p.Arg287Gln)
GMPPB 281875334 38553 NM_001101.3(ACTB): c.587G > A (p.Arg196His) ACTB 386134249
45185 NM_000244.3(MEN1): c.1277G > A (p.Cys426Tyr) MEN1 387906623 38652
NM_000138.4(FBN1): c.5284G > A (p.Gly1762Ser) FBN1 387906905 39430
NM_021625.4(TRPV4): c.947G > A (p.Arg316His) TRPV4 397507479 48850
NM_004333.5(BRAF): c.1595G > A (p.Cys532Tyr) BRAF 397514494 48018
NM_021625.4(TRPV4): c.557G > A (p.Arg186Gln) TRPV4 397515854 51599
NM_000138.4(FBN1): c.7606G > A (p.Gly2536Arg) FBN1 397515982 51820
NM_000256.3(MYBPC3): c.2670G > A (p.Trp890Ter) MYBPC3 397516031 51898
NM_000256.3(MYBPC3): c.3627 + 1G > A MYBPC3 397516471 52818
NM_001001430.2(TNNT2): c.518G > A (p.Arg173Gln) TNNT2 398122853 38917
NM_004006.2(DMD): c.9G > A (p.Trp3Ter) DMD 483352809 65656 NM_006087.3(TUBB4A):
c.745G > A (p.Asp249Asn) TUBB4A 515726205 40114 NM_001031726.3(C19orf12): c.205G > A
(p.Gly69Arg) C19orf12 564069299 200114 NM_000255.3(MMUT): c.1106G > A (p.Arg369His)
MMUT 574673404 182906 NM_002485.4(NBN): c.37 + 1G > A NBN 587780345 134590
NM_000162.3(GCK): c.544G > A (p.Val182Met) GCK 606231324 136674
NM_000257.3(MYH7): c.1573G > A (p.Glu525Lys) MYH7 727504382 49283
NM_030662.3(MAP2K2): c.619G > A (p.Glu207Lys) MAP2K2 730880850 29166
NM_000257.3(MYH7): c.732 + 1G > A MYH7 730882175 181517 NM_002238.3(KCNH1):
c.1405G > A (p.Gly469Arg) KCNH1 751604696 425943 NM_001360.2(DHCR7): c.1337G > A
(p.Arg446Gln) DHCR7 753288303 216044 NM_000255.3(MMUT): c.1280G > A (p.Gly427Asp)
MMUT 767399782 213656 NM_006087.3(TUBB4A): c.763G > A (p.Val255Ile) TUBB4A
794728208 197723 NM_000138.4(FBN1): c.3712G > A (p.Asp1238Asn) FBN1 796756333
410338 NM_024422.4(DSC2): c.943-1G > A DSC2 797044872 205316 NM_004977.2(KCNC3):
c.1268G > A (p.Arg423His) KCNC3 797045586 207083 NM_032682.5(FOXP1): c.1541G > A
(p.Arg514His) FOXP1 863223403 209408 NM_002140.4(HNRNPK): c.257G > A (p.Arg86His)
HNRNPK 876658367 232176 NM_003000.2(SDHB): c.587G > A (p.Cys196Tyr) SDIIB
1057517585 358911 NM_024675.3(PALB2): c.3G > A (p.Met1Ile) PALB2 1555582065 431537
NM_014233.3(UBTF): c.628G > A (p.Glu210Lys) UBTF 140630 197685 NM_000138.4(FBN1):
c.4930C > T (p.Arg1644Ter) FBN1 28940869 19031 NM_017739.3(POMGNT1): c.1324C > T
(p.Arg442Cys) POMGNT1 34451549 30497 NM_000518.5(HBB): c.316-197C > T HBB
41556519 31832 NM_000400.3(ERCC2): c.2047C > T (p.Arg683Trp) ERCC2 45611033 175462

NM_000257.4(MYH7): c.3133C > T (p.Arg1045Cys) MYH7 55832599 151478
NM_000546.5(TP53): c.799C > T (p.Arg267Trp) TP53 59616921 18036 NM_000226.3(KRT9):
c.487C > T (p.Arg163Trp) KRT9 60399023 29651 NM_000526.4(KRT14): c.373C > T
(p.Arg125Cys) KRT14 61749409 104973 NM_000350.2(ABCA4): c.1804C > T (p.Arg602Trp)
ABCA4 61749423 105003 NM_000350.2(ABCA4): c.2041C > T (p.Arg681Ter) ABCA4
61750645 105327 NM_000350.2(ABCA4): c.6229C > T (p.Arg2077Trp) ABCA4 61751383 22946
NM_000350.2(ABCA4): c.6088C > T (p.Arg2030Ter) ABCA4 61752871 28154
NM_000329.2(RPE65): c.271C > T (p.Arg91Trp) RPE65 61757582 21827
NM_001360.2(DHCR7): c.1210C > T (p.Arg404Cys) DHCR7 61816761 31358
NM_002016.1(FLG): c.1501C > T (p.Arg501Ter) FLG 62507344 15662 NM_000277.2(PAH):
c.1066-3C > T PAH 72559722 186816 NM_001287174.1(ABCC8): c.2509C > T (p.Arg837Ter)
ABCC8 72646846 56340 NM_001256850.1(TTN): c.56953C > T (p.Arg18985Ter) TTN 72648250
225057 NM_001256850.1(TTN): c.88243C > T (p.Arg29415Ter) TTN 72650700 39295
NM_001171.5(ABCC6): c.1552C > T (p.Arg518Ter) ABCC6 72651642 271557
NM_000088.3(COL1A1): c.2089C > T (p.Arg697Ter) COL1A1 72653170 32386
NM_000088.3(COL1A1): c.3040C > T (p.Arg1014Cys) COL1A1 74315348 20408
NM_014625.3(NPHS2): c.871C > T (p.Arg291Trp) NPHS2 74315391 22425
NM_172107.3(KCNQ2): c.619C > T (p.Arg207Trp) KCNQ2 74315442 23435
NM_000100.3(CSTB): c.202C > T (p.Arg68Ter) CSTB 74315472 18114 NM_000487.5(ARSA):
c.827C > T (p.Thr276Met) ARSA 75166491 108429 NM_000277.3(PAH): c.472C > T
(p.Arg158Trp) PAH 75949023 39947 NM_144612.6(LOXHD1): c.4714C > T (p.Arg1572Ter)
LOXHD1 78635798 16299 NM_032193.3(RNASEH2C): c.205C > T (p.Arg69Trp) RNASEH2C
80338652 18848 NM_000081.3(LYST): c.3310C > T (p.Arg1104Ter) LYST 80338826 29117
NM_002473.5(MYH9): c.2104C > T (p.Arg702Cys) MYH9 80338934 17522
NM_024577.3(SH3TC2): c.3325C > T (p.Arg1109Ter) SH3TC2 80338957 20935
NM_000334.4(SCN4A): c.2111C > T (p.Thr704Met) SCN4A 80356680 29580
NM_000228.2(LAMB3): c.124C > T (p.Arg42Ter) LAMB3 80356779 76552
NM_001876.3(CPT1A): c.1436C > T (p.Pro479Leu) CPT1A 80356973 69370
NM_007294.3(BRCA1): c.2869C > T (p.Gln957Ter) BRCA1 80356982 69227
NM_007294.3(BRCA1): c.2410C > T (p.Gln804Ter) BRCA1 80357067 69840
NM_007294.3(BRCA1): c.4339C > T (p.Gln1447Ter) BRCA1 80357089 69512
NM_007294.3(BRCA1): c.3331C > T (p.Gln1111Ter) BRCA1 80357352 69958
NM_007294.3(BRCA1): c.4810C > T (p.Gln1604Ter) BRCA1 80357485 69485
NM_007294.3(BRCA1): c.3286C > T (p.Gln1096Ter) BRCA1 80359818 31157
NM_006516.3(SLC2A1): c.376C > T (p.Arg126Cys) SLC2A1 80359826 201142
NM_006516.3(SLC2A1): c.988C > T (p.Arg330Ter) SLC2A1 104894003 33314
NM_001101.4(ACTB): c.547C > T (p.Arg183Trp) ACTB 104894261 31727
NM_130799.2(MEN1): c.1579C > T (p.Arg527Ter) MEN1 104894267 31731
NM_130799.2(MEN1): c.1378C > T (p.Arg460Ter) MEN1 104894364 27627
NM_004985.4(KRAS): c.173C > T (p.Thr58Ile) KRAS 104894621 23472 NM_000304.3(PMP22):
c.215C > T (p.Ser72Leu) PMP22 104894714 19826 NM_181882.2(PRX): c.2857C > T
(p.Arg953Ter) PRX 104894797 26321 NM_004006.2(DMD): c.9568C > T (p.Arg3190Ter) DMD
111033297 53892 NM_004004.5(GJB2): c.169C > T (p.Gln57Ter) GJB2 111033538 17382
NM_032601.3(MCEE): c.139C > T (p.Arg47Ter) MCEE 111687884 51571 NM_000138.4(FBN1):
c.643C > T (p.Arg215Ter) FBN1 112901682 76366 NM_001141945.2(ACTA2): c.115C > T
(p.Arg39Cys) ACTA2 114368325 38634 NM_000782.4(CYP24A1): c.1186C > T (p.Arg396Trp)
CYP24A1 118192226 34614 NM_172107.3(KCNQ2): c.1342C > T (p.Arg448Ter) KCNQ2
118192251 34269 NM_004519.3(KCNQ3): c.988C > T (p.Arg330Cys) KCNQ3 118203427 58245
NM_000368.4(TSC1): c.682C > T (p.Arg228Ter) TSC1 118203434 58253 NM_000368.4(TSC1):
c.733C > T (p.Arg245Ter) TSC1 118203542 57958 NM_000368.4(TSC1): c.1525C > T

(p.Arg509Ter) TSC1 118203999 16285 NM_024675.3(PALB2): c.2962C > T (p.Gln988Ter)
PALB2 118204429 15511 NM_000035.4(ALDOB): c.178C > T (p.Arg60Ter) ALDOB 121907916
18505 NM_000280.4(PAX6): c.607C > T (p.Arg203Ter) PAX6 121908212 23527
NM_001127221.1(CACNA1A): c.1997C > T (p.Thr666Met) CACNA1A 121908427 20365
NM_133647.1(SLC12A6): c.3031C > T (p.Arg1011Ter) SLC12A6 121908489 20807
NM_003919.2(SGCE): c.289C > T (p.Arg97Ter) SGCE 121912708 33034
NM_001182.4(ALDH7A1): c.328C > T (p.Arg110Ter) ALDH7A1 121913344 151858
NM_000546.5(TP53): c.916C > T (p.Arg306Ter) TP53 121917784 27085
NM_000136.2(FANCC): c.37C > T (p.Gln13Ter) FANCC 121918167 15995
NM_000275.2(OCA2): c.2228C > T (p.Pro743Leu) OCA2 121918244 16869 NM_001023
570.3(IQCB1): c.1381C > T (p.Arg461Ter) IQCB1 121918257 16926 NM_000255.3(MMUT):
c.322C > T (p.Arg108Cys) MMUT 122445105 26774 NM_000489.4(ATRX): c.736C > T
(p.Arg246Cys) ATRX 122445108 26781 NM_000489.4(ATRX): c.109C > T (p.Arg37Ter) ATRX
122453121 26733 NM_004484.3(GPC3): c.1159C > T (p.Arg387Ter) GPC3 128626235 26264
NM_004006.2(DMD): c.433C > T (p.Arg145Ter) DMD 137852897 17803
NM_024312.4(GNPTAB): c.3565C > T (p.Arg1189Ter) GNPTAB 137852994 19999
NM_018136.4(ASPM): c.9178C > T (p.Gln3060Ter) ASPM 137853229 21102
NM_004260.3(RECQL4): c.2269C > T (p.Gln757Ter) RECQL4 138049878 171163
NM_000257.4(MYH7): c.2608C > T (p.Arg870Cys) MYH7 138119149 39897
NM_020745.3(AARS2): c.1774C > T (p.Arg592Trp) AARS2 139675596 40180
NM_023073.3(CPLANE1): c.7477C > T (p.Arg2493Ter) CPLANE1 140511594 39892
NM_024753.4(TTC21B): c.626C > T (p.Pro209Leu) TTC21B 143343083 169011
NM_004004.5(GJB2): c.298C > T (p.His100Tyr) GJB2 148865119 210450 NM_000071.2(CBS):
c.146C > T (p.Pro49Leu) CBS 180177091 132277 NM_024675.3(PALB2): c.751C > T
(p.Gln251Ter) PALB2 199422209 33004 NM_001127701.1(SERPINA1): c.1178C > T
(p.Pro393Leu) SERPINA1 199473556 78702 NM_198056.2(SCN5A): c.361C > T (p.Arg121Trp)
SCN5A 200075782 39327 NM_003560.3(PLA2G6): c.109C > T (p.Arg37Ter) PLA2G6
200287925 151917 NM_002485.4(NBN): c.127C > T (p.Arg43Ter) NBN 200309328 176122
NM_000138.4(FBN1): c.8080C > T (p.Arg2694Ter) FBN1 200440128 205749
NM_012160.4(FBXL4): c.64C > T (p.Arg22Ter) FBXL4 201632198 55279
NM_024022.2(TMPRSS3): c.325C > T (p.Arg109Trp) TMPRSS3 267606919 21912
NM_004646.3(NPHS1): c.3478C > T (p.Arg1160Ter) NPHS1 267607143 20038
NM_021625.4(TRPV4): c.943C > T (p.Arg315Trp) TRPV4 267607258 46918
NM_002437.5(MPV17): c.293C > T (p.Pro98Leu) MPV17 375699023 223602
NM_024675.3(PALB2): c.1042C > T (p.Gln348Ter) PALB2 387906799 39125
NM_001244008.1(KIF1A): c.296C > T (p.Thr99Met) KIF1A 387906904 39429
NM_021625.4(TRPV4): c.694C > T (p.Arg232Cys) TRPV4 387907329 51081
NM_007075.3(WDR45): c.700C > T (p.Arg234Ter) WDR45 397507215 46080
NM_007294.3(BRCA1): c.3352C > T (p.Gln1118Ter) BRCA1 397507447 47625
NM_024312.4(GNPTAB): c.1123C > T (p.Arg375Ter) GNPTAB 397509002 69322
NM_007294.3(BRCA1): c.2713C > T (p.Gln905Ter) BRCA1 397509151 69806
NM_007294.3(BRCA1): c.4201C > T (p.Gln1401Ter) BRCA1 397509330 70405
NM_007294.3(BRCA1): c.850C > T (p.Gln284Ter) BRCA1 397514477 40113
NM_001031726.3(C19orf12): c.32C > T (p.Thr11Met) C19orf12 397515848 51592
NM_000138.4(FBN1): c.7180C > T (p.Arg2394Ter) FBN1 397516463 52805
NM_001001430.2(TNNT2): c.388C > T (p.Arg130Cys) TNNT2 398123061 76995
NM_012160.4(FBXL4): c.1444C > T (p.Arg482Trp) FBXL4 398123168 98367
NM_000143.3(FH): c.952C > T (p.His318Tyr) FH 398123832 100328 NM_004006.2(DMD):
c.10171C > T (p.Arg3391Ter) DMD 398123929 100476 NM_004006.2(DMD): c.3151C > T
(p.Arg1051Ter) DMD 398124478 102281 NM_138694.3(PKHD1): c.2341C > T (p.Arg781Ter)

PKHD1 536907995 137626 NM_007194.4(CHEK2): c.58C > T (p.Gln20Ter) CHEK2 587776407
 153707 NM_024675.3(PALB2): c.451C > T (p.Gln151Ter) PALB2 587776935 48413
 NM_005465.4(AKT3): c.1393C > T (p.Arg465Trp) AKT3 587780062 133253
 NM_000535.5(PMS2): c.823C > T (p.Gln275Ter) PMS2 587780226 133611
 NM_032043.2(BRIP1): c.1315C > T (p.Arg439Ter) BRIP1 587781948 151416
 NM_000465.3(BARD1): c.1921C > T (p.Arg641Ter) BARD1 587783685 168920
 NM_003482.3(KMT2D): c.12592C > T (p.Arg4198Ter) KMT2D 587784339 169779
 NM_003560.3(PLA2G6): c.1903C > T (p.Arg635Ter) PLA2G6 724159971 172085
 NM_152778.2(MFSD8): c.1444C > T (p.Arg482Ter) MFSD8 727503504 176073
 NM_000363.4(TNNI3): c.508C > T (p.Arg170Trp) TNNI3 727503513 172503
 NM_001001430.2(TNNT2): c.280C > T (p.Arg94Cys) TNNT2 727504136 177069
 NM_001165963.1(SCN1A): c.3733C > T (p.Arg1245Ter) SCN1A 730881422 179951
 NM_000465.3(BARD1): c.1996C > T (p.Gln666Ter) BARD1 730882029 180988
 NM_000546.5(TP53): c.1024C > T (p.Arg342Ter) TP53 747604569 185305
 NM_032043.2(BRIP1): c.484C > T (p.Arg162Ter) BRIP1 750621215 184806
 NM_002878.3(RAD51D): c.898C > T (p.Arg300Ter) RAD51D 753330544 195505
 NM_206933.2(USH2A): c.13316C > T (p.Thr4439Ile) USH2A 761494650 185659
 NM_007194.4(CHEK2): c.85C > T (p.Gln29Ter) CHEK2 763091520 197655
 NM_000138.4(FBN1): c.6169C > T (p.Arg2057Ter) FBN1 768933093 226933
 NM_024685.4(BBS10): c.145C > T (p.Arg49Trp) BBS10 773770609 264863
 NM_177550.4(SLC13A5): c.997C > T (p.Arg333Ter) SLC13A5 778989252 236615
 NM_007194.4(CHEK2): c.1315C > T (p.Gln439Ter) CHEK2 786202064 184902
 NM_007294.3(BRCA1): c.4834C > T (p.Gln1612Ter) BRCA1 786203821 184272
 NM_024675.3(PALB2): c.940C > T (p.Gln314Ter) PALB2 794726710 187772
 NM_001165963.1(SCN1A): c.3637C > T (p.Arg1213Ter) SCN1A 794726730 187817
 NM_001165963.1(SCN1A): c.2134C > T (p.Arg712Ter) SCN1A 794728195 197755
 NM_000138.4(FBN1): c.2645C > T (p.Ala882Val) FBN1 796051885 199890
 NM_003239.4(TGFB3): c.898C > T (p.Arg300Trp) TGFB3 797044883 205286
 NM_019066.4(MAGEL2): c.1912C > T (p.Gln638Ter) MAGEL2 869312892 226683
 NM_139276.2(STAT3): c.2147C > T (p.Thr716Met) STAT3 876658461 232175
 NM_003000.2(SDHB): c.640C > T (p.Gln214Ter) SDHB 886037684 248861
 NM_177438.2(DICER1): c.2062C > T (p.R688*) DICER1 886038001 249129
 NM_007294.3(BRCA1): c.2599C > T (p.Gln867Ter) BRCA1 886039480 260102
 NM_024675.3(PALB2): c.2368C > T (p.Gln790Ter) PALB2 886040218 261660
 NM_007294.3(BRCA1): c.4225C > T (p.Gln1409Ter) BRCA1 886041222 264422
 NM_000280.4(PAX6): c.781C > T (p.Arg261Ter) PAX6 1057521083 366251
 NM_015265.3(SATB2): c.1165C > T (p.Arg389Cys) SATB2

Example 6: Demonstration of Gene Editing Activity in Plant Cells

(302) Base-editing activity of an RGN-deaminase fusion protein of the invention is demonstrated in plant cells using protocols adapted from Li, et al., 2013 (*Nat. Biotech.* 31:688-691). Briefly, an expression vector comprising an expression cassette capable of expressing in plant cells an RGN-deaminase fusion protein operably linked to a SV40 nuclear localization signal (SEQ ID NO: 43) and a second expression cassette encoding a guide RNA targeting one or more sites in the plant PDS gene that flank an appropriate PAM sequence are introduced into *Nicotiana benthamiana* mesophyll protoplasts using PEG-mediated transformation. The transformed protoplasts are incubated in the dark for up to 36 hr. Genomic DNA is isolated from the protoplasts using a DNeasy Plant Mini Kit (Qiagen). The genomic region flanking the RGN target site is PCR amplified, products are purified, and the purified PCR products are analyzed using Next Generation Sequencing on Illumina MiSeq. Typically, 100,000 of 250 bp paired-end reads (2×100,000 reads) are generated per amplicon. The reads are analyzed using CRISPResso (Pinello, et al. 2016 *Nature*

Biotech, 34:695-697) to calculate the rates of editing. Output alignments are analyzed for INDEL formation or introduction of specific adenine mutations.

Example 7: Testing mRNA Delivery

(303) To determine if the base editors are capable of delivery in different formats, mRNA delivery was tested with primary T-cells. Purified CD3⁺ T-cells or PBMCs were thawed, activated using CD3/CD28 beads (ThermoFisher) for 3 days, then nucleofected using the Lonza 4D-Nucleofector X unit and Nucleocuvette strips. The P3 Primary Cell kit was used for both mRNA and RNP delivery. Cells were transfected using the EO-115 and EH-115 programs for mRNA and RNP delivery respectively. Cells were cultured in CTS OpTimizer T cell expansion medium (ThermoFisher) containing IL-2, IL-7, and IL-15 (Miltenyi Biotec) for 4 days post nucleofection before being harvested using a Nucleospin Tissue genomic DNA isolation kit (Machery Nagel).

(304) Amplicons surrounding the editing sites were generated by PCR using primers identified in Table 35 and subjected to NGS sequencing using the Illumina Nexterra platform using 2×250 bp paired end sequencing. The estimated base editing rate was determined by calculating the overall substitution rate for each sample. The average and number of samples for each guide tested are shown below.

(305) TABLE-US-00035 TABLE 35 Average Editing rate for LPG50148- nAPG07433.1 via mRNA delivery

Average %	SGN	Edit	N	SGN002352	7.84	2	SGN002364	29.79	2	SGN002367	0.1
2	SGN001061	0.37	1	SGN001062	71.81	1	SGN001064	3.99	1	SGN002254	8.92
2	SGN002255	5.26	2	SGN002256	8.32	2	SGN002290	2.88	2	SGN002293	9.68
2	SGN002299	27.05	2	SGN002132	29.11	2	SGN002137	7.77	2	SGN002139	6.00
2	SGN001770	1.22	2	SGN001773	0.49	2	SGN002212	29.63	2	SGN002216	2.58
2	SGN002218	36.13	2	SGN002230	14.32	2	SGN002231	33.18	2	SGN000753	6.84
2	SGN000754	26.41	1	SGN001856	0.5	2	SGN002248	9.91	2	SGN002249	40.19

Claims

1. A fusion protein comprising a Type II CRISPR-Cas protein nickase and a deaminase, wherein said deaminase comprises an amino acid sequence having at least 90% sequence identity to SEQ ID NO: 430 and has adenine deaminase activity, and wherein said nickase (a) is a Cas9 nickase; or (b) comprises an amino acid sequence having at least 95% sequence identity to SEQ ID NO: 42, 52, 53, 55-59, 61, 397, or 398.
2. The fusion protein of claim 1, wherein the nickase has the amino acid sequence of SEQ ID NO: 42, 53, 55-59, 61, 397, or 398.
3. The fusion protein of claim 1, wherein the fusion protein further comprises at least one nuclear localization signal (NLS).
4. The fusion protein of claim 1, wherein said deaminase comprises an amino acid sequence having at least 95% sequence identity to the amino acid sequence of SEQ ID NO: 430 and has adenine deaminase activity.
5. The fusion protein of claim 1, wherein said deaminase comprises the amino acid sequence of SEQ ID NO: 430, 419, or 434.
6. The fusion protein of claim 1, wherein said fusion protein comprises the amino acid sequence of SEQ ID NO: 519, 508, or 523.
7. The fusion protein of claim 1, wherein the fusion protein further comprises a protein tag or a cell penetrating domain.
8. A polypeptide having adenine deaminase activity and comprising an amino acid sequence having at least 90% sequence identity to the amino acid sequence of SEQ ID NO: 430.
9. The polypeptide of claim 8, wherein said polypeptide comprises an amino acid sequence having at least 95% sequence identity to the amino acid sequence of SEQ ID NO: 430.
10. The polypeptide of claim 8, wherein said polypeptide has the amino acid sequence of SEQ ID

NO: 430, 419, or 434.

11. The polypeptide of claim 8, wherein said polypeptide further comprises at least one nuclear localization signal.

12. An adenine base editor comprising the polypeptide of claim 8 and a Type II CRISPR-Cas nickase, wherein said nickase (a) is a Cas9 nickase; or (b) comprises an amino acid sequence having at least 95% sequence identity to of SEQ ID NO: 42, 52, 53, 55-59, 61, 397, or 398.

13. The adenine base editor of claim 12, wherein the adenine base editor introduces an A>G mutation in a DNA molecule.

14. A cell comprising the fusion protein of claim 1, wherein the cell further comprises a guide RNA.

15. A system for modifying a target DNA molecule, said system comprising: a) the fusion protein of claim 1; and b) one or more guide RNAs (gRNAs) capable of hybridizing to said target DNA molecule, or one or more nucleic acids encoding the one or more gRNAs; and wherein the one or more gRNAs are capable of forming a complex with the fusion protein in order to direct said fusion protein to bind to and modify the target DNA molecule.

16. The system of claim 15, wherein the nickase has the amino acid sequence of any one of SEQ ID NO: 42, 52, 53, 55-59, 61, 397, or 398.
